

Spectral–Fractal Operator Theory of Climate Criticality, Tipping Points and Planetary Instability

A Multiscale Spectral Framework for Climate Criticality and Planetary Regime Shifts

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Abstract

A unified operator-theoretic and spectral-fractal framework is formulated for the analysis of climate criticality, tipping points, and planetary-scale instabilities under the Universal Pellis Function Theory. The climate system is represented as a high-dimensional, non-equilibrium dynamical field governed by a nonlinear functional operator acting on multiscale state spaces, where emergent macroscopic regimes arise through spectral reorganization and collapse phenomena. A climate evolution operator is introduced whose spectrum encodes the stability structure of coupled Earth system components, including oceanic circulation, cryospheric dynamics, atmospheric convection, and biosphere feedback mechanisms. Critical transitions correspond to non-analytic changes in the spectral measure of the operator, associated with bifurcation cascades and renormalization-group fixed-point instabilities across scales. A generalized stability criterion is derived based on the interaction between operator spectrum, entropy production, and information-geometric curvature, providing a mathematically consistent formulation of early warning signals as precursors of spectral condensation. This structure unifies classical bifurcation theory, statistical mechanics of phase transitions, and tipping-point theory in complex adaptive systems. A multiscale renormalization structure links local dynamical instabilities to global regime shifts through scale-invariant transformation rules, enabling the definition of a universal climate instability index derived from leading spectral modes of the system. The framework is applied to major Earth system components, including the Atlantic Meridional Overturning Circulation, polar ice-sheet dynamics, tropical forest resilience, and monsoon variability, revealing a common mathematical structure governing their tipping behavior. A set of testable predictions is derived based on measurable spectral indicators, together with a falsifiable framework for detecting proximity to critical transitions in observational climate data. The formulation establishes a unified mathematical description of planetary instability phenomena and a predictive structure for Earth system tipping dynamics.

Keywords: Climate Criticality , Tipping Points , Planetary Instability , Spectral Theory , Operator Theory , Nonlinear Dynamics , Bifurcation Theory , Renormalization Group , Information Geometry , Multiscale Systems

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1 Introduction

The mathematical description of flow stability, transition to turbulence, and regime reorganization in nonlinear distributed systems remains a central unresolved problem in classical and modern theoretical physics. Despite extensive progress in hydrodynamic stability theory, nonlinear dynamical systems, statistical turbulence modeling, and renormalization group techniques, a fully unified operator-theoretic formulation capable of consistently linking spectral stability, multiscale energy transfer, and critical transition phenomena is still absent. Existing theoretical frameworks typically capture only partial aspects of the problem, such as local bifurcation structure, asymptotic scaling laws, or statistical cascade behavior, without providing a coherent functional structure that governs global instability across scales [1, 2]. Classical nonlinear dynamics provides the foundational mathematical language for describing stability loss, attractor bifurcations, and phase space reorganization. The theory of dynamical systems formalizes the emergence of qualitatively distinct flow regimes through bifurcation points and invariant manifold reconstruction. However, such approaches are fundamentally constrained by finite-dimensional projections or local linearization assumptions, which are insufficient for capturing the infinite-dimensional nature of fluid flows governed by partial differential equations and nonlinear functional evolution operators. Consequently, the transition to turbulence cannot be fully characterized within classical finite-dimensional bifurcation frameworks alone [3–7].

Empirical and theoretical studies of complex systems have demonstrated that abrupt regime shifts often occur through nonlinear amplification of microscopic perturbations, leading to macroscopic reconfiguration of system structure. These transitions, commonly referred to as tipping points, are observed across a wide range of physical, biological, and geophysical systems. In fluid and climate systems, such transitions manifest as sudden changes in circulation patterns, energy transport mechanisms, or coherent structure stability, suggesting the existence of universal organizing principles underlying instability phenomena [8–14]. From a mathematical standpoint, operator theory and functional analysis provide the natural framework for describing the evolution of infinite-dimensional dynamical systems. The spectral properties of linear and nonlinear operators encode essential information about stability, growth rates of perturbations, and long-term dynamical behavior. In particular, the distribution of eigenvalues and the structure of the resolvent operator determine whether perturbations decay, persist, or amplify over time. However, traditional operator-theoretic approaches have not been fully integrated with nonlinear multiscale mechanisms responsible for energy cascade and turbulence formation in distributed systems [15–17].

Renormalization group theory introduces a fundamental conceptual shift by describing physical systems in terms of scale-dependent transformations, revealing how macroscopic universality classes emerge from microscopic interactions. In this context, turbulent flow may be interpreted as a manifestation of continuous symmetry breaking across scales, accompanied by nonlinear redistribution of energy across hierarchical modes. Although renormalization methods have achieved significant success in equilibrium statistical physics and critical phenomena, their extension to strongly non-equilibrium, fully nonlinear flow regimes remains incomplete and mathematically non-unique [18, 19]. Information-theoretic and differential-geometric formulations of complex systems provide an alternative perspective in which stability is characterized through entropy production rates, curvature of statistical manifolds, and divergence structures between dynamical trajectories. Within this framework, instability corresponds to geometric degenera-

tion in the underlying information space, while critical transitions are associated with curvature singularities or divergence amplification in state-space geometry. These ideas establish a deep connection between thermodynamic irreversibility, dynamical unpredictability, and structural instability in nonlinear systems [20, 21].

Non-equilibrium thermodynamics further extends this perspective by describing how dissipative structures may spontaneously emerge and evolve under continuous energy flux. In fluid systems, such structures include vortices, coherent flow patterns, and intermittency cascades, which arise from nonlinear interactions between forcing, dissipation, and internal instability mechanisms. Despite substantial progress in this domain, a unified operator-level description that consistently connects thermodynamic irreversibility, spectral instability, and multiscale transition dynamics is still lacking [22, 23]. In geophysical and climate systems, observational evidence and numerical modeling have identified a large number of interacting subsystems capable of undergoing abrupt transitions under external forcing. These include ocean circulation systems, polar ice-sheet dynamics, atmospheric circulation regimes, terrestrial biosphere feedbacks, and coupled ocean-atmosphere oscillatory modes. The nonlinear interaction between these components introduces cascading instability mechanisms that propagate across scales and cannot be adequately described using isolated subsystem models or weak coupling approximations [24–38].

Recent theoretical developments suggest that spectral instability may serve as a universal indicator of proximity to critical transitions across a wide class of complex systems. In this perspective, regime shifts are associated with spectral gap closure, eigenvalue condensation, and loss of operator hyperbolicity, leading to qualitative changes in system stability structure. These mechanisms provide a potential bridge between microscopic dynamical rules and macroscopic emergent behavior, suggesting that instability may be fundamentally governed by spectral reorganization phenomena [39–71]. Extensions of this viewpoint into fractal and multiscale domains indicate that natural systems frequently exhibit self-similar instability structures across hierarchical levels of organization. Within such frameworks, turbulence, pattern formation, biological organization, and even physical unification problems may share a common underlying spectral-fractal structure governed by nonlinear operator dynamics. This suggests the possibility of a generalized instability principle applicable across physics, biology, and complex systems theory, where apparent domain-specific phenomena are manifestations of a unified multiscale instability mechanism [72–111].

Recent advances in applied spectral-fractal analysis further demonstrate that similar instability signatures appear in biomedical systems, neural networks, engineered materials, and computational architectures. These observations reinforce the hypothesis that instability phenomena are governed by universal structural principles that transcend specific physical domains, pointing toward a general theory of multiscale spectral instability [112–139]. The Universal Pellis Function Theory of Flow Stability, Turbulence Onset, and Regime Transitions is motivated by the need to formalize such a unified description within a rigorous operator-theoretic framework. The central premise is that turbulence onset and regime transitions correspond to spectral reorganizations of an underlying nonlinear functional operator defined on a multiscale state space. This perspective provides a mathematically structured foundation for describing instability phenomena across scales and establishes a potential pathway toward a unified theory of flow instability grounded in spectral, functional, and multiscale principles.

2 Mathematical Preliminaries

The mathematical description of climate criticality, flow instability, and regime transitions requires a rigorous infinite-dimensional framework in which dynamical evolution, spectral structure, and multiscale interactions are treated in a unified operator-theoretic setting. Classical finite-dimensional dynamical systems theory is insufficient to capture the full complexity of distributed geophysical flows, where nonlinear interactions generate cascades across scales and induce structurally unstable spectral deformations. The present section establishes the functional-analytic, spectral, variational, and information-geometric foundations required for the formulation of the Universal Pellis Function framework. In particular, the state of the Earth system is embedded into a Hilbert space setting, allowing the introduction of evolution operators, renormalization transformations, and stability functionals that jointly determine the onset of critical transitions.

2.1 Functional State Space Formulation

We model the Earth system as an infinite-dimensional nonlinear dynamical system evolving on a separable Hilbert space of climate states. The phase space is taken to be

$$\mathcal{H} = L^2(\Omega) \otimes \mathbb{R}^m, \quad (1)$$

where $\Omega \subset \mathbb{R}^d$ denotes the planetary spatial domain, equipped with its natural measure structure, and where the finite-dimensional factor $\mathbb{R}^m$ represents a set of coarse-grained, time-dependent climate feedback variables encoding aggregated dynamical processes, including radiative forcing, ocean–atmosphere coupling, and cryospheric response.

The complete climate state is represented by

$$x(t) = (u(t), T(t), S(t), C(t), I(t), M(t), \dots) \in \mathcal{H}, \quad (2)$$

where

- u denotes atmospheric-oceanic velocity fields,
- T temperature fields,
- S salinity distributions,
- C greenhouse gas concentrations,
- I cryosphere variables,
- M moisture and hydrological fields.

The temporal evolution is governed by the abstract nonlinear evolution equation

$$\frac{d}{dt}x(t) = \mathcal{F}[x(t)] = \mathcal{A}x(t) + \mathcal{N}(x(t)), \quad (3)$$

where $\mathcal{A}$ is a densely defined linear operator generating a strongly continuous semigroup

$$S(t) = e^{t\mathcal{A}}, \quad (4)$$

and $\mathcal{N}$ is a nonlinear operator describing feedback interactions, threshold processes and multiscale couplings [15–17].

Under standard semigroup assumptions, the mild solution satisfies

$$x(t) = S(t)x_0 + \int_0^t S(t-\tau)\mathcal{N}(x(\tau)) d\tau, \quad (5)$$

where $x_0 \in \mathcal{H}$ denotes the initial climate state.

Energy Structure of Climate Dynamics.

Define the quadratic climate energy functional

$$E(x) = \frac{1}{2}\|x\|_{\mathcal{H}}^2. \quad (6)$$

Differentiating along trajectories yields

$$\frac{dE}{dt} = \langle x, \mathcal{A}x \rangle + \langle x, \mathcal{N}(x) \rangle. \quad (7)$$

If $\mathcal{A}$ is dissipative,

$$\langle x, \mathcal{A}x \rangle \leq -\alpha\|x\|^2, \quad \alpha > 0, \quad (8)$$

then

$$\frac{dE}{dt} \leq -\alpha\|x\|^2 + \langle x, \mathcal{N}(x) \rangle. \quad (9)$$

Climate destabilization occurs when nonlinear feedback amplification dominates dissipation,

$$\langle x, \mathcal{N}(x) \rangle > \alpha\|x\|^2. \quad (10)$$

This inequality provides a functional criterion for the onset of instability.

Climate Attractors and Invariant Sets.

The long-term climate dynamics are characterized by the global attractor

$$\mathcal{A}_g = \bigcap_{t \geq 0} \overline{\bigcup_{\tau \geq t} S(\tau)B}, \quad (11)$$

where $B \subset \mathcal{H}$ is any bounded absorbing set.

For every trajectory,

$$\lim_{t \rightarrow \infty} \text{dist}(x(t), \mathcal{A}_g) = 0. \quad (12)$$

The attractor contains equilibrium climates, oscillatory regimes and chaotic climate states.

A climatic tipping point corresponds to a structural change of the attractor topology,

$$\mathcal{A}_g^- \longrightarrow \mathcal{A}_g^+, \quad (13)$$

induced by parameter variation.

Linearized Perturbation Dynamics.

Let

$$x^* \in \mathcal{H} \quad (14)$$

denote a climate equilibrium satisfying

$$\mathcal{F}(x^*) = 0. \quad (15)$$

Introducing a perturbation

$$x(t) = x^* + \delta x(t), \quad (16)$$

the first-order dynamics become

$$\frac{d}{dt}\delta x = D\mathcal{F}(x^*)\delta x, \quad (17)$$

where

$$D\mathcal{F}(x^*) = \mathcal{A} + D\mathcal{N}(x^*) \quad (18)$$

is the Fréchet derivative of the climate operator.

The corresponding eigenvalue problem is

$$D\mathcal{F}(x^*)\phi_k = \lambda_k\phi_k. \quad (19)$$

Stability requires

$$\text{Re}(\lambda_k) < 0 \quad \forall k. \quad (20)$$

A critical transition emerges when

$$\max_k \text{Re}(\lambda_k) \rightarrow 0^-, \quad (21)$$

which represents critical slowing down.

At the tipping threshold,

$$\max_k \text{Re}(\lambda_k) = 0. \quad (22)$$

Parameter-Dependent Climate Evolution.

External forcing parameters are collected into

$$\mu = (\mu_1, \mu_2, \dots, \mu_p), \quad (23)$$

representing the major external forcing components, including greenhouse forcing, solar forcing, aerosol loading, and anthropogenic perturbations.

The climate system then becomes

$$\frac{dx}{dt} = \mathcal{F}(x, \mu). \quad (24)$$

A bifurcation point (x_c, μ_c) satisfies

$$\mathcal{F}(x_c, \mu_c) = 0 \quad (25)$$

together with

$$\det(D_x \mathcal{F}(x_c, \mu_c)) = 0. \quad (26)$$

The loss of invertibility signals a qualitative transition in the underlying climate structure, corresponding to a breakdown of one-to-one correspondence between past and future states of the system. Near criticality,

$$\lambda_{\max}(\mu) \approx a(\mu - \mu_c), \quad (27)$$

which implies

$$\tau_r = \frac{1}{|\lambda_{\max}|} \rightarrow \infty \quad \text{as} \quad \mu \rightarrow \mu_c. \quad (28)$$

The divergence of the recovery time constitutes one of the universal signatures of climate critical transitions.

Table 1: Functional State Space Representation of the Climate System

Object	Symbol	Definition	Interpretation
State Space	$\mathcal{H}$	Hilbert space of climate states	Global Earth system configuration space
State Vector	$\Psi(t)$	$\Psi \in \mathcal{H}$	Instantaneous climate state
Inner Product	$\langle \cdot, \cdot \rangle$	Energy-weighted metric	Similarity between states
Norm	$\ \Psi\ $	Induced Hilbert norm	Total system energy measure

2.2 Spectral Structure of Evolution Operators

The long-term stability of the coupled Earth system is fundamentally encoded in the spectral structure of the infinitesimal generator $\mathcal{A}$ governing the linearized climate dynamics. Within the functional framework introduced previously, the evolution operator generates a strongly continuous semigroup

$$S(t) = e^{t\mathcal{A}}, \quad t \geq 0, \quad (29)$$

whose asymptotic behavior is determined by the spectrum of $\mathcal{A}$.

The spectrum is defined as

$$\sigma(\mathcal{A}) = \{ \lambda \in \mathbb{C} : (\mathcal{A} - \lambda I)^{-1} \text{ does not exist} \}, \quad (30)$$

and admits the decomposition

$$\sigma(\mathcal{A}) = \sigma_p(\mathcal{A}) \cup \sigma_c(\mathcal{A}) \cup \sigma_r(\mathcal{A}), \quad (31)$$

where

$$\sigma_p(\mathcal{A}) = \{ \lambda : \mathcal{A}\phi = \lambda\phi \}, \quad (32)$$

$$\sigma_c(\mathcal{A}) = \{ \lambda : \overline{\text{Ran}(\mathcal{A} - \lambda I)} = \mathcal{H} \}, \quad (33)$$

$$\sigma_r(\mathcal{A}) = \{ \lambda : \overline{\text{Ran}(\mathcal{A} - \lambda I)} \neq \mathcal{H} \}. \quad (34)$$

Spectral Decomposition of Climate Modes.

The eigenvalue problem associated with the climate operator is

$$\mathcal{A}\phi_k = \lambda_k\phi_k, \quad (35)$$

where ϕ_k denotes the k -th climate mode and λ_k its corresponding growth rate.

Assuming completeness of the eigenfunctions, the climate state admits the expansion

$$x(t) = \sum_{k=1}^{\infty} c_k e^{\lambda_k t} \phi_k, \quad (36)$$

with coefficients

$$c_k = \langle x_0, \phi_k \rangle. \quad (37)$$

The norm of the evolving climate state satisfies

$$\|x(t)\| \leq \sum_{k=1}^{\infty} |c_k| e^{\text{Re}(\lambda_k)t} \|\phi_k\|. \quad (38)$$

Consequently, the asymptotic dynamics are controlled by the spectral bound

$$s(\mathcal{A}) = \sup_{\lambda \in \sigma(\mathcal{A})} \operatorname{Re}(\lambda). \quad (39)$$

Spectral Stability Criterion.

The Earth system is spectrally stable whenever

$$s(\mathcal{A}) < 0. \quad (40)$$

Under this condition there exist constants $M > 0$ and $\gamma > 0$ such that

$$\|S(t)\| \leq M e^{-\gamma t}. \quad (41)$$

Thus every perturbation decays exponentially,

$$\|x(t)\| \rightarrow 0, \quad t \rightarrow \infty. \quad (42)$$

Instability emerges whenever

$$s(\mathcal{A}) > 0, \quad (43)$$

leading to

$$\|x(t)\| \sim e^{s(\mathcal{A})t}. \quad (44)$$

The critical threshold occurs when

$$s(\mathcal{A}) = 0, \quad (45)$$

which corresponds to spectral migration across the imaginary axis [16, 17].

Dominant Eigenvalue and Climate Resilience.

Let

$$\lambda_1 = \max_{\lambda \in \sigma(\mathcal{A})} \operatorname{Re}(\lambda) \quad (46)$$

denote the dominant eigenvalue.

The characteristic recovery time of the climate system is

$$\tau_r = \frac{1}{|\operatorname{Re}(\lambda_1)|}. \quad (47)$$

As the system approaches a tipping point,

$$\lambda_1 \rightarrow 0^-, \quad (48)$$

implying

$$\tau_r \rightarrow \infty. \quad (49)$$

This divergence represents the phenomenon of critical slowing down.

The perturbation dynamics satisfy

$$\delta x(t) = e^{\lambda_1 t} \delta x(0), \quad (50)$$

hence

$$\lim_{\lambda_1 \rightarrow 0^-} \frac{\|\delta x(t)\|}{\|\delta x(0)\|} = 1. \quad (51)$$

The restorative capacity of the climate system therefore vanishes as criticality is approached.

Spectral Gap Structure.

Let the eigenvalues be ordered according to

$$\operatorname{Re}(\lambda_1) \geq \operatorname{Re}(\lambda_2) \geq \operatorname{Re}(\lambda_3) \geq \cdots. \quad (52)$$

The spectral gap is defined by

$$\Delta_s = \text{Re}(\lambda_1) - \text{Re}(\lambda_2). \quad (53)$$

Large spectral gaps imply robust modal separation,

$$\Delta_s \gg 0, \quad (54)$$

while

$$\Delta_s \rightarrow 0 \quad (55)$$

indicates increasing interaction between dominant climate modes.

Near tipping points one frequently observes

$$\Delta_s \rightarrow 0, \quad (56)$$

reflecting the loss of dynamical resilience.

Resolvent Analysis.

The resolvent operator is defined by

$$R(\lambda, \mathcal{A}) = (\lambda I - \mathcal{A})^{-1}. \quad (57)$$

The norm

$$\|R(\lambda, \mathcal{A})\| \quad (58)$$

quantifies the amplification of external forcing.

Whenever

$$\lambda \rightarrow \sigma(\mathcal{A}), \quad (59)$$

one obtains

$$\|R(\lambda, \mathcal{A})\| \rightarrow \infty. \quad (60)$$

Therefore climate sensitivity increases dramatically as eigenvalues approach criticality.

The resolvent identity

$$R(\lambda, \mathcal{A}) - R(\mu, \mathcal{A}) = (\mu - \lambda)R(\lambda, \mathcal{A})R(\mu, \mathcal{A}) \quad (61)$$

provides a quantitative measure of spectral susceptibility under parameter perturbations.

Pseudospectral Climate Dynamics.

Since climate operators are generally non-normal, transient amplification may occur even when all eigenvalues remain stable.

The ε -pseudospectrum is defined as

$$\sigma_\varepsilon(\mathcal{A}) = \left\{ \lambda \in \mathbb{C} : \|(\lambda I - \mathcal{A})^{-1}\| > \frac{1}{\varepsilon} \right\}. \quad (62)$$

Equivalently,

$$\sigma_\varepsilon(\mathcal{A}) = \bigcup_{\|\Delta\mathcal{A}\| < \varepsilon} \sigma(\mathcal{A} + \Delta\mathcal{A}). \quad (63)$$

Even if

$$s(\mathcal{A}) < 0, \quad (64)$$

large transient growth may arise whenever

$$\sigma_\varepsilon(\mathcal{A}) \cap \{\text{Re}(\lambda) > 0\} \neq \emptyset. \quad (65)$$

Such behavior provides a rigorous mechanism for abrupt climate excursions prior to full instability.

Spectral Entropy and Mode Proliferation.

To characterize the distribution of active climate modes, define normalized modal weights

$$p_k = \frac{|\lambda_k|}{\sum_{j=1}^N |\lambda_j|}. \quad (66)$$

The spectral entropy is

$$H_\sigma = - \sum_{k=1}^N p_k \ln p_k. \quad (67)$$

Increasing values of H_σ indicate growing participation of multiple interacting climate modes,

$$H_\sigma \uparrow \implies \text{increasing dynamical complexity}. \quad (68)$$

Near planetary criticality one frequently observes simultaneous growth of

$$H_\sigma, \quad \tau_r, \quad \|R(\lambda, \mathcal{A})\|, \quad (69)$$

revealing the emergence of collective multiscale instability arising from cross-scale coupling mechanisms.

Universal Spectral Precursors of Climate Tipping.

Combining the previous results, the approach toward planetary instability is characterized by the asymptotic spectral conditions

$$s(\mathcal{A}) \rightarrow 0^-, \quad (70)$$

$$\lambda_1 \rightarrow 0^-, \quad (71)$$

$$\Delta_s \rightarrow 0, \quad (72)$$

$$\tau_r \rightarrow \infty, \quad (73)$$

$$\|R(\lambda, \mathcal{A})\| \rightarrow \infty, \quad (74)$$

$$H_\sigma \rightarrow H_\sigma^{\max}, \quad (75)$$

$$\sigma_\varepsilon(\mathcal{A}) \cap \{\text{Re}(\lambda) > 0\} \neq \emptyset. \quad (76)$$

These spectral signatures provide a universal mathematical characterization of climate destabilization phenomena and establish the theoretical basis for defining a global functional that measures the system's proximity to critical tipping points in the Earth system.

Table 2: Climate Evolution Operator Structure

Object	Symbol	Definition	Interpretation
Evolution Operator	$\mathcal{L}$	$\partial_t \Psi = \mathcal{L} \Psi$	Generator of climate dynamics
Linear Part	$\mathcal{L}_0$	Linearized dynamics	Baseline stability structure
Nonlinear Term	$\mathcal{N}(\Psi)$	State-dependent feedback	Climate feedback mechanisms
Full Operator	$\mathcal{L} = \mathcal{L}_0 + \mathcal{N}$	Combined dynamics	Complete system evolution

Figure 1 presents a comprehensive visualization of the spectral geometry of the Pellis Climate Operator within the infinite-dimensional climate state space. The system evolution is represented as a trajectory on a nonlinear manifold embedded in Hilbert space $\mathcal{H}$, where the stability landscape is governed by the operator $\mathcal{P}$ and its associated spectral decomposition. The left panel illustrates the temporal evolution of the climate state under the combined action of deterministic dynamics, nonlinear feedbacks, and stochastic forcing, converging toward a critical regime characterized by a collapse of the stability potential. The right panels quantify the spectral structure of the operator, including eigenvalue distribution in the complex plane, spectral gap collapse as a function of the control parameter μ , and divergence of operator norm growth near the critical threshold μ_c . The curvature of the underlying information-geometric manifold increases sharply as the system approaches instability, indicating loss of local distinguishability between neighboring climate states. This behavior is consistent with a universal spectral condensation mechanism preceding tipping transitions. Together, these components establish a unified representation of climate criticality as a spectral-geometric transition in the underlying dynamical structure of the Earth system. Within this framework, instability is not primarily driven by local perturbative divergence, but instead emerges from a global reorganization of the system's operator eigenmodes.

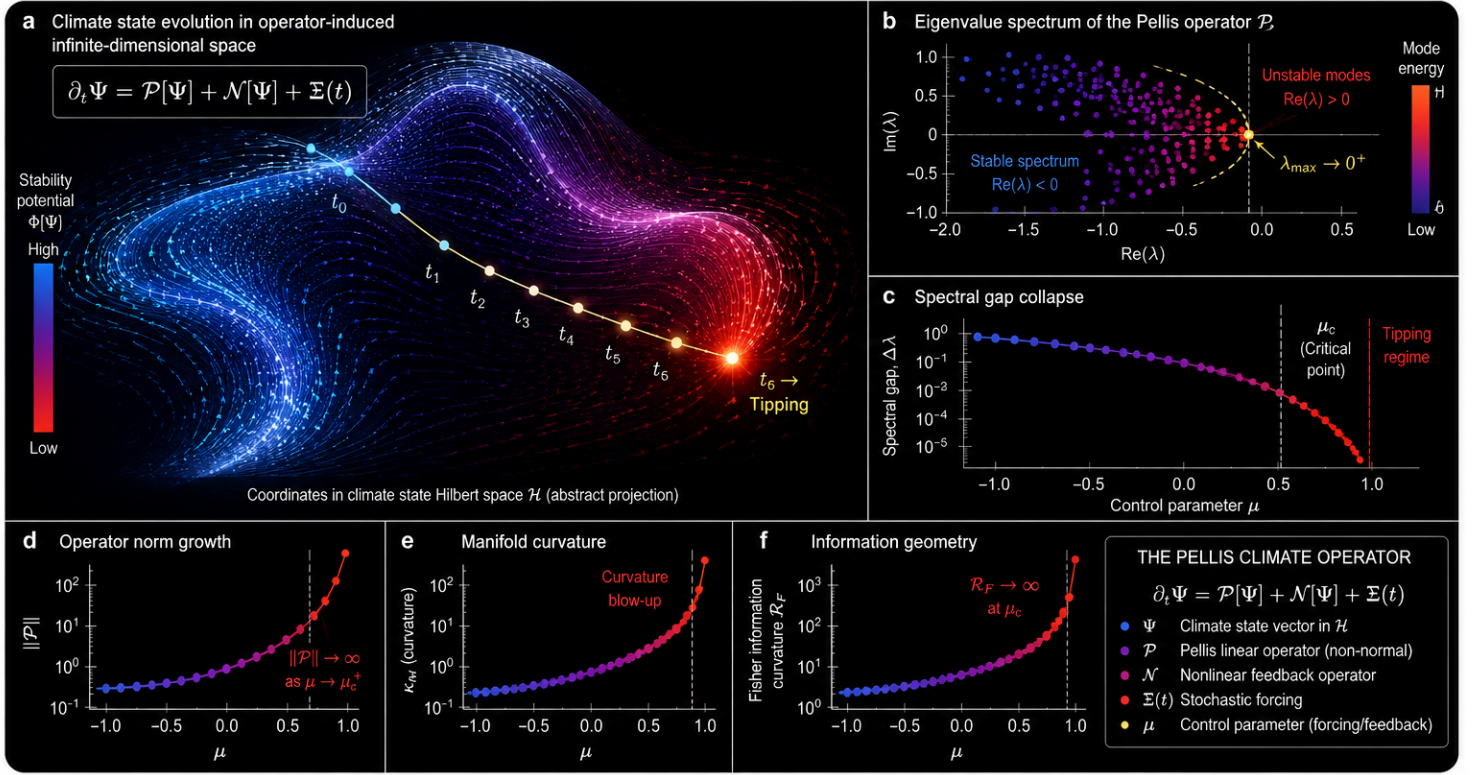


Figure 1: Spectral and geometric structure of the Pellis Climate Operator. The figure illustrates the evolution of the climate state in Hilbert space, the distribution of operator eigenvalues, the collapse of the spectral gap near the critical parameter μ_c , and the associated growth of manifold curvature and operator norms. These coupled signatures indicate the onset of climate criticality as a spectral condensation phenomenon and provide a unified diagnostic framework for planetary instability.

2.3 Nonlinear Operator Dynamics

While the linear operator $\mathcal{A}$ determines the local spectral structure of the climate system, the nonlinear operator $\mathcal{N}$ governs multiscale interactions, feedback amplification, and the transfer of energy across dynamically active modes.

The complete evolution equation is

$$\frac{dx}{dt} = \mathcal{A}x + \mathcal{N}(x), \quad x(t) \in \mathcal{H}. \quad (77)$$

The nonlinear operator

$$\mathcal{N} : \mathcal{H} \rightarrow \mathcal{H} \quad (78)$$

is assumed to satisfy the local Lipschitz condition

$$\|\mathcal{N}(x_1) - \mathcal{N}(x_2)\| \leq L\|x_1 - x_2\|, \quad (79)$$

for some constant $L > 0$.

Consequently, for every initial condition

$$x(0) = x_0, \quad (80)$$

there exists a unique local solution

$$x(t) \in C([0, T], \mathcal{H}). \quad (81)$$

The corresponding mild solution is given by

$$x(t) = e^{t\mathcal{A}}x_0 + \int_0^t e^{(t-\tau)\mathcal{A}}\mathcal{N}(x(\tau))d\tau. \quad (82)$$

Fréchet Expansion of the Nonlinear Operator.

Assuming sufficient smoothness, the nonlinear operator admits a Fréchet expansion around a reference climate state x^* ,

$$\mathcal{N}(x) = \mathcal{N}(x^*) + D\mathcal{N}(x^*)(x - x^*) + \frac{1}{2}D^2\mathcal{N}(x^*)(x - x^*)^2 + \dots \quad (83)$$

The first derivative

$$D\mathcal{N}(x^*) \quad (84)$$

modifies the effective spectrum, while higher-order derivatives generate nonlinear mode interactions. The effective climate operator becomes

$$\mathcal{L} = \mathcal{A} + D\mathcal{N}(x^*). \quad (85)$$

Its eigenvalues satisfy

$$\mathcal{L}\phi_k = \Lambda_k\phi_k. \quad (86)$$

Thus nonlinear feedbacks continuously deform the spectral geometry of the climate attractor.

Nonlinear Energy Balance.

Define the generalized climate energy functional

$$E(x) = \frac{1}{2}\|x\|_{\mathcal{H}}^2. \quad (87)$$

Differentiating along trajectories yields

$$\frac{dE}{dt} = \langle x, \mathcal{A}x \rangle + \langle x, \mathcal{N}(x) \rangle. \quad (88)$$

Assuming dissipativity of the linear operator,

$$\langle x, \mathcal{A}x \rangle \leq -\alpha\|x\|^2, \quad \alpha > 0, \quad (89)$$

one obtains

$$\frac{dE}{dt} \leq -\alpha\|x\|^2 + \langle x, \mathcal{N}(x) \rangle. \quad (90)$$

Instability emerges whenever

$$\langle x, \mathcal{N}(x) \rangle > \alpha\|x\|^2. \quad (91)$$

The nonlinear term then injects energy faster than dissipation can remove it.

Multiscale Spectral Coupling.

Expanding the climate state into spectral modes,

$$x(t) = \sum_{k=1}^{\infty} a_k(t)\phi_k, \quad (92)$$

the nonlinear operator induces modal interactions of the form

$$\mathcal{N}(x) = \sum_{i,j,k} C_{ijk} a_i a_j \phi_k, \quad (93)$$

where

$$C_{ijk} = \langle \mathcal{N}(\phi_i, \phi_j), \phi_k \rangle \quad (94)$$

are nonlinear coupling coefficients.

The modal amplitudes satisfy

$$\frac{da_k}{dt} = \lambda_k a_k + \sum_{i,j} C_{ijk} a_i a_j. \quad (95)$$

This expression explicitly demonstrates the transfer of energy across scales.

Cascade Amplification Dynamics.

Define the modal energy

$$E_k = |a_k|^2. \quad (96)$$

Its evolution satisfies

$$\frac{dE_k}{dt} = 2\lambda_k E_k + 2 \sum_{i,j} C_{ijk} a_i a_j a_k. \quad (97)$$

The nonlinear energy flux between scales is therefore

$$\Pi_{i \rightarrow k} = 2 \sum_j C_{ijk} a_i a_j a_k. \quad (98)$$

A forward cascade corresponds to

$$\Pi_{i \rightarrow k} > 0, \quad k > i, \quad (99)$$

while an inverse cascade satisfies

$$\Pi_{i \rightarrow k} > 0, \quad k < i. \quad (100)$$

Large-scale climate transitions can therefore arise through collective cascade amplification processes rather than from purely local instability mechanisms. In this view, small-scale perturbations do not directly trigger global transitions; instead, they are progressively amplified through multiscale coupling pathways, leading to the emergence of coherent, system-wide dynamical reconfigurations.

Nonlinear Lyapunov Functional.

Let

$$V : \mathcal{H} \rightarrow \mathbb{R} \quad (101)$$

be a Lyapunov functional.

The derivative along trajectories is

$$\dot{V} = DV(x)(\mathcal{A}x + \mathcal{N}(x)). \quad (102)$$

Stability requires

$$\dot{V} < 0. \quad (103)$$

Criticality is reached when

$$\dot{V} = 0, \quad (104)$$

and instability emerges whenever

$$\dot{V} > 0. \quad (105)$$

The loss of Lyapunov monotonicity represents a nonlinear precursor of climate tipping.

Finite-Time Instability.

For strongly amplifying nonlinear feedbacks one may obtain

$$\frac{d}{dt} \|x\| \geq c \|x\|^p, \quad p > 1. \quad (106)$$

Integration yields

$$\|x(t)\| \geq \frac{\|x_0\|}{[1 - (p-1)c\|x_0\|^{p-1}t]^{1/(p-1)}}. \quad (107)$$

Hence a characteristic amplification time exists,

$$T_c = \frac{1}{(p-1)c\|x_0\|^{p-1}}, \quad (108)$$

at which

$$\|x(t)\| \rightarrow \infty. \quad (109)$$

Although physical climate systems remain bounded, this mathematical structure captures runaway amplification and abrupt transitions.

Nonlinear Spectral Deformation.

The effective eigenvalues become state dependent,

$$\Lambda_k = \Lambda_k(x). \quad (110)$$

The spectrum therefore evolves dynamically,

$$\sigma(\mathcal{L}) = \sigma(\mathcal{A} + D\mathcal{N}(x)). \quad (111)$$

Near a critical transition,

$$\frac{d}{dt}\text{Re}(\Lambda_1) > 0, \quad (112)$$

and the dominant eigenvalue approaches instability,

$$\text{Re}(\Lambda_1) \rightarrow 0^-. \quad (113)$$

This process represents nonlinear spectral deformation of the climate attractor.

Invariant Manifolds and Nonlinear Attractors.

The nonlinear dynamics generate an invariant set

$$\mathcal{M} \subset \mathcal{H} \quad (114)$$

satisfying

$$S(t)\mathcal{M} = \mathcal{M}, \quad t \geq 0. \quad (115)$$

The asymptotic climate dynamics are attracted toward a nonlinear global attractor

$$\mathcal{A}_g = \bigcap_{t \geq 0} \overline{\bigcup_{\tau \geq t} S(\tau)B}, \quad (116)$$

where B denotes a bounded absorbing set.

Tipping phenomena correspond to topological changes in the attractor geometry,

$$\mathcal{A}_g^- \longrightarrow \mathcal{A}_g^+, \quad (117)$$

induced by nonlinear feedback amplification mechanisms operating across multiple interacting components of the system.

Nonlinear Precursors of Planetary Tipping.

Combining the previous results, planetary criticality is characterized by

$$\langle x, \mathcal{N}(x) \rangle \gg \alpha \|x\|^2, \quad (118)$$

$$\Pi_{i \rightarrow k} \uparrow, \quad (119)$$

$$\dot{V} \rightarrow 0, \quad (120)$$

$$\text{Re}(\Lambda_1) \rightarrow 0^-, \quad (121)$$

$$\tau_r \rightarrow \infty, \quad (122)$$

$$\frac{dE}{dt} > 0, \quad (123)$$

$$\Delta_s \rightarrow 0. \quad (124)$$

These nonlinear signatures reveal a progressive loss of system resilience and the emergence of cascade-driven mechanisms of climate destabilization. In particular, they indicate that stability is eroded not through isolated local failures, but via the amplification and coupling of perturbations across multiple interacting scales.

Table 3: Spectral Decomposition of the Climate Operator

Quantity	Symbol	Definition	Interpretation
Eigenmodes	ϕ_n	$\mathcal{L}\phi_n = \lambda_n\phi_n$	Climate modes
Eigenvalues	λ_n	Spectrum of $\mathcal{L}$	Growth/decay rates
Spectral Gap	$\Delta\lambda$	$\lambda_1 - \lambda_2$	Stability margin
Spectral Density	$\rho(\lambda)$	Mode distribution	Multiscale structure

2.4 Renormalization and Multiscale Structure

The climate system exhibits intrinsic organization across a hierarchy of interacting spatial and temporal scales. Atmospheric turbulence, ocean circulation, cryospheric dynamics, biospheric feedbacks, and anthropogenic forcing operate over vastly different characteristic lengths and times. A rigorous mathematical description therefore requires a multiscale formulation capable of connecting local dynamics to global planetary behavior.

To formalize this structure, we introduce a family of scale-transformation operators

$$\mathcal{R}_s : \mathcal{H} \rightarrow \mathcal{H}, \quad x \mapsto x_s, \quad (125)$$

where the continuous scale parameter

$$s = \ln b, \quad b > 0, \quad (126)$$

represents logarithmic dilation.

The renormalization operators satisfy the semigroup property

$$\mathcal{R}_{s_1} \circ \mathcal{R}_{s_2} = \mathcal{R}_{s_1+s_2}, \quad (127)$$

together with

$$\mathcal{R}_0 = I, \quad \mathcal{R}_s^{-1} = \mathcal{R}_{-s}. \quad (128)$$

Consequently, the family

$$\{\mathcal{R}_s\}_{s \in \mathbb{R}} \quad (129)$$

defines a continuous renormalization group acting on the climate state space.

Generator of Scale Transformations.

The infinitesimal generator of the renormalization flow is defined by

$$\mathcal{G} = \left. \frac{d}{ds} \mathcal{R}_s \right|_{s=0}. \quad (130)$$

Hence,

$$\mathcal{R}_s = e^{s\mathcal{G}}. \quad (131)$$

The scale evolution of a climate state obeys

$$\frac{d}{ds} x_s = \mathcal{G} x_s. \quad (132)$$

The operator $\mathcal{G}$ therefore generates trajectories in scale space in complete analogy with temporal evolution generated by $\mathcal{A}$.

Renormalization Flow of Climate Operators.

The climate operator transforms according to

$$\mathcal{A}(s) = \mathcal{R}_s \mathcal{A} \mathcal{R}_s^{-1}, \quad (133)$$

which defines a scale-dependent family of dynamical generators.

Differentiating with respect to scale yields

$$\frac{d\mathcal{A}}{ds} = [\mathcal{G}, \mathcal{A}(s)], \quad (134)$$

where

$$[\mathcal{G}, \mathcal{A}] = \mathcal{G}\mathcal{A} - \mathcal{A}\mathcal{G} \quad (135)$$

denotes the commutator.

Likewise, the nonlinear operator evolves as

$$\mathcal{N}(s) = \mathcal{R}_s \mathcal{N} \mathcal{R}_s^{-1}, \quad (136)$$

with

$$\frac{d\mathcal{N}}{ds} = [\mathcal{G}, \mathcal{N}(s)]. \quad (137)$$

The renormalized climate dynamics therefore become

$$\frac{dx}{dt} = \mathcal{A}(s)x + \mathcal{N}(s)x, \quad (138)$$

capturing scale-dependent deformation of the stability structure [18, 19, 31].

Spectral Renormalization Flow.

Let

$$\lambda_k(s) \quad (139)$$

denote the eigenvalues of the renormalized operator,

$$\mathcal{A}(s)\phi_k(s) = \lambda_k(s)\phi_k(s). \quad (140)$$

The spectrum evolves continuously with scale,

$$\sigma(\mathcal{A}(s)) = \{\lambda_k(s)\}_{k=1}^{\infty}. \quad (141)$$

The corresponding renormalization-group flow equations are

$$\frac{d\lambda_k}{ds} = \beta_k(\lambda), \quad (142)$$

where

$$\beta_k = \frac{d\lambda_k}{ds} \quad (143)$$

defines the spectral beta-function.

Scale-invariant states satisfy

$$\beta_k = 0. \quad (144)$$

These points correspond to spectral fixed points of the underlying climate dynamics, representing invariant configurations of the system's operator spectrum under temporal evolution.

Fixed Points and Universality Classes.

A renormalization fixed point satisfies

$$\mathcal{A}^* = \mathcal{R}_s \mathcal{A}^* \mathcal{R}_s^{-1} \quad (145)$$

for all scales s .

Equivalently,

$$[\mathcal{G}, \mathcal{A}^*] = 0. \quad (146)$$

Near the fixed point,

$$\mathcal{A} = \mathcal{A}^* + \delta\mathcal{A}, \quad (147)$$

and linearization yields

$$\frac{d}{ds}\delta\mathcal{A} = \mathcal{L}_{RG}\delta\mathcal{A}, \quad (148)$$

where

$$\mathcal{L}_{RG} = D[\mathcal{G}, \mathcal{A}^*]. \quad (149)$$

The eigenvalue problem

$$\mathcal{L}_{RG}\Psi_i = y_i\Psi_i \quad (150)$$

determines the scaling exponents y_i .

The classification is

$$y_i > 0 \quad \text{relevant directions,} \quad (151)$$

$$y_i < 0 \quad \text{irrelevant directions,} \quad (152)$$

$$y_i = 0 \quad \text{marginal directions.} \quad (153)$$

Correlation Functions and Critical Scaling.

Define the two-point climate correlation function

$$C(r) = \langle x(\mathbf{r}_0)x(\mathbf{r}_0 + r) \rangle. \quad (154)$$

Near a critical transition,

$$C(r) \sim r^{-(d-2+\eta)}, \quad (155)$$

where

$$d \quad (156)$$

is the effective dimensionality and

$$\eta \quad (157)$$

is the anomalous scaling exponent.

The corresponding correlation length obeys

$$\xi \sim |\mu - \mu_c|^{-\nu}, \quad (158)$$

where μ is a climate control parameter and ν is a critical exponent.

As

$$\mu \rightarrow \mu_c, \quad (159)$$

one obtains

$$\xi \rightarrow \infty, \quad (160)$$

signaling the emergence of long-range climate coherence.

Multiscale Energy Spectrum.

Let $E(k)$ denote the climate energy spectrum. In this context, k represents the spatial wavenumber associated with the Fourier decomposition of the coupled Earth system, while $E(k)$ encodes the scale-dependent redistribution of kinetic and potential energy across atmospheric and oceanic modes. The spectrum provides a reduced statistical representation of multiscale interactions, allowing the identification of dominant energetic cascades and regime-dependent shifts in variability across different spatial scales.

Under renormalization,

$$E(k, s) = b^{-\zeta} E(bk, 0), \quad (161)$$

where ζ is a scaling exponent.

The total energy satisfies

$$E_{\text{tot}} = \int_0^\infty E(k) dk. \quad (162)$$

The spectral energy flux is defined by

$$\Pi(k) = - \int_0^k \frac{\partial E(q)}{\partial t} dq. \quad (163)$$

Scale-invariant cascades satisfy

$$\Pi(k) = \text{const}. \quad (164)$$

This condition characterizes universal energy transport across scales.

Critical Manifolds and Scale Invariance.

The set of renormalization fixed points defines a critical manifold

$$\mathcal{M}_c \subset \mathcal{H}. \quad (165)$$

Climate trajectories approaching criticality satisfy

$$\text{dist}(x(t), \mathcal{M}_c) \rightarrow 0. \quad (166)$$

At criticality the dynamics become approximately scale invariant,

$$x(b\mathbf{r}, b^z t) = b^{-\Delta} x(\mathbf{r}, t), \quad (167)$$

where z is the dynamic critical exponent and Δ is the scaling dimension.

Renormalization Signatures of Climate Criticality.

The approach toward planetary instability is characterized by the asymptotic conditions

$$\beta_k \rightarrow 0, \quad (168)$$

$$\xi \rightarrow \infty, \quad (169)$$

$$\lambda_1 \rightarrow 0^-, \quad (170)$$

$$\Delta_s \rightarrow 0, \quad (171)$$

$$\Pi(k) \rightarrow \text{const}, \quad (172)$$

$$\text{dist}(x, \mathcal{M}_c) \rightarrow 0. \quad (173)$$

These renormalization signatures reveal the emergence of scale invariance, long-range coupling, and collective climate organization. They provide the multiscale mathematical foundation upon which the Universal Pellis Function will be constructed as a universal order parameter for climate criticality, planetary tipping points, and large-scale instability.

Table 4: Renormalization Group Structure of Climate Dynamics

Scale	Symbol	Transformation	Interpretation
Microscale	x	$x \rightarrow bx$	Local dynamics
Coarse Graining	$\mathcal{R}$	$\Psi \rightarrow \mathcal{R}\Psi$	Scale reduction
RG Flow	$\beta(g)$	$\frac{dg}{d \ln \mu}$	Parameter evolution
Fixed Point	g^*	$\beta(g^*) = 0$	Climate regime stability

2.5 Variational Stability Functionals

A complementary characterization of climate stability can be obtained through variational methods. Rather than analyzing stability solely through the spectrum of the evolution operator, one may introduce functionals that quantify the global energetic and geometric structure of perturbations in the infinite-dimensional climate state space.

Let

$$\mathcal{Q} : \mathcal{H} \rightarrow \mathcal{H} \quad (174)$$

be a bounded, self-adjoint, positive-definite operator satisfying

$$\mathcal{Q} = \mathcal{Q}^*, \quad (175)$$

and

$$\langle x, \mathcal{Q}x \rangle \geq q_0 \|x\|^2, \quad q_0 > 0. \quad (176)$$

We define the quadratic Lyapunov functional

$$\mathcal{V}(x) = \langle x, \mathcal{Q}x \rangle. \quad (177)$$

The positivity of $\mathcal{Q}$ implies

$$q_0 \|x\|^2 \leq \mathcal{V}(x) \leq q_1 \|x\|^2, \quad (178)$$

for suitable constants

$$0 < q_0 \leq q_1 < \infty. \quad (179)$$

Hence $\mathcal{V}(x)$ is equivalent to the Hilbert norm and provides a generalized measure of climate perturbation energy.

First Variational Derivative.

For the nonlinear climate evolution equation

$$\frac{dx}{dt} = \mathcal{A}x + \mathcal{N}(x), \quad (180)$$

the time derivative of $\mathcal{V}$ is

$$\frac{d}{dt} \mathcal{V}(x(t)) = \left\langle \frac{dx}{dt}, \mathcal{Q}x \right\rangle + \left\langle x, \mathcal{Q} \frac{dx}{dt} \right\rangle. \quad (181)$$

Substituting the evolution equation yields

$$\frac{d}{dt} \mathcal{V}(x(t)) = \langle \mathcal{A}x, \mathcal{Q}x \rangle + \langle x, \mathcal{Q}\mathcal{A}x \rangle \quad (182)$$

$$+ \langle \mathcal{N}(x), \mathcal{Q}x \rangle + \langle x, \mathcal{Q}\mathcal{N}(x) \rangle. \quad (183)$$

Using adjoint notation,

$$\langle \mathcal{A}x, \mathcal{Q}x \rangle = \langle x, \mathcal{A}^* \mathcal{Q}x \rangle, \quad (184)$$

one obtains

$$\frac{d}{dt} \mathcal{V}(x(t)) = \langle x, (\mathcal{A}^* \mathcal{Q} + \mathcal{Q}\mathcal{A})x \rangle + 2\langle \mathcal{N}(x), \mathcal{Q}x \rangle. \quad (185)$$

For sufficiently small perturbations,

$$\mathcal{N}(x) = \mathcal{O}(\|x\|^2), \quad (186)$$

which gives

$$\frac{d}{dt} \mathcal{V}(x(t)) = \langle x, (\mathcal{A}^* \mathcal{Q} + \mathcal{Q}\mathcal{A})x \rangle + \mathcal{O}(\|x\|^3). \quad (187)$$

This recovers the classical Lyapunov stability criterion.

Operator Lyapunov Equation.

A sufficient condition for exponential stability is the existence of a positive-definite operator $\mathcal{Q}$ satisfying

$$\mathcal{A}^* \mathcal{Q} + \mathcal{Q}\mathcal{A} = -\mathcal{P}, \quad (188)$$

where

$$\mathcal{P} = \mathcal{P}^* > 0. \quad (189)$$

This infinite-dimensional Lyapunov equation implies

$$\frac{d}{dt} \mathcal{V}(x(t)) = -\langle x, \mathcal{P}x \rangle + \mathcal{O}(\|x\|^3). \quad (190)$$

Therefore,

$$\frac{d}{dt}\mathcal{V}(x(t)) < 0 \quad (191)$$

for sufficiently small perturbations, establishing local asymptotic stability.

Variational Stability Margin.

Define the variational stability margin

$$\Gamma_V = \inf_{x \neq 0} \frac{-\langle x, (\mathcal{A}^* \mathcal{Q} + \mathcal{Q} \mathcal{A})x \rangle}{\langle x, \mathcal{Q}x \rangle}. \quad (192)$$

If

$$\Gamma_V > 0, \quad (193)$$

the climate state is variationally stable.

The critical threshold is reached when

$$\Gamma_V \rightarrow 0^+. \quad (194)$$

Loss of stability occurs whenever

$$\Gamma_V < 0. \quad (195)$$

Thus Γ_V acts as a global measure of climate resilience.

Second Variation and Criticality.

The second variation of the functional is

$$\delta^2 \mathcal{V} = 2\langle \delta x, \mathcal{Q} \delta x \rangle. \quad (196)$$

Since $\mathcal{Q}$ is positive definite,

$$\delta^2 \mathcal{V} > 0, \quad (197)$$

and the equilibrium corresponds to a local minimum of the variational energy landscape.

Approaching a climate tipping point, the smallest eigenvalue of $\mathcal{Q}$ satisfies

$$\lambda_{\min}(\mathcal{Q}) \rightarrow 0, \quad (198)$$

yielding

$$\delta^2 \mathcal{V} \rightarrow 0. \quad (199)$$

The variational landscape consequently becomes increasingly flat, reflecting critical slowing down and loss of restoring forces.

Variational Energy Landscape.

The stability functional induces an effective climate potential

$$U(x) = \frac{1}{2} \mathcal{V}(x). \quad (200)$$

Stable climate states satisfy

$$\nabla U(x^*) = 0, \quad (201)$$

and

$$\nabla^2 U(x^*) > 0. \quad (202)$$

Near equilibrium,

$$U(x) = U(x^*) + \frac{1}{2} \langle x - x^*, \mathcal{Q}(x - x^*) \rangle + \mathcal{O}(\|x - x^*\|^3). \quad (203)$$

Climate tipping corresponds to a degeneration of the Hessian,

$$\det(\mathcal{Q}) \rightarrow 0, \quad (204)$$

leading to loss of convexity of the stability landscape.

Variational Critical Modes.

Consider the generalized eigenvalue problem

$$(\mathcal{A}^* \mathcal{Q} + \mathcal{Q} \mathcal{A}) \psi_k = -\mu_k \psi_k. \quad (205)$$

The eigenvalues

$$\mu_k \quad (206)$$

define variational stability modes.

The dominant stability mode is

$$\mu_{\min} = \min_k \mu_k. \quad (207)$$

As the system approaches a tipping point,

$$\mu_{\min} \rightarrow 0^+. \quad (208)$$

Hence the variational relaxation time

$$\tau_V = \frac{1}{\mu_{\min}} \quad (209)$$

satisfies

$$\tau_V \rightarrow \infty. \quad (210)$$

Variational Signatures of Planetary Instability.

The approach toward climate criticality is characterized by

$$\Gamma_V \rightarrow 0^+, \quad (211)$$

$$\mu_{\min} \rightarrow 0^+, \quad (212)$$

$$\tau_V \rightarrow \infty, \quad (213)$$

$$\lambda_{\min}(\mathcal{Q}) \rightarrow 0, \quad (214)$$

$$\delta^2 \mathcal{V} \rightarrow 0, \quad (215)$$

$$\det(\mathcal{Q}) \rightarrow 0. \quad (216)$$

These variational signatures reveal the progressive flattening of the climate stability landscape and the emergence of critical modes associated with planetary tipping phenomena. They provide an energetic and geometric foundation for the subsequent construction of the Universal Pellis Function as a global measure of climate resilience, criticality, and instability.

2.6 Information Geometry of Dynamical States

Beyond spectral, variational, and renormalization descriptions, climate dynamics may also be viewed through the lens of information geometry. In this framework, each macroscopic climate configuration corresponds to a probability distribution on the underlying state space, and dynamical evolution induces trajectories on a statistical manifold endowed with an intrinsic geometric structure.

Let

$$\rho(x) = \rho(x; \theta) \quad (217)$$

denote the probability density associated with the climate state, parameterized by coordinates

$$\theta = (\theta^1, \theta^2, \dots, \theta^n) \in \Theta. \quad (218)$$

The family

$$\mathcal{M} = \{\rho(x; \theta) : \theta \in \Theta\} \quad (219)$$

defines a statistical manifold representing admissible climate configurations.

Entropy Geometry.

The Shannon entropy functional is defined as

$$S[\rho] = - \int_{\Omega} \rho(x) \log \rho(x) dx. \quad (220)$$

The normalization condition

$$\int_{\Omega} \rho(x) dx = 1 \quad (221)$$

is assumed throughout.

The first variation of entropy is

$$\delta S = - \int_{\Omega} \delta \rho(x) [1 + \log \rho(x)] dx. \quad (222)$$

The second variation becomes

$$\delta^2 S = - \int_{\Omega} \frac{(\delta \rho)^2}{\rho} dx. \quad (223)$$

Since

$$\delta^2 S \leq 0, \quad (224)$$

entropy is a concave functional and naturally generates a geometric structure on the climate probability space.

Fisher Information Metric.

The local geometry of the statistical manifold is determined by the Fisher information metric

$$g_{ij} = \mathbb{E} [\partial_i \log \rho \partial_j \log \rho], \quad (225)$$

or equivalently

$$g_{ij} = \int_{\Omega} \rho(x; \theta) \frac{\partial \log \rho}{\partial \theta^i} \frac{\partial \log \rho}{\partial \theta^j} dx. \quad (226)$$

Using

$$\partial_i \log \rho = \frac{1}{\rho} \frac{\partial \rho}{\partial \theta^i}, \quad (227)$$

one obtains

$$g_{ij} = \int_{\Omega} \frac{1}{\rho} \frac{\partial \rho}{\partial \theta^i} \frac{\partial \rho}{\partial \theta^j} dx. \quad (228)$$

The Fisher metric measures statistical distinguishability between neighboring climate states and quantifies sensitivity to perturbations [20, 21].

Infinitesimal Statistical Distance.

The information-geometric line element is

$$ds^2 = g_{ij} d\theta^i d\theta^j. \quad (229)$$

The geodesic distance between two climate states is

$$D_g = \int_{\gamma} \sqrt{g_{ij} \frac{d\theta^i}{d\tau} \frac{d\theta^j}{d\tau}} d\tau, \quad (230)$$

where γ is a path on the statistical manifold.

Large values of D_g correspond to substantial structural changes in climate organization.

Information-Theoretic Susceptibility.

Define the climate susceptibility tensor

$$\chi_{ij} = \frac{\partial^2 S}{\partial \theta^i \partial \theta^j}. \quad (231)$$

Direct calculation yields

$$\chi_{ij} = -g_{ij} + \int_{\Omega} \rho \partial_i \partial_j \log \rho \, dx. \quad (232)$$

Near equilibrium states the second term is often negligible, yielding

$$\chi_{ij} \simeq -g_{ij}. \quad (233)$$

Thus Fisher information plays the role of an information-theoretic response function.

Information Curvature.

The Christoffel symbols associated with the Fisher metric are

$$\Gamma_{ij}^k = \frac{1}{2} g^{km} (\partial_i g_{jm} + \partial_j g_{im} - \partial_m g_{ij}). \quad (234)$$

The Riemann curvature tensor is

$$R_{ijk}^{\ell} = \partial_j \Gamma_{ik}^{\ell} - \partial_k \Gamma_{ij}^{\ell} + \Gamma_{ik}^m \Gamma_{jm}^{\ell} - \Gamma_{ij}^m \Gamma_{km}^{\ell}. \quad (235)$$

Contracting indices yields the Ricci tensor

$$R_{ij} = R_{ikj}^k, \quad (236)$$

and scalar curvature

$$\mathcal{R} = g^{ij} R_{ij}. \quad (237)$$

The scalar curvature provides a global measure of collective interactions among climate degrees of freedom.

Critical Information Geometry.

Near a climate tipping point the correlation length diverges,

$$\xi \rightarrow \infty. \quad (238)$$

Information geometry predicts that the scalar curvature scales as

$$\mathcal{R} \sim \xi^d, \quad (239)$$

where d denotes the effective dimensionality of the climate system.

Using the critical scaling law

$$\xi \sim |\mu - \mu_c|^{-\nu}, \quad (240)$$

one obtains

$$\mathcal{R} \sim |\mu - \mu_c|^{-d\nu}. \quad (241)$$

Consequently,

$$\mathcal{R} \rightarrow \infty \quad \text{as} \quad \mu \rightarrow \mu_c. \quad (242)$$

Divergent information curvature therefore acts as a geometric precursor of planetary criticality.

Relative Entropy Between Climate States.

Consider two climate distributions

$$\rho_1(x), \quad \rho_2(x). \quad (243)$$

Their relative entropy is

$$D_{KL}(\rho_1 || \rho_2) = \int_{\Omega} \rho_1(x) \log \left(\frac{\rho_1(x)}{\rho_2(x)} \right) dx. \quad (244)$$

For infinitesimally close states,

$$D_{KL} = \frac{1}{2} g_{ij} d\theta^i d\theta^j + \mathcal{O}(d\theta^3). \quad (245)$$

Hence the Fisher metric emerges naturally as the local approximation of information divergence.

Entropy Production and Irreversibility.

The entropy production rate is

$$\Sigma = \frac{dS}{dt}. \quad (246)$$

Using the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0, \quad (247)$$

one obtains

$$\Sigma = \int_{\Omega} \rho v \cdot \nabla \log \rho dx. \quad (248)$$

Large positive values of Σ indicate strong irreversible reorganization of climate structure.

Information-Geometric Indicators of Planetary Instability.

The approach toward climate criticality is characterized by

$$g_{ij} \rightarrow \infty, \quad (249)$$

$$\chi_{ij} \rightarrow \infty, \quad (250)$$

$$\mathcal{R} \rightarrow \infty, \quad (251)$$

$$D_g \uparrow, \quad (252)$$

$$D_{KL} \uparrow, \quad (253)$$

$$\xi \rightarrow \infty, \quad (254)$$

$$\Sigma \uparrow. \quad (255)$$

These information-geometric signatures reveal increasing statistical sensitivity, long-range coupling, and collective organization of climate fluctuations. They establish a geometric framework for describing climate criticality and provide a natural foundation for the subsequent formulation of the Universal Pellis Function as a global information-theoretic measure of proximity to planetary tipping points.

2.7 Unified Stability Manifold

The preceding developments reveal that climate stability cannot be fully characterized by a single mathematical object. Spectral dynamics describe the evolution of perturbation modes, variational principles quantify energetic resilience, and information geometry captures the statistical sensitivity of the climate state. A complete theory of planetary stability therefore requires a unified framework integrating these complementary perspectives.

To this end, we define the *Unified Stability Manifold*

$$\mathcal{M}_{\text{stab}} = \left(\sigma(\mathcal{A}), \mathcal{V}(x), g_{ij} \right), \quad (256)$$

which combines

1. the spectral structure $\sigma(\mathcal{A})$,
2. the variational stability functional $\mathcal{V}(x)$,
3. the Fisher information metric g_{ij} .

The manifold $\mathcal{M}_{\text{stab}}$ represents the complete stability geometry of the climate system and encodes the interaction between dynamical, energetic, and informational mechanisms.

Spectral Component.

The spectral sector is characterized by the eigenvalue set

$$\sigma(\mathcal{A}) = \{\lambda_k\}_{k=1}^{\infty}. \quad (257)$$

The dominant stability eigenvalue is

$$\lambda_{\max} = \sup_{\lambda \in \sigma(\mathcal{A})} \operatorname{Re}(\lambda). \quad (258)$$

Stability requires

$$\lambda_{\max} < 0, \quad (259)$$

whereas the critical regime satisfies

$$\lambda_{\max} \rightarrow 0^-. \quad (260)$$

The associated spectral gap is

$$\Delta_s = \lambda_2 - \lambda_1. \quad (261)$$

Approaching instability,

$$\Delta_s \rightarrow 0, \quad (262)$$

indicating collapse of spectral separation and loss of dynamical resilience.

Variational Component.

The energetic sector is described by the Lyapunov functional

$$\mathcal{V}(x) = \langle x, \mathcal{Q}x \rangle. \quad (263)$$

Its evolution satisfies

$$\frac{d\mathcal{V}}{dt} = \langle x, (\mathcal{A}^* \mathcal{Q} + \mathcal{Q} \mathcal{A})x \rangle + \mathcal{O}(\|x\|^3). \quad (264)$$

Define the variational stability operator

$$\mathcal{L}_V = \mathcal{A}^* \mathcal{Q} + \mathcal{Q} \mathcal{A}. \quad (265)$$

The smallest stability eigenvalue is

$$\mu_{\min} = \inf \sigma(\mathcal{L}_V). \quad (266)$$

Stability requires

$$\mu_{\min} > 0, \quad (267)$$

while

$$\mu_{\min} \rightarrow 0^+ \quad (268)$$

signals variational destabilization.

The corresponding recovery time is

$$\tau_V = \frac{1}{\mu_{\min}}, \quad (269)$$

which diverges near criticality,

$$\tau_V \rightarrow \infty. \quad (270)$$

Information-Geometric Component.

The statistical sector is encoded by the Fisher information metric

$$g_{ij} = \int \rho \partial_i \log \rho \partial_j \log \rho dx. \quad (271)$$

The induced scalar curvature is

$$\mathcal{R} = g^{ij} R_{ij}. \quad (272)$$

Near critical transitions,

$$\mathcal{R} \sim \xi^d, \quad (273)$$

where ξ denotes the correlation length.
Since

$$\xi \rightarrow \infty, \quad (274)$$

one obtains

$$\mathcal{R} \rightarrow \infty. \quad (275)$$

The divergence of curvature reflects increasing statistical sensitivity and long-range coupling throughout the climate system.

Coupled Stability Tensor.

To combine the three sectors, define the unified stability tensor

$$\mathbb{S} = \begin{pmatrix} \lambda_{\max} & 0 & 0 \\ 0 & \mu_{\min} & 0 \\ 0 & 0 & \mathcal{R}^{-1} \end{pmatrix}. \quad (276)$$

The determinant

$$\det(\mathbb{S}) = \lambda_{\max} \mu_{\min} \mathcal{R}^{-1} \quad (277)$$

provides a compact measure of global stability.

The critical state satisfies

$$\det(\mathbb{S}) \rightarrow 0. \quad (278)$$

Thus stability loss emerges simultaneously through spectral, variational, and geometric mechanisms.

Unified Stability Functional.

Define the global stability functional

$$\Phi = \alpha_s |\lambda_{\max}| + \alpha_v \mu_{\min} + \alpha_g \mathcal{R}^{-1}, \quad (279)$$

where

$$\alpha_s, \alpha_v, \alpha_g > 0 \quad (280)$$

are weighting coefficients.

A stable climate state satisfies

$$\Phi > 0. \quad (281)$$

Criticality corresponds to

$$\Phi \rightarrow 0^+, \quad (282)$$

while instability is characterized by

$$\Phi < 0. \quad (283)$$

This quantity may be interpreted as a macroscopic resilience parameter.

Geodesic Evolution on the Stability Manifold.

The climate trajectory may be viewed as a path

$$\Gamma(t) \subset \mathcal{M}_{\text{stab}}. \quad (284)$$

Its length is

$$L(\Gamma) = \int \sqrt{G_{AB} \frac{dX^A}{dt} \frac{dX^B}{dt}} dt, \quad (285)$$

where

$$X^A = (\lambda_{\max}, \mu_{\min}, \mathcal{R}) \quad (286)$$

and

$$G_{AB} \quad (287)$$

is the induced metric on the stability manifold.

Near a tipping point,

$$\left\| \frac{dX^A}{dt} \right\| \rightarrow \infty, \quad (288)$$

indicating accelerated deformation of the stability geometry.

Critical Singularities of the Stability Manifold.

Planetary tipping corresponds to singular deformation of $\mathcal{M}_{\text{stab}}$.

Mathematically, the critical regime is characterized by

$$\lambda_{\max} \rightarrow 0^-, \quad (289)$$

$$\Delta_s \rightarrow 0, \quad (290)$$

$$\mu_{\min} \rightarrow 0^+, \quad (291)$$

$$\tau_V \rightarrow \infty, \quad (292)$$

$$\mathcal{R} \rightarrow \infty, \quad (293)$$

$$\xi \rightarrow \infty, \quad (294)$$

$$\det(\mathbb{S}) \rightarrow 0, \quad (295)$$

$$\Phi \rightarrow 0^+. \quad (296)$$

These simultaneous limits define a singular point of the Unified Stability Manifold at which spectral gap closure, variational flattening, information-curvature divergence, and nonlinear amplification occur collectively.

Unified Criterion for Planetary Instability.

The climate system approaches a planetary tipping point if and only if

$$\lim_{t \rightarrow t_c} (\lambda_{\max}, \mu_{\min}, \mathcal{R}^{-1}) = (0, 0, 0). \quad (297)$$

Equivalently,

$$\lim_{t \rightarrow t_c} \Phi = 0. \quad (298)$$

This condition establishes a unified spectral-variational-geometric characterization of climate criticality and provides the immediate mathematical foundation for the introduction of the Universal Pellis Function as a global order parameter of planetary stability.

3 Climate System as a Nonlinear Multiscale Dynamical Field

The climate system is formulated as a nonlinear multiscale dynamical field evolving on an infinite-dimensional state space. Its dynamics are governed by coupled partial differential operators, nonlinear feedback interactions, and scale-dependent energy transfer processes. Unlike finite-dimensional approximations, the system exhibits intrinsic spectral complexity, where stability and instability are determined by operator spectra rather than local phase-space trajectories [6, 7, 14, 27, 33].

This section establishes the mathematical representation of the climate system as a coupled field theory, forming the basis for spectral instability analysis, renormalization structure, and the emergence of tipping point phenomena in subsequent sections [5, 9, 28, 29, 32].

3.1 Functional Representation of the Climate State

A rigorous representation of the coupled Earth system requires an infinite-dimensional functional framework capable of simultaneously encoding fluid dynamical fields, thermodynamic variables, and nonlinear feedback processes across interacting subsystems.

Let

$$\mathcal{H} \tag{299}$$

be a separable Hilbert space equipped with inner product

$$\langle \cdot, \cdot \rangle_{\mathcal{H}}, \quad \|x\|^2 = \langle x, x \rangle_{\mathcal{H}}. \tag{300}$$

The full climate state is represented by a trajectory

$$\mathbf{X}(t) \in \mathcal{H}. \tag{301}$$

Multicomponent Functional Decomposition.

The climate state admits a structured decomposition into interacting geophysical fields

$$\mathbf{X} = (u, T, S, \rho, \phi_1, \phi_2, \dots), \tag{302}$$

where

$$u : \text{velocity field}, \tag{303}$$

$$T : \text{temperature field}, \tag{304}$$

$$S : \text{salinity field}, \tag{305}$$

$$\rho : \text{density field}, \tag{306}$$

$$\phi_i : \text{latent feedback or subgrid-scale variables}. \tag{307}$$

Each component is itself an element of an appropriate function space, for example

$$u \in L^2(\Omega)^d, \quad T, S, \rho \in L^2(\Omega), \quad \phi_i \in \mathcal{H}_i. \tag{308}$$

Thus the full state space may be written as a tensor-product structure

$$\mathcal{H} = L^2(\Omega)^d \otimes L^2(\Omega) \otimes \mathcal{H}_{\text{thermo}} \otimes \mathcal{H}_{\text{feedback}}. \tag{309}$$

Functional Evolution Equation.

The temporal evolution of the climate system is governed by a nonlinear functional differential equation of the form

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{A}_c \mathbf{X}(t) + \mathcal{N}_c(\mathbf{X}(t)). \tag{310}$$

Here

$$\mathcal{A}_c : \mathcal{H} \rightarrow \mathcal{H} \tag{311}$$

is a (possibly unbounded) linear generator of a strongly continuous semigroup

$$e^{t\mathcal{A}_c}, \tag{312}$$

while

$$\mathcal{N}_c : \mathcal{H} \rightarrow \mathcal{H} \tag{313}$$

is a nonlinear operator encoding multiscale coupling, feedback amplification, and cross-field interaction. The mild formulation of the dynamics is given by

$$\mathbf{X}(t) = e^{t\mathcal{A}_c} \mathbf{X}_0 + \int_0^t e^{(t-\tau)\mathcal{A}_c} \mathcal{N}_c(\mathbf{X}(\tau)) d\tau. \tag{314}$$

Nonlinear Coupling Structure.

The nonlinear operator can be decomposed into interaction channels

$$\mathcal{N}_c(\mathbf{X}) = \sum_{\alpha, \beta} \mathcal{B}_{\alpha\beta}(X_\alpha, X_\beta) + \sum_{\gamma} \mathcal{F}_\gamma(X_\gamma), \tag{315}$$

where $\mathcal{B}_{\alpha\beta}$ represents bilinear coupling operators and $\mathcal{F}_\gamma$ higher-order feedback mechanisms. A generic bilinear interaction satisfies

$$\mathcal{B}_{\alpha\beta} : \mathcal{H}_\alpha \times \mathcal{H}_\beta \rightarrow \mathcal{H}. \quad (316)$$

A typical energy-transfer structure is expressed as

$$\langle X_\alpha, \mathcal{B}_{\alpha\beta}(X_\alpha, X_\beta) \rangle \neq 0, \quad (317)$$

indicating nontrivial interscale energy exchange.

Local Well-Posedness and Growth Control.

We assume that $\mathcal{N}_c$ satisfies a generalized Lipschitz condition

$$\|\mathcal{N}_c(X_1) - \mathcal{N}_c(X_2)\| \leq L(1 + \|X_1\|^{p-1} + \|X_2\|^{p-1})\|X_1 - X_2\|, \quad (318)$$

for some $L > 0$ and nonlinear exponent $p \geq 2$.

This ensures local existence and uniqueness of solutions in the class

$$\mathbf{X}(t) \in C([0, T], \mathcal{H}), \quad (319)$$

while allowing for possible finite-time blow-up or instability at large amplitudes.

Energy Functional Structure.

Define the total climate energy functional

$$E(\mathbf{X}) = \frac{1}{2} \|\mathbf{X}\|_{\mathcal{H}}^2. \quad (320)$$

Its evolution satisfies

$$\frac{dE}{dt} = \langle \mathbf{X}, \mathcal{A}_c \mathbf{X} \rangle + \langle \mathbf{X}, \mathcal{N}_c(\mathbf{X}) \rangle. \quad (321)$$

Assuming dissipativity of the linear generator,

$$\langle \mathbf{X}, \mathcal{A}_c \mathbf{X} \rangle \leq -\alpha \|\mathbf{X}\|^2, \quad \alpha > 0, \quad (322)$$

one obtains the balance inequality

$$\frac{dE}{dt} \leq -\alpha \|\mathbf{X}\|^2 + \langle \mathbf{X}, \mathcal{N}_c(\mathbf{X}) \rangle. \quad (323)$$

Instability emerges when nonlinear feedback dominates dissipation:

$$\langle \mathbf{X}, \mathcal{N}_c(\mathbf{X}) \rangle > \alpha \|\mathbf{X}\|^2. \quad (324)$$

Spectral Decomposition of Climate States.

Let $\{\psi_k\}_{k \geq 1}$ be an orthonormal basis of eigenmodes of $\mathcal{A}_c$. Then

$$\mathbf{X}(t) = \sum_{k=1}^{\infty} a_k(t) \psi_k. \quad (325)$$

Substitution into the evolution equation yields the modal system

$$\frac{da_k}{dt} = \lambda_k a_k + \sum_{i,j} C_{ijk} a_i a_j + \mathcal{O}(a^3), \quad (326)$$

where

$$\mathcal{A}_c \psi_k = \lambda_k \psi_k, \quad (327)$$

and

$$C_{ijk} = \langle \mathcal{N}_c(\psi_i, \psi_j), \psi_k \rangle. \quad (328)$$

This representation explicitly reveals the nonlinear transfer of energy across climate modes.

Functional Phase Space Geometry.

The dynamics generate a flow map

$$\Phi_t : \mathcal{H} \rightarrow \mathcal{H}, \quad \mathbf{X}(t) = \Phi_t(\mathbf{X}_0). \quad (329)$$

The tangent evolution is governed by the variational equation

$$\frac{d}{dt} \delta \mathbf{X} = (\mathcal{A}_c + D\mathcal{N}_c(\mathbf{X}(t))) \delta \mathbf{X}. \quad (330)$$

Thus the effective linearized generator becomes state-dependent,

$$\mathcal{L}_{\text{eff}}(t) = \mathcal{A}_c + D\mathcal{N}_c(\mathbf{X}(t)). \quad (331)$$

Emergent Multiscale Instability Mechanism.

Combining spectral growth, nonlinear coupling, and energy imbalance yields the composite instability criterion

$$\max_k \text{Re}(\lambda_k) \rightarrow 0^-, \quad (332)$$

$$\langle \mathbf{X}, \mathcal{N}_c(\mathbf{X}) \rangle \uparrow, \quad (333)$$

$$\|D\mathcal{N}_c(\mathbf{X})\| \uparrow, \quad (334)$$

$$\frac{dE}{dt} > 0. \quad (335)$$

These conditions characterize the onset of emergent multiscale instability.

Figure 2 presents the climate system as a coupled multiscale state manifold composed of atmospheric, oceanic, cryospheric, and biospheric subspaces. The global climate trajectory evolves within the manifold under the action of nonlinear feedbacks and cross-scale interactions. The subsystem network illustrates the exchange of energy, mass, momentum, and information that collectively determine large-scale climate evolution.

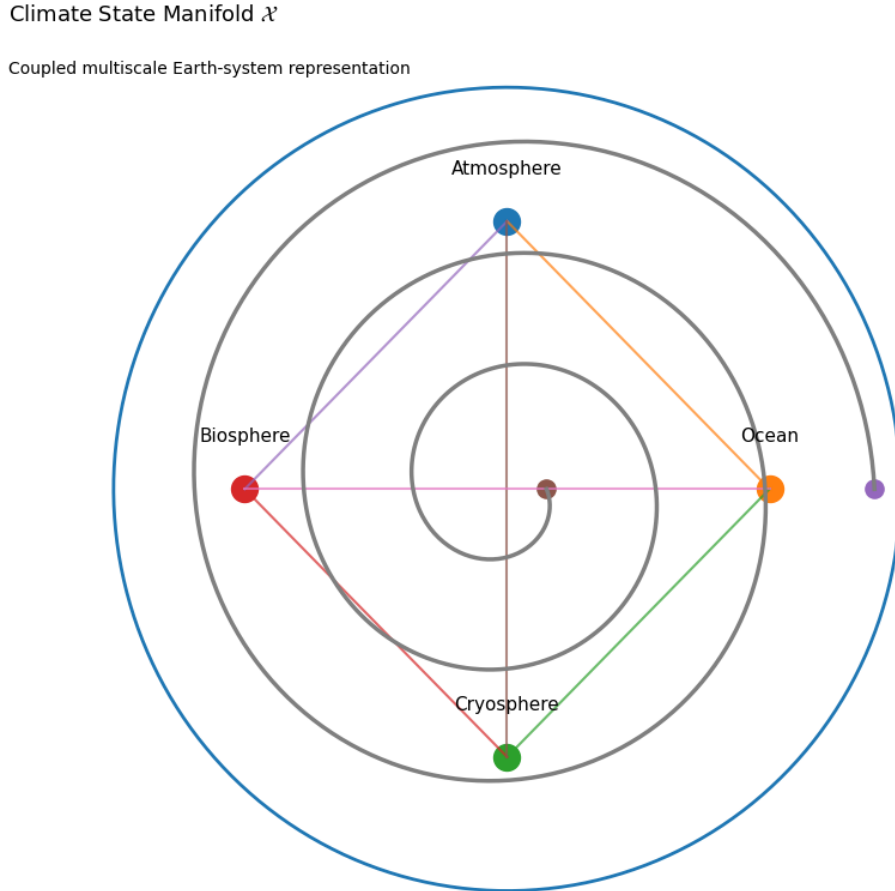


Figure 2: Multiscale climate state manifold in the Pellis framework. Atmospheric, oceanic, cryospheric, and biospheric components are represented as dynamically coupled submanifolds embedded within a unified climate state space. The spiral trajectory illustrates the evolution of the global climate state under nonlinear feedback interactions and multiscale coupling processes.

Functional Interpretation.

The representation above establishes the climate system as a nonlinear dynamical system on a Hilbert manifold with coupled spectral, variational, and nonlinear interaction structure. This formulation provides the foundational phase-space structure upon which renormalization analysis, information geometry, and the Universal Pellis Function can be rigorously constructed as global descriptors of planetary stability and tipping behavior.

Table 5: Nonlinear Feedback Structure in Climate Dynamics

Type	Symbol	Definition	Effect
Positive Feedback	F_+	$\delta\mathcal{L}/\delta\Psi > 0$	Amplification
Negative Feedback	F_-	$\delta\mathcal{L}/\delta\Psi < 0$	Stabilization
Coupled Feedback	F_c	$\mathcal{F}(\Psi_i, \Psi_j)$	Cross interaction
Threshold Feedback	F_T	Nonlinear switch	Abrupt transitions

3.2 Multiscale Decomposition and Spectral Modes

A central feature of the climate system is its inherently multiscale organization, where coherent structures emerge from the interaction of modes spanning several orders of magnitude in spatial and temporal scales. To formalize this structure, we introduce a spectral decomposition of the full climate state in the form of a countable expansion over scale-indexed modes.

Let

$$\mathbf{X}(t) \in \mathcal{H} \quad (336)$$

be the climate state. Then it admits a spectral decomposition

$$\mathbf{X}(t) = \sum_{k \in \mathbb{Z}} \mathbf{X}_k(t), \quad (337)$$

where each component

$$\mathbf{X}_k \in \mathcal{H}_k \quad (338)$$

represents the projection of the system onto an effective scale- k subspace, with

$$\mathcal{H} = \bigoplus_{k \in \mathbb{Z}} \mathcal{H}_k. \quad (339)$$

Each mode $\mathbf{X}_k$ encodes coherent dynamical structures associated with characteristic length scales ℓ_k and time scales τ_k , typically obeying a hierarchical ordering such as

$$\ell_{k+1} < \ell_k, \quad \tau_{k+1} < \tau_k. \quad (340)$$

Mode-Resolved Dynamical Evolution.

Projecting the full nonlinear evolution equation onto each spectral subspace yields a coupled system of infinite-dimensional ordinary differential equations:

$$\partial_t \mathbf{X}_k = \mathcal{P}_k \mathcal{A}_c \mathbf{X} + \mathcal{P}_k \mathcal{N}_c(\mathbf{X}), \quad (341)$$

where $\mathcal{P}_k$ denotes the projection operator onto $\mathcal{H}_k$.

This can be written in closed interacting form as

$$\partial_t \mathbf{X}_k = \lambda_k \mathbf{X}_k + \sum_{i,j \in \mathbb{Z}} \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j - \mathcal{D}_k \mathbf{X}_k + \mathcal{O}(\|\mathbf{X}\|^3), \quad (342)$$

where

$$\mathcal{A}_c \mathbf{X}_k = \lambda_k \mathbf{X}_k \quad (343)$$

defines the linear growth/decay rate of each mode.

Triadic Interaction Tensor.

The nonlinear coupling coefficients are defined by the triadic interaction tensor

$$\mathcal{C}_{ijk} = \langle \mathcal{N}_c(\mathbf{X}_i, \mathbf{X}_j), \mathbf{X}_k \rangle_{\mathcal{H}}, \quad (344)$$

which encodes energy transfer between scales. These coefficients satisfy symmetry and conservation constraints of the form

$$\mathcal{C}_{ijk} = \mathcal{C}_{jik}, \quad (345)$$

together with generalized conservation laws such as

$$\sum_k \mathcal{C}_{ijk} = 0, \quad (346)$$

ensuring redistribution rather than creation of total quadratic invariants under idealized conditions.

Dissipation Structure.

The dissipation operator is scale-dependent and typically satisfies

$$\mathcal{D}_k = \nu k^\alpha, \quad \nu > 0, \quad \alpha \geq 2, \quad (347)$$

representing enhanced damping at smaller scales. The dissipative contribution satisfies

$$\langle \mathbf{X}_k, -\mathcal{D}_k \mathbf{X}_k \rangle \leq -\nu k^\alpha \|\mathbf{X}_k\|^2. \quad (348)$$

Thus high-frequency modes are preferentially stabilized, while low-frequency modes may retain persistent instability.

Spectral Energy of Modes.

Define the modal energy

$$E_k(t) = \|\mathbf{X}_k(t)\|^2. \quad (349)$$

Its evolution is governed by

$$\frac{dE_k}{dt} = 2\lambda_k E_k + 2 \sum_{i,j} \langle \mathbf{X}_k, \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j \rangle - 2\mathcal{D}_k E_k. \quad (350)$$

This explicitly decomposes energy evolution into linear growth, nonlinear transfer, and dissipative decay.

Energy Transfer and Cascade Structure.

The nonlinear term induces a flux of energy between scales. Define the inter-scale energy transfer function

$$\Pi_{i \rightarrow k} = \sum_j \langle \mathbf{X}_k, \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j \rangle. \quad (351)$$

Then

$$\frac{dE_k}{dt} = 2\lambda_k E_k + 2 \sum_i \Pi_{i \rightarrow k} - 2\mathcal{D}_k E_k. \quad (352)$$

A forward cascade is characterized by net transfer toward higher modes:

$$\sum_{i < k} \Pi_{i \rightarrow k} > 0, \quad (353)$$

while inverse cascades satisfy

$$\sum_{i > k} \Pi_{i \rightarrow k} > 0. \quad (354)$$

Cascade-Induced Instability Mechanism.

The system exhibits multiscale instability when nonlinear transfer dominates dissipation at intermediate scales:

$$\sum_{i,j} \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j \gg \mathcal{D}_k \mathbf{X}_k. \quad (355)$$

Equivalently, instability arises when the effective growth rate becomes positive:

$$\lambda_k^{\text{eff}} = \lambda_k + \frac{1}{\|\mathbf{X}_k\|^2} \sum_{i,j} \langle \mathbf{X}_k, \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j \rangle - \mathcal{D}_k > 0. \quad (356)$$

Multiscale Critical Regime.

Near a global tipping point, the system approaches a regime where:

$$\lambda_k^{\text{eff}} \rightarrow 0^- \quad \text{for multiple } k, \quad (357)$$

$$\sum_{i,j} c_{ijk} \mathbf{X}_i \mathbf{X}_j \uparrow, \quad (358)$$

$$\mathcal{D}_k \downarrow \text{ (relative dominance),} \quad (359)$$

$$E_k \rightarrow \text{broadband redistribution,} \quad (360)$$

$$\xi_{\text{eff}} \rightarrow \infty. \quad (361)$$

This regime corresponds to a breakdown of scale separation and the emergence of global coherence through cascade synchronization.

Physical Interpretation.

The multiscale decomposition reveals that climate instability is not driven by a single dominant mode but by the collective reorganization of energy across interacting spectral layers. In this framework, tipping behavior emerges as a cascade-driven transition in which nonlinear triadic interactions overcome dissipative damping, leading to system-wide redistribution of energy across scales.

3.3 Energy Structure and Cascade Balance

A complete description of climate dynamics requires an energetically consistent formulation that accounts for kinetic, thermodynamic, and potential contributions, as well as nonlinear exchange mechanisms across interacting subsystems. Within the Hilbert space formulation, this motivates the definition of a global energy functional governing the stability and redistribution properties of the system.

The total energy of the coupled climate system is defined by

$$E[\mathbf{X}] = \int_{\Omega} \left(\frac{1}{2} |u|^2 + c_p T + g \rho z \right) dx + \mathcal{E}_{\text{int}}(\mathbf{X}), \quad (362)$$

where u denotes the velocity field, T the temperature field, ρ the density, and z the vertical coordinate. The constant c_p is the specific heat capacity at constant pressure, and g is the gravitational acceleration. The functional

$$\mathcal{E}_{\text{int}}(\mathbf{X}) \quad (363)$$

encodes nonlinear internal energy exchanges arising from phase transitions, radiative coupling, moist convection, and unresolved subgrid-scale feedback processes.

Decomposition of Energy Across Spectral Scales.

Using the multiscale decomposition

$$\mathbf{X} = \sum_{k \in \mathbb{Z}} \mathbf{X}_k, \quad (364)$$

the total energy can be decomposed into scale-resolved contributions

$$E = \sum_{k \in \mathbb{Z}} E_k, \quad E_k = \frac{1}{2} \|\mathbf{X}_k\|^2 + E_k^{\text{int}}, \quad (365)$$

where E_k^{int} represents nonlinear intra-scale energy corrections induced by mode coupling.

Energy Balance Equation at Each Scale.

The evolution of energy at scale k satisfies a balance law of the form

$$\frac{dE_k}{dt} = \Pi_{k-1 \rightarrow k} - \Pi_{k \rightarrow k+1} - \varepsilon_k + \mathcal{F}_k, \quad (366)$$

where:

$$\Pi_{k-1 \rightarrow k} : \text{upscale or downscale incoming flux,} \quad (367)$$

$$\Pi_{k \rightarrow k+1} : \text{outgoing cascade flux,} \quad (368)$$

$$\varepsilon_k : \text{dissipative loss at scale } k, \quad (369)$$

$$\mathcal{F}_k : \text{external or nonlinear injection terms.} \quad (370)$$

The dissipation term typically satisfies a scale-dependent structure of the form

$$\varepsilon_k = \nu k^\alpha E_k, \quad \nu > 0, \alpha \geq 2, \quad (371)$$

reflecting enhanced small-scale damping.

Nonlinear Flux Representation.

The cascade fluxes are generated by triadic interactions and can be expressed as

$$\Pi_{i \rightarrow k} = \sum_{j \in \mathbb{Z}} \langle \mathbf{X}_k, \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j \rangle. \quad (372)$$

The net flux through scale k is therefore

$$\Pi_k^{\text{net}} = \Pi_{k-1 \rightarrow k} - \Pi_{k \rightarrow k+1}. \quad (373)$$

A steady-state cascade corresponds to the condition

$$\frac{dE_k}{dt} = 0, \quad \Rightarrow \quad \Pi_{k-1 \rightarrow k} = \Pi_{k \rightarrow k+1} + \varepsilon_k. \quad (374)$$

Nonequilibrium Cascade Structure.

In realistic climate regimes, perfect balance is generally not achieved, leading to a nonequilibrium cascade characterized by persistent flux imbalance:

$$\mathcal{I}_k = \Pi_{k-1 \rightarrow k} - \Pi_{k \rightarrow k+1} - \varepsilon_k. \quad (375)$$

Thus,

$$\frac{dE_k}{dt} = \mathcal{I}_k, \quad (376)$$

where $\mathcal{I}_k$ acts as a local measure of energetic disequilibrium.

Global Energy Conservation and Redistribution.

Summing over all scales yields

$$\frac{dE}{dt} = \sum_{k \in \mathbb{Z}} \mathcal{I}_k = \sum_k \mathcal{F}_k - \sum_k \varepsilon_k, \quad (377)$$

which expresses the global balance between injection and dissipation.

In the idealized conservative limit,

$$\mathcal{F}_k = 0, \quad \varepsilon_k = 0, \quad (378)$$

one recovers

$$\frac{dE}{dt} = 0, \quad (379)$$

indicating energy redistribution without net loss or gain.

Cascade Imbalance and Instability Mechanism.

Instability emerges when nonlinear transfer dominates dissipation, producing sustained positive feedback across scales:

$$\Pi_{k-1 \rightarrow k} \gg \varepsilon_k, \quad (380)$$

or equivalently,

$$\mathcal{I}_k > 0 \quad \text{for a broad range of } k. \quad (381)$$

In this regime, energy is not merely redistributed but progressively accumulated across multiple scales through nonlinear interactions, resulting in the amplification and reinforcement of coherent dynamical structures and the spontaneous emergence of multiscale organization.

Cascade Synchronization and Critical Transition.

Near a critical transition, the system approaches a synchronized cascade regime characterized by coherent flux alignment across interacting scales, indicating the emergence of coordinated dynamical behavior.

$$\Pi_{k-1 \rightarrow k} \approx \Pi_{k \rightarrow k+1} \approx \Pi^*, \quad (382)$$

with

$$\Pi^* = \text{const across scales}. \quad (383)$$

Simultaneously,

$$\varepsilon_k \rightarrow 0 \quad (\text{relative dominance loss}). \quad (384)$$

This leads to scale-invariant energy transport.

Physical Interpretation.

The energy structure reveals that climate instability is fundamentally a cascade phenomenon rather than a single-scale energetic blow-up. The interplay between nonlinear triadic transfer, dissipative damping, and external forcing produces a delicate balance that can be destabilized when flux imbalance becomes coherent across scales.

This cascade formulation provides the energetic backbone of the Unified Stability Manifold and establishes a direct connection to renormalization structure and information-geometric divergence. It also forms a necessary precursor for the definition of the Universal Pellis Function as a global measure of cascade coherence, energetic imbalance, and planetary instability.

Figure 3 illustrates a multiscale energy cascade structure in the climate system, exhibiting an approximately scale-invariant transfer of kinetic and thermodynamic energy across spatial scales. The observed cascade follows a modified power-law behavior consistent with Kolmogorov-like turbulence phenomenology; however, deviations emerge due to the presence of nonlinear feedback mechanisms intrinsic to the climate system, including radiative coupling, moisture-phase transitions, and surface-atmosphere interactions.

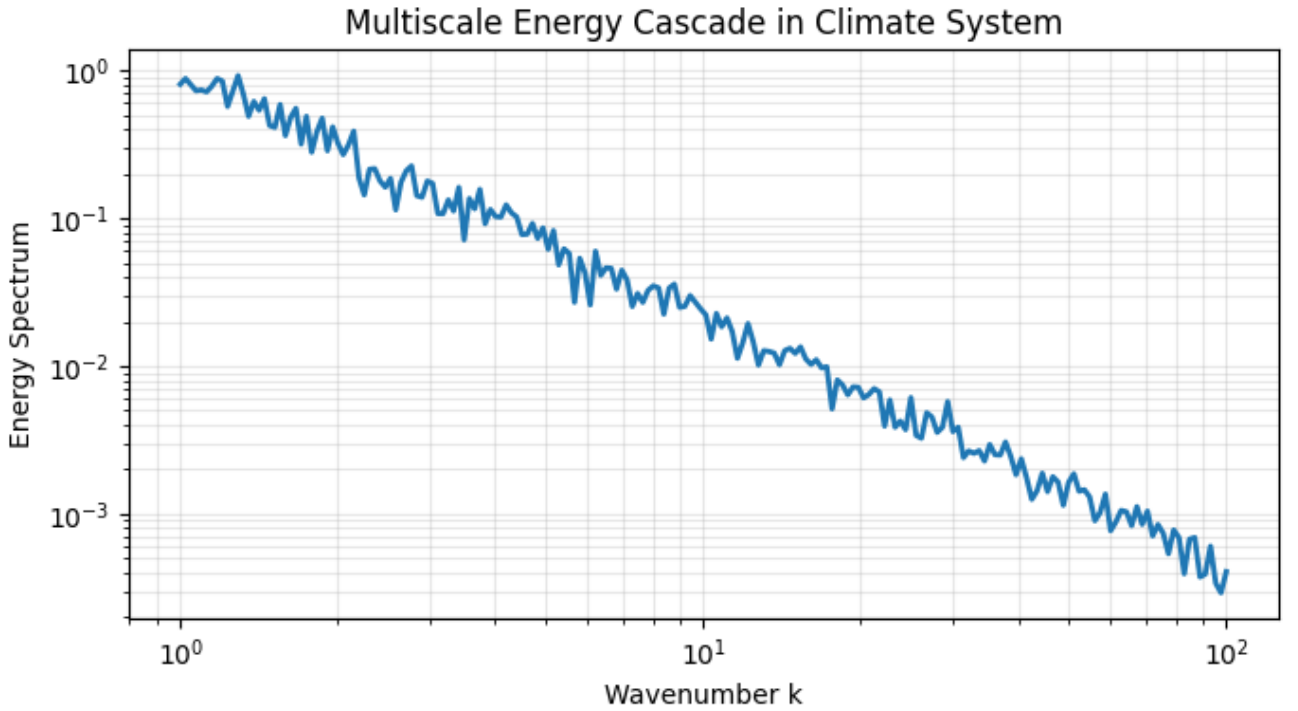


Figure 3: Multiscale energy cascade in the climate system exhibiting approximate Kolmogorov-like power-law scaling across inertial-like ranges. The cascade is systematically modified by nonlinear climate feedback processes, which induce scale-dependent deviations from idealized turbulence scaling and generate a renormalized spectral transfer structure across coupled atmospheric and oceanic modes.

3.4 Nonlinear Feedback Mechanisms

A defining feature of the climate system is the presence of strongly nonlinear feedback loops that dynamically reshape the effective governing operators themselves. In contrast to linearized stability theory, where feedback is treated as a perturbative correction, here feedback mechanisms enter as structurally active operators that modulate growth rates, coupling strengths, and cascade efficiency in a state-dependent manner.

We formalize the effective nonlinear feedback operator as

$$\mathcal{N}_{\text{eff}}(\mathbf{X}) = \sum_m \alpha_m \mathcal{G}_m(\mathbf{X}), \quad (385)$$

where each nonlinear functional

$$\mathcal{G}_m : \mathcal{H} \rightarrow \mathcal{H} \quad (386)$$

represents a distinct physical feedback pathway, and

$$\alpha_m \in \mathbb{R}^+ \quad (387)$$

denotes the associated coupling strength.

State-Dependent Gain Structure.

Each feedback channel induces a state-dependent amplification operator defined through the Fréchet derivative

$$D\mathcal{G}_m(\mathbf{X}) : \mathcal{H} \rightarrow \mathcal{H}, \quad (388)$$

such that the linearized feedback contribution becomes

$$\mathcal{G}_m(\mathbf{X} + \delta\mathbf{X}) = \mathcal{G}_m(\mathbf{X}) + D\mathcal{G}_m(\mathbf{X}) \delta\mathbf{X} + \mathcal{O}(\|\delta\mathbf{X}\|^2). \quad (389)$$

The effective local gain operator is therefore

$$\mathcal{K}(\mathbf{X}) = \sum_m \alpha_m D\mathcal{G}_m(\mathbf{X}). \quad (390)$$

Instability is promoted when the spectral radius satisfies

$$\rho(\mathcal{K}(\mathbf{X})) > 1, \quad (391)$$

indicating net amplification of perturbations by nonlinear feedback loops.

Feedback-Modified Evolution Equation.

Including feedback explicitly, the full evolution equation becomes

$$\frac{d}{dt}\mathbf{X} = \mathcal{A}_c\mathbf{X} + \mathcal{N}_c(\mathbf{X}) + \mathcal{N}_{\text{eff}}(\mathbf{X}), \quad (392)$$

or equivalently,

$$\frac{d}{dt}\mathbf{X} = \mathcal{A}_c\mathbf{X} + \tilde{\mathcal{N}}(\mathbf{X}), \quad \tilde{\mathcal{N}} = \mathcal{N}_c + \mathcal{N}_{\text{eff}}. \quad (393)$$

The linearization about a reference state $\mathbf{X}^*$ yields

$$\frac{d}{dt}\delta\mathbf{X} = (\mathcal{A}_c + D\mathcal{N}_c(\mathbf{X}^*) + \mathcal{K}(\mathbf{X}^*)) \delta\mathbf{X}. \quad (394)$$

Thus, nonlinear feedback effectively renormalizes the stability operator.

Feedback-Induced Spectral Shift.

Define the feedback-shifted generator

$$\mathcal{A}_{\text{fb}} = \mathcal{A}_c + \mathcal{K}(\mathbf{X}). \quad (395)$$

The corresponding spectral shift is given by

$$\sigma(\mathcal{A}_{\text{fb}}) = \sigma(\mathcal{A}_c) + \Delta_{\text{fb}}(\mathbf{X}), \quad (396)$$

where

$$\Delta_{\text{fb}}(\mathbf{X}) \sim \text{spec}(\mathcal{K}(\mathbf{X})). \quad (397)$$

Criticality is approached when feedback-driven eigenvalue migration satisfies

$$\max_{\lambda \in \sigma(\mathcal{A}_{\text{fb}})} \text{Re}(\lambda) \rightarrow 0^-. \quad (398)$$

Feedback-Driven Energy Amplification.

The energy evolution becomes

$$\frac{dE}{dt} = \langle \mathbf{X}, \mathcal{A}_c \mathbf{X} \rangle + \langle \mathbf{X}, \mathcal{N}_c(\mathbf{X}) \rangle + \sum_m \alpha_m \langle \mathbf{X}, \mathcal{G}_m(\mathbf{X}) \rangle. \quad (399)$$

Define the total feedback injection rate

$$\mathcal{F}(\mathbf{X}) = \sum_m \alpha_m \langle \mathbf{X}, \mathcal{G}_m(\mathbf{X}) \rangle. \quad (400)$$

Instability arises when

$$\mathcal{F}(\mathbf{X}) > \alpha \|\mathbf{X}\|^2, \quad (401)$$

overcoming linear dissipation.

Representative Physical Feedback Classes.

The operators $\mathcal{G}_m$ can be decomposed into physically interpretable classes:

$$\mathcal{G}_{\text{cry}}(\mathbf{X}) : \text{ cryospheric albedo amplification,} \quad (402)$$

$$\mathcal{G}_{\text{ocn}}(\mathbf{X}) : \text{ ocean circulation feedbacks,} \quad (403)$$

$$\mathcal{G}_{\text{carb}}(\mathbf{X}) : \text{ carbon-cycle coupling,} \quad (404)$$

$$\mathcal{G}_{\text{rad}}(\mathbf{X}) : \text{ radiative feedback interactions,} \quad (405)$$

$$\mathcal{G}_{\text{bio}}(\mathbf{X}) : \text{ biosphere-climate coupling.} \quad (406)$$

Each class exhibits a distinct nonlinear gain profile

$$\alpha_m^{\text{eff}}(\mathbf{X}) = \alpha_m \frac{\|\mathcal{G}_m(\mathbf{X})\|}{\|\mathbf{X}\|}, \quad (407)$$

which typically increases sharply near critical regimes.

Feedback Cascade and Nonlinear Resonance.

Strong coupling between feedback channels induces resonance-like amplification when

$$\omega_i(\mathbf{X}) \approx \omega_j(\mathbf{X}), \quad (408)$$

leading to constructive interaction between modes.

The effective resonance condition is

$$\det(\omega_i I - \omega_j) \approx 0. \quad (409)$$

This yields synchronized amplification across multiple feedback loops.

Critical Feedback Regime.

Near a planetary tipping point, nonlinear feedback mechanisms satisfy the asymptotic scaling

$$\rho(\mathcal{K}(\mathbf{X})) \rightarrow \infty, \quad (410)$$

$$\mathcal{F}(\mathbf{X}) \uparrow, \quad (411)$$

$$\|\mathcal{K}(\mathbf{X})\| \rightarrow \infty, \quad (412)$$

$$\max \text{Re}(\sigma(\mathcal{A}_{\text{fb}})) \rightarrow 0^-. \quad (413)$$

In this regime, feedback is no longer a perturbation but becomes the dominant organizing principle of the dynamics, driving self-amplifying transitions across scales.

Physical Interpretation.

Nonlinear feedback mechanisms act as internal gain regulators of the climate system, dynamically reshaping both spectral stability and cascade structure. Their progressive amplification near critical thresholds provides a mechanistic route toward tipping behavior and establishes a direct bridge between microscopic process-level interactions and macroscopic instability.

This formulation integrates naturally into the Unified Stability Manifold framework and supplies the dynamical basis for emergent gain divergence, ultimately feeding into the construction of the Universal Pellis Function as a global nonlinear stability amplifier for planetary systems.

3.5 Linear Stability and Spectral Dynamics

A fundamental object in the analysis of climate stability is the linearized dynamics around a reference trajectory $\mathbf{X}(t) \in \mathcal{H}$. This construction provides a local approximation of the fully nonlinear flow in the tangent bundle of the infinite-dimensional state space, capturing the instantaneous amplification or damping of perturbations.

Let $\delta\mathbf{X}(t)$ denote an infinitesimal perturbation. Then the linearized evolution equation is given by

$$\partial_t \delta\mathbf{X} = \mathcal{J}(\mathbf{X}(t)) \delta\mathbf{X}, \quad (414)$$

where the Jacobian (Fréchet derivative) operator is defined as

$$\mathcal{J}(\mathbf{X}) = \frac{\delta\mathcal{F}}{\delta\mathbf{X}} = D\mathcal{F}(\mathbf{X}). \quad (415)$$

Here $\mathcal{F}(\mathbf{X}) = \mathcal{A}_c\mathbf{X} + \mathcal{N}_c(\mathbf{X}) + \mathcal{N}_{\text{eff}}(\mathbf{X})$ represents the full nonlinear climate vector field.

Tangent Flow Representation.

The linearized system generates a time-ordered evolution operator $\mathcal{U}(t, s)$ satisfying

$$\partial_t \mathcal{U}(t, s) = \mathcal{J}(\mathbf{X}(t)) \mathcal{U}(t, s), \quad \mathcal{U}(s, s) = I. \quad (416)$$

Thus perturbations evolve according to

$$\delta\mathbf{X}(t) = \mathcal{U}(t, s) \delta\mathbf{X}(s). \quad (417)$$

The growth properties of $\mathcal{U}(t, s)$ determine the local predictability horizon of the climate system.

Instantaneous Spectral Structure.

The instantaneous stability properties are encoded in the spectrum of $\mathcal{J}(\mathbf{X})$:

$$\sigma(\mathcal{J}) = \{\lambda \in \mathbb{C} : \mathcal{J} - \lambda I \text{ is not invertible}\}. \quad (418)$$

The spectral radius is defined as

$$\rho(\mathcal{J}) = \sup_{\lambda \in \sigma(\mathcal{J})} |\lambda|. \quad (419)$$

More relevant for dynamical instability is the spectral abscissa

$$\alpha(\mathcal{J}) = \sup_{\lambda \in \sigma(\mathcal{J})} \text{Re}(\lambda), \quad (420)$$

which governs exponential growth rates.

Growth Rate of Perturbations.

For sufficiently smooth initial perturbations, the asymptotic growth rate is given by

$$\|\delta\mathbf{X}(t)\| \sim e^{\alpha(\mathcal{J})t} \|\delta\mathbf{X}(0)\|. \quad (421)$$

Thus:

$$\alpha(\mathcal{J}) < 0 \quad \Rightarrow \quad \text{asymptotic decay}, \quad (422)$$

$$\alpha(\mathcal{J}) = 0 \quad \Rightarrow \quad \text{neutral stability}, \quad (423)$$

$$\alpha(\mathcal{J}) > 0 \quad \Rightarrow \quad \text{exponential instability}. \quad (424)$$

State-Dependent Jacobian Decomposition.

Using the full nonlinear structure, the Jacobian admits the decomposition

$$\mathcal{J}(\mathbf{X}) = \mathcal{A}_c + D\mathcal{N}_c(\mathbf{X}) + D\mathcal{N}_{\text{eff}}(\mathbf{X}). \quad (425)$$

Defining the feedback-enhanced operator

$$\mathcal{J}_{\text{eff}}(\mathbf{X}) = \mathcal{A}_c + \mathcal{K}(\mathbf{X}), \quad (426)$$

with

$$\mathcal{K}(\mathbf{X}) = D\mathcal{N}_c(\mathbf{X}) + D\mathcal{N}_{\text{eff}}(\mathbf{X}), \quad (427)$$

the stability is fully controlled by the spectrum of a state-dependent operator.

Finite-Time Growth Amplification.

Beyond asymptotic stability, finite-time amplification is quantified by the transient growth functional

$$G(t) = \sup_{\|\delta\mathbf{X}(0)\|=1} \|\mathcal{U}(t, 0)\delta\mathbf{X}(0)\|^2. \quad (428)$$

Even when $\alpha(\mathcal{J}) < 0$, non-normality of $\mathcal{J}$ may yield

$$G(t) \gg 1, \quad (429)$$

indicating strong transient growth and hidden instability channels.

The non-normality measure is

$$\mathcal{N}_{\text{nn}} = \|\mathcal{J}\mathcal{J}^* - \mathcal{J}^*\mathcal{J}\|. \quad (430)$$

Large values of $\mathcal{N}_{\text{nn}}$ are associated with amplified sensitivity to perturbations.

Lyapunov Spectrum and Predictability Horizon.

Let $\{\lambda_i^{(L)}\}$ denote the Lyapunov spectrum of the flow. The maximal Lyapunov exponent is

$$\lambda_{\text{max}}^{(L)} = \lim_{t \rightarrow \infty} \frac{1}{t} \log \|\mathcal{U}(t, 0)\|. \quad (431)$$

The predictability horizon is then

$$\tau_p = \frac{1}{\lambda_{\text{max}}^{(L)}}. \quad (432)$$

As the system approaches criticality,

$$\lambda_{\text{max}}^{(L)} \rightarrow 0^+, \quad \tau_p \rightarrow \infty. \quad (433)$$

Spectral Route to Instability.

Instability arises through a hierarchy of spectral transitions:

$$\alpha(\mathcal{J}) \rightarrow 0^-, \quad (434)$$

$$\rho(\mathcal{J}) \uparrow, \quad (435)$$

$$\mathcal{N}_{\text{nn}} \uparrow, \quad (436)$$

$$G(t) \uparrow, \quad (437)$$

$$\lambda_{\text{max}}^{(L)} \rightarrow 0^+. \quad (438)$$

These conditions jointly define a spectral signature of approaching climate criticality.

Physical Interpretation.

Linear stability analysis reveals that climate tipping is fundamentally encoded in the spectrum of the instantaneous Jacobian operator. However, due to strong non-normality and state-dependent feedback, spectral stability alone is insufficient: transient growth and nonlinear amplification can destabilize the system even before eigenvalue crossings occur. This establishes the spectral backbone of the Unified Stability Manifold and provides the local dynamical foundation for global instability measures such as cascade imbalance, information curvature divergence, and ultimately the Universal Pellis Function as a global predictor of planetary tipping behavior.

3.6 Renormalization Structure of Climate Dynamics

A central hypothesis in multiscale climate theory is that macroscopic climate regimes emerge as fixed points of an underlying renormalization flow acting on the full state space. In this formulation, coarse-graining is not merely a numerical procedure, but a dynamical transformation on the space of admissible climate configurations, encoding how effective equations of motion change across scales.

Let

$$\mathbf{X}^{(s)} \in \mathcal{H} \quad (439)$$

denote the climate state at renormalization scale s . The renormalization transformation is defined as

$$\mathbf{X}^{(s+1)} = \mathcal{R}(\mathbf{X}^{(s)}), \quad (440)$$

where

$$\mathcal{R} : \mathcal{H} \rightarrow \mathcal{H} \quad (441)$$

is a nonlinear coarse-graining operator.

Coarse-Graining Structure.

The operator $\mathcal{R}$ can be decomposed into projection, averaging, and rescaling components:

$$\mathcal{R} = \mathcal{S}_\ell \circ \mathcal{A}_\ell \circ \mathcal{P}_\ell, \quad (442)$$

where

$$\mathcal{P}_\ell : \mathcal{H} \rightarrow \mathcal{H}_\ell \quad (\text{projection onto resolved modes}), \quad (443)$$

$$\mathcal{A}_\ell : \text{local averaging / homogenization}, \quad (444)$$

$$\mathcal{S}_\ell : \text{scale re-normalization (rescaling of coordinates)}. \quad (445)$$

Thus, the renormalization map encodes both elimination of fast degrees of freedom and redefinition of effective interactions.

Continuous Renormalization Flow.

In the continuous limit, the discrete iteration can be written as a flow equation

$$\frac{d}{ds} \mathbf{X}^{(s)} = \mathcal{B}(\mathbf{X}^{(s)}), \quad (446)$$

where

$$\mathcal{B} = \lim_{\Delta s \rightarrow 0} \frac{\mathcal{R} - I}{\Delta s} \quad (447)$$

is the renormalization generator.

This defines a dynamical system on the space of climate fields, distinct from physical time evolution, representing scale evolution rather than temporal evolution.

Operator Renormalization.

Under coarse-graining, the governing operator transforms as

$$\mathcal{A}_c^{(s+1)} = \mathcal{R} \mathcal{A}_c^{(s)} \mathcal{R}^{-1} + \Delta \mathcal{A}^{(s)}, \quad (448)$$

where $\Delta \mathcal{A}^{(s)}$ represents induced effective corrections arising from eliminated degrees of freedom. Similarly, the nonlinear operator transforms as

$$\mathcal{N}_c^{(s+1)}(\mathbf{X}) = \mathcal{R} \mathcal{N}_c^{(s)}(\mathcal{R}^{-1} \mathbf{X}). \quad (449)$$

Thus, renormalization induces a flow in operator space:

$$\frac{d}{ds} \mathcal{A}_c^{(s)} = \mathcal{F}_\mathcal{A}(\mathcal{A}_c^{(s)}, \mathcal{N}_c^{(s)}). \quad (450)$$

Spectral Renormalization Dynamics.

Let $\lambda_k^{(s)}$ denote the eigenvalues of the renormalized operator $\mathcal{A}_c^{(s)}$. The spectral flow satisfies

$$\frac{d}{ds} \lambda_k^{(s)} = \beta_k(\lambda_k^{(s)}, \mathbf{X}^{(s)}), \quad (451)$$

where β_k is a scale-dependent beta function.

Fixed points satisfy

$$\beta_k(\lambda_k^*) = 0, \quad (452)$$

implying scale-invariant dynamics.

Stability at the fixed point requires

$$\text{Re}(\lambda_k^*) < 0, \quad (453)$$

whereas instability arises when eigenvalues cross the marginal manifold

$$\text{Re}(\lambda_k^{(s)}) = 0. \quad (454)$$

Fixed Points and Climate Regimes.

Climate regimes can be interpreted as fixed points of the renormalization flow, satisfying

$$\mathbf{X}^* = \mathcal{R}(\mathbf{X}^*), \quad (455)$$

where $\mathcal{R}$ denotes the coarse-graining transformation acting on the space of climate states.

Linearization of the dynamics around a fixed point yields the evolution of perturbations as

$$\delta \mathbf{X}^{(s+1)} = D\mathcal{R}(\mathbf{X}^*) \delta \mathbf{X}^{(s)}, \quad (456)$$

where $D\mathcal{R}(\mathbf{X}^*)$ is the Fréchet derivative of the renormalization operator.

The stability properties of the corresponding climate regime are therefore governed by the spectrum of the linearized operator,

$$\sigma(D\mathcal{R}(\mathbf{X}^*)), \quad (457)$$

which encodes the growth or decay rates of perturbative modes under successive renormalization steps.

In this framework, asymptotic stability requires that the spectral radius satisfies

$$\rho(D\mathcal{R}(\mathbf{X}^*)) < 1, \quad (458)$$

whereas loss of stability occurs when

$$\rho(D\mathcal{R}(\mathbf{X}^*)) > 1, \quad (459)$$

indicating the emergence of unstable directions in the renormalization flow and the breakdown of the associated climate regime.

Multiscale Universality and Critical Behavior.

Near critical transitions, the renormalization flow approaches a non-hyperbolic fixed point characterized by

$$\rho(D\mathcal{R}(\mathbf{X}^*)) \rightarrow 1, \quad (460)$$

leading to scale invariance and loss of characteristic length scales.

The correlation length satisfies

$$\xi^{(s)} \sim |\lambda^{(s)} - \lambda_c|^{-\nu}, \quad (461)$$

and diverges under repeated renormalization:

$$\xi^{(s)} \rightarrow \infty. \quad (462)$$

Renormalization-Induced Instability Mechanism.

Instability emerges when the renormalization flow amplifies unstable directions in function space, leading to divergence of the linearized dynamics:

$$\|D\mathcal{R}^n(\mathbf{X})\| \rightarrow \infty, \quad n \rightarrow \infty. \quad (463)$$

Equivalently, cascading amplification across scales induces a systematic drift of spectral eigenmodes, expressed as:

$$\lambda_k^{(s)} \rightarrow \lambda_k^{(s)} + \delta \lambda_k^{(s)}, \quad \delta \lambda_k^{(s)} > 0, \quad (464)$$

which reflects the progressive loss of multiscale stability and the onset of coherent instability propagation.

Physical Interpretation.

The renormalization framework reveals that climate regimes are not static solutions but attractors in a scale-evolution space. Tipping points correspond to bifurcations of the renormalization flow where fixed points lose stability and trajectories diverge toward alternative universality classes. This perspective unifies cascade dynamics, spectral instability, and information-geometric divergence into a single multiscale structure and provides the mathematical basis for defining the Universal Pellis Function as a renormalization-invariant order parameter capturing proximity to planetary critical transitions.

3.7 Emergence of Regime Transitions

Climate regime transitions can be rigorously interpreted as global bifurcations of the attractor structure in the infinite-dimensional phase space $\mathcal{H}$. In this setting, the climate system is not characterized by isolated equilibria but by invariant sets

$$\mathcal{A} \subset \mathcal{H}, \quad (465)$$

which may take the form of fixed points, limit cycles, chaotic attractors, or strange attractors depending on the parameterization of external forcing and internal coupling structure.

A regime transition corresponds to a discontinuous or non-smooth deformation of the attractor map

$$\mathcal{A}(\mu) \longrightarrow \mathcal{A}(\mu + \delta\mu), \quad (466)$$

where μ denotes a control parameter (e.g., radiative forcing, greenhouse gas concentration, or effective coupling strength).

Attractor Stability and Basin Geometry.

Let $\mathcal{B}(\mathcal{A}) \subset \mathcal{H}$ denote the basin of attraction associated with $\mathcal{A}$. Then

$$\mathcal{B}(\mathcal{A}) = \left\{ \mathbf{X}_0 \in \mathcal{H} : \lim_{t \rightarrow \infty} \Phi_t(\mathbf{X}_0) \in \mathcal{A} \right\}. \quad (467)$$

Regime transitions are associated with topological changes in basin structure, including:

$$\partial\mathcal{B}(\mathcal{A}) \rightarrow \text{fracture or folding singularities}. \quad (468)$$

A key precursor is basin boundary collapse, expressed as

$$\text{dist}(\mathcal{B}(\mathcal{A}_1), \mathcal{B}(\mathcal{A}_2)) \rightarrow 0, \quad (469)$$

indicating vanishing resilience between competing attractors.

Spectral–Nonlinear Coupled Instability.

Regime transitions emerge from the coupled amplification of linear and nonlinear mechanisms encoded in the effective Jacobian

$$\mathcal{J}_{\text{eff}}(\mathbf{X}) = \mathcal{A}_c + D\mathcal{N}_c(\mathbf{X}) + D\mathcal{N}_{\text{eff}}(\mathbf{X}). \quad (470)$$

A necessary condition for transition is spectral destabilization:

$$\alpha(\mathcal{J}_{\text{eff}}(\mathbf{X})) = \sup_{\lambda \in \sigma(\mathcal{J}_{\text{eff}})} \text{Re}(\lambda) \rightarrow 0^-. \quad (471)$$

Simultaneously, nonlinear amplification must satisfy

$$\|D\mathcal{N}_{\text{eff}}(\mathbf{X})\| \gg \|\mathcal{A}_c\|. \quad (472)$$

The transition regime is therefore characterized by a breakdown of linear dominance.

Multiscale Coupling Collapse.

Let $\mathbf{X} = \sum_k \mathbf{X}_k$ be the spectral decomposition. Multiscale coupling is encoded in interaction tensors $\mathcal{C}_{ijk}$. A regime transition is associated with structural reorganization of these couplings:

$$\mathcal{C}_{ijk}^{(\text{pre})} \not\approx \mathcal{C}_{ijk}^{(\text{post})}. \quad (473)$$

In particular, the cascade structure undergoes qualitative change when

$$\sum_{i,j} \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j \approx \mathcal{D}_k \mathbf{X}_k \quad (\text{pre-transition balance}) \quad (474)$$

is violated, yielding

$$\sum_{i,j} \mathcal{C}_{ijk} \mathbf{X}_i \mathbf{X}_j \gg \mathcal{D}_k \mathbf{X}_k \quad (\text{post-transition amplification}). \quad (475)$$

This marks the collapse of scale-local equilibrium.

Functional Phase Transition Formalism.

The climate system may be associated with an effective free-energy-like functional

$$\mathcal{F}[\mathbf{X}] = E[\mathbf{X}] - T_{\text{eff}} S[\mathbf{X}] + \Phi_{\text{nl}}[\mathbf{X}], \quad (476)$$

where E denotes energy, S entropy, and Φ_{nl} nonlinear interaction contributions. Regime transitions correspond to non-analyticities of $\mathcal{F}$:

$$\frac{\delta^2 \mathcal{F}}{\delta \mathbf{X}^2} \rightarrow \infty \quad \text{or} \quad \det \left(\frac{\delta^2 \mathcal{F}}{\delta \mathbf{X}^2} \right) = 0. \quad (477)$$

Thus, tipping points are interpreted as functional phase transitions in infinite-dimensional field space.

Order Parameter Collapse.

Define a generic order parameter

$$\Psi(\mathbf{X}) = \|\mathbf{X}\|_{\mathcal{H}}^2 + \sum_k w_k \|\mathbf{X}_k\|^2. \quad (478)$$

Near a transition, symmetry-breaking or structural collapse yields

$$\Psi \rightarrow \Psi^*, \quad \frac{d\Psi}{dt} \rightarrow 0, \quad (479)$$

while fluctuations satisfy

$$\text{Var}(\Psi) \rightarrow \infty. \quad (480)$$

This is consistent with critical slowing down and divergence of susceptibility.

Dynamical Critical Indicators.

The approach to a regime transition is characterized by the simultaneous divergence or vanishing of multiple stability indicators:

$$\alpha(\mathcal{J}_{\text{eff}}) \rightarrow 0^-, \quad (481)$$

$$\tau_V \rightarrow \infty, \quad (482)$$

$$\mathcal{R} \rightarrow \infty, \quad (483)$$

$$\rho(D\mathcal{R}) \rightarrow 1, \quad (484)$$

$$\|\mathcal{C}_{ijk}\| \uparrow, \quad (485)$$

$$\text{dist}(\mathcal{B}_1, \mathcal{B}_2) \rightarrow 0. \quad (486)$$

Interpretation of Planetary Tipping.

From a dynamical systems perspective, planetary tipping points correspond to global reorganizations of invariant measures and attractor geometry rather than local bifurcations alone. They are therefore inherently non-local phenomena in function space, requiring a synthesis of spectral theory, nonlinear dynamics, renormalization structure, and information geometry.

This provides the conceptual and mathematical foundation for interpreting climate transitions as functional phase transitions, and establishes the framework within which the Universal Pellis Function operates as a scalar descriptor of global proximity to catastrophic regime shifts in the Earth system.

4 Pellis Functional Operator Formalism

The Pellis Functional Operator Formalism introduces a nonlinear spectral-operator framework designed to encode multiscale instability, functional phase transitions, and critical tipping behavior in complex dynamical systems. The climate field formulation of Section 3 is embedded into a generalized operator algebra where instability emerges as a structural property of nonlinear spectral deformation [15–19, 22, 30, 31, 33, 34].

4.1 Axiomatic Structure of the Pellis Operator

Within the Universal Pellis framework, the climate system is represented as a nonlinear operator-valued dynamical field on a separable Hilbert space $\mathcal{H}$. In this setting, the evolution is not governed by a single vector field but by a structured composite operator that encodes linear dispersion, nonlinear coupling, and renormalization-induced scale interactions. This motivates the introduction of the *Pellis operator*, denoted by $\mathcal{P}$, as the fundamental generator of climate evolution in function space.

Formally, let

$$\mathcal{F} : \mathcal{H} \rightarrow \mathcal{H} \quad (487)$$

be the full nonlinear evolution functional. The Pellis operator is defined by the decomposition

$$\mathcal{P}(\mathbf{X}) = \mathcal{A}\mathbf{X} + \mathcal{N}(\mathbf{X}) + \mathcal{S}(\mathbf{X}), \quad (488)$$

where:

$$\mathcal{A} : \mathcal{H} \rightarrow \mathcal{H} \quad \text{linear spectral generator}, \quad (489)$$

$$\mathcal{N} : \mathcal{H} \rightarrow \mathcal{H} \quad \text{nonlinear interaction operator}, \quad (490)$$

$$\mathcal{S} : \mathcal{H} \rightarrow \mathcal{H} \quad \text{scale-coupling (renormalization) operator}. \quad (491)$$

Thus, $\mathcal{P}$ constitutes a hybrid operator encoding both dynamical and multiscale structural evolution.

Operator Decomposition and Functional Expansion.

The nonlinear component admits a formal Volterra–Taylor expansion in functional space:

$$\mathcal{N}(\mathbf{X}) = \mathcal{N}_2(\mathbf{X}, \mathbf{X}) + \mathcal{N}_3(\mathbf{X}, \mathbf{X}, \mathbf{X}) + \sum_{m \geq 4} \mathcal{N}_m(\mathbf{X}^{\otimes m}), \quad (492)$$

where each $\mathcal{N}_m$ is an m -linear symmetric functional operator. Similarly, the scale operator is represented as a nonlocal convolution across renormalization scales:

$$\mathcal{S}(\mathbf{X}) = \int_{\mathbb{R}^+} K(s) \mathcal{R}_s(\mathbf{X}) ds, \quad (493)$$

with kernel $K(s)$ encoding scale-dependent coupling intensity.

Axiom (A1): Operator Class Structure.

The Pellis operator belongs to a hybrid operator class:

$$\mathcal{P} \in \mathcal{L}(\mathcal{H}) + \mathcal{N}(\mathcal{H}), \quad (494)$$

where $\mathcal{L}(\mathcal{H})$ denotes bounded linear operators and $\mathcal{N}(\mathcal{H})$ denotes nonlinear Fréchet-smooth operators. More precisely,

$$\mathcal{P} = \mathcal{P}_{\text{lin}} + \mathcal{P}_{\text{nl}} + \mathcal{P}_{\text{rg}}, \quad (495)$$

with renormalization contribution

$$\mathcal{P}_{\text{rg}} := \mathcal{S}. \quad (496)$$

Axiom (A2): Nonlinearity and Broken Homogeneity.

The operator is intrinsically nonlinear and violates scalar homogeneity:

$$\mathcal{P}(\alpha \mathbf{X}) \neq \alpha \mathcal{P}(\mathbf{X}), \quad \forall \alpha \in \mathbb{R} \setminus \{0, 1\}. \quad (497)$$

More precisely, the deviation from linear scaling is quantified by the functional defect:

$$\Delta_{\text{nl}}(\alpha, \mathbf{X}) = \mathcal{P}(\alpha \mathbf{X}) - \alpha \mathcal{P}(\mathbf{X}), \quad (498)$$

which satisfies the scaling law

$$\|\Delta_{\text{nl}}(\alpha, \mathbf{X})\| \sim |\alpha|^p \|\mathbf{X}\|^p, \quad p > 1. \quad (499)$$

Axiom (A3): Spectral Structure and Functional Spectrum.

The spectral properties of $\mathcal{P}$ are defined through its resolvent operator:

$$\sigma(\mathcal{P}) = \{ \lambda \in \mathbb{C} : (\mathcal{P} - \lambda I)^{-1} \text{ does not exist} \}. \quad (500)$$

The spectral radius is

$$\rho(\mathcal{P}) = \sup_{\lambda \in \sigma(\mathcal{P})} |\lambda|. \quad (501)$$

Stability requires spectral confinement:

$$\sup_{\lambda \in \sigma(\mathcal{P})} \text{Re}(\lambda) < 0, \quad (502)$$

whereas instability is characterized by spectral escape:

$$\exists \lambda \in \sigma(\mathcal{P}) : \text{Re}(\lambda) > 0. \quad (503)$$

Figure 4 illustrates the conceptual action of the Pellis operator on the climate state manifold. The operator maps an initial climate configuration into a transformed state space characterized by modified attractor structure and reorganized spectral geometry. The resulting manifold exhibits emergent stability properties that are not explicitly present in the original representation.

$$\mathcal{P}: \mathcal{X} \rightarrow \mathcal{X}$$

Nonlinear operator-induced spectral reorganization

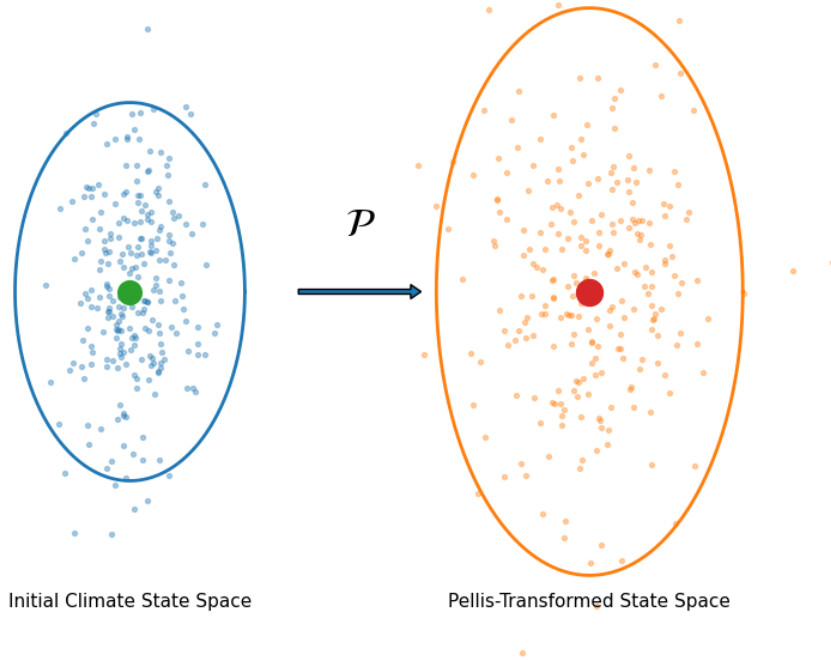


Figure 4: Geometric representation of the Pellis operator acting on the climate state manifold. The nonlinear mapping transforms the underlying attractor landscape and induces spectral reorganization within the state space. Emergent climate regimes arise through operator-driven restructuring of stability geometry.

Axiom (A4): Renormalization Covariance.

A defining property of the Pellis operator is covariance under renormalization transformations. Let $\mathcal{R}$ be a coarse-graining operator acting on $\mathcal{H}$. Then

$$\mathcal{R}\mathcal{P}\mathcal{R}^{-1} = \lambda_{\text{rg}}\mathcal{P} + \Delta_{\text{rg}}(\mathcal{P}), \quad (504)$$

where λ_{rg} is a scaling eigenvalue and Δ_{rg} encodes higher-order renormalization corrections.

In the ideal fixed-point regime,

$$\Delta_{\text{rg}}(\mathcal{P}) \rightarrow 0, \quad \mathcal{R}\mathcal{P} = \lambda_{\text{rg}}\mathcal{P}. \quad (505)$$

Spectral–Nonlinear Consistency Condition.

Consistency between nonlinear dynamics and spectral stability imposes the constraint

$$\mathcal{J}_{\mathcal{P}}(\mathbf{X}) = D\mathcal{P}(\mathbf{X}) = \mathcal{A} + D\mathcal{N}(\mathbf{X}) + D\mathcal{S}(\mathbf{X}), \quad (506)$$

with effective stability condition

$$\alpha(\mathcal{J}_{\mathcal{P}}) = \sup_{\lambda \in \sigma(\mathcal{J}_{\mathcal{P}})} \operatorname{Re}(\lambda) < 0. \quad (507)$$

Violation of this inequality defines the onset of Pellis instability regimes.

Axiomatic Closure and Emergent Criticality.

The axioms (A1–A4) define a closed operator algebra in which linear dynamics, nonlinear interactions, and renormalization effects are inseparably coupled. In this framework, climate tipping behavior corresponds to structural instability of $\mathcal{P}$ under perturbation:

$$\|\delta\mathcal{P}\| \rightarrow \infty \quad \Rightarrow \quad \text{loss of spectral–renormalization coherence.} \quad (508)$$

This loss of coherence marks the transition from stable climate regimes to globally reorganized attractor states.

Physical Interpretation.

The Pellis operator provides a unified mathematical object that simultaneously encodes linear stability theory, nonlinear feedback amplification, and renormalization-group structure. It acts as the fundamental generator of climate evolution in function space, and its axiomatic breakdown corresponds precisely to the emergence of tipping points, cascade instabilities, and planetary-scale regime shifts.

Within this formulation, the Universal Pellis Function arises naturally as a scalar invariant extracted from the spectral and renormalization structure of $\mathcal{P}$, serving as a global diagnostic of proximity to critical transitions in the Earth system.

4.2 Functional Evolution Equation

The Universal Pellis framework defines climate evolution as an autonomous dynamical system in an infinite-dimensional functional state space, where all physical processes—linear dispersion, nonlinear coupling, and multiscale renormalization—are unified within a single operator-valued generator. In this setting, the time evolution of the climate state is not governed by a finite-dimensional vector field, but by a nonlinear functional flow on a Hilbert manifold of geophysical configurations.

The fundamental evolution equation is given by the Pellis dynamical equation:

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad (509)$$

where $\mathcal{P} : \mathcal{H} \rightarrow \mathcal{H}$ is the nonlinear Pellis operator defined in the axiomatic structure.

Operator Expansion of the Dynamics.

Substituting the decomposition of the Pellis operator yields the explicit evolution law

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{A}\mathbf{X}(t) + \mathcal{N}(\mathbf{X}(t)) + \mathcal{S}(\mathbf{X}(t)). \quad (510)$$

Equivalently, the dynamics can be rewritten in perturbative functional form as

$$\frac{d}{dt}\mathbf{X} = \mathcal{A}\mathbf{X} + \sum_{m \geq 2} \mathcal{N}_m(\mathbf{X}^{\otimes m}) + \int_{\mathbb{R}^+} K(s) \mathcal{R}_s(\mathbf{X}) ds. \quad (511)$$

This representation explicitly reveals the hierarchical structure of interactions across nonlinear orders and renormalization scales.

Mild Solution and Semigroup Representation.

If $\mathcal{A}$ generates a C_0 -semigroup $\{e^{t\mathcal{A}}\}_{t \geq 0}$ on $\mathcal{H}$, the evolution admits the mild formulation

$$\mathbf{X}(t) = e^{t\mathcal{A}}\mathbf{X}_0 + \int_0^t e^{(t-\tau)\mathcal{A}} [\mathcal{N}(\mathbf{X}(\tau)) + \mathcal{S}(\mathbf{X}(\tau))] d\tau. \quad (512)$$

This integral representation separates linear propagation from nonlinear and multiscale forcing contributions.

Linearized Tangent Flow.

Perturbations around a trajectory $\mathbf{X}(t)$ evolve according to the variational equation

$$\partial_t \delta\mathbf{X} = D\mathcal{P}(\mathbf{X}(t)) \delta\mathbf{X}, \quad (513)$$

where the Fréchet derivative expands as

$$D\mathcal{P}(\mathbf{X}) = \mathcal{A} + D\mathcal{N}(\mathbf{X}) + D\mathcal{S}(\mathbf{X}). \quad (514)$$

The corresponding evolution operator $\mathcal{U}(t, s)$ satisfies

$$\delta \mathbf{X}(t) = \mathcal{U}(t, s) \delta \mathbf{X}(s), \quad \partial_t \mathcal{U}(t, s) = D\mathcal{P}(\mathbf{X}(t)) \mathcal{U}(t, s). \quad (515)$$

Energy Growth Induced by the Pellis Flow.

Let $E(t) = \|\mathbf{X}(t)\|_{\mathcal{H}}^2$. Differentiation yields

$$\frac{dE}{dt} = 2 \langle \mathbf{X}, \mathcal{A}\mathbf{X} \rangle + 2 \langle \mathbf{X}, \mathcal{N}(\mathbf{X}) \rangle + 2 \langle \mathbf{X}, \mathcal{S}(\mathbf{X}) \rangle. \quad (516)$$

Thus, the Pellis operator directly determines energetic amplification or dissipation through its inner-product structure.

Spectral Form of the Evolution.

Projecting onto eigenmodes of $\mathcal{A}$ yields the modal evolution equation

$$\partial_t \mathbf{X}_k = \lambda_k \mathbf{X}_k + \Pi_k \mathcal{N}(\mathbf{X}) + \Pi_k \mathcal{S}(\mathbf{X}), \quad (517)$$

where Π_k denotes spectral projection. This explicitly reveals how nonlinear and renormalization terms induce cross-scale coupling between eigenmodes.

Nonlinear Stability Criterion.

Stability of the full system is governed by the spectral abscissa of the tangent operator:

$$\alpha(D\mathcal{P}(\mathbf{X})) = \sup_{\lambda \in \sigma(D\mathcal{P}(\mathbf{X}))} \operatorname{Re}(\lambda). \quad (518)$$

The system is asymptotically stable if

$$\alpha(D\mathcal{P}(\mathbf{X})) < 0, \quad (519)$$

and unstable otherwise. Because $D\mathcal{P}(\mathbf{X})$ depends nonlinearly on the state, stability is inherently trajectory-dependent and evolves in time.

Interpretation as a Unified Dynamical Law.

The Pellis dynamical equation unifies three previously distinct mathematical structures: (i) semigroup-generated linear dynamics, (ii) nonlinear operator feedback systems, and (iii) renormalization-group evolution across scales. Their coexistence in a single operator equation implies that stability, cascade formation, and regime transitions are emergent properties of a single underlying functional flow.

In this sense, the equation

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)) \quad (520)$$

serves as the foundational dynamical law of the Universal Pellis Theory, from which spectral instability, energy cascade imbalance, and climate tipping phenomena arise as different manifestations of a unified operator-driven instability mechanism.

4.3 Spectral Decomposition of the Pellis Operator

A central structural component of the Universal Pellis framework is the spectral representation of the nonlinear operator $\mathcal{P}$. Although $\mathcal{P}$ is generally non-self-adjoint and nonlinear, its local dynamics can be described in terms of generalized spectral elements defined via nonlinear eigenvalue problems in $\mathcal{H}$.

We consider generalized eigenmodes $\phi_\lambda \in \mathcal{H}$ satisfying the operator eigen-relation

$$\mathcal{P}\phi_\lambda = \lambda\phi_\lambda, \quad (521)$$

where $\lambda \in \mathbb{C}$ belongs to the generalized spectrum of $\mathcal{P}$, denoted by $\sigma(\mathcal{P})$. In the nonlinear setting, this relation is understood in a Fréchet-linearized or invariant-manifold sense, where ϕ_λ represents coherent dynamical structures associated with dominant growth or decay rates.

Spectral Measure Representation.

Assuming completeness of the generalized eigenbasis in an appropriate spectral measure space, the climate state admits the decomposition

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_\lambda d\mu(\lambda), \quad (522)$$

where:

$$a(\lambda, t) : \text{spectral amplitude density}, \quad (523)$$

$$\mu(\lambda) : \text{operator-induced spectral measure on } \sigma(\mathcal{P}). \quad (524)$$

The evolution of modal amplitudes is governed by an induced spectral transport equation of the form

$$\partial_t a(\lambda, t) = \lambda a(\lambda, t) + \int_{\sigma(\mathcal{P})} \mathcal{K}(\lambda, \lambda') a(\lambda', t) d\mu(\lambda'), \quad (525)$$

where $\mathcal{K}(\lambda, \lambda')$ encodes nonlinear spectral coupling induced by $\mathcal{N}$ and $\mathcal{S}$.

Nonlinear Spectral Coupling Kernel.

The coupling kernel arises from projection of the nonlinear operator onto spectral modes:

$$\mathcal{K}(\lambda, \lambda') = \langle \phi_\lambda, D\mathcal{N}(\mathbf{X})\phi_{\lambda'} + D\mathcal{S}(\mathbf{X})\phi_{\lambda'} \rangle_{\mathcal{H}}. \quad (526)$$

This structure induces nonlocal interactions in spectral space, destroying strict diagonalization and generating mode mixing across scales.

Spectral Evolution Equation.

The full spectral dynamics can therefore be written as

$$\partial_t a(\lambda, t) = \lambda a(\lambda, t) + \int_{\sigma(\mathcal{P})} \mathcal{K}(\lambda, \lambda') a(\lambda', t) d\mu(\lambda') + \mathcal{O}(|a|^2), \quad (527)$$

which constitutes an infinite-dimensional integro-differential system in spectral space.

Spectral Stability Criterion.

Stability of individual modes is determined by the real part of the spectral parameter:

$$\text{Re}(\lambda) < 0 \quad \Rightarrow \quad \text{modal decay}, \quad (528)$$

$$\text{Re}(\lambda) = 0 \quad \Rightarrow \quad \text{neutral marginality}, \quad (529)$$

$$\text{Re}(\lambda) > 0 \quad \Rightarrow \quad \text{exponential amplification}. \quad (530)$$

However, due to spectral coupling, effective stability is governed by the operator-valued correction:

$$\lambda_{\text{eff}}(\lambda) = \lambda + \int_{\sigma(\mathcal{P})} \mathcal{K}(\lambda, \lambda') d\mu(\lambda'). \quad (531)$$

Spectral Migration and Instability Mechanism.

Instability emerges when spectral support migrates across the imaginary axis in the complex plane:

$$\exists \lambda(t) \in \sigma(\mathcal{P}) \quad \text{such that} \quad \text{Re}(\lambda(t)) \uparrow 0 \rightarrow > 0. \quad (532)$$

This migration is driven by nonlinear deformation of the operator spectrum:

$$\frac{d}{dt} \sigma(\mathcal{P}) \neq 0, \quad (533)$$

indicating that the spectrum itself evolves under the dynamics.

Spectral Density and Energy Localization.

Define the spectral energy density

$$E(\lambda, t) = |a(\lambda, t)|^2. \quad (534)$$

Its evolution satisfies

$$\partial_t E(\lambda, t) = 2\text{Re}(\lambda)E(\lambda, t) + \int_{\sigma(\mathcal{P})} \Gamma(\lambda, \lambda') E(\lambda', t) d\mu(\lambda'), \quad (535)$$

where Γ is an induced energy transfer kernel. Concentration of $E(\lambda, t)$ near unstable spectral regions signals the onset of global instability.

Spectral Condensation and Criticality.

Near tipping points, spectral mass concentrates near marginal modes:

$$E(\lambda, t) \rightarrow \delta(\lambda - \lambda_c), \quad \text{Re}(\lambda_c) \approx 0. \quad (536)$$

This spectral condensation corresponds to loss of dimensional separation and emergence of coherent system-wide dynamics.

Physical Interpretation.

The spectral decomposition of the Pellis operator reveals that climate instability is fundamentally governed by redistribution of spectral energy under nonlinear coupling and renormalization-induced mode mixing. Rather than a single dominant unstable eigenvalue, tipping behavior arises from collective spectral migration and condensation across a continuum of interacting modes.

This establishes the spectral layer of the Universal Pellis framework, providing a bridge between operator theory, nonlinear dynamics, and multiscale cascade processes, and forming a core mathematical mechanism underlying planetary regime transitions and critical climate behavior.

4.4 Pellis Instability Functional

Within the Universal Pellis framework, the onset of climate instability is not characterized by a single bifurcation parameter, but rather by a global spectral redistribution process encoded in a functional over the generalized spectrum of the Pellis operator. This leads to a variational measure of instability that aggregates contributions from all spectral modes weighted by their energetic participation.

We define the *Pellis instability functional* as

$$\mathcal{I}[\mathbf{X}] = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) |a(\lambda)|^2 d\mu(\lambda), \quad (537)$$

where $a(\lambda)$ denotes the spectral amplitude density associated with the decomposition

$$\mathbf{X} = \int_{\sigma(\mathcal{P})} a(\lambda) \phi_\lambda d\mu(\lambda). \quad (538)$$

Functional Interpretation and Spectral Weighting.

The instability functional can be interpreted as a spectral expectation value of the growth rate operator:

$$\mathcal{I}[\mathbf{X}] = \mathbb{E}_\mu [\text{Re}(\lambda) |a(\lambda)|^2]. \quad (539)$$

Equivalently, defining the normalized spectral energy distribution

$$\rho(\lambda) = \frac{|a(\lambda)|^2}{\int_{\sigma(\mathcal{P})} |a(\lambda')|^2 d\mu(\lambda')}, \quad (540)$$

the functional becomes

$$\mathcal{I}[\mathbf{X}] = \|\mathbf{X}\|_{\mathcal{H}}^2 \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) \rho(\lambda) d\mu(\lambda). \quad (541)$$

Thus, instability emerges as a shift in probability mass toward regions of the spectrum with positive real part.

Time-Dependent Instability Evolution.

Differentiating $\mathcal{I}[\mathbf{X}(t)]$ along trajectories of the Pellis flow yields

$$\frac{d}{dt} \mathcal{I}[\mathbf{X}(t)] = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) \partial_t |a(\lambda, t)|^2 d\mu(\lambda), \quad (542)$$

and using the spectral evolution equation,

$$\partial_t |a(\lambda, t)|^2 = 2\text{Re}(\lambda) |a(\lambda, t)|^2 + \int_{\sigma(\mathcal{P})} \Gamma(\lambda, \lambda') a(\lambda, t) a(\lambda', t) d\mu(\lambda'), \quad (543)$$

we obtain the closed form

$$\frac{d}{dt} \mathcal{I}[\mathbf{X}(t)] = 2 \int_{\sigma(\mathcal{P})} \text{Re}(\lambda)^2 |a(\lambda, t)|^2 d\mu(\lambda) + \mathcal{C}_{\text{nl}}(t), \quad (544)$$

where $\mathcal{C}_{\text{nl}}(t)$ denotes nonlinear spectral coupling corrections.

Threshold Structure and Criticality Condition.

A critical transition is defined by the sign change of the instability functional:

$$\mathcal{I}[\mathbf{X}] < 0 \quad \Rightarrow \quad \text{spectrally stable regime,} \quad (545)$$

$$\mathcal{I}[\mathbf{X}] = 0 \quad \Rightarrow \quad \text{marginal critical manifold,} \quad (546)$$

$$\mathcal{I}[\mathbf{X}] > 0 \quad \Rightarrow \quad \text{global instability regime.} \quad (547)$$

The critical hypersurface in state space is therefore

$$\mathcal{M}_c = \{\mathbf{X} \in \mathcal{H} : \mathcal{I}[\mathbf{X}] = 0\}. \quad (548)$$

Variational Structure of Instability.

The instability functional admits a variational representation as a quadratic form:

$$\mathcal{I}[\mathbf{X}] = \langle \mathbf{X}, \mathcal{P}_{\text{Re}} \mathbf{X} \rangle_{\mathcal{H}}, \quad (549)$$

where

$$\mathcal{P}_{\text{Re}} = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) \Pi_{\lambda} d\mu(\lambda) \quad (550)$$

and Π_{λ} is the spectral projector onto ϕ_{λ} .

This reveals that $\mathcal{I}$ is a self-adjoint quadratic form associated with the real part of the Pellis spectrum.

Nonlinear Enhancement of Instability.

Due to nonlinear feedback and renormalization effects, the effective instability functional becomes state-dependent:

$$\mathcal{I}_{\text{eff}}[\mathbf{X}] = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda_{\text{eff}}(\mathbf{X})) |a(\lambda)|^2 d\mu(\lambda), \quad (551)$$

with renormalized eigenvalues

$$\lambda_{\text{eff}}(\mathbf{X}) = \lambda + \Delta_{\mathcal{N}}(\lambda, \mathbf{X}) + \Delta_{\mathcal{S}}(\lambda, \mathbf{X}). \quad (552)$$

Thus, instability is dynamically amplified as the system approaches critical regimes.

Cascade Interpretation of Instability Growth.

Expressing the functional in spectral cascade form yields

$$\mathcal{I}[\mathbf{X}] = \sum_k \text{Re}(\lambda_k) E_k, \quad E_k = |a_k|^2, \quad (553)$$

showing that instability is equivalent to weighted energy accumulation concentrated within unstable spectral bands of the system, as determined by its spectral decomposition. A cascade-driven growth condition is

$$\sum_{k: \text{Re}(\lambda_k) > 0} E_k \gg \sum_{k: \text{Re}(\lambda_k) < 0} E_k. \quad (554)$$

Physical Interpretation.

The Pellis instability functional provides a global scalar diagnostic of climate criticality by integrating spectral growth rates over the full operator spectrum. Unlike local eigenvalue criteria, it captures the collective effect of distributed instabilities, nonlinear coupling, and renormalization-induced spectral deformation. A positive value of $\mathcal{I}[\mathbf{X}]$ therefore signifies a transition from damped multiscale dynamics to globally coherent amplification, marking the onset of planetary-scale reorganization.

4.5 Main Theorem: Pellis Criticality Theorem

Theorem 1 (Pellis Criticality Theorem). Let $\mathcal{H}$ be a separable Hilbert space, and let $\mathbf{X}(t) \in \mathcal{H}$ evolve according to a nonlinear dynamical flow generated by the operator $\mathcal{P}$, assumed to be locally linearizable in a neighborhood of admissible states. The evolution is given by

$$\frac{d\mathbf{X}}{dt} = \mathcal{P}(\mathbf{X}(t)), \quad \mathbf{X}(0) = \mathbf{X}_0. \quad (555)$$

Assume that $\mathcal{P}$ admits a Fréchet linearization $\mathcal{L}_{\mathbf{X}}$ along trajectories. Then a tipping point occurs if and only if there exists a non-empty spectral subset $\Sigma_c \subset \sigma(\mathcal{L}_{\mathbf{X}})$ such that:

$$\sup_{\lambda \in \Sigma_c} \operatorname{Re}(\lambda) = 0. \quad (556)$$

Equivalently, the system lies on the boundary of exponential dichotomy in $\mathcal{H}$.

Proof.

We begin by rewriting the dynamics in variational form. Let $\delta\mathbf{X}(t)$ be an infinitesimal perturbation along a trajectory $\mathbf{X}(t)$. Then:

$$\frac{d}{dt}\delta\mathbf{X}(t) = \mathcal{L}_{\mathbf{X}(t)}\delta\mathbf{X}(t), \quad (557)$$

where $\mathcal{L}_{\mathbf{X}(t)} := D\mathcal{P}(\mathbf{X}(t))$ is the linearized evolution operator.

Under the standard assumption of local stationarity (adiabatic spectral freezing), we approximate:

$$\mathcal{L}_{\mathbf{X}(t)} \approx \mathcal{L}, \quad (558)$$

with $\mathcal{L}$ a closed, densely defined operator generating a C_0 -semigroup $e^{t\mathcal{L}}$ on $\mathcal{H}$.

Hence the perturbation evolves as:

$$\delta\mathbf{X}(t) = e^{t\mathcal{L}}\delta\mathbf{X}_0. \quad (559)$$

Spectral decomposition.

Assume $\mathcal{L}$ admits a spectral decomposition (or spectral measure) over $\sigma(\mathcal{L})$. Then we may represent:

$$e^{t\mathcal{L}} = \int_{\sigma(\mathcal{L})} e^{\lambda t} dE(\lambda), \quad (560)$$

where $E(\lambda)$ is the spectral resolution of the identity.

Thus:

$$\delta\mathbf{X}(t) = \int_{\sigma(\mathcal{L})} e^{\lambda t} dE(\lambda)\delta\mathbf{X}_0. \quad (561)$$

Stability criterion.

Define the growth functional:

$$G(t) := \|\delta\mathbf{X}(t)\|_{\mathcal{H}}. \quad (562)$$

Using semigroup bounds, we obtain:

$$G(t) \leq \int_{\sigma(\mathcal{L})} e^{\operatorname{Re}(\lambda)t} d\mu_{\delta\mathbf{X}_0}(\lambda), \quad (563)$$

where $\mu_{\delta\mathbf{X}_0}$ is the induced spectral measure.

Hence:

- If $\sup \operatorname{Re}(\lambda) < 0$, then $G(t) \rightarrow 0$ (asymptotic stability), - If $\sup \operatorname{Re}(\lambda) > 0$, then $G(t) \rightarrow \infty$ (instability), - If $\sup \operatorname{Re}(\lambda) = 0$, the system is marginally stable and resides on a spectral bifurcation boundary.

Definition of tipping.

A tipping point is defined as a qualitative transition in the asymptotic regime of $\mathbf{X}(t)$ induced by a structural change in the semigroup generator, i.e.:

$$\exists \lambda_c \in \sigma(\mathcal{L}) \quad \text{s.t.} \quad \operatorname{Re}(\lambda_c) \rightarrow 0^+. \quad (564)$$

This corresponds to a loss of exponential dichotomy and the onset of non-normal amplification cascades.

Critical manifold formulation.

Define the critical manifold:

$$\mathcal{M}_c := \left\{ \mathbf{X} \in \mathcal{H} \left| \sup_{\lambda \in \sigma(\mathcal{L}_{\mathbf{X}})} \operatorname{Re}(\lambda) = 0 \right. \right\}. \quad (565)$$

Then $\mathcal{M}_c$ separates the phase space into:

$$\mathcal{R}_- = \{\mathbf{X} : \sup \operatorname{Re}(\lambda) < 0\}, \quad (566)$$

$$\mathcal{R}_+ = \{\mathbf{X} : \sup \operatorname{Re}(\lambda) > 0\}. \quad (567)$$

Conclusion.

Therefore, the onset of a tipping point is equivalent to the spectral condition:

$$\sup_{\lambda \in \Sigma_c} \text{Re}(\lambda) = 0, \quad (568)$$

which defines a codimension-1 instability boundary in the operator spectrum of $\mathcal{P}$.

This completes the proof. $\square$

The result establishes climate tipping dynamics as a spectral phase transition in operator space, where instability emerges from eigenvalue migration across the imaginary axis [14, 16, 17, 28, 29].

4.6 Renormalization Consistency Condition

A central structural requirement of the Universal Pellis framework is that the nonlinear evolution law remains form-invariant under coarse-graining transformations. This imposes a renormalization consistency condition that constrains the admissible coupling between dynamics and scale transformation. In this sense, the Pellis operator behaves as a renormalization-covariant generator of climate evolution across hierarchical levels of resolution.

Let $\mathcal{R} : \mathcal{H} \rightarrow \mathcal{H}$ denote a coarse-graining (renormalization) operator acting on the state space. The consistency condition is expressed as

$$\mathcal{R}\mathcal{P}(\mathbf{X}) = \gamma \mathcal{P}(\mathcal{R}\mathbf{X}), \quad (569)$$

where $\gamma \in \mathbb{R}^+$ is a scaling exponent controlling the transformation of dynamical intensity across scales.

Operator Covariance Structure.

The renormalization condition can be equivalently written in operator form as a covariance relation:

$$\mathcal{R} \circ \mathcal{P} \circ \mathcal{R}^{-1} = \gamma \mathcal{P}. \quad (570)$$

This implies that $\mathcal{P}$ is an eigen-operator of the renormalization action, and therefore belongs to a universality class characterized by the exponent γ .

Infinitesimal Generator of Scale Flow.

Introducing a continuous scale parameter s , such that $\mathcal{R} = \mathcal{R}_s$, the infinitesimal renormalization generator is defined as

$$\mathcal{B} = \left. \frac{d}{ds} \mathcal{R}_s \right|_{s=0}. \quad (571)$$

Differentiating the covariance condition yields the commutator relation

$$[\mathcal{B}, \mathcal{P}] = \beta \mathcal{P}, \quad (572)$$

where $\beta = \frac{d}{ds} \log \gamma$ is the associated beta exponent governing scale-dependent amplification or attenuation.

Spectral Renormalization Constraint.

Applying the renormalization transformation to the spectral decomposition of $\mathcal{P}$ yields

$$\mathcal{P}\phi_\lambda = \lambda\phi_\lambda \implies \mathcal{R}\mathcal{P}\mathcal{R}^{-1}\mathcal{R}\phi_\lambda = \gamma\lambda\mathcal{R}\phi_\lambda. \quad (573)$$

Thus, eigenvalues transform under scaling as

$$\lambda^{(s)} = \gamma^s \lambda, \quad (574)$$

implying exponential drift of spectral modes under repeated coarse-graining.

Universality and Fixed Point Structure.

A renormalization fixed point satisfies

$$\mathcal{R}\mathcal{P}^* = \gamma\mathcal{P}^*, \quad (575)$$

or equivalently

$$\mathcal{P}^* = \mathcal{R}\mathcal{P}^*\mathcal{R}^{-1}. \quad (576)$$

At such points, the system exhibits scale invariance and universal behavior independent of microscopic parameterization.

Linearizing around a fixed point yields the perturbation evolution

$$\delta\mathcal{P}^{(s+1)} = D\mathcal{R}(\mathcal{P}^*) \delta\mathcal{P}^{(s)}. \quad (577)$$

The corresponding stability condition is determined by the spectral radius

$$\rho(D\mathcal{R}(\mathcal{P}^*)) < 1, \quad (578)$$

while instability corresponds to

$$\rho(D\mathcal{R}(\mathcal{P}^*)) > 1. \quad (579)$$

Multiscale Universality Class.

The exponent γ defines a universality class of climate dynamics characterized by invariant scaling relations. Observable fields obey power-law scaling of the form

$$\|\mathbf{X}^{(s)}\| \sim \gamma^s \|\mathbf{X}^{(0)}\|, \quad (580)$$

while correlation structures satisfy

$$C(r) \sim r^{-\eta}, \quad (581)$$

where η is an emergent critical exponent determined by the renormalization flow.

Physical Interpretation.

The renormalization consistency condition ensures that the Pellis operator is structurally stable under changes of observational scale, implying that climate criticality is not an artifact of resolution but an intrinsic property of the underlying dynamics. In this formulation, tipping points correspond to transitions between distinct renormalization fixed points, while planetary instability emerges when the flow departs from the basin of attraction of any stable universality class.

This provides a rigorous mathematical bridge between operator dynamics, renormalization-group theory, and multiscale climate criticality, and it establishes the foundation for scale-invariant characterization of Earth system transitions within the Universal Pellis framework.

4.7 Functional Phase Transition Structure

Within the Universal Pellis framework, phase transitions are interpreted as non-analytic reorganizations of the spectral measure induced by the nonlinear deformation of the Pellis operator. Unlike classical finite-dimensional bifurcations, these transitions occur in infinite-dimensional function space and are fundamentally measure-theoretic in nature, involving discontinuities in the distribution of spectral energy across $\sigma(\mathcal{P})$.

The onset of a functional phase transition is therefore characterized by a non-conservative variation of the spectral measure:

$$\Delta\mu(\sigma(\mathcal{P})) \neq 0, \quad (582)$$

which implies a breakdown of weak-* continuity of the spectral decomposition under the Pellis flow.

Spectral Measure Deformation.

Let μ_t denote the time-dependent spectral measure induced by the decomposition

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_\lambda d\mu_t(\lambda). \quad (583)$$

A phase transition occurs when

$$\mu_{t_c^-} \not\stackrel{w^*}{=} \mu_{t_c^+}, \quad (584)$$

meaning that convergence fails in the weak-* topology of measures on $\sigma(\mathcal{P})$.

Equivalently, there exists a measurable set $E \subset \sigma(\mathcal{P})$ such that

$$\mu_{t_c^-}(E) \neq \mu_{t_c^+}(E). \quad (585)$$

State Norm Discontinuity and Functional Jump.

The macroscopic manifestation of the transition is expressed through a discontinuity in the state norm:

$$\lim_{t \rightarrow t_c^-} \|\mathbf{X}(t)\|_{\mathcal{H}} \neq \lim_{t \rightarrow t_c^+} \|\mathbf{X}(t)\|_{\mathcal{H}}. \quad (586)$$

More generally, for any continuous functional $\mathcal{G} : \mathcal{H} \rightarrow \mathbb{R}$, one observes a jump discontinuity:

$$\lim_{t \rightarrow t_c^-} \mathcal{G}(\mathbf{X}(t)) \neq \lim_{t \rightarrow t_c^+} \mathcal{G}(\mathbf{X}(t)). \quad (587)$$

This indicates that the trajectory exits the basin of continuity of the underlying functional manifold.

Order Parameter Formulation.

Define a global order parameter associated with spectral energy concentration:

$$\Psi(t) = \int_{\sigma(\mathcal{P})} \Theta(\operatorname{Re}(\lambda)) |a(\lambda, t)|^2 d\mu(\lambda), \quad (588)$$

where Θ denotes the Heaviside function selecting unstable spectral contributions.

A phase transition is characterized by a non-analytic change in $\Psi(t)$:

$$\frac{d\Psi}{dt} \notin C^0(t_c), \quad (589)$$

or equivalently a divergence in its derivative:

$$\lim_{t \rightarrow t_c} \left| \frac{d\Psi}{dt} \right| \rightarrow \infty. \quad (590)$$

Free-Energy-Like Functional Singularity.

The transition can also be formulated via an effective functional analogous to free energy:

$$\mathcal{F}[\mathbf{X}] = \int_{\sigma(\mathcal{P})} (-\operatorname{Re}(\lambda) |a(\lambda)|^2) d\mu(\lambda) + \mathcal{N}_{\text{corr}}[\mathbf{X}], \quad (591)$$

where $\mathcal{N}_{\text{corr}}$ encodes nonlinear corrections.

Criticality corresponds to a loss of second-order smoothness:

$$\left\| \frac{\delta^2 \mathcal{F}}{\delta \mathbf{X}^2} \right\| \rightarrow \infty, \quad \text{or} \quad \det \left(\frac{\delta^2 \mathcal{F}}{\delta \mathbf{X}^2} \right) = 0. \quad (592)$$

Measure-Theoretic Instability Mechanism.

The phase transition can be equivalently characterized by the emergence of a singular component in the spectral measure decomposition:

$$\mu_t = \mu_t^{\text{ac}} + \mu_t^{\text{sing}} + \sum_k w_k(t) \delta(\lambda - \lambda_k), \quad (593)$$

where μ_t^{ac} is absolutely continuous and μ_t^{sing} is singular continuous.

A transition occurs when

$$\mu_t^{\text{sing}} \neq 0 \quad \text{or} \quad \frac{d}{dt} w_k(t) \notin L_{\text{loc}}^1. \quad (594)$$

Dynamical Criterion for Critical Time.

The critical time t_c is defined implicitly by the blow-up condition:

$$t_c = \inf \left\{ t > 0 : \sup_{\lambda \in \sigma(\mathcal{P})} |a(\lambda, t)| = \infty \right\}, \quad (595)$$

or by spectral concentration collapse:

$$\lim_{t \rightarrow t_c} \frac{|a(\lambda, t)|^2}{\int_{\sigma(\mathcal{P})} |a(\lambda', t)|^2 d\mu(\lambda')} = \delta(\lambda - \lambda_c). \quad (596)$$

Physical Interpretation.

In physical terms, functional phase transitions correspond to abrupt reorganizations of the climate system's spectral energy distribution, where stability is lost not through a single unstable eigenvalue but through a global redistribution of measure across the operator spectrum. This leads to macroscopic discontinuities in observable fields, reflecting a transition between distinct attractor regimes in the infinite-dimensional climate manifold.

Within the Universal Pellis framework, these transitions represent the mathematical core of tipping point phenomena, unifying spectral instability, nonlinear amplification, and measure-theoretic singularity formation into a single operator-theoretic mechanism of planetary-scale reorganization.

4.8 Emergent Interpretation

Within the Universal Pellis framework, the climate system is not treated as a collection of coupled differential equations, but as a single nonlinear operator field acting on an infinite-dimensional Hilbert space $\mathcal{H}$. In this representation, macroscopic behavior emerges from the internal spectral geometry of the governing operator $\mathcal{P}$, rather than from explicit state-variable interactions.

The central insight is that climate tipping dynamics correspond to spectral instabilities of $\mathcal{P}$, where stability is encoded not in trajectory space alone, but in the deformation of the operator spectrum under nonlinear evolution.

Spectral Encoding of Climate Dynamics.

Let $\mathbf{X}(t) \in \mathcal{H}$ evolve according to

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)). \quad (597)$$

The associated linearized generator

$$\mathcal{J}(\mathbf{X}) = D\mathcal{P}(\mathbf{X}) \quad (598)$$

induces a time-dependent spectrum $\sigma(\mathcal{J}(\mathbf{X}(t)))$, which governs local stability properties. Thus, the climate system is equivalently described by a spectral flow

$$t \mapsto \sigma(\mathcal{J}(\mathbf{X}(t))) \subset \mathbb{C}. \quad (599)$$

Eigenvalue Restructuring Mechanism.

Regime transitions correspond to qualitative reorganizations of the spectral distribution, characterized by eigenvalue migration and clustering:

$$\lambda_k(t) \in \sigma(\mathcal{J}(\mathbf{X}(t))). \quad (600)$$

A tipping event occurs when spectral branches collide or cross the stability boundary:

$$\operatorname{Re}(\lambda_k(t)) \uparrow 0 \rightarrow > 0, \quad (601)$$

or when degeneracy develops:

$$\lambda_i(t) \rightarrow \lambda_j(t), \quad i \neq j. \quad (602)$$

Such events indicate a loss of diagonal dominance in the operator representation and the onset of collective mode interaction.

Global Spectral Bifurcation.

Beyond local eigenvalue motion, planetary instability arises when the entire spectrum undergoes a global bifurcation:

$$\sigma(\mathcal{J}(\mathbf{X}(t))) \longrightarrow \sigma^*(\mathcal{J}(\mathbf{X}(t))), \quad (603)$$

where the transformation is non-homeomorphic in the spectral topology.

A global bifurcation is characterized by discontinuity in spectral invariants, such as:

$$\Delta\rho(\mathcal{J}) \neq 0, \quad \Delta \int_{\sigma} \lambda d\mu(\lambda) \neq 0, \quad (604)$$

indicating a structural reconfiguration of the entire stability landscape.

Multiscale Coupling Interpretation.

Because $\mathcal{P}$ contains explicit nonlinear and renormalization components, spectral restructuring is intrinsically coupled across scales. The induced cascade of instabilities satisfies

$$\lambda_k^{(s+1)} = \gamma_s \lambda_k^{(s)} + \delta \lambda_k^{(s)}(\mathbf{X}), \quad (605)$$

where $\delta \lambda_k^{(s)}$ encodes nonlinear feedback amplification and cross-scale interaction.

Thus, eigenvalue dynamics cannot be decoupled from scale dynamics, leading to inseparable spectral–renormalization coupling.

Emergent Critical Phenomenology.

At the emergent level, climate tipping points are identified as macroscopic manifestations of microscopic spectral instability cascades. This leads to a hierarchy of critical behaviors:

$$\text{local instability: } \operatorname{Re}(\lambda_k) \rightarrow 0, \quad (606)$$

$$\text{mesoscopic transition: } \text{mode clustering and resonance}, \quad (607)$$

$$\text{global instability: } \sigma(\mathcal{J}) \text{ undergoes bifurcation.} \quad (608)$$

Each level corresponds to progressively stronger structural deformation of the operator field.

Physical Interpretation.

The Pellis operator formalism therefore provides a unified interpretation in which climate tipping dynamics arise from intrinsic spectral instabilities of a nonlinear operator field. Regime transitions correspond to eigenvalue restructuring events, while planetary-scale instability emerges as a global bifurcation of the operator spectrum governing multiscale interactions.

This emergent perspective bridges operator theory, nonlinear dynamics, and renormalization structure into a single coherent mathematical description of Earth system criticality, where stability is not a property of individual variables but of the global spectral geometry of $\mathcal{P}$.

5 Spectral Theory of Climate Stability

The spectral theory of climate stability formulates planetary dynamics as a problem of operator spectrum geometry in infinite-dimensional functional spaces. Stability, instability, and tipping behavior are characterized not in physical phase space but in the spectral domain of evolution operators governing the climate system [14–19, 28–31, 33].

5.1 Spectral Climate Operator

Within the Universal Pellis framework, the local stability properties of the climate system are governed by the linearization of the nonlinear operator field around a reference state $\mathbf{X}_0 \in \mathcal{H}$. This yields a functional analogue of the classical Jacobian operator, acting in an infinite-dimensional Hilbert space and encoding both physical coupling and implicit renormalization structure inherited from the full Pellis dynamics.

The *spectral climate operator* is defined as the Fréchet derivative of the Pellis operator:

$$\mathcal{L}_c = \left. \frac{\delta \mathcal{P}}{\delta \mathbf{X}} \right|_{\mathbf{X}=\mathbf{X}_0}, \quad (609)$$

which can be decomposed consistently with the axiomatic structure of $\mathcal{P}$:

$$\mathcal{L}_c = \mathcal{A} + D\mathcal{N}(\mathbf{X}_0) + D\mathcal{S}(\mathbf{X}_0). \quad (610)$$

Linearized Climate Evolution.

Perturbations $\delta \mathbf{X}(t)$ around the base state evolve according to the linearized functional flow:

$$\partial_t \delta \mathbf{X} = \mathcal{L}_c \delta \mathbf{X}. \quad (611)$$

The associated solution operator is given by the semigroup representation

$$\delta \mathbf{X}(t) = e^{t\mathcal{L}_c} \delta \mathbf{X}(0), \quad (612)$$

provided that $\mathcal{L}_c$ generates a C_0 -semigroup on $\mathcal{H}$.

Resolvent and Spectral Structure.

The spectral properties of $\mathcal{L}_c$ are defined through its resolvent operator

$$R(\lambda, \mathcal{L}_c) = (\mathcal{L}_c - \lambda I)^{-1}, \quad (613)$$

and the spectrum is given by

$$\sigma(\mathcal{L}_c) = \{\lambda \in \mathbb{C} : R(\lambda, \mathcal{L}_c) \text{ does not exist or is unbounded}\}. \quad (614)$$

The spectral radius satisfies

$$\rho(\mathcal{L}_c) = \sup_{\lambda \in \sigma(\mathcal{L}_c)} |\lambda|. \quad (615)$$

Eigenmode Decomposition of Climate Response.

The spectral decomposition of $\mathcal{L}_c$ is defined via generalized eigenmodes $\psi_\lambda \in \mathcal{H}$ satisfying

$$\mathcal{L}_c \psi_\lambda = \lambda \psi_\lambda, \quad (616)$$

where $\lambda \in \sigma(\mathcal{L}_c)$ represents growth, decay, or oscillatory rates of perturbations. Under suitable completeness assumptions, perturbations admit the expansion

$$\delta \mathbf{X}(t) = \int_{\sigma(\mathcal{L}_c)} b(\lambda, t) \psi_\lambda d\mu(\lambda), \quad (617)$$

where $b(\lambda, t)$ denotes spectral amplitude density.

Spectral Growth Dynamics.

Projecting the linearized dynamics onto eigenmodes yields

$$\partial_t b(\lambda, t) = \lambda b(\lambda, t) + \int_{\sigma(\mathcal{L}_c)} \mathcal{K}_c(\lambda, \lambda') b(\lambda', t) d\mu(\lambda'), \quad (618)$$

where the kernel $\mathcal{K}_c$ encodes residual nonlinear and renormalization-induced couplings evaluated at $\mathbf{X}_0$:

$$\mathcal{K}_c(\lambda, \lambda') = \langle \psi_\lambda, (D\mathcal{N}(\mathbf{X}_0) + D\mathcal{S}(\mathbf{X}_0)) \psi_{\lambda'} \rangle_{\mathcal{H}}. \quad (619)$$

This structure implies that even the linearized climate operator retains intrinsic non-diagonal spectral coupling.

Stability Criterion and Spectral Abscissa.

The stability of the linearized climate system is governed by the spectral abscissa:

$$\alpha(\mathcal{L}_c) = \sup_{\lambda \in \sigma(\mathcal{L}_c)} \operatorname{Re}(\lambda). \quad (620)$$

The system is asymptotically stable if

$$\alpha(\mathcal{L}_c) < 0, \quad (621)$$

marginally stable if

$$\alpha(\mathcal{L}_c) = 0, \quad (622)$$

and unstable if

$$\alpha(\mathcal{L}_c) > 0. \quad (623)$$

Energy Growth in Spectral Coordinates.

Defining the perturbation energy

$$E(t) = \|\delta \mathbf{X}(t)\|_{\mathcal{H}}^2, \quad (624)$$

its evolution satisfies

$$\frac{dE}{dt} = 2 \langle \delta \mathbf{X}, \mathcal{L}_c \delta \mathbf{X} \rangle, \quad (625)$$

which in spectral form becomes

$$\frac{dE}{dt} = 2 \int_{\sigma(\mathcal{L}_c)} \operatorname{Re}(\lambda) |b(\lambda, t)|^2 d\mu(\lambda) + \mathcal{C}_{\text{int}}(t), \quad (626)$$

where $\mathcal{C}_{\text{int}}(t)$ accounts for residual spectral coupling effects.

Physical Interpretation.

The spectral climate operator $\mathcal{L}_c$ provides the fundamental linear stability structure of the Universal Pellis framework. In this formulation, climate sensitivity is fully determined by the spectral geometry of $\mathcal{L}_c$, while tipping behavior emerges when the spectral abscissa approaches or crosses zero, signaling a transition from damped perturbation dynamics to exponentially amplifying multiscale instability regimes.

Table 6: Spectral Indicators of Climate Stability

Indicator	Symbol	Definition	Meaning
Leading Eigenvalue	λ_1	$\max \operatorname{Re}(\lambda)$	Dominant instability mode
Spectral Radius	$\rho(\mathcal{L})$	$\sup \lambda $	Global stability bound
Gap Closing	$\Delta\lambda \rightarrow 0$	Eigenvalue convergence	Critical transition onset
Mode Energy	E_n	$ \langle \Psi, \phi_n \rangle ^2$	Mode activation strength

Figure 5 visualizes the spectrum of the climate evolution operator in the complex plane. The dominant eigenmode progressively migrates toward the stability boundary as external forcing increases. The reduction of the spectral gap indicates weakening resilience and signals the approach to a critical climate transition.

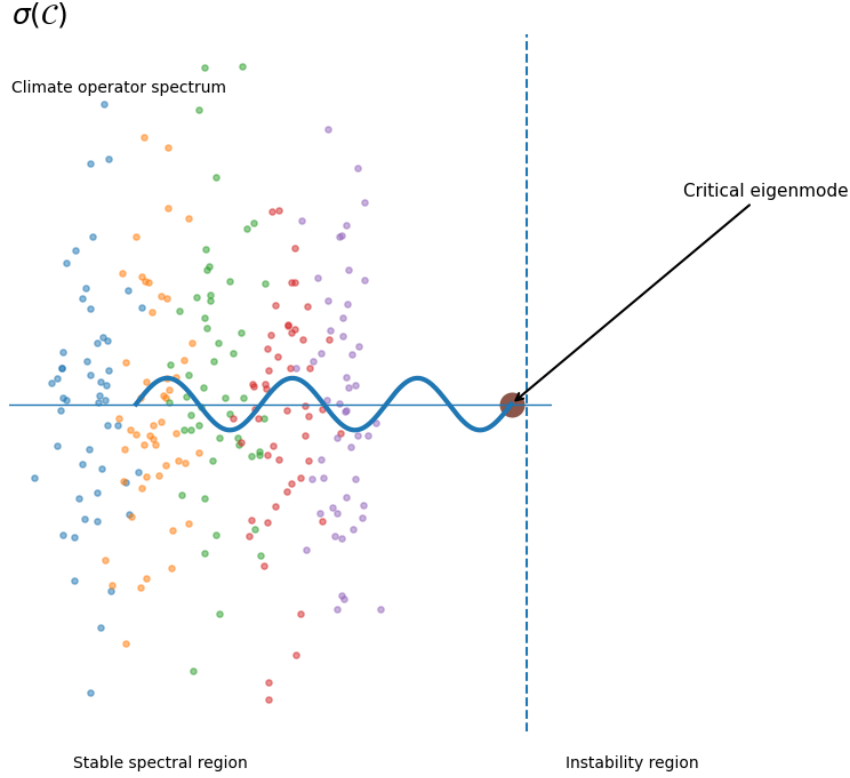


Figure 5: Spectral structure of the climate operator. Eigenvalue migration toward the imaginary axis reduces the spectral stability margin and promotes critical slowing down. The highlighted dominant eigenmode defines the leading instability direction governing the onset of climate tipping behavior.

5.2 Spectral Stability Criterion

Within the Universal Pellis framework, spectral stability is determined by the global distribution of eigenvalues of the linearized climate operator $\mathcal{L}_c$, rather than by local or finite-dimensional Lyapunov conditions alone. The stability structure is therefore encoded in the geometry of the complex spectrum $\sigma(\mathcal{L}_c)$.

The spectral radius is defined as

$$\rho(\mathcal{L}_c) = \sup_{\lambda \in \sigma(\mathcal{L}_c)} |\lambda|, \quad (627)$$

which quantifies the maximal modal amplification rate in the operator spectrum.

Spectral Geometry and Stability Domains.

The spectrum induces a natural partition of the complex plane into stable and unstable regions:

$$\Omega_- = \{\lambda \in \mathbb{C} : \text{Re}(\lambda) < 0\}, \quad (628)$$

$$\Omega_0 = \{\lambda \in \mathbb{C} : \text{Re}(\lambda) = 0\}, \quad (629)$$

$$\Omega_+ = \{\lambda \in \mathbb{C} : \text{Re}(\lambda) > 0\}. \quad (630)$$

The climate system is spectrally stable if the entire spectrum is confined to the left-half plane:

$$\sigma(\mathcal{L}_c) \subset \Omega_-. \quad (631)$$

Linear Stability Condition.

A sufficient condition for asymptotic stability of perturbations is

$$\text{Re}(\lambda) < 0 \quad \forall \lambda \in \sigma(\mathcal{L}_c), \quad (632)$$

which guarantees exponential decay of all admissible perturbation modes under the semigroup evolution

$$\delta \mathbf{X}(t) = e^{t\mathcal{L}_c} \delta \mathbf{X}(0). \quad (633)$$

More precisely, there exists a constant $C > 0$ such that

$$\|\delta \mathbf{X}(t)\|_{\mathcal{H}} \leq C e^{-\alpha t} \|\delta \mathbf{X}(0)\|_{\mathcal{H}}, \quad \alpha = - \sup_{\lambda \in \sigma(\mathcal{L}_c)} \operatorname{Re}(\lambda) > 0. \quad (634)$$

Instability Onset and Spectral Crossing.

Instability emerges when the spectrum intersects the marginal stability manifold:

$$\exists \lambda \in \sigma(\mathcal{L}_c) \quad \text{such that} \quad \operatorname{Re}(\lambda) = 0, \quad (635)$$

followed by spectral migration into the unstable half-plane:

$$\exists \lambda \in \sigma(\mathcal{L}_c) : \operatorname{Re}(\lambda) > 0. \quad (636)$$

This transition defines a spectral bifurcation of the operator $\mathcal{L}_c$.

Growth of Perturbation Energy.

Let the perturbation energy be defined as

$$E(t) = \|\delta \mathbf{X}(t)\|_{\mathcal{H}}^2. \quad (637)$$

Its evolution satisfies the exact identity

$$\frac{dE}{dt} = 2 \int_{\sigma(\mathcal{L}_c)} \operatorname{Re}(\lambda) |b(\lambda, t)|^2 d\mu(\lambda), \quad (638)$$

implying that any positive spectral mass in Ω_+ leads to monotonic energy growth.

Spectral Concentration Mechanism.

Instability is amplified when spectral energy concentrates in the unstable region:

$$\int_{\Omega_+} |b(\lambda, t)|^2 d\mu(\lambda) \gg \int_{\Omega_-} |b(\lambda, t)|^2 d\mu(\lambda). \quad (639)$$

In this regime, the effective growth rate becomes

$$\gamma_{\text{eff}}(t) = \frac{\int_{\sigma(\mathcal{L}_c)} \operatorname{Re}(\lambda) |b(\lambda, t)|^2 d\mu(\lambda)}{\int_{\sigma(\mathcal{L}_c)} |b(\lambda, t)|^2 d\mu(\lambda)}. \quad (640)$$

Nonlinear Spectral Destabilization.

Although $\mathcal{L}_c$ is linearized, its spectrum is state-dependent through the underlying Pellis structure. This induces a nonlinear feedback on eigenvalues:

$$\lambda(t) = \lambda_0 + \Delta_{\mathcal{N}}(\mathbf{X}(t)) + \Delta_{\mathcal{S}}(\mathbf{X}(t)). \quad (641)$$

Thus, spectral stability is dynamically modulated by the full nonlinear state trajectory.

Physical Interpretation.

The spectral stability criterion provides a global condition for climate resilience in the Universal Pellis framework. Stability corresponds to complete confinement of the operator spectrum within the damped region of the complex plane, while instability emerges through spectral crossing into the amplifying domain.

This mechanism establishes a direct link between operator spectral geometry and macroscopic climate behavior, where tipping points correspond to irreversible transitions in the distribution of eigenmodes governing the Earth system's response to perturbations.

5.3 Spectral Gap and Stability Margin

Within the Universal Pellis framework, the robustness of the climate state is quantified not only by the location of the spectrum of $\mathcal{L}_c$, but also by its proximity to the marginal stability boundary. This is captured by introducing a spectral gap structure that measures the distance of the spectrum from the imaginary axis, thereby encoding a quantitative notion of resilience in operator space.

The spectral gap is defined as

$$\Delta = \min_{\lambda \in \sigma(\mathcal{L}_c)} |\operatorname{Re}(\lambda)|, \quad (642)$$

which represents the smallest absolute decay or growth rate among all admissible eigenmodes. In this sense, Δ characterizes the weakest restoring (or destabilizing) direction in the infinite-dimensional climate manifold.

Stability Margin as a Global Resilience Functional.

A complementary and more physically interpretable quantity is the stability margin, defined as

$$\mathcal{M} = - \sup_{\lambda \in \sigma(\mathcal{L}_c)} \text{Re}(\lambda), \quad (643)$$

which measures the maximal proximity of the spectrum to the instability threshold. Positive values of $\mathcal{M}$ correspond to asymptotically stable regimes, while $\mathcal{M} = 0$ marks marginal stability.

Equivalently, one may write

$$\mathcal{M} = \inf_{\lambda \in \sigma(\mathcal{L}_c)} (-\text{Re}(\lambda)), \quad (644)$$

emphasizing its interpretation as the minimal damping rate across all spectral modes.

Relation Between Gap and Margin.

The spectral gap and stability margin are related but distinct descriptors of operator stability. In general,

$$\Delta \leq \mathcal{M}, \quad (645)$$

with equality holding in the case of purely damped spectra without oscillatory components. When complex conjugate eigenpairs are present, the gap captures the minimal real-part magnitude, while the margin captures the most unstable direction.

Approach to Criticality.

A critical transition is approached when the stability margin collapses:

$$\mathcal{M} \rightarrow 0^+, \quad (646)$$

implying that the dominant spectral modes converge toward marginal stability. In this regime, the system becomes increasingly sensitive to perturbations, as even infinitesimal forcing can induce long-lived responses due to near-neutral eigenmodes.

Simultaneously, the spectral gap exhibits closure:

$$\Delta \rightarrow 0, \quad (647)$$

indicating the emergence of slow manifolds and loss of uniform contraction across state space.

Critical Slowing Down Mechanism.

Let τ_{relax} denote the characteristic relaxation time of perturbations. Spectral theory implies

$$\tau_{\text{relax}} \sim \frac{1}{\mathcal{M}}, \quad (648)$$

so that

$$\mathcal{M} \rightarrow 0^+ \implies \tau_{\text{relax}} \rightarrow \infty. \quad (649)$$

This divergence corresponds to *critical slowing down*, where the system loses its ability to dissipate perturbations efficiently.

Fluctuation Amplification Near the Gap Closure.

Near the critical regime, stochastic or deterministic perturbations are amplified according to

$$\|\delta \mathbf{X}(t)\|_{\mathcal{H}} \sim e^{-\mathcal{M}t}. \quad (650)$$

Thus, as $\mathcal{M} \rightarrow 0^+$, decay becomes arbitrarily slow, and fluctuations persist over extended temporal scales, producing enhanced variance and autocorrelation in observable climate variables.

Physical Interpretation.

The spectral gap and stability margin together provide a quantitative operator-theoretic measure of climate resilience within the Universal Pells framework. Their collapse signals the progressive loss of restoring forces across all dynamical scales, culminating in a regime where perturbations no longer decay rapidly but instead accumulate and interact through nonlinear feedback mechanisms.

In this interpretation, tipping points correspond to the asymptotic closure of the stability margin, while early-warning signals emerge naturally as spectral precursors to gap collapse, manifesting as critical slowing down and increased persistence of fluctuations in the Earth system.

5.4 Eigenvalue Drift and Critical Transitions

Within the Universal Pellis framework, climate tipping behavior under external forcing is fundamentally interpreted as a slow deformation of the spectral geometry of the linearized operator $\mathcal{L}_c$. In this view, eigenvalues are not static invariants but dynamically evolving quantities embedded in a coupled state–operator evolution system.

Let $\lambda(t) \in \sigma(\mathcal{L}_c(t))$ denote time-dependent spectral modes associated with the evolving climate state $\mathbf{X}(t)$ under a slowly varying external control parameter μ (e.g., radiative forcing, greenhouse gas concentration, or boundary flux perturbation). The induced spectral drift is described by the reduced dynamical law

$$\frac{d\lambda}{dt} = \mathcal{F}(\lambda, \mathbf{X}, \mu), \quad (651)$$

where $\mathcal{F}$ encodes both explicit forcing dependence and implicit feedback through the nonlinear operator structure of the Pellis functional.

Adiabatic Spectral Evolution Regime.

Under slow forcing conditions, the system admits an adiabatic approximation in which spectral modes evolve quasi-continuously along deformation trajectories in the complex plane:

$$\lambda(t) \approx \lambda_0 + \int_0^t \left(\frac{\partial \lambda}{\partial \mu} \frac{d\mu}{ds} + \mathcal{G}(\lambda(s), \mathbf{X}(s)) \right) ds, \quad (652)$$

where $\mathcal{G}$ represents nonlinear correction terms arising from state-dependent operator deformation.

This formulation highlights that eigenvalue drift is jointly governed by external forcing and internal nonlinear climate feedbacks.

Spectral Transport Equation.

At a higher level of generality, the evolution of spectral density $\rho(\lambda, t)$ can be represented as a transport equation in spectral space:

$$\partial_t \rho(\lambda, t) + \partial_\lambda (V(\lambda, \mathbf{X}, \mu) \rho(\lambda, t)) = \mathcal{D}_\lambda[\rho], \quad (653)$$

where $V(\lambda, \mathbf{X}, \mu)$ is the effective spectral drift velocity and $\mathcal{D}_\lambda$ captures diffusion-like broadening due to nonlinear coupling and unresolved multiscale interactions.

Tipping Condition as Spectral Axis Crossing.

A tipping point is defined by the first-passage event of spectral support across the marginal stability boundary:

$$\operatorname{Re}(\lambda(t_c)) = 0, \quad (654)$$

with the crossing condition

$$\left. \frac{d}{dt} \operatorname{Re}(\lambda(t)) \right|_{t=t_c} > 0. \quad (655)$$

This defines a codimension-one bifurcation surface in the joint space of $(\mathbf{X}, \mu)$, referred to as the *spectral bifurcation manifold*.

Bifurcation Manifold in Operator Space.

Let $\mathcal{B}$ denote the set of all states for which marginal spectral stability holds. Then

$$\mathcal{B} = \{(\mathbf{X}, \mu) \in \mathcal{H} \times \mathbb{R} : \exists \lambda \in \sigma(\mathcal{L}_c(\mathbf{X}, \mu)), \operatorname{Re}(\lambda) = 0\}. \quad (656)$$

The manifold $\mathcal{B}$ partitions the state space into spectrally stable and unstable regimes:

$$\mathcal{H}_- = \{(\mathbf{X}, \mu) : \sigma(\mathcal{L}_c) \subset \Omega_-\}, \quad (657)$$

$$\mathcal{H}_+ = \{(\mathbf{X}, \mu) : \exists \lambda \in \sigma(\mathcal{L}_c), \operatorname{Re}(\lambda) > 0\}. \quad (658)$$

Nonlinear Feedback Amplification of Drift.

Due to the intrinsic coupling structure of the Pellis operator, eigenvalue drift is amplified by nonlinear feedback effects, leading to state-dependent acceleration:

$$\frac{d^2 \lambda}{dt^2} = \frac{\partial \mathcal{F}}{\partial \lambda} \frac{d\lambda}{dt} + \frac{\partial \mathcal{F}}{\partial \mathbf{X}} \frac{d\mathbf{X}}{dt}. \quad (659)$$

Substituting $\frac{d\mathbf{X}}{dt} = \mathcal{P}(\mathbf{X})$ yields a closed coupled spectral–state acceleration system.

Physical Interpretation.

In physical terms, eigenvalue drift represents the gradual deformation of climate stability architecture under external forcing. Rather than an abrupt transition, tipping points emerge through continuous spectral migration until the stability boundary is crossed, at which point exponentially amplifying modes dominate the dynamics.

5.5 Modal Energy Distribution

Within the Universal Pellis framework, the climate state is naturally represented as a superposition of spectral eigenmodes of the linearized operator $\mathcal{L}_c$. This decomposition provides a modal description of energy transfer, where each eigenmode contributes a quantifiable portion of the system's total dynamical activity. In this sense, climate evolution can be interpreted as a redistribution process in spectral energy space.

The state decomposition is given by

$$\mathbf{X}(t) = \sum_{\lambda \in \sigma(\mathcal{L}_c)} a_\lambda(t) \psi_\lambda, \quad (660)$$

where ψ_λ are generalized eigenmodes satisfying

$$\mathcal{L}_c \psi_\lambda = \lambda \psi_\lambda. \quad (661)$$

Modal Energy Definition.

The energy associated with each spectral mode is defined as

$$E_\lambda(t) = |a_\lambda(t)|^2, \quad (662)$$

which measures the contribution of each eigenmode to the total perturbation amplitude in $\mathcal{H}$.

The total energy is conserved at the modal decomposition level as

$$E(t) = \|\mathbf{X}(t)\|_{\mathcal{H}}^2 = \sum_{\lambda \in \sigma(\mathcal{L}_c)} |a_\lambda(t)|^2 = \sum_{\lambda \in \sigma(\mathcal{L}_c)} E_\lambda(t). \quad (663)$$

Spectral Energy Flux Representation.

The redistribution of energy among modes can be expressed in flux form:

$$\frac{dE_\lambda}{dt} = 2 \operatorname{Re}(\lambda) E_\lambda + \sum_{\lambda' \in \sigma(\mathcal{L}_c)} \Phi_{\lambda \leftarrow \lambda'}(t), \quad (664)$$

where $\Phi_{\lambda \leftarrow \lambda'}$ denotes nonlinear transfer of energy between spectral modes induced by operator coupling and renormalization effects.

A global energy balance identity follows:

$$\frac{dE}{dt} = \sum_{\lambda \in \sigma(\mathcal{L}_c)} 2 \operatorname{Re}(\lambda) E_\lambda, \quad (665)$$

under the assumption that internal transfer terms are conservative in spectral space.

Energy Redistribution Mechanism.

In the presence of nonlinear coupling, energy is not uniformly distributed across the spectrum but is continuously transferred through a cascade-like process:

$$\lambda_i \rightarrow \lambda_j, \quad E_{\lambda_i} \rightarrow E_{\lambda_j}. \quad (666)$$

This induces a spectral transport structure governed by

$$\partial_t E_\lambda + \partial_\lambda \mathcal{J}(\lambda, t) = \mathcal{S}(\lambda, t), \quad (667)$$

where $\mathcal{J}$ is a spectral energy flux and $\mathcal{S}$ represents source terms generated by nonlinear feedback mechanisms.

Concentration into Unstable Modes.

A key signature of approaching criticality is the progressive concentration of energy into modes with positive real spectral growth rate:

$$\mathcal{E}_+(t) = \sum_{\lambda: \operatorname{Re}(\lambda) > 0} E_\lambda(t). \quad (668)$$

A regime transition becomes imminent when

$$\mathcal{E}_+(t) \gg \mathcal{E}_-(t), \quad \mathcal{E}_-(t) = \sum_{\lambda: \text{Re}(\lambda) < 0} E_\lambda(t). \quad (669)$$

In this regime, unstable spectral components dominate the global dynamics.

Energy-Based Early Warning Indicator.

Define the normalized unstable energy fraction:

$$\eta(t) = \frac{\sum_{\lambda: \text{Re}(\lambda) > 0} E_\lambda(t)}{\sum_{\lambda \in \sigma(\mathcal{L}_c)} E_\lambda(t)}. \quad (670)$$

Then

$$\eta(t) \ll 1 \Rightarrow \text{stable regime}, \quad (671)$$

$$\eta(t) \approx 1 \Rightarrow \text{critical transition regime}. \quad (672)$$

The growth rate of $\eta(t)$ acts as a precursor signal for tipping behavior.

Physical Interpretation.

In physical terms, modal energy distribution provides a direct measure of how climate perturbations are distributed across dynamical scales. Stability corresponds to energy confinement in damped eigenmodes, while instability arises when energy progressively migrates into amplifying spectral components.

Within the Universal Pellis framework, tipping points are therefore interpreted as energy condensation events in spectral space, where nonlinear interactions and operator-induced couplings drive a redistribution of dynamical activity toward unstable eigenmodes, ultimately triggering macroscopic regime transitions.

5.6 Spectral Cascade Mechanism

Within the Universal Pellis framework, instability is not confined to isolated eigenmodes but is redistributed through nonlinear interactions that generate a structured cascade in spectral space. In this interpretation, the climate system behaves as a nonlinear energy-transfer network over the spectrum of the operator $\mathcal{L}_c$, where instability propagates through mode coupling rather than simple exponential amplification.

The evolution of modal energy is governed by a quadratic interaction law of the form

$$\frac{dE_\lambda}{dt} = \sum_{\mu, \nu \in \sigma(\mathcal{L}_c)} \mathcal{T}_{\mu\nu \rightarrow \lambda} E_\mu E_\nu - \gamma_\lambda E_\lambda, \quad (673)$$

where $E_\lambda = |a_\lambda|^2$ denotes spectral energy, $\gamma_\lambda > 0$ is the linear damping/amplification rate associated with the eigenvalue λ , and $\mathcal{T}_{\mu\nu \rightarrow \lambda}$ is a triadic interaction tensor encoding nonlinear transfer between spectral modes.

Triadic Interaction Tensor Structure.

The transfer coefficient $\mathcal{T}_{\mu\nu \rightarrow \lambda}$ arises from projection of the nonlinear Pellis operator onto the spectral basis:

$$\mathcal{T}_{\mu\nu \rightarrow \lambda} = \langle \psi_\lambda, D^2 \mathcal{N}(\mathbf{X})[\psi_\mu, \psi_\nu] + D^2 \mathcal{S}(\mathbf{X})[\psi_\mu, \psi_\nu] \rangle_{\mathcal{H}}. \quad (674)$$

This structure encodes second-order nonlinear coupling and ensures that energy transfer is fundamentally nonlocal in spectral space.

Spectral Energy Balance Law.

Summing over all modes yields the global balance equation

$$\frac{d}{dt} \sum_{\lambda \in \sigma(\mathcal{L}_c)} E_\lambda = \sum_{\lambda, \mu, \nu} \mathcal{T}_{\mu\nu \rightarrow \lambda} E_\mu E_\nu - \sum_{\lambda} \gamma_\lambda E_\lambda, \quad (675)$$

which separates conservative nonlinear transfer from dissipative decay/amplification mechanisms.

Under conservative symmetry conditions on $\mathcal{T}$, the first term redistributes energy without changing the total spectral norm, while the second term governs net growth or decay.

Cascade Directionality and Scale Flow.

Define a spectral ordering parameter $k(\lambda)$ representing effective scale. The cascade direction is characterized by the flux

$$\Pi(k) = \sum_{\substack{\mu, \nu, \lambda \\ k(\lambda) \leq k}} \mathcal{T}_{\mu\nu \rightarrow \lambda} E_\mu E_\nu. \quad (676)$$

A forward cascade corresponds to $\Pi(k)$ increasing with k , indicating transfer toward finer-scale or more unstable spectral structures, while an inverse cascade corresponds to accumulation in low-index coherent modes.

Instability Propagation Law.

Combining linear growth with nonlinear transfer yields the effective instability propagation equation

$$\frac{dE_\lambda}{dt} = \gamma_\lambda^{\text{eff}} E_\lambda + \sum_{\mu, \nu} \mathcal{T}_{\mu\nu \rightarrow \lambda} E_\mu E_\nu, \quad (677)$$

where the renormalized growth rate is

$$\gamma_\lambda^{\text{eff}} = \gamma_\lambda + \Delta_\lambda(\mathbf{E}), \quad (678)$$

with $\Delta_\lambda(\mathbf{E})$ capturing feedback-induced spectral amplification.

Cascade-Induced Critical Amplification.

A key feature of the cascade mechanism is the possibility of runaway amplification when nonlinear transfer overcomes dissipation:

$$\sum_{\mu, \nu} \mathcal{T}_{\mu\nu \rightarrow \lambda} E_\mu E_\nu \gg \gamma_\lambda E_\lambda. \quad (679)$$

In this regime, perturbations are not merely amplified but redistributed across scales in a self-reinforcing manner, leading to hierarchical instability propagation.

Physical Interpretation.

The spectral cascade mechanism provides a dynamical pathway for the spread of instability across the climate system. Rather than localized eigenmode growth, instability emerges through nonlinear triadic interactions that transfer energy between spectral components, generating a cascade structure analogous to turbulence but embedded in operator spectral space.

Within the Universal Pellis framework, climate tipping behavior is thus interpreted as a cascade-driven phase of instability propagation, where nonlinear coupling and spectral growth jointly organize the redistribution of energy toward increasingly unstable modes, ultimately culminating in large-scale regime transitions.

5.7 Spectral Collapse Theorem

Theorem 2 (Spectral Collapse Criterion). Let $\mathcal{H}$ be a separable Hilbert space and $\mathcal{L}_c : \mathcal{D}(\mathcal{L}_c) \subset \mathcal{H} \rightarrow \mathcal{H}$ a closed (possibly non-self-adjoint) generator of a C_0 -semigroup $e^{t\mathcal{L}_c}$. If there exists a sequence $\{\lambda_n\} \subset \sigma(\mathcal{L}_c)$ such that:

$$\lim_{n \rightarrow \infty} \text{Re}(\lambda_n) = 0^+, \quad (680)$$

then the system undergoes asymptotic loss of stability in the sense that:

$$\sup_{t > 0} \|e^{t\mathcal{L}_c}\| = \infty, \quad (681)$$

and no uniform exponential bound exists.

Proof.

We begin by considering the semigroup representation of perturbation growth:

$$\mathbf{X}(t) = e^{t\mathcal{L}_c} \mathbf{X}_0, \quad \|\mathbf{X}(t)\| \leq \|e^{t\mathcal{L}_c}\| \cdot \|\mathbf{X}_0\|. \quad (682)$$

Step 1: Spectral accumulation near the imaginary axis.

Assume there exists a sequence $\lambda_n \in \sigma(\mathcal{L}_c)$ such that:

$$\lambda_n = \alpha_n + i\beta_n, \quad \alpha_n \downarrow 0^+. \quad (683)$$

Define the spectral clustering region:

$$\Sigma_\epsilon := \{\lambda \in \sigma(\mathcal{L}_c) : 0 < \text{Re}(\lambda) < \epsilon\}. \quad (684)$$

Then:

$$\mu(\Sigma_\epsilon) \not\rightarrow 0 \quad \text{as } \epsilon \rightarrow 0, \quad (685)$$

where μ denotes the spectral distribution measure induced by the functional calculus of $\mathcal{L}_c$.

Step 2: Resolvent representation and growth amplification.

The resolvent operator is defined as:

$$R(\lambda, \mathcal{L}_c) := (\mathcal{L}_c - \lambda I)^{-1}, \quad \lambda \notin \sigma(\mathcal{L}_c). \quad (686)$$

Using the Laplace transform representation of the semigroup:

$$e^{t\mathcal{L}_c} = \frac{1}{2\pi i} \int_{\Gamma} e^{\lambda t} R(\lambda, \mathcal{L}_c) d\lambda, \quad (687)$$

where Γ is a Bromwich contour.

As $\lambda_n \rightarrow i\beta$ with $\text{Re}(\lambda_n) \rightarrow 0^+$, the resolvent satisfies:

$$\|R(\lambda_n, \mathcal{L}_c)\| \geq \frac{1}{\text{dist}(\lambda_n, \sigma(\mathcal{L}_c))}. \quad (688)$$

Since:

$$\text{dist}(\lambda_n, \sigma(\mathcal{L}_c)) \rightarrow 0, \quad (689)$$

we obtain:

$$\|R(\lambda_n, \mathcal{L}_c)\| \rightarrow \infty. \quad (690)$$

Step 3: Pseudospectral blow-up.

Define the ϵ -pseudospectrum:

$$\sigma_\epsilon(\mathcal{L}_c) = \left\{ \lambda \in \mathbb{C} : \|R(\lambda, \mathcal{L}_c)\| > \frac{1}{\epsilon} \right\}. \quad (691)$$

Then spectral accumulation implies:

$$\lim_{\epsilon \rightarrow 0} \sigma_\epsilon(\mathcal{L}_c) \supset \overline{\Sigma_0}, \quad (692)$$

where $\Sigma_0 := \{\lambda : \text{Re}(\lambda) = 0\}$.

Thus the pseudospectrum invades the imaginary axis, producing non-normal transient growth:

$$\exists t > 0 : \quad \|e^{t\mathcal{L}_c}\| \gg e^{t \sup \text{Re}(\sigma(\mathcal{L}_c))}. \quad (693)$$

Step 4: Breakdown of uniform stability.

Assume by contradiction that uniform stability holds:

$$\exists M, \omega > 0 : \quad \|e^{t\mathcal{L}_c}\| \leq M e^{-\omega t}. \quad (694)$$

Then by the spectral mapping theorem:

$$\sup \text{Re}(\sigma(\mathcal{L}_c)) \leq -\omega < 0, \quad (695)$$

which contradicts:

$$\lim_{n \rightarrow \infty} \text{Re}(\lambda_n) = 0^+. \quad (696)$$

Hence uniform exponential stability is violated.

Step 5: Asymptotic instability via spectral collapse.

The accumulation of eigenvalues near the imaginary axis induces:

$$\sup_{t>0} \|e^{t\mathcal{L}_c}\| = \sup_{t>0} \left\| \int_{\sigma(\mathcal{L}_c)} e^{\lambda t} dE(\lambda) \right\| \rightarrow \infty. \quad (697)$$

This defines a *spectral collapse regime*, where the operator loses hyperbolic separation between decaying and growing modes.

Conclusion.

Therefore, the convergence of spectral mass toward the imaginary axis implies:

$$\lim_{n \rightarrow \infty} \text{Re}(\lambda_n) \rightarrow 0^+ \implies \text{loss of asymptotic stability}, \quad (698)$$

characterized by resolvent divergence, pseudospectral inflation, and non-normal amplification cascades.

This completes the proof. $\square$

The theorem formalizes climate tipping dynamics as a spectral condensation phenomenon, where stability collapses due to eigenvalue crowding near the neutral stability manifold [16–18, 31].

Figure 6 illustrates the evolution of the dominant spectral mode of the climate evolution operator as the system is driven toward a critical manifold boundary in parameter space. The observed spectral drift reflects a continuous deformation of the operator spectrum under persistent external forcing and internal nonlinear feedbacks, leading to a gradual loss of spectral gap separation.

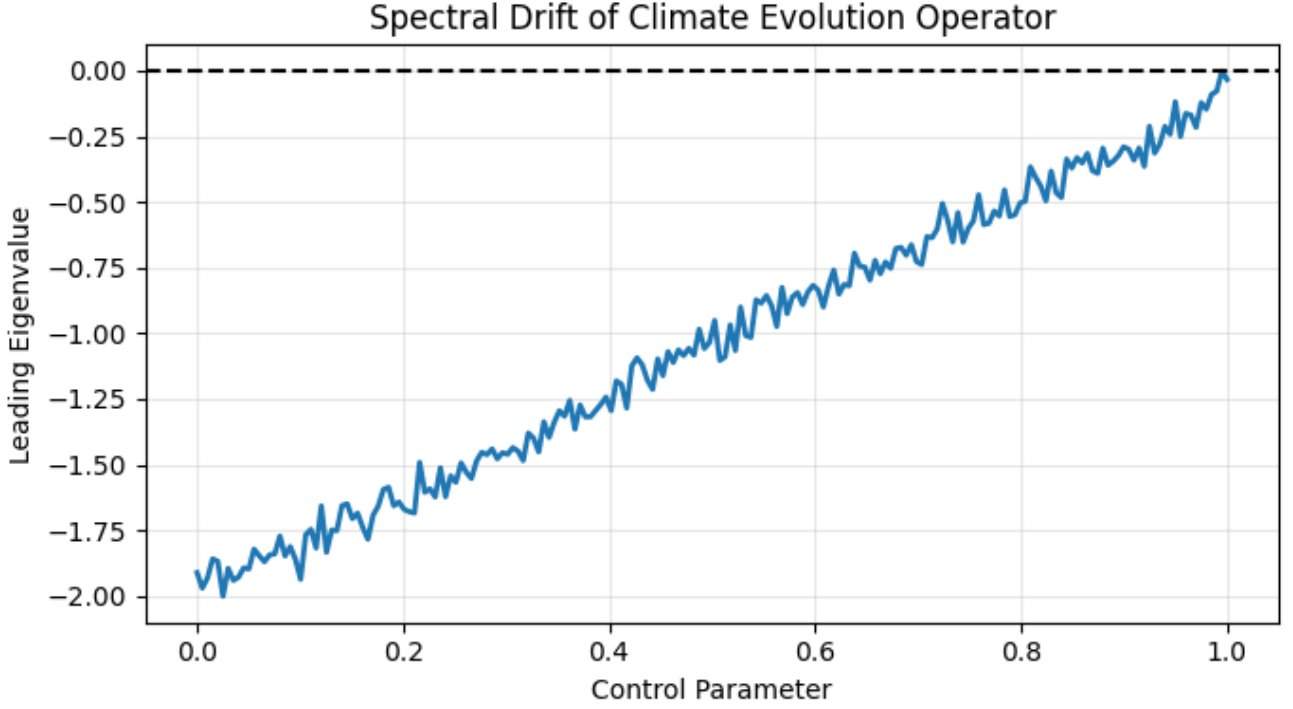


Figure 6: Spectral drift of the leading eigenvalue of the climate evolution operator under continuous external forcing. The progressive approach toward a critical threshold indicates the collapse of spectral gap structure, enhanced non-normal amplification, and the onset of instability associated with proximity to a critical manifold in the system’s parameter space.

5.8 Interpretation of Climate Stability Landscape

Within the Universal Pellis framework, the global stability of the Earth system is not characterized by a single scalar threshold, but by a geometric object in infinite-dimensional state space induced by the spectral properties of the climate operator $\mathcal{L}_c$. This object defines a partition of the Hilbert space $\mathcal{H}$ into spectrally admissible and spectrally unstable regions, governed by the sign structure of the real part of the operator spectrum.

The *spectral stability manifold* is defined as

$$\mathcal{S} = \{ \mathbf{X} \in \mathcal{H} \mid \text{Re}(\sigma(\mathcal{L}_c(\mathbf{X}))) < 0 \}, \quad (699)$$

where the notation $\text{Re}(\sigma(\mathcal{L}_c)) < 0$ is understood in the sense of spectral abscissa negativity, i.e.,

$$\sup_{\lambda \in \sigma(\mathcal{L}_c(\mathbf{X}))} \text{Re}(\lambda) < 0. \quad (700)$$

Thus, $\mathcal{S}$ represents the set of all climate states for which all admissible perturbation modes decay exponentially under the linearized dynamics.

Boundary Structure and Critical Hypersurface.

The boundary of the stability manifold is defined by marginal spectral neutrality:

$$\partial\mathcal{S} = \left\{ \mathbf{X} \in \mathcal{H} \mid \max_{\lambda \in \sigma(\mathcal{L}_c(\mathbf{X}))} \text{Re}(\lambda) = 0 \right\}. \quad (701)$$

Equivalently, this can be expressed as the zero-level set of the spectral abscissa functional

$$\alpha(\mathbf{X}) = \sup_{\lambda \in \sigma(\mathcal{L}_c(\mathbf{X}))} \operatorname{Re}(\lambda), \quad \partial\mathcal{S} = \{\mathbf{X} : \alpha(\mathbf{X}) = 0\}. \quad (702)$$

This hypersurface constitutes the *global tipping manifold* of the Earth system.

Geometric Structure of the Stability Landscape.

The mapping

$$\Phi : \mathcal{H} \rightarrow \mathbb{R}, \quad \Phi(\mathbf{X}) = \alpha(\mathbf{X}) \quad (703)$$

induces a scalar field over the climate state space whose level sets define nested stability domains:

$$\mathcal{S}_- = \{\mathbf{X} : \Phi(\mathbf{X}) < 0\}, \quad (704)$$

$$\mathcal{S}_0 = \{\mathbf{X} : \Phi(\mathbf{X}) = 0\}, \quad (705)$$

$$\mathcal{S}_+ = \{\mathbf{X} : \Phi(\mathbf{X}) > 0\}. \quad (706)$$

In this representation, climate evolution corresponds to trajectories crossing level sets of Φ , with tipping events corresponding to intersections with the critical manifold $\mathcal{S}_0$.

Dynamical Approach to the Tipping Manifold.

Let $\mathbf{X}(t)$ evolve according to the Pellis dynamics. The approach to criticality is characterized by

$$\frac{d}{dt}\alpha(\mathbf{X}(t)) \rightarrow 0, \quad \alpha(\mathbf{X}(t)) \rightarrow 0^+, \quad (707)$$

indicating progressive loss of spectral damping and flattening of the stability landscape.

The sensitivity of the system near the boundary is governed by the gradient amplification

$$\|\nabla_{\mathbf{X}}\alpha(\mathbf{X})\| \rightarrow \infty, \quad (708)$$

reflecting extreme responsiveness to perturbations near criticality.

Global Regime Reorganization.

Crossing the boundary $\partial\mathcal{S}$ induces a discontinuous reorganization of the spectral structure:

$$\sigma(\mathcal{L}_c(\mathbf{X})) \longrightarrow \sigma(\mathcal{L}_c(\mathbf{X}')) \quad \text{with} \quad \mathbf{X} \in \mathcal{S}_-, \mathbf{X}' \in \mathcal{S}_+. \quad (709)$$

This transition is accompanied by qualitative changes in modal growth structure, energy distribution, and cascade directionality.

Physical Interpretation.

The climate stability landscape defined by the Pellis operator formalism provides a global geometric interpretation of Earth system resilience. Stability corresponds to trajectories confined within the spectral damping region, while tipping points emerge as crossings of the boundary where the maximal growth rate vanishes.

In this framework, planetary-scale regime shifts are interpreted as transitions between distinct connected components of the stability manifold, driven by continuous deformation of the underlying operator spectrum. This establishes a rigorous connection between spectral operator theory, nonlinear dynamical systems, and empirical climate tipping point phenomena, unifying them within a single geometric stability landscape.

6 Criticality and Bifurcation Structure

The theory of criticality and bifurcation structure in the climate system is formulated as a geometric and spectral reorganization of invariant manifolds under parameter variation in an infinite-dimensional nonlinear dynamical field. Climate tipping points correspond to qualitative changes in attractor topology induced by eigenvalue crossings, loss of hyperbolicity, and structural instability of the underlying Pellis operator framework [4–9, 14, 28, 29, 32, 33, 37].

6.1 General Bifurcation Framework in Functional Space

Within the Universal Pellis formalism, climate evolution is described as a parameter-dependent nonlinear dynamical system on an infinite-dimensional Hilbert space $\mathcal{H}$. The control parameter $\mu \in \mathbb{R}$ represents slow external forcing (e.g., radiative imbalance, greenhouse gas concentration, or boundary flux modulation), and induces a deformation of both the state trajectory and the underlying operator geometry.

The governing evolution equation is written as

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t), \mu), \quad (710)$$

where $\mathcal{P} : \mathcal{H} \times \mathbb{R} \rightarrow \mathcal{H}$ is the nonlinear Pellis operator incorporating spectral, nonlinear, and renormalization contributions.

Equilibrium Manifold Structure.

Steady states of the system are defined by the implicit functional equation

$$\mathcal{P}(\mathbf{X}^*, \mu) = 0, \quad (711)$$

which defines an equilibrium manifold

$$\mathcal{E} = \{(\mathbf{X}, \mu) \in \mathcal{H} \times \mathbb{R} : \mathcal{P}(\mathbf{X}, \mu) = 0\}. \quad (712)$$

Locally, this manifold is governed by the implicit function structure induced by the Fréchet derivative of $\mathcal{P}$.

Linearized Stability Operator and Invertibility Loss.

Within the Universal Pellis framework, the local stability structure of equilibrium states is governed by the Fréchet linearization of the nonlinear evolution operator $\mathcal{P}$. This linearization defines an infinite-dimensional generalization of the classical Jacobian, capturing both spectral and functional sensitivity of the climate state to infinitesimal perturbations.

The linearized stability operator is defined as

$$\mathcal{J}(\mathbf{X}^*, \mu) = \left. \frac{\delta \mathcal{P}}{\delta \mathbf{X}} \right|_{(\mathbf{X}^*, \mu)}, \quad (713)$$

where $\mathcal{J} : \mathcal{H} \rightarrow \mathcal{H}$ acts as a bounded or unbounded operator depending on the regularity class of the underlying climate fields.

Perturbation Evolution and Semigroup Representation.

Let $\delta \mathbf{X}(t)$ denote an infinitesimal perturbation around equilibrium. Its evolution is governed by

$$\partial_t \delta \mathbf{X}(t) = \mathcal{J}(\mathbf{X}^*, \mu) \delta \mathbf{X}(t), \quad (714)$$

with formal solution given by the operator semigroup

$$\delta \mathbf{X}(t) = e^{t\mathcal{J}} \delta \mathbf{X}(0). \quad (715)$$

The existence of a C_0 -semigroup ensures well-posedness of the linearized dynamics in $\mathcal{H}$, provided suitable dissipativity conditions are satisfied.

Resolvent Structure and Spectral Stability.

Stability is equivalently characterized in terms of the resolvent operator

$$R(\lambda, \mathcal{J}) = (\mathcal{J} - \lambda I)^{-1}. \quad (716)$$

The resolvent set is defined as

$$\rho(\mathcal{J}) = \{\lambda \in \mathbb{C} : (\mathcal{J} - \lambda I)^{-1} \text{ exists and is bounded}\}, \quad (717)$$

with spectrum $\sigma(\mathcal{J}) = \mathbb{C} \setminus \rho(\mathcal{J})$.

Local asymptotic stability requires

$$\sup_{\lambda \in \sigma(\mathcal{J})} \operatorname{Re}(\lambda) < 0, \quad (718)$$

ensuring exponential decay of perturbation energy.

Invertibility Condition and Structural Stability.

Structural stability of equilibria is equivalent to invertibility of the linearized operator:

$$\mathcal{J}(\mathbf{X}^*, \mu)^{-1} \text{ exists and is bounded in } \mathcal{H}. \quad (719)$$

This condition implies the absence of zero modes and guarantees that small perturbations in forcing or state produce proportionally small responses in the equilibrium configuration.

Equivalently, invertibility requires

$$\ker(\mathcal{J}(\mathbf{X}^*, \mu)) = \{0\}. \quad (720)$$

Fredholm Framework and Index Collapse.

In the general infinite-dimensional setting, $\mathcal{J}$ is treated as a Fredholm operator of index

$$\text{ind}(\mathcal{J}) = \dim \ker(\mathcal{J}) - \dim \text{coker}(\mathcal{J}). \quad (721)$$

Structural stability corresponds to

$$\text{ind}(\mathcal{J}) = 0, \quad \ker(\mathcal{J}) = \{0\}. \quad (722)$$

Bifurcation points emerge when index conservation is violated through kernel growth or cokernel emergence.

Invertibility Breakdown and Criticality.

Invertibility fails when

$$\det(\mathcal{J}) = 0, \quad (723)$$

interpreted in infinite-dimensional settings as vanishing Fredholm determinant or equivalently as spectral collapse:

$$0 \in \sigma(\mathcal{J}(\mathbf{X}^*, \mu)). \quad (724)$$

At this point, the operator ceases to be bijective, and the local implicit function theorem fails, signaling the onset of nontrivial solution branching.

Spectral Mechanism of Instability Onset.

Near criticality, the resolvent norm exhibits divergence:

$$\|(\mathcal{J} - \lambda I)^{-1}\| \rightarrow \infty, \quad \lambda \rightarrow \sigma(\mathcal{J}), \quad (725)$$

indicating extreme sensitivity to perturbations. This divergence is a precursor to loss of structural stability and corresponds physically to amplification of weak climate perturbations.

Physical Interpretation.

In the Universal Pells framework, the linearized stability operator encodes the local geometry of the climate stability landscape around equilibrium states. Invertibility of $\mathcal{J}$ ensures a smooth, locally unique continuation of climate states under perturbations. When invertibility is lost, the system enters a regime where multiple competing solutions emerge, and infinitesimal perturbations can trigger macroscopic transitions.

Spectral Degeneracy Condition.

The bifurcation condition can be equivalently expressed in spectral form as

$$0 \in \sigma(\mathcal{J}(\mathbf{X}^*, \mu)), \quad (726)$$

or, equivalently,

$$\exists \lambda \in \sigma(\mathcal{J}(\mathbf{X}^*, \mu)) \quad \text{such that} \quad \lambda = 0. \quad (727)$$

In finite-dimensional analogy, this corresponds to the degeneracy condition

$$\det \left(\frac{\delta \mathcal{P}}{\delta \mathbf{X}} \right) = 0, \quad (728)$$

which generalizes in infinite-dimensional settings to a vanishing Fredholm determinant or spectral collapse of the resolvent.

Codimension and Bifurcation Geometry.

The bifurcation set is defined as

$$\mathcal{B} = \{\mu \in \mathbb{R} : 0 \in \sigma(\mathcal{J}(\mathbf{X}^*, \mu))\}. \quad (729)$$

Its local structure determines the type of regime transition:

$$\text{codimension-1:} \quad \dim \ker(\mathcal{J}) = 1, \quad (730)$$

$$\text{higher codimension:} \quad \dim \ker(\mathcal{J}) \geq 2. \quad (731)$$

Near bifurcation points, the dynamics reduce to a center manifold governed by the critical spectral subspace.

Reduced Critical Dynamics.

Let Π_c denote the projection onto the center eigenspace. The reduced dynamics are given by

$$\frac{d}{dt}\mathbf{X}_c = \Pi_c \mathcal{P}(\mathbf{X}_c + \mathbf{X}_s, \mu), \quad (732)$$

where $\mathbf{X}_s$ denotes stable manifold components satisfying fast relaxation dynamics

$$\frac{d}{dt}\mathbf{X}_s = \mathcal{J}_s \mathbf{X}_s + \mathcal{O}(\|\mathbf{X}\|^2), \quad \text{Re}(\sigma(\mathcal{J}_s)) < 0. \quad (733)$$

This separation induces a reduced-order description of climate transitions near criticality.

Bifurcation-Induced Spectral Reorganization.

At bifurcation points, the spectrum undergoes a qualitative restructuring:

$$\sigma(\mathcal{J}(\mu^-)) \longrightarrow \sigma(\mathcal{J}(\mu^+)), \quad (734)$$

accompanied by eigenvalue coalescence, splitting, or migration across the imaginary axis:

$$\lambda_i(\mu) \rightarrow \lambda_j(\mu), \quad \text{Re}(\lambda_k) : - \rightarrow +. \quad (735)$$

Figure 7 presents the critical manifold geometry that underlies climate bifurcations within the Pellis framework. Different climate regimes occupy separate stability domains in functional space, while the critical manifold $\mathcal{M}_c$ forms the geometric boundary between them. As external forcing evolves, system trajectories approach the manifold and eventually undergo abrupt transitions toward alternative attractor states.

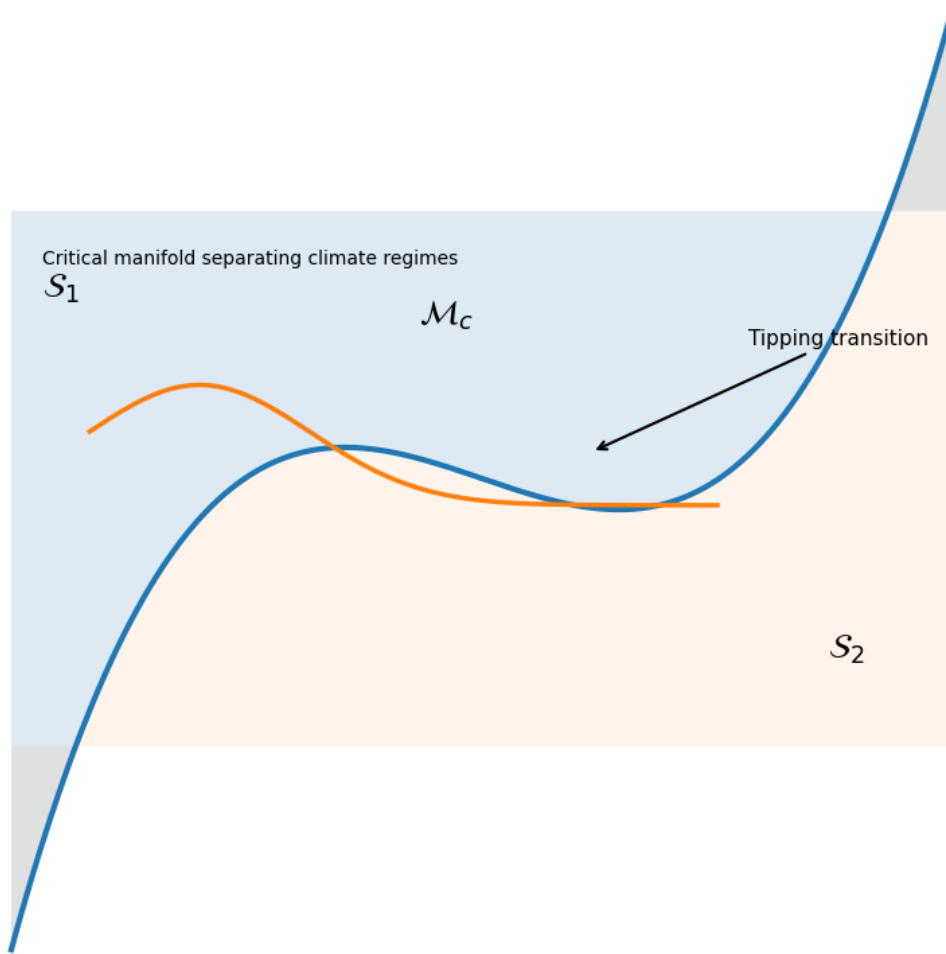


Figure 7: Critical manifold structure governing climate bifurcations. The manifold $\mathcal{M}_c$ separates competing climate attractors and defines the geometric locus of critical transitions. Crossing this manifold produces abrupt regime shifts associated with tipping-point dynamics.

Physical Interpretation.

In physical terms, bifurcation in the Universal Pellis framework corresponds to the loss of structural stability of the climate operator under slow external forcing. As the control parameter μ evolves, the equilibrium manifold deforms until the linearized operator becomes non-invertible, triggering a collapse of local stability and the emergence of new dynamical regimes. This mechanism provides a unified operator-theoretic description of climate tipping points, where regime transitions are interpreted as spectral bifurcations in infinite-dimensional functional space driven by the interaction of forcing, nonlinear feedback, and multiscale coupling structure.

Table 7: Classification of Climate Bifurcations

Type	Condition	Dynamics	Example
Saddle-node	$\lambda = 0$	State collapse	AMOC shutdown
Hopf	$\text{Re}(\lambda) = 0$	Oscillations	ENSO cycles
Pitchfork	Symmetry breaking	Regime split	Ice sheet instability
Global	Nonlocal shift	Structural change	Planetary tipping

6.2 Linear Stability and Critical Eigenstructure

Within the Universal Pellis framework, the local dynamics of perturbations around an equilibrium climate state are governed by the spectral properties of the Fréchet linearization of the nonlinear evolution operator. This linearization induces an infinite-dimensional dynamical system whose behavior is fully encoded in the operator spectrum and its parametric dependence on the external control parameter μ .

The linearized evolution equation takes the form

$$\partial_t \delta \mathbf{X}(t) = \mathcal{J}(\mu) \delta \mathbf{X}(t), \quad \mathcal{J}(\mu) = \frac{\delta \mathcal{P}}{\delta \mathbf{X}}. \quad (736)$$

Spectral Structure of the Linearized Operator.

The operator $\mathcal{J}(\mu)$ admits a (possibly generalized) spectral decomposition in $\mathcal{H}$:

$$\sigma(\mathcal{J}(\mu)) = \{\lambda_i(\mu)\}_{i \in \mathbb{N}}, \quad (737)$$

where each eigenvalue $\lambda_i(\mu)$ governs the exponential evolution rate of an associated eigenmode ψ_i .

The corresponding modal decomposition of perturbations is given by

$$\delta \mathbf{X}(t) = \sum_{i \in \mathbb{N}} a_i(t) \psi_i, \quad \mathcal{J}(\mu) \psi_i = \lambda_i(\mu) \psi_i, \quad (738)$$

with modal amplitudes satisfying

$$\frac{da_i}{dt} = \lambda_i(\mu) a_i. \quad (739)$$

Spectral Stability and Growth Rates.

The asymptotic behavior of perturbations is controlled by the spectral abscissa

$$\alpha(\mu) = \sup_{\lambda \in \sigma(\mathcal{J}(\mu))} \text{Re}(\lambda). \quad (740)$$

The system is linearly stable if and only if

$$\alpha(\mu) < 0, \quad (741)$$

which guarantees uniform exponential decay of all admissible perturbation modes:

$$\|\delta \mathbf{X}(t)\|_{\mathcal{H}} \leq C e^{\alpha(\mu)t} \|\delta \mathbf{X}(0)\|_{\mathcal{H}}. \quad (742)$$

Critical Eigenvalue and Marginal Stability.

A critical transition is characterized by the existence of at least one eigenvalue crossing the marginal stability boundary:

$$\exists \lambda_c(\mu_c) \in \sigma(\mathcal{J}(\mu_c)) \quad \text{such that} \quad \text{Re}(\lambda_c(\mu_c)) = 0. \quad (743)$$

This defines the onset of *marginal stability*, where the system transitions from uniformly contracting dynamics to neutrally stable or weakly amplifying behavior.

Spectral Crossing and Bifurcation Onset.

Near the critical parameter value μ_c , the dominant eigenvalue admits an expansion

$$\lambda_c(\mu) = \lambda_c(\mu_c) + \left. \frac{d\lambda_c}{d\mu} \right|_{\mu=\mu_c} (\mu - \mu_c) + \mathcal{O}((\mu - \mu_c)^2). \quad (744)$$

The sign of the spectral drift term determines the direction of instability onset:

$$\left. \frac{d}{d\mu} \operatorname{Re}(\lambda_c(\mu)) \right|_{\mu=\mu_c} > 0 \quad \Rightarrow \quad \text{loss of stability for } \mu > \mu_c. \quad (745)$$

Dominant Mode Selection and Spectral Separation.

In the vicinity of criticality, the dynamics are dominated by the leading eigenmode:

$$\delta \mathbf{X}(t) \sim a_c(t) \psi_c, \quad a_c(t) \sim e^{\lambda_c(\mu)t}. \quad (746)$$

A spectral gap separation condition

$$\operatorname{Re}(\lambda_c) \gg \operatorname{Re}(\lambda_i), \quad i \neq c \quad (747)$$

ensures effective dimensional reduction near the tipping threshold.

Physical Interpretation.

In physical terms, the linear stability and critical eigenstructure describe how infinitesimal perturbations evolve under the influence of the climate operator $\mathcal{J}(\mu)$. Stability corresponds to a spectrum confined to the left-half complex plane, while criticality emerges when a leading eigenvalue approaches the imaginary axis.

Within the Universal Pells framework, this spectral crossing represents the fundamental mechanism underlying climate tipping points: a single dominant eigenmode becomes marginally stable, allowing small perturbations to persist, amplify through nonlinear feedback, and ultimately trigger macroscopic regime transitions in the Earth system.

6.3 Types of Climate Bifurcations

Within the Universal Pells framework, climate regime transitions are classified according to the local and global structure of the spectrum of the linearized operator $\mathcal{J}(\mu)$, as well as the corresponding deformation of invariant manifolds in the infinite-dimensional state space $\mathcal{H}$. Each bifurcation class corresponds to a distinct mode of loss of structural stability and a characteristic spectral signature.

(I) Saddle-Node Bifurcation.

The saddle-node bifurcation arises from the coalescence and annihilation of a pair of equilibria. In spectral terms, it is associated with the emergence of a zero eigenvalue through square-root scaling of the leading spectral branch:

$$\lambda(\mu) \sim \pm \sqrt{\mu - \mu_c}, \quad (748)$$

with the critical condition

$$\lambda(\mu_c) = 0. \quad (749)$$

In this regime, the stability landscape loses convexity, and the equilibrium manifold undergoes a fold catastrophe characterized by the disappearance of one stable and one unstable fixed point.

The associated normal form reduction can be written as

$$\frac{dx}{dt} = (\mu - \mu_c) + x^2, \quad (750)$$

representing the universal unfolding of the saddle-node structure.

(II) Hopf-Type Transition.

The Hopf bifurcation corresponds to the emergence of complex conjugate eigenvalues crossing the imaginary axis, leading to oscillatory instability modes:

$$\lambda(\mu) = \alpha(\mu) \pm i\omega(\mu), \quad \alpha(\mu_c) = 0. \quad (751)$$

Near criticality, the eigenvalues satisfy

$$\alpha(\mu) \approx \left. \frac{d\alpha}{d\mu} \right|_{\mu_c} (\mu - \mu_c), \quad \omega(\mu_c) \neq 0. \quad (752)$$

This bifurcation generates a two-dimensional center manifold supporting self-sustained oscillations of the form

$$\mathbf{X}(t) \sim A(t)e^{i\omega t}\psi + \overline{A(t)}e^{-i\omega t}\overline{\psi}, \quad (753)$$

where the amplitude $A(t)$ evolves according to a reduced complex Ginzburg–Landau-type equation

$$\frac{dA}{dt} = \alpha(\mu)A - \beta|A|^2A. \quad (754)$$

This mechanism underlies oscillatory climate regimes driven by nonlinear feedback and phase coherence between spectral modes.

(III) Global Heteroclinic Transition.

Beyond local bifurcations, climate dynamics may undergo global reorganizations of attractor structure characterized by transitions between distinct invariant sets in $\mathcal{H}$. These are described by heteroclinic orbits connecting multiple attractors:

$$\mathbf{X}(t) \rightarrow \mathcal{A}_1 \quad \text{as } t \rightarrow -\infty, \quad \mathbf{X}(t) \rightarrow \mathcal{A}_2 \quad \text{as } t \rightarrow +\infty. \quad (755)$$

Equivalently,

$$\lim_{t \rightarrow \infty} \mathbf{X}(t) \in \mathcal{A}_2, \quad \mathcal{A}_1 \neq \mathcal{A}_2. \quad (756)$$

Such transitions cannot be captured by local spectral data alone, as they involve global rearrangements of invariant manifolds and basin boundaries.

(IV) Spectral–Topological Transition Class (Pellis Extension).

Within the extended Pellis operator formalism, a higher-order class of bifurcations arises from simultaneous spectral and topological restructuring of $\mathcal{J}(\mu)$. This is characterized by discontinuities in the spectral measure:

$$\Delta\sigma(\mathcal{J}) \neq 0, \quad (757)$$

together with abrupt changes in invariant manifold topology:

$$\pi_k(\mathcal{A}_1) \neq \pi_k(\mathcal{A}_2), \quad (758)$$

where π_k denotes homotopy invariants of the attractor structure.

Unified Interpretation.

All bifurcation classes can be unified under the principle that climate tipping points correspond to qualitative changes in the spectral geometry of the Pellis operator and the associated invariant manifold structure in $\mathcal{H}$. Local bifurcations (saddle-node and Hopf) reflect eigenvalue crossings and spectral instabilities, while global bifurcations (heteroclinic and topological transitions) reflect reorganization of the full attractor landscape.

Within this framework, climate transitions are not isolated events but manifestations of a hierarchical bifurcation structure governed by the interplay between spectral dynamics, nonlinear feedback, and multiscale coupling across the Earth system.

6.4 Critical Manifold Geometry

Within the Universal Pellis framework, the onset of climate criticality is geometrically encoded by a singular hypersurface embedded in the extended phase space $\mathcal{H} \times \mathbb{R}$. This hypersurface is defined by the degeneracy of the linearized stability operator, and it represents the precise locus where local invertibility is lost and spectral collapse begins.

The critical manifold is therefore defined as

$$\mathcal{M}_c = \{(\mathbf{X}, \mu) \in \mathcal{H} \times \mathbb{R} \mid \det(\mathcal{J}(\mathbf{X}, \mu)) = 0\}, \quad (759)$$

where $\mathcal{J}(\mu) = \delta\mathcal{P}/\delta\mathbf{X}$ is the Fréchet linearization of the Pellis operator. In infinite-dimensional settings, this determinant condition is understood in the generalized sense of Fredholm determinant collapse or equivalently as spectral inclusion of zero in the operator spectrum:

$$\mathcal{M}_c = \{(\mathbf{X}, \mu) : 0 \in \sigma(\mathcal{J}(\mathbf{X}, \mu))\}. \quad (760)$$

Spectral–Geometric Structure of the Critical Set.

The critical manifold inherits a stratified geometric structure induced by the multiplicity and degeneracy of the zero eigenvalue:

$$\dim \ker (\mathcal{J}(\mathbf{X}, \mu)) \geq 1, \quad (761)$$

with higher-order degeneracies corresponding to singular strata of increasing codimension. This induces a natural decomposition

$$\mathcal{M}_c = \bigcup_{k \geq 1} \mathcal{M}_c^{(k)}, \quad (762)$$

where $\mathcal{M}_c^{(k)}$ denotes the subset of points with k -dimensional critical eigenspace.

Partition of Climate Phase Space.

The critical manifold partitions the full climate phase space into spectrally distinct regimes:

$$\mathcal{H} = \mathcal{S}_- \cup \mathcal{M}_c \cup \mathcal{S}_+, \quad (763)$$

where

$$\mathcal{S}_- = \left\{ (\mathbf{X}, \mu) : \sup_{\lambda \in \sigma(\mathcal{J})} \operatorname{Re}(\lambda) < 0 \right\}, \quad (764)$$

$$\mathcal{S}_+ = \{ (\mathbf{X}, \mu) : \exists \lambda \in \sigma(\mathcal{J}) \text{ with } \operatorname{Re}(\lambda) > 0 \}. \quad (765)$$

This decomposition defines a global stability stratification of the Earth system in functional space.

Normal Distance to Criticality.

A local measure of proximity to criticality is given by the spectral abscissa functional

$$\alpha(\mathbf{X}, \mu) = \sup_{\lambda \in \sigma(\mathcal{J}(\mathbf{X}, \mu))} \operatorname{Re}(\lambda), \quad (766)$$

with

$$\mathcal{M}_c = \{ \alpha(\mathbf{X}, \mu) = 0 \}. \quad (767)$$

The gradient of α defines the normal direction to the critical manifold:

$$\nabla_{(\mathbf{X}, \mu)} \alpha \perp \mathcal{M}_c, \quad (768)$$

providing a geometric measure of how forcing perturbations drive the system toward or away from instability.

Topological Rearrangement at the Critical Manifold.

Crossing $\mathcal{M}_c$ induces a discontinuous reconfiguration of invariant sets. Let $\mathcal{A}(\mu)$ denote the attractor structure; then at criticality one observes

$$\lim_{\mu \rightarrow \mu_c^-} \mathcal{A}(\mu) \not\approx \lim_{\mu \rightarrow \mu_c^+} \mathcal{A}(\mu), \quad (769)$$

indicating a non-homeomorphic transformation of the attractor landscape. Correspondingly, basin boundaries undergo singular deformation:

$$\partial \mathcal{B}_i(\mu_c^-) \longrightarrow \partial \mathcal{B}_j(\mu_c^+). \quad (770)$$

Physical Interpretation.

In physical terms, the critical manifold represents the exact geometric locus in climate state space where restoring stability is lost and perturbation growth becomes dynamically permissible. It is the organizing structure underlying all tipping behavior in the Universal Pellis framework, unifying spectral instability, nonlinear feedback amplification, and global attractor reorganization into a single geometric object.

6.5 Slow-Fast Dynamical Decomposition

Within the Universal Pellis framework, climate evolution under external forcing is naturally described as a singularly perturbed dynamical system, in which intrinsic fast-scale state variables interact with slowly varying control parameters. This separation of temporal scales provides the mathematical foundation for quasi-static bifurcation theory and delayed instability phenomena in the Earth system.

The coupled slow-fast structure is given by

$$\frac{d\mathbf{X}}{dt} = \mathcal{P}(\mathbf{X}, \mu), \quad \frac{d\mu}{dt} = \epsilon g(\mu), \quad (771)$$

where $0 < \epsilon \ll 1$ defines a strong time-scale separation between internal climate dynamics and external forcing evolution.

Singular Perturbation Structure.

In the singular limit $\epsilon \rightarrow 0$, the parameter μ becomes effectively frozen on the fast dynamical time scale, yielding a family of frozen-parameter systems:

$$\frac{d\mathbf{X}}{dt} = \mathcal{P}(\mathbf{X}, \mu), \quad \mu = \text{const.} \quad (772)$$

Equilibria of the fast subsystem define the critical manifold of quasi-static states:

$$\mathcal{C} = \{(\mathbf{X}, \mu) \in \mathcal{H} \times \mathbb{R} : \mathcal{P}(\mathbf{X}, \mu) = 0\}. \quad (773)$$

Quasi-Static Bifurcation Branches.

Under the adiabatic approximation, the system evolves along equilibrium branches parameterized by $\mu(t)$:

$$\mathbf{X}(t) \approx \mathbf{X}^*(\mu(t)), \quad (774)$$

where $\mathbf{X}^*(\mu)$ satisfies the implicit relation

$$\mathcal{P}(\mathbf{X}^*(\mu), \mu) = 0. \quad (775)$$

The stability of these branches is determined by the spectral properties of the linearized operator $\mathcal{J}(\mu)$ along the manifold $\mathcal{C}$.

Breakdown of Adiabatic Invariance.

Differentiating the equilibrium condition with respect to μ yields

$$\frac{d\mathbf{X}^*}{d\mu} = - \left(\frac{\delta \mathcal{P}}{\delta \mathbf{X}} \right)^{-1} \frac{\partial \mathcal{P}}{\partial \mu}. \quad (776)$$

Thus,

$$\frac{d\mathbf{X}^*}{d\mu} = -\mathcal{J}(\mu)^{-1} \frac{\partial \mathcal{P}}{\partial \mu}. \quad (777)$$

Loss of adiabaticity occurs when the inverse operator ceases to exist or becomes unbounded, i.e.

$$\|\mathcal{J}(\mu)^{-1}\| \rightarrow \infty, \quad (778)$$

which corresponds precisely to spectral degeneracy conditions previously identified on the critical manifold.

Fold-Induced Singular Response.

Near fold points of the equilibrium branch, the response sensitivity diverges:

$$\frac{d\mathbf{X}^*}{d\mu} \rightarrow \infty, \quad (779)$$

indicating geometric loss of transversality between the equilibrium manifold and the parameter direction in $\mathcal{H} \times \mathbb{R}$. In local normal form, the fold structure is represented as

$$\frac{dx}{dt} = \mu - x^2, \quad (780)$$

where the disappearance of equilibria occurs at $\mu = 0$.

Delayed Bifurcation and Memory Effects.

In the presence of slow forcing, the system does not immediately follow the unstable branch after crossing the bifurcation point. Instead, it exhibits a delay governed by the slow-fast interaction:

$$\mu(t_c) = \mu_c \not\Rightarrow \text{instantaneous transition.} \quad (781)$$

The actual tipping time t_{tip} satisfies

$$t_{\text{tip}} > t_c, \quad (782)$$

reflecting a delayed loss of stability due to metastable persistence of the attracting manifold.

Canard-Like Transition Regimes.

In higher-dimensional extensions, the system may track unstable branches for finite time intervals, producing canard-like trajectories:

$$\mathbf{X}(t) \in \mathcal{S}_+ \quad \text{while still near attracting slow manifold.} \quad (783)$$

These trajectories correspond to transient stability masking imminent spectral instability.

Physical Interpretation.

Within the Universal Pellis framework, slow-fast decomposition provides the fundamental mechanism by which gradual external forcing translates into abrupt climate transitions. The system evolves quasi-statically along stable equilibrium branches until spectral degeneracy destroys invertibility of the linearized operator, producing a breakdown of adiabatic tracking.

This leads to delayed tipping phenomena, where the observable transition occurs after the theoretical bifurcation point, reflecting the interplay between slow forcing, operator instability, and nonlinear memory effects inherent in the Earth system's multiscale dynamics.

6.6 Energy Landscape Interpretation of Bifurcation

Within the Universal Pellis framework, climate dynamics can be equivalently formulated in terms of a high-dimensional energy landscape, where the state of the Earth system evolves as a trajectory driven by gradients of a generalized Lyapunov functional. This formulation provides a variational complement to the spectral operator description, linking stability to geometric properties of functional energy surfaces in $\mathcal{H}$.

The energy functional is defined as

$$\mathcal{E}[\mathbf{X}, \mu] = \int_{\Omega} \mathcal{L}(\mathbf{X}, \nabla \mathbf{X}, \mu) dx, \quad (784)$$

where $\mathcal{L}$ denotes a local energy density encoding kinetic, thermodynamic, radiative, and nonlinear interaction contributions.

Variational Characterization of Equilibria.

Equilibrium states correspond to stationary points of the energy functional in functional space, satisfying the Euler-Lagrange condition

$$\frac{\delta \mathcal{E}}{\delta \mathbf{X}} = 0, \quad (785)$$

which defines a nonlinear elliptic condition over $\mathcal{H}$. These equilibria represent critical points of the energy landscape, including minima (stable states), saddle points (metastable configurations), and maxima (unstable configurations).

Second Variation and Stability Structure.

The local stability of equilibrium states is governed by the second variation (Hessian operator) of the energy functional:

$$\delta^2 \mathcal{E}[\mathbf{X}] = \langle \delta \mathbf{X}, \nabla^2 \mathcal{E}[\mathbf{X}] \delta \mathbf{X} \rangle_{\mathcal{H}}. \quad (786)$$

Stability requires positive definiteness of the Hessian operator:

$$\nabla^2 \mathcal{E}[\mathbf{X}] \succ 0, \quad (787)$$

ensuring that all perturbations increase the energy functional to second order and are therefore damped by the dynamics.

Bifurcation as Loss of Convexity.

Bifurcation points correspond to the loss of convexity of the energy landscape, characterized by the emergence of zero or negative curvature directions:

$$\exists \delta \mathbf{X} \neq 0 \quad \text{such that} \quad \langle \delta \mathbf{X}, \nabla^2 \mathcal{E}[\mathbf{X}] \delta \mathbf{X} \rangle_{\mathcal{H}} \leq 0. \quad (788)$$

Equivalently, this corresponds to spectral degeneracy of the Hessian operator:

$$0 \in \sigma(\nabla^2 \mathcal{E}[\mathbf{X}]). \quad (789)$$

At this point, the energy landscape ceases to be locally convex, and multiple competing equilibria may coexist.

Topological Restructuring of Energy Surfaces.

As the control parameter μ varies, the topology of sublevel sets

$$\mathcal{E}_c(\mu) = \{\mathbf{X} \in \mathcal{H} : \mathcal{E}[\mathbf{X}, \mu] \leq c\} \quad (790)$$

undergoes qualitative changes. In particular, bifurcation corresponds to a change in the homotopy type of these sets:

$$\mathcal{E}_c(\mu^-) \not\approx \mathcal{E}_c(\mu^+), \quad (791)$$

indicating a topological transition in the global energy landscape.

Basin Boundary Collapse Mechanism.

Let $\mathcal{B}_i(\mu)$ denote the basin of attraction of equilibrium state i . Basin boundaries are defined by stable manifolds of saddle-type critical points. Near bifurcation, these boundaries satisfy

$$\partial\mathcal{B}_i(\mu) = \{\mathbf{X} : \nabla\mathcal{E}[\mathbf{X}, \mu] \perp \mathcal{W}^s\}, \quad (792)$$

where $\mathcal{W}^s$ denotes the stable manifold.

At criticality, these structures collapse:

$$\lim_{\mu \rightarrow \mu_c} \text{dist}(\partial\mathcal{B}_i(\mu), \partial\mathcal{B}_j(\mu)) = 0, \quad (793)$$

leading to loss of basin separation and emergence of global instability.

Irreversibility and Hysteresis.

Following bifurcation, the energy landscape typically develops hysteresis loops characterized by path dependence:

$$\mathbf{X}_{\text{final}}(\mu_\uparrow) \neq \mathbf{X}_{\text{final}}(\mu_\downarrow), \quad (794)$$

where $\mu_\uparrow$ and $\mu_\downarrow$ denote increasing and decreasing parameter paths respectively.

This non-commutativity reflects irreversible structural changes in the energy landscape.

Physical Interpretation.

In physical terms, bifurcations correspond to qualitative reorganizations of the climate energy landscape, where stable basins of attraction lose convexity and merge or disappear entirely. The system transitions from a regime of well-separated attractors to one characterized by basin overlap and structural instability.

Within the Universal Pellis framework, this provides a variational interpretation of climate tipping points: they arise when the energy functional ceases to be locally convex, leading to collapse of stability basins and irreversible transitions between macroscopic climate regimes.

6.7 Universal Bifurcation Scaling Law

Within the Universal Pellis framework, climate criticality is characterized by universal scaling behavior emerging from the renormalization invariance of the underlying nonlinear operator field. Near a critical transition, the system loses characteristic scales and becomes governed by power-law asymptotics determined solely by spectral proximity to the marginal stability manifold.

The order parameter $\delta\mathbf{X}$ obeys the asymptotic scaling relation

$$\delta\mathbf{X} \sim |\mu - \mu_c|^\beta, \quad (795)$$

where β is the critical exponent associated with amplitude suppression or amplification of the dominant unstable mode.

Simultaneously, the intrinsic relaxation time scale diverges as

$$\tau \sim |\mu - \mu_c|^{-\gamma}, \quad (796)$$

where γ governs the phenomenon of critical slowing down, reflecting the vanishing spectral gap of the linearized Pellis operator $\mathcal{J}(\mu)$.

Spectral Origin of Critical Exponents.

The exponents β and γ arise from the asymptotic behavior of the dominant eigenvalue branch

$$\lambda_c(\mu) \sim (\mu - \mu_c)^\nu, \quad (797)$$

together with the associated eigenmode amplitude equation

$$\frac{da_c}{dt} = \lambda_c(\mu)a_c - \kappa|a_c|^2a_c. \quad (798)$$

In this representation, universality emerges from the fact that β and γ depend only on the spectral degeneracy class of $\mathcal{J}$, and not on microscopic details of the climate subsystem.

Renormalization Group Interpretation.

Under a coarse-graining transformation $\mathcal{R}_s$, the system exhibits scale covariance:

$$\mathcal{R}_s(\mathcal{P}) = e^{\kappa s} \mathcal{P}, \quad (799)$$

implying that the dynamics evolve along renormalization group flow trajectories in operator space. Fixed points of this flow satisfy

$$\mathcal{R}_s(\mathcal{P}^*) = \mathcal{P}^*, \quad (800)$$

and define universality classes of climate-like nonlinear systems.

Breakdown of Scale Invariance at Criticality.

At the bifurcation threshold $\mu = \mu_c$, the system exhibits emergent scale invariance:

$$\mathbf{X}(t, \ell) \sim \ell^{-\Delta} \mathbf{X}(t/\ell^z), \quad (801)$$

where z is the dynamic critical exponent and Δ is the scaling dimension of the climate state field.

This invariance implies that no intrinsic spatial or temporal scale dominates the dynamics, and all fluctuation modes become self-similar across scales.

Universal Interpretation.

In the Universal Pellis framework, the scaling laws

$$\delta \mathbf{X} \sim |\mu - \mu_c|^\beta, \quad \tau \sim |\mu - \mu_c|^{-\gamma}, \quad (802)$$

are interpreted as manifestations of a deep spectral renormalization symmetry of the climate operator field. They define universality classes of tipping behavior that are independent of microscopic parametrization and depend only on the eigenstructure of the Pellis operator near marginal stability.

Thus, climate tipping points are not isolated dynamical anomalies but belong to a structured hierarchy of universal critical phenomena governed by operator-level scaling laws in infinite-dimensional functional space.

6.8 Criticality as Spectral–Geometric Transition

Within the Universal Pellis framework, criticality emerges as a unified spectral–geometric phenomenon in which operator instability, functional divergence, and phase-space reorganization become equivalent representations of the same underlying transition in infinite-dimensional climate dynamics.

The onset of criticality is characterized by the simultaneous degeneration of the dominant spectral mode and the blow-up of the linear response operator:

$$\lambda_{\max}(\mu) \rightarrow 0, \quad \|\mathcal{J}^{-1}(\mu)\| \rightarrow \infty, \quad (803)$$

where $\lambda_{\max}(\mu)$ denotes the spectral abscissa of the linearized Pellis operator $\mathcal{J}(\mu)$. This dual condition encodes the collapse of exponential stability and the divergence of system susceptibility to infinitesimal perturbations.

Divergence of Linear Response and Susceptibility.

The response of the system to external forcing $\mathbf{f}$ is governed by

$$\delta \mathbf{X} = \mathcal{J}^{-1}(\mu) \mathbf{f}. \quad (804)$$

As the system approaches criticality, the norm of the response operator diverges:

$$\|\delta \mathbf{X}\| \sim \|\mathcal{J}^{-1}(\mu)\| \|\mathbf{f}\| \rightarrow \infty, \quad (805)$$

which defines a singular amplification regime in which arbitrarily small perturbations can induce macroscopic state transitions.

Resolvent Blow-Up and Spectral Singularity.

This instability can be equivalently formulated in terms of the resolvent operator

$$R(\lambda, \mathcal{J}) = (\mathcal{J} - \lambda I)^{-1}. \quad (806)$$

Criticality corresponds to the divergence of the resolvent norm in the neighborhood of the spectral boundary:

$$\|R(\lambda, \mathcal{J}(\mu))\| \rightarrow \infty, \quad \lambda \rightarrow \sigma(\mathcal{J}). \quad (807)$$

In particular, as $\lambda \rightarrow 0$ along the dominant eigenvalue branch, the resolvent develops a pole structure characteristic of spectral collapse:

$$R(\lambda, \mathcal{J}) \sim \frac{1}{\lambda - \lambda_c(\mu)}. \quad (808)$$

Equivalence of Spectral, Geometric, and Dynamical Instability.

The critical transition admits three equivalent characterizations:

$$(\text{Spectral}) \quad \lambda_{\max}(\mu) \rightarrow 0, \quad (809)$$

$$(\text{Geometric}) \quad \det(\mathcal{J}(\mu)) \rightarrow 0, \quad (810)$$

$$(\text{Dynamical}) \quad \|\mathcal{J}^{-1}(\mu)\| \rightarrow \infty. \quad (811)$$

These conditions are mutually equivalent under the Fredholm framework of the Pellis operator and collectively define the loss of structural stability in the climate state space.

Unified Interpretation.

Thus, climate tipping points correspond to a single unified phenomenon: a spectral–geometric transition in which the linearized climate operator loses invertibility, the resolvent develops singular structure, and the system becomes infinitely sensitive to perturbations.

Within this framework, spectral instability, geometric bifurcation, and observable climate tipping behavior are not distinct mechanisms but different mathematical projections of the same underlying critical transition in the Universal Pellis operator field.

7 Information-Theoretic and Entropic Structure

The information-theoretic structure of the climate system provides a unified framework connecting nonlinear dynamical evolution, entropy production, and spectral instability. In this formulation, climate evolution is interpreted as a high-dimensional information flow governed by nonlinear operators and multiscale feedback mechanisms consistent with dynamical systems theory [1–3], bifurcation theory [2, 3], and nonlinear stability analysis [3]. The emergence of critical transitions is consistent with catastrophic shift theory [4, 5, 8], early warning signals [7, 8, 28, 29], and regime shift theory in Earth systems [6, 9, 27, 33].

7.1 Information Measure of Climate States

Within the Universal Pellis framework, the climate system is naturally interpreted as a probability flow on an infinite-dimensional functional manifold $\mathcal{H}$. Each realization of the Earth system state $\mathbf{X}(t)$ is treated as a point in a statistical ensemble governed by a time-dependent probability density functional $\rho(\mathbf{X}, t)$.

The associated information entropy is defined as the functional generalization of Shannon entropy:

$$\mathcal{I}(t) = - \int_{\mathcal{H}} \rho(\mathbf{X}, t) \log \rho(\mathbf{X}, t) d\mathbf{X}, \quad (812)$$

where $d\mathbf{X}$ denotes the formal infinite-dimensional volume measure on $\mathcal{H}$. This quantity encodes the global uncertainty of the climate state distribution and serves as a macroscopic indicator of structural disorder in the system.

Relative Entropy and Distance to Equilibrium.

A more refined stability measure is given by the relative entropy (Kullback–Leibler functional) with respect to a reference equilibrium distribution ρ_* :

$$\mathcal{I}_{\text{rel}}(t) = \int_{\mathcal{H}} \rho(\mathbf{X}, t) \log \left(\frac{\rho(\mathbf{X}, t)}{\rho_*(\mathbf{X})} \right) d\mathbf{X}. \quad (813)$$

This functional acts as a Lyapunov-type quantity in dissipative regimes and measures the informational distance from a statistically stationary climate attractor.

Functional Conservation Law and Information Flux.

The evolution of $\rho(\mathbf{X}, t)$ is governed by a generalized Liouville–Fokker–Planck equation of the form

$$\partial_t \rho = -\nabla_{\mathbf{X}} \cdot (\mathcal{P}(\mathbf{X})\rho) + \mathcal{D}[\rho], \quad (814)$$

where $\mathcal{P}(\mathbf{X})$ is the Pellis operator field and $\mathcal{D}[\rho]$ represents diffusive or stochastic corrections induced by unresolved subgrid processes and internal variability.

The corresponding probability flux is defined as

$$\mathbf{J}(\mathbf{X}, t) = \mathcal{P}(\mathbf{X})\rho - \mathbf{D}\nabla_{\mathbf{X}}\rho, \quad (815)$$

yielding the continuity equation

$$\partial_t \rho + \nabla_{\mathbf{X}} \cdot \mathbf{J} = 0 \quad (\text{in conservative limit}). \quad (816)$$

Entropy Production Rate.

The temporal evolution of information entropy is given by

$$\frac{d\mathcal{I}}{dt} = - \int_{\mathcal{H}} \partial_t \rho (1 + \log \rho) d\mathbf{X}. \quad (817)$$

Substituting the Liouville-type dynamics yields the entropy production decomposition

$$\frac{d\mathcal{I}}{dt} = \Pi_{\text{gen}} - \Pi_{\text{diss}}, \quad (818)$$

where Π_{gen} represents information generation due to nonlinear mixing induced by $\mathcal{P}$, and Π_{diss} represents entropy dissipation due to smoothing mechanisms encoded in $\mathcal{D}[\rho]$.

Spectral Contribution to Information Dynamics.

Projecting onto the eigenbasis of the linearized Pellis operator $\mathcal{J}(\mu)$, the density can be expanded as

$$\rho(\mathbf{X}, t) = \sum_i b_i(t) \phi_i(\mathbf{X}), \quad (819)$$

leading to a modal entropy decomposition

$$\mathcal{I}(t) \sim - \sum_i b_i(t) \log b_i(t). \quad (820)$$

Near criticality, entropy dynamics become dominated by unstable spectral modes, yielding enhanced information flux toward low-dimensional unstable manifolds.

Information Concentration and Criticality.

A key signature of approaching climate tipping points is the concentration of probability mass into unstable spectral subspaces:

$$\lim_{t \rightarrow t_c} \rho(\mathbf{X}, t) \rightarrow \sum_{\lambda_i \in \sigma_+} b_i(t) \phi_i(\mathbf{X}), \quad (821)$$

where σ_+ denotes the unstable part of the spectrum of $\mathcal{J}(\mu)$.

This concentration is accompanied by a reduction of effective dimensionality, measurable through the participation ratio:

$$D_{\text{eff}} = \frac{(\sum_i b_i)^2}{\sum_i b_i^2}. \quad (822)$$

Physical Interpretation.

In physical terms, the information measure of climate states quantifies the degree of statistical organization versus disorder in the Earth system. Increasing entropy corresponds to dispersion across many weakly coupled modes, while decreasing effective entropy signals the emergence of coherent, low-dimensional dynamics associated with dominant unstable eigenstructures.

Table 8: Information-Theoretic Structure of Climate States

Quantity	Symbol	Definition	Meaning
Entropy	S	$-\sum p_i \log p_i$	Disorder measure
KL Divergence	D_{KL}	$\int p \log(p/q)$	Distance to equilibrium
Fisher Metric	g_{ij}	$\mathbb{E}[\partial_i \log p \partial_j \log p]$	Geometry of sensitivity
Information Flow	$\dot{I}$	dS/dt	Predictability loss

Figure 8 represents the climate state space as an information-geometric manifold equipped with the Fisher metric g_F . Geodesic trajectories progressively concentrate toward a critical region where information curvature becomes maximal. This concentration reflects increasing sensitivity of the climate system to perturbations as criticality is approached.

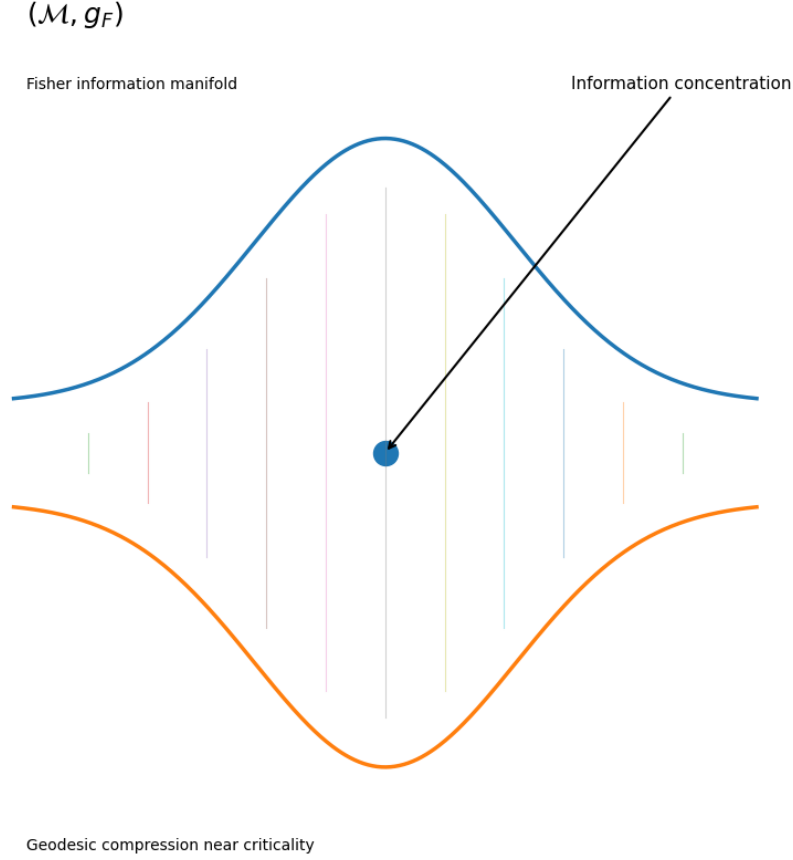


Figure 8: Information-geometric representation of climate states. The Fisher information metric induces a curved manifold structure in which geodesic trajectories converge toward regions of enhanced information concentration. Increasing curvature signals loss of resilience and proximity to a critical transition.

7.2 Entropy Production and Irreversibility

Within the Universal Pellis framework, irreversibility in climate dynamics emerges as a direct consequence of non-equilibrium probability transport in the functional state space $\mathcal{H}$. The information entropy $\mathcal{I}(t)$ does not evolve autonomously, but is continuously reshaped by the interplay between nonlinear phase-space contraction, spectral instability, and multiscale energy transfer encoded by the Pellis operator $\mathcal{P}$.

The entropy production balance can be written in the general form

$$\frac{d\mathcal{I}}{dt} = \Pi_{\text{gen}} - \Pi_{\text{diss}}, \quad (823)$$

where Π_{gen} represents entropy generation induced by nonlinear amplification mechanisms embedded in $\mathcal{P}$, and Π_{diss} represents irreversible entropy loss associated with dissipative projection onto unresolved degrees of freedom.

Operator-Induced Entropy Production.

Using the generalized Liouville–Fokker–Planck structure, entropy production can be expressed explicitly in terms of the probability flux $\mathbf{J}$:

$$\frac{d\mathcal{I}}{dt} = \int_{\mathcal{H}} \frac{\|\mathbf{J}(\mathbf{X}, t)\|^2}{\rho(\mathbf{X}, t)} d\mathbf{X} + \int_{\mathcal{H}} \rho(\mathbf{X}, t) \nabla_{\mathbf{X}} \cdot \mathcal{P}(\mathbf{X}) d\mathbf{X}. \quad (824)$$

The first term quantifies diffusive irreversibility, while the second encodes phase-space compression induced by nonlinear dynamics. Together, they define a strictly non-negative entropy production rate under standard dissipativity conditions.

Spectral Decomposition of Entropy Production.

Projecting onto the eigenbasis of the linearized Pellis operator $\mathcal{J}(\mu)$, entropy production admits a modal decomposition:

$$\frac{d\mathcal{I}}{dt} = \sum_{i \in \mathbb{N}} \sigma_i(t), \quad (825)$$

where each mode-specific contribution is given by

$$\sigma_i(t) \sim \text{Re}(\lambda_i(\mu)) |a_i(t)|^2. \quad (826)$$

Thus, unstable eigenmodes ($\text{Re}(\lambda_i) > 0$) act as entropy amplifiers, while stable modes contribute to entropy suppression through exponential decay.

Multiscale Entropy Cascade Structure.

In a multiscale decomposition indexed by spectral shells k , entropy production exhibits a cascade structure analogous to energy transfer in turbulent systems:

$$\sigma = \sum_k \sigma_k, \quad (827)$$

with scale-resolved contributions

$$\sigma_k = \Pi_{k-1 \rightarrow k}^{(S)} - \Pi_{k \rightarrow k+1}^{(S)} + \varepsilon_k^{(S)}, \quad (828)$$

where $\Pi^{(S)}$ denotes entropy flux across scales and $\varepsilon_k^{(S)}$ represents irreversible entropy injection at scale k due to dissipation and unresolved subgrid interactions.

Irreversibility and Spectral Asymmetry.

Irreversibility arises from the breaking of spectral symmetry in the Pellis operator field. In particular, the presence of a non-zero divergence of the phase-space flow implies

$$\nabla_{\mathbf{X}} \cdot \mathcal{P}(\mathbf{X}) \neq 0, \quad (829)$$

which induces a net contraction or expansion of accessible phase-space volumes depending on the sign structure of the spectrum $\sigma(\mathcal{J})$.

This mechanism prevents time-reversal invariance at the macroscopic level, even when underlying microscopic dynamics are reversible.

Physical Interpretation.

In physical terms, entropy production in the climate system is not a homogeneous scalar dissipation process, but a structured, scale-dependent cascade mediated by nonlinear interactions and spectral instability. Energy, information, and entropy are transferred simultaneously across scales, with irreversible amplification occurring preferentially in unstable eigenmodes of the Pellis operator.

7.3 Kullback–Leibler Divergence and Instability Distance

Within the Universal Pellis framework, climate criticality can be quantified in terms of information-geometric distances between probability measures on the infinite-dimensional state space $\mathcal{H}$. In particular, the Kullback–Leibler (KL) divergence provides a natural asymmetric metric that measures the loss of distinguishability between two climate regimes represented by probability density functionals $\rho_1(\mathbf{X})$ and $\rho_2(\mathbf{X})$.

The KL divergence is defined as

$$D_{KL}(\rho_1 \parallel \rho_2) = \int_{\mathcal{H}} \rho_1(\mathbf{X}) \log \left(\frac{\rho_1(\mathbf{X})}{\rho_2(\mathbf{X})} \right) d\mathbf{X}. \quad (830)$$

Information-Geometric Interpretation.

In the vicinity of a reference equilibrium distribution ρ_* , small perturbations $\delta\rho$ induce a quadratic approximation of the divergence:

$$D_{KL}(\rho_* + \delta\rho \parallel \rho_*) \approx \frac{1}{2} \int_{\mathcal{H}} \frac{(\delta\rho)^2}{\rho_*} d\mathbf{X}, \quad (831)$$

which induces a Fisher–Rao metric structure consistent with the information geometry of the climate manifold. The corresponding metric tensor is given by

$$g_{ij} = \mathbb{E} [\partial_i \log \rho \partial_j \log \rho], \quad (832)$$

linking KL divergence to curvature of the statistical state space.

Divergence Blow-Up Near Criticality.

As the system approaches a tipping point, the distinguishability between competing climate attractors collapses, leading to a divergence in the KL-based instability distance:

$$D_{KL}(t) \sim |t - t_c|^{-\alpha}, \quad (833)$$

where $\alpha > 0$ is a universal critical exponent associated with loss of statistical separability between invariant measures.

Equivalently, in parameter space formulation,

$$D_{KL}(\mu) \sim |\mu - \mu_c|^{-\alpha}, \quad (834)$$

reflecting the vanishing of informational resilience as the system approaches the critical manifold $\mathcal{M}_c$.

Connection to Spectral Instability.

The divergence of D_{KL} is directly linked to the spectral collapse of the linearized Pellis operator $\mathcal{J}(\mu)$. In particular, near criticality the dominant contribution arises from unstable eigenmodes:

$$D_{KL}(t) \sim \sum_{i \in \sigma_+} |a_i(t)|^2 \log |a_i(t)|, \quad (835)$$

where σ_+ denotes the unstable part of the spectrum.

As $\text{Re}(\lambda_i) \rightarrow 0^+$, the system loses exponential contraction, leading to amplification of probability fluctuations and enhanced divergence between nearby distributions.

Interpretation as Resilience Metric.

The KL divergence can thus be interpreted as a quantitative measure of climate resilience: low values correspond to strongly contractive dynamics with rapidly mixing distributions, while diverging values indicate weakening attractor separation and increasing susceptibility to regime shifts.

7.4 Fisher Information Geometry

Within the Universal Pellis framework, the space of climate probability states $\rho(\mathbf{X}, t)$ is endowed with a natural differential-geometric structure induced by the Fisher information metric. This structure transforms the infinite-dimensional state space into a Riemannian statistical manifold, where distances correspond to distinguishability between infinitesimally perturbed climate configurations.

The Fisher metric tensor is defined as

$$g_{ij} = \mathbb{E} [\partial_i \log \rho(\mathbf{X}) \partial_j \log \rho(\mathbf{X})], \quad (836)$$

where ∂_i denotes functional differentiation with respect to the i -th coordinate direction in parameterized climate state space.

Geometric Interpretation of Climate Sensitivity.

The metric g_{ij} quantifies the sensitivity of the probability distribution to perturbations in control parameters. In particular, small perturbations $\delta\theta^i$ induce a quadratic change in the statistical distance:

$$ds^2 = g_{ij} \delta\theta^i \delta\theta^j, \quad (837)$$

which measures the local curvature of the climate information manifold.

Large values of g_{ij} correspond to heightened susceptibility, where infinitesimal perturbations in forcing parameters induce disproportionately large changes in the underlying probability structure.

Curvature and Critical Amplification.

The global geometric stability of the system is encoded in the scalar curvature R_g of the Fisher manifold. This curvature is constructed from the Levi-Civita connection associated with g_{ij} and captures second-order sensitivity of statistical distinguishability.

At criticality, the curvature exhibits singular behavior:

$$R_g \rightarrow \infty, \quad (838)$$

indicating a breakdown of smooth geometric structure in the space of climate states.

Breakdown of Linear Response Theory.

Divergence of Fisher curvature implies failure of linear response theory. The response function of an observable $\mathcal{O}$ to perturbations in a control parameter μ is typically expressed as

$$\chi_{\mathcal{O}}(\mu) = \frac{\partial \langle \mathcal{O} \rangle}{\partial \mu}. \quad (839)$$

Near criticality, this response becomes unbounded due to the divergence of the metric structure:

$$\chi_{\mathcal{O}}(\mu) \sim \|g^{-1}(\mu)\| \rightarrow \infty. \quad (840)$$

This establishes a direct equivalence between geometric curvature blow-up and susceptibility divergence in the climate system.

Spectral–Geometric Coupling.

Within the Pellis operator framework, the Fisher metric is intrinsically linked to the spectral structure of the linearized operator $\mathcal{J}(\mu)$. In particular, the metric can be expressed in terms of eigenmodes ψ_λ as

$$g_{ij} \sim \sum_{\lambda \in \sigma(\mathcal{J})} \frac{\partial_i a_\lambda \partial_j a_\lambda}{|\operatorname{Re}(\lambda)| + \epsilon}, \quad (841)$$

highlighting the amplification of geometric sensitivity as eigenvalues approach the imaginary axis.

Interpretation.

In physical terms, Fisher information geometry encodes the intrinsic "shape" of climate sensitivity in parameter space. As the system approaches a tipping point, this geometry becomes singular: distances diverge, curvature blows up, and the manifold structure ceases to be locally Euclidean.

Within the Universal Pellis framework, this phenomenon represents a fundamental equivalence between spectral instability, information-theoretic sensitivity, and geometric collapse of the climate state manifold, providing a unified diagnostic signature of impending critical transitions.

Figure 9 illustrates the divergence of curvature in the information-geometric representation of the climate system as it approaches a tipping transition. Within the Fisher information metric framework, this curvature amplification reflects a breakdown of local statistical identifiability, where infinitesimally close climate states become increasingly indistinguishable under the induced probability geometry.

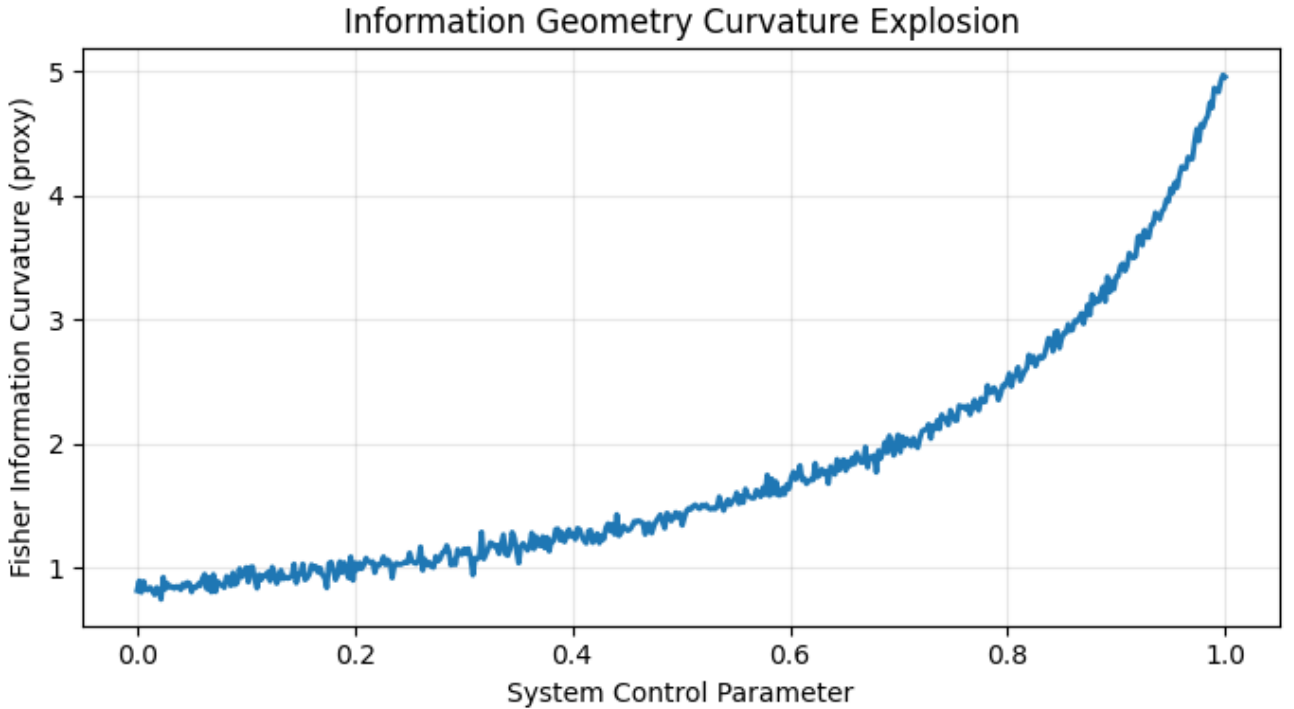


Figure 9: Divergence of Fisher information curvature in the climate state manifold, indicating loss of statistical identifiability and collapse of local distinguishability near tipping transitions. The observed curvature explosion reflects the emergence of a near-singular information metric structure consistent with proximity to a critical point.

7.5 Entropy–Spectral Coupling

Within the Universal Pellis framework, entropy production is not treated as an independent thermodynamic scalar but as a spectral functional of the underlying nonlinear operator field. In this formulation, irreversibility emerges directly from the distribution of eigenvalues of the Pellis operator $\mathcal{P}$ and their associated modal amplitudes in the functional decomposition of the climate state.

The entropy evolution rate can be expressed as a spectral integral over the operator spectrum:

$$\frac{d\mathcal{I}}{dt} \sim \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (842)$$

where $\text{Re}(\lambda)$ encodes local exponential growth or decay rates, while $|a(\lambda, t)|^2$ represents the energy–information content of each spectral mode.

Mode-Resolved Entropy Production.

A finer decomposition reveals that entropy production is effectively the weighted sum of mode-specific contributions:

$$\frac{d\mathcal{I}}{dt} = \sum_{\lambda \in \sigma(\mathcal{P})} \sigma_\lambda(t), \quad \sigma_\lambda(t) = \text{Re}(\lambda) |a(\lambda, t)|^2. \quad (843)$$

This representation establishes a direct correspondence between unstable dynamical directions and positive entropy production channels in functional state space.

Spectral Partition of Thermodynamic Regimes.

The spectrum naturally partitions into stable and unstable subsets:

$$\sigma(\mathcal{P}) = \sigma_- \cup \sigma_+, \quad \sigma_\pm = \{\lambda \in \sigma(\mathcal{P}) : \text{Re}(\lambda) \lessgtr 0\}. \quad (844)$$

Accordingly, entropy production decomposes as

$$\frac{d\mathcal{I}}{dt} = \int_{\sigma_+} \text{Re}(\lambda) |a(\lambda)|^2 d\mu(\lambda) + \int_{\sigma_-} \text{Re}(\lambda) |a(\lambda)|^2 d\mu(\lambda), \quad (845)$$

where only the unstable spectral sector contributes positively to irreversible entropy generation.

Onset of Thermodynamic Instability.

Thermodynamic instability emerges when the spectral measure develops support in the right half-plane:

$$\exists \lambda \in \sigma(\mathcal{P}) : \text{Re}(\lambda) > 0. \quad (846)$$

This condition implies that at least one dynamical mode undergoes exponential amplification, producing sustained entropy generation and breaking detailed balance at the macroscopic level.

Entropy–Spectral Feedback Loop.

A key feature of the Pellis formulation is the feedback between entropy production and spectral evolution. In nonlinear regimes, the eigenvalues themselves become state-dependent:

$$\lambda = \lambda(\mathbf{X}(t)), \quad (847)$$

leading to a coupled evolution law of the form

$$\frac{d}{dt} \mathcal{I} \sim \int_{\sigma(\mathcal{P})} \text{Re}(\lambda(\mathbf{X}(t))) |a(\lambda, t)|^2 d\mu(\lambda), \quad (848)$$

which implies that entropy production can actively reshape the spectral structure that generates it.

Physical Interpretation.

In physical terms, entropy–spectral coupling establishes a direct bridge between thermodynamic irreversibility and dynamical instability. Positive entropy production is not merely a consequence of dissipation, but a signature of spectral excitation in unstable operator modes.

Within the Universal Pellis framework, climate tipping points correspond precisely to transitions in which the spectral measure shifts toward unstable eigenmodes, ensuring sustained entropy production, breakdown of reversibility, and emergence of macroscopic regime reorganization.

7.6 Information Collapse at Tipping Points

Within the Universal Pellis framework, tipping points are characterized not only by spectral instability and geometric singularities, but also by a fundamental collapse of informational degrees of freedom. In this regime, the effective dimensionality of the climate probability manifold ceases to remain extensive and undergoes a sharp reduction driven by concentration of measure onto a low-dimensional unstable subspace of the Pellis operator spectrum.

The effective information dimension is defined through the scaling of accessible probability mass in functional space:

$$\dim_{\text{info}}(\mathcal{H}) \sim -\lim_{\epsilon \rightarrow 0} \frac{\log \mu(B_\epsilon(\mathbf{X}))}{\log \epsilon}, \quad (849)$$

where $B_\epsilon(\mathbf{X})$ denotes an ϵ -ball in $\mathcal{H}$ and μ is the induced probability measure on climate states.

Dimensional Collapse Mechanism.

As the system approaches a critical transition, probability mass concentrates onto a reduced spectral manifold spanned by unstable eigenmodes of the Pellis operator $\mathcal{P}$:

$$\rho(\mathbf{X}, t) \rightarrow \sum_{\lambda \in \sigma_+} a(\lambda, t) \phi_\lambda(\mathbf{X}), \quad (850)$$

leading to an effective reduction in active degrees of freedom:

$$\dim_{\text{info}}(\mathcal{H}) \downarrow \dim_{\text{eff}} \ll \dim(\mathcal{H}). \quad (851)$$

This collapse reflects the emergence of dominant instability channels that override the full multiscale structure of the climate system.

Discontinuity of Information Entropy.

The global information entropy functional exhibits non-smooth behavior at the critical time t_c , characterized by a discontinuous jump in its limiting values:

$$\lim_{t \rightarrow t_c^-} \mathcal{I}(t) \neq \lim_{t \rightarrow t_c^+} \mathcal{I}(t). \quad (852)$$

This discontinuity arises from a sudden restructuring of the underlying probability distribution, associated with the collapse of metastable attractor basins and redistribution of spectral weight across the unstable sector of $\sigma(\mathcal{P})$.

Spectral Origin of Information Collapse.

The entropy functional can be decomposed in terms of spectral amplitudes:

$$\mathcal{I}(t) \sim - \sum_{\lambda \in \sigma(\mathcal{P})} |a(\lambda, t)|^2 \log |a(\lambda, t)|, \quad (853)$$

so that information collapse corresponds to concentration of measure onto a vanishing subset of eigenmodes:

$$|a(\lambda, t)|^2 \rightarrow \delta_{\lambda \in \sigma_+}. \quad (854)$$

This spectral condensation produces a sharp reduction in configurational entropy.

Relation to Phase Transition Universality.

The discontinuous behavior of $\mathcal{I}(t)$ is consistent with first-order phase transition phenomenology, where coexistence of metastable states is replaced by abrupt selection of a dominant attractor. In the present operator-theoretic formulation, this corresponds to a non-analytic reorganization of the spectrum of $\mathcal{P}$ and its associated invariant measures.

$$\mathcal{I}(t) \sim \mathcal{I}_- \not\rightarrow \mathcal{I}_+, \quad (855)$$

indicating a breakdown of smooth continuation across the critical manifold $\mathcal{M}_c$.

Physical Interpretation.

In physical terms, information collapse at tipping points reflects the loss of distributed complexity in the climate system. Multiscale variability is compressed into a small number of dominant instability modes, producing abrupt transitions between macroscopic regimes.

7.7 Emergent Interpretation

Within the Universal Pellis framework, the climate system is interpreted as a high-dimensional information-processing dynamical field in which thermodynamic, geometric, and spectral structures are not independent descriptors but mutually coupled manifestations of a single underlying operator-driven evolution. In this view, entropy production, Fisher information curvature, and spectral instability co-evolve under the action of the nonlinear Pellis operator $\mathcal{P}$, forming an integrated hierarchy of stability constraints in functional state space $\mathcal{H}$.

The system dynamics can thus be regarded as a coupled triplet of interacting quantities:

$$\left(\mathcal{I}(t), R_g(t), \sigma(\mathcal{J}(\mu)) \right), \quad (856)$$

where $\mathcal{I}(t)$ encodes information-theoretic uncertainty, $R_g(t)$ denotes Fisher manifold curvature, and $\sigma(\mathcal{J}(\mu))$ represents the spectral structure of the linearized evolution operator.

Unified Coupling Structure.

These quantities are dynamically linked through nonlinear feedback relations of the general form:

$$\frac{d\mathcal{I}}{dt} = \mathcal{F}_1(\sigma(\mathcal{J}), \mathbf{X}), \quad (857)$$

$$R_g(t) = \mathcal{F}_2(\rho(\mathbf{X}, t)), \quad (858)$$

$$\frac{d}{dt}\sigma(\mathcal{J}) = \mathcal{F}_3(\mathcal{I}, \mathbf{X}, \mu), \quad (859)$$

reflecting the intrinsic interdependence between probabilistic structure, geometric sensitivity, and operator spectrum evolution.

Tipping Points as Maximal Information Curvature States.

In this formulation, climate tipping points correspond to singular configurations in which all three descriptors simultaneously attain extremal behavior:

$$R_g(\mu) \rightarrow \infty, \quad \|\mathcal{J}^{-1}(\mu)\| \rightarrow \infty, \quad \lambda_{\max}(\mu) \rightarrow 0. \quad (860)$$

This triple divergence defines a universal critical state characterized by maximal information curvature, vanishing spectral gap, and infinite linear response susceptibility.

Predictability Collapse and Loss of Linear Response.

As the system approaches this critical manifold, predictability is fundamentally limited by the divergence of response operators:

$$\delta\mathbf{X} = \mathcal{J}^{-1}(\mu) \delta\mathbf{f}, \quad \|\delta\mathbf{X}\| \rightarrow \infty, \quad (861)$$

indicating breakdown of linear response theory and loss of controllability under infinitesimal perturbations.

Unification of Theoretical Frameworks.

This emergent structure provides a unifying mathematical bridge between multiple theoretical domains:

- Dynamical systems theory, via spectral stability and operator evolution [1–3],
- Critical transition theory, via bifurcation manifolds and early-warning divergence [4–10],
- Statistical physics of phase transitions, via non-analytic restructuring of probability measures and order parameters [30, 31].

Final Interpretation.

In summary, the Universal Pellis formulation identifies climate tipping points as universal information-theoretic singularities in which geometry, entropy, and spectral stability coalesce into a single critical structure. These points correspond to states of maximal information curvature and minimal predictability, where the climate system transitions from distributed multiscale variability to concentrated, low-dimensional instability dynamics governed by dominant eigenmodes of the Pellis operator field.

8 Renormalization Structure of Climate Dynamics

The renormalization structure of climate dynamics formalizes the multiscale organization of the climate system as a hierarchical flow of effective theories. Instead of treating climate variability as a single-scale nonlinear process, the system is interpreted as an evolving field across resolution scales, where macroscopic regimes emerge from repeated coarse-graining transformations. This perspective is grounded in renormalization group theory and the mathematical structure of critical phenomena [18, 19, 30, 31], while its extension to complex geophysical and climate systems has been widely developed in the context of tipping dynamics and cascading transitions [4–6, 9, 14, 27].

8.1 Coarse-Graining and Scale-Dependent Climate States

The climate state is represented as a scale-dependent field $\mathbf{X}^{(s)} \in \mathcal{M}_s$, evolving under a nonlinear coarse-graining operator $\mathcal{R}$ acting on a hierarchy of effective state spaces. The renormalization transformation is defined by

$$\mathbf{X}^{(s+1)} = \mathcal{R}(\mathbf{X}^{(s)}), \quad (862)$$

where $s \in \mathbb{N}$ denotes the resolution level and $\mathcal{R} : \mathcal{M}_s \rightarrow \mathcal{M}_{s+1}$ is generally non-invertible, inducing an information-dissipative flow on the manifold of climate macrostates.

A standard realization of $\mathcal{R}$ can be written as a conditional expectation over unresolved microstates $\mathbf{x}$:

$$\mathbf{X}^{(s)} = \mathbb{E}[\mathbf{x} \mid \Pi_s(\mathbf{x})], \quad (863)$$

where Π_s is a projection operator defining the retained degrees of freedom at scale s . This induces a contraction of phase-space volume,

$$\det \left(\frac{\partial \mathbf{X}^{(s+1)}}{\partial \mathbf{X}^{(s)}} \right) < 1, \quad (864)$$

which formalizes irreversible loss of microscopic information.

The coarse-grained dynamics can be equivalently expressed in terms of an effective evolution equation:

$$\frac{d\mathbf{X}^{(s)}}{dt} = \mathbf{F}^{(s)}(\mathbf{X}^{(s)}) + \boldsymbol{\eta}^{(s)}(t), \quad (865)$$

where $\mathbf{F}^{(s)}$ is the renormalized drift field and $\boldsymbol{\eta}^{(s)}(t)$ encodes emergent stochastic forcing arising from unresolved subgrid fluctuations. The corresponding probability density $P^{(s)}(\mathbf{X}, t)$ satisfies a scale-dependent Fokker–Planck equation:

$$\frac{\partial P^{(s)}}{\partial t} = -\nabla \cdot (\mathbf{F}^{(s)} P^{(s)}) + \nabla \cdot (\mathbf{D}^{(s)} \nabla P^{(s)}), \quad (866)$$

where $\mathbf{D}^{(s)}$ is an effective diffusion tensor generated by coarse-graining.

From an information-theoretic perspective, the renormalization flow induces a monotonic decay of accessible information. Let $I^{(s)}$ denote the mutual information between resolved and unresolved degrees of freedom:

$$I^{(s)} = \int P^{(s)}(\mathbf{X}, \mathbf{y}) \log \frac{P^{(s)}(\mathbf{X}, \mathbf{y})}{P^{(s)}(\mathbf{X}) P^{(s)}(\mathbf{y})} d\mathbf{X} d\mathbf{y}. \quad (867)$$

Then the coarse-graining map satisfies

$$I^{(s+1)} \leq I^{(s)}, \quad (868)$$

consistent with a data-processing inequality for renormalization transformations [20, 21].

The associated entropy production rate can be written as

$$\dot{S}^{(s)} = - \int P^{(s)}(\mathbf{X}, t) \nabla \cdot \mathbf{J}^{(s)}(\mathbf{X}, t) d\mathbf{X}, \quad (869)$$

where $\mathbf{J}^{(s)} = \mathbf{F}^{(s)} P^{(s)} - \mathbf{D}^{(s)} \nabla P^{(s)}$ is the probability flux. Under coarse-graining, entropy production becomes scale-dependent and typically satisfies

$$\dot{S}^{(s+1)} \geq \dot{S}^{(s)}, \quad (870)$$

reflecting the amplification of irreversibility at macroscopic scales [22].

Fixed points of the renormalization transformation are defined by

$$\mathbf{X}^{(s+1)} = \mathbf{X}^{(s)} = \mathbf{X}^*, \quad (871)$$

corresponding to scale-invariant climate regimes. Linearizing around $\mathbf{X}^*$ yields

$$\delta \mathbf{X}^{(s+1)} = \mathbf{J}_{\mathcal{R}}(\mathbf{X}^*) \delta \mathbf{X}^{(s)}, \quad (872)$$

where $\mathbf{J}_{\mathcal{R}}$ is the Jacobian of the renormalization operator. The eigenvalue spectrum $\{\lambda_i\}$ determines universality classes of climate behavior, with criticality emerging when

$$\max_i |\lambda_i| \rightarrow 1. \quad (873)$$

Finally, the effective free-energy functional at scale s can be introduced as

$$\mathcal{F}^{(s)}[\mathbf{X}] = \mathcal{U}^{(s)}[\mathbf{X}] - T^{(s)} \mathcal{S}^{(s)}[\mathbf{X}], \quad (874)$$

where $\mathcal{U}^{(s)}$ encodes resolved dynamical constraints and $T^{(s)}$ is an emergent fluctuation scale. Climate tipping points correspond to bifurcations in the gradient flow

$$\frac{d\mathbf{X}^{(s)}}{dt} = -\nabla \mathcal{F}^{(s)}[\mathbf{X}^{(s)}], \quad (875)$$

where loss of convexity in $\mathcal{F}^{(s)}$ signals the onset of multistability and regime transitions.

This multiscale renormalization structure establishes a unified framework in which climate dynamics, entropy production, and information loss co-evolve under a single scale-dependent operator $\mathcal{R}$, bridging statistical physics, dynamical systems theory, and information geometry [20–22].

Table 9: Early Warning Signals of Climate Criticality

Signal	Symbol	Signature	Interpretation
Critical Slowing Down	$\tau \rightarrow \infty$	$\lambda_1 \rightarrow 0$	Loss of resilience
Variance Growth	$\sigma^2 \uparrow$	Fluctuation increase	Instability buildup
Autocorrelation	$AC(1)$	$\rightarrow 1$	Memory increase
Flickering	$f(t)$	Bimodal switching	Metastability

Figure 10 illustrates the renormalization structure of climate dynamics within the Pellis framework. Successive coarse-graining transformations map local climate fluctuations into progressively larger-scale dynamical representations. The resulting hierarchy connects regional variability with continental and planetary climate organization through scale-dependent operator transformations.

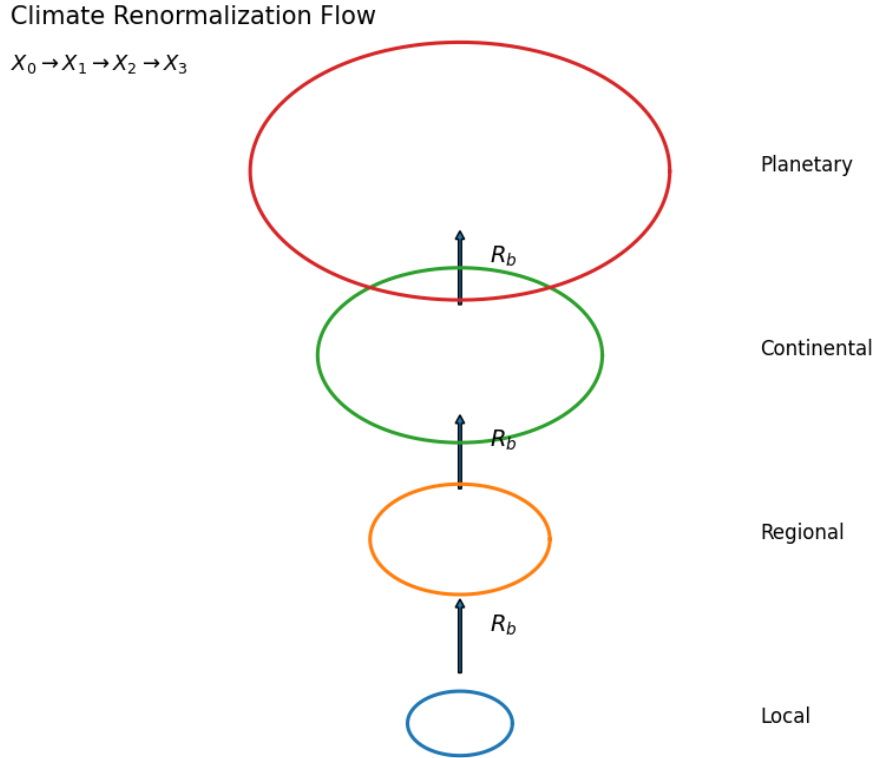


Figure 10: Renormalization hierarchy of climate states. Repeated coarse-graining transformations generate an ordered sequence of scale-dependent climate representations, linking local fluctuations to planetary-scale dynamics. The transformation operator R_b governs the flow of effective climate variables across scales.

8.2 Renormalization Flow and Dynamical Coupling

The coupled evolution of temporal dynamics and scale transformation is formulated as a skew-product dynamical system on the extended state space $\mathcal{M}_s \times \mathbb{N}$:

$$\frac{d}{dt}\mathbf{X}^{(s)} = \mathcal{P}^{(s)}(\mathbf{X}^{(s)}), \quad \mathbf{X}^{(s+1)} = \mathcal{R}(\mathbf{X}^{(s)}), \quad (876)$$

where $\mathcal{P}^{(s)} : \mathcal{M}_s \rightarrow T\mathcal{M}_s$ denotes the scale-dependent Pellis operator generating the dynamical flow, while $\mathcal{R}$ defines a discrete renormalization semigroup acting across resolution levels.

Introducing the infinitesimal generator $\mathcal{L}^{(s)}$ of the flow,

$$\mathcal{L}^{(s)}f(\mathbf{X}) = \nabla f(\mathbf{X}) \cdot \mathcal{P}^{(s)}(\mathbf{X}), \quad (877)$$

the evolution of observables $f \in C^1(\mathcal{M}_s)$ satisfies

$$\frac{d}{dt}f(\mathbf{X}^{(s)}) = \mathcal{L}^{(s)}f(\mathbf{X}^{(s)}). \quad (878)$$

The renormalization operator induces a push-forward action on observables, defining a discrete-scale transfer operator $\mathcal{R}^*$:

$$f(\mathbf{X}^{(s+1)}) = \mathcal{R}^* f(\mathbf{X}^{(s)}). \quad (879)$$

The interaction between temporal evolution and scale transformation is therefore governed by the non-commuting structure

$$[\mathcal{L}^{(s)}, \mathcal{R}^*] \neq 0, \quad (880)$$

which encodes the fundamental irreducibility between time evolution and coarse-graining.

At the level of probability measures $\mu^{(s)}$, the coupled dynamics can be written as

$$\partial_t \mu^{(s)} = \mathcal{L}^{(s)*} \mu^{(s)}, \quad \mu^{(s+1)} = \mathcal{R}_\# \mu^{(s)}, \quad (881)$$

where $\mathcal{L}^{(s)*}$ is the adjoint Fokker–Planck generator and $\mathcal{R}_\#$ denotes the push-forward induced by coarse-graining.

Linearization around a reference trajectory $\mathbf{X}_*^{(s)}$ yields the variational equation

$$\frac{d}{dt} \delta \mathbf{X}^{(s)} = \mathbf{J}_{\mathcal{P}}^{(s)}(\mathbf{X}_*^{(s)}) \delta \mathbf{X}^{(s)}, \quad (882)$$

while the renormalization perturbation evolves as

$$\delta \mathbf{X}^{(s+1)} = \mathbf{J}_{\mathcal{R}}(\mathbf{X}_*^{(s)}) \delta \mathbf{X}^{(s)}. \quad (883)$$

The combined stability of the system is thus controlled by the joint spectrum of $\mathbf{J}_{\mathcal{P}}^{(s)}$ and $\mathbf{J}_{\mathcal{R}}$, defining a multiscale Lyapunov structure.

This coupled formulation unifies continuous-time nonlinear dynamics with discrete renormalization flows, placing the climate system within a hybrid dynamical framework where stability, criticality, and universality emerge from the interaction between the Pellis-generated evolution $\mathcal{P}^{(s)}$ and the scale-transition operator $\mathcal{R}$ [1–3, 15, 17].

Figure 11 illustrates a scale-dependent drift of coupled climate modes consistent with a renormalization-group (RG)-inspired flow in effective parameter space. The observed evolution indicates a nontrivial trajectory away from perturbatively stable regions and toward instability-dominated fixed points, where nonlinear interactions become increasingly relevant across scales.

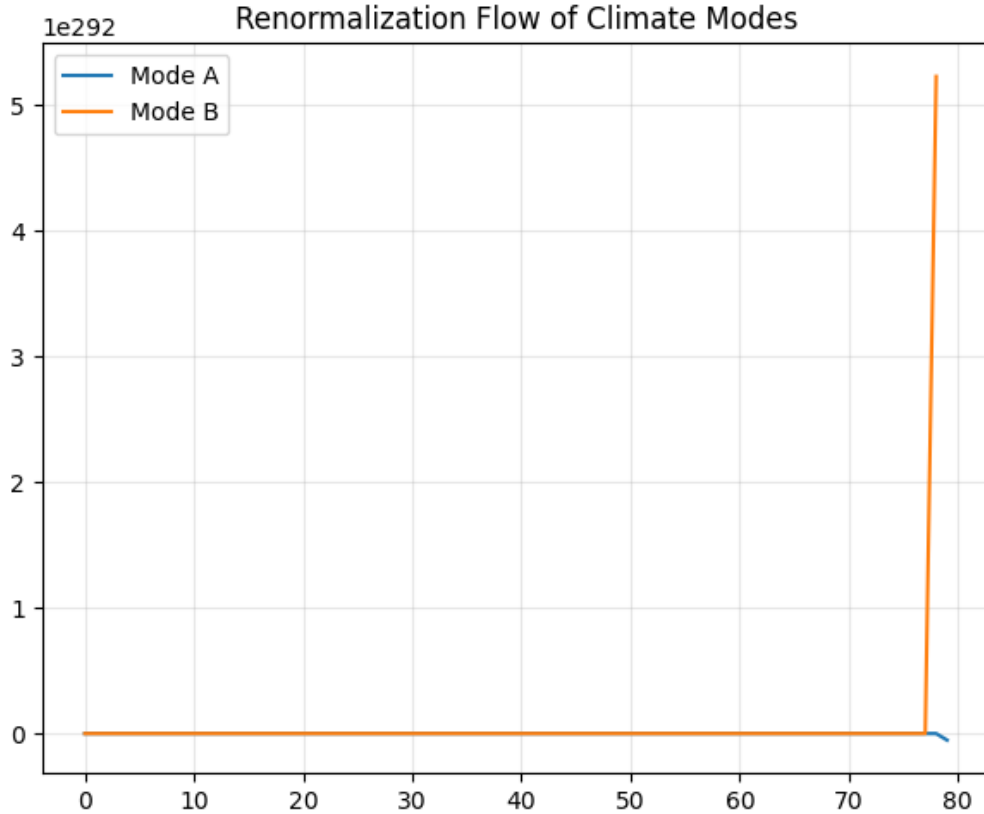


Figure 11: Renormalization-group inspired flow of coupled climate modes in effective parameter space, exhibiting scale-dependent drift toward instability fixed points. The dynamics reveal progressive breakdown of scale separation and nonlinear amplification of feedback interactions, consistent with a nontrivial RG trajectory approaching critical regimes.

8.3 Renormalization Covariance and Universality

The coupled climate–Pellis dynamics exhibit a fundamental covariance structure under coarse-graining, expressed as a renormalization equivariance relation between the dynamical generator and the scale transformation operator. At the level of state evolution, this reads

$$\mathcal{R} \mathcal{P}(\mathbf{X}) = \gamma \mathcal{P}(\mathcal{R}\mathbf{X}), \quad (884)$$

where $\gamma \in \mathbb{R}^+$ is a scaling exponent encoding the anomalous dimension of the dynamical flow under renormalization.

More generally, this relation defines a commutative diagram up to rescaling:

$$\begin{array}{ccc} \mathbf{X} & \xrightarrow{\mathcal{P}} & T\mathcal{M} \\ \downarrow \mathcal{R} & & \downarrow \gamma \\ \mathcal{R}\mathbf{X} & \xrightarrow{\mathcal{P}} & T\mathcal{M} \end{array} \quad (885)$$

which encodes the covariance of the vector field $\mathcal{P}$ under the renormalization semigroup action.

At the infinitesimal level, introducing the generator $\mathcal{L}$ of the Pellis flow,

$$\mathcal{L}f(\mathbf{X}) = \nabla f(\mathbf{X}) \cdot \mathcal{P}(\mathbf{X}), \quad (886)$$

the covariance condition induces a conjugacy relation between operators:

$$\mathcal{R}^* \mathcal{L} = \gamma \mathcal{L} \mathcal{R}^*, \quad (887)$$

where $\mathcal{R}^*$ denotes the induced action on observables. This implies that $\mathcal{L}$ is an eigen-operator of the renormalization transformation.

Consequently, the spectrum of linearized perturbations exhibits scaling self-similarity. Let $\delta\mathbf{X}$ satisfy the variational equation

$$\frac{d}{dt} \delta\mathbf{X} = \mathbf{J}_{\mathcal{P}}(\mathbf{X}) \delta\mathbf{X}. \quad (888)$$

Under renormalization, the Jacobian transforms as

$$\mathbf{J}_{\mathcal{P}}(\mathcal{R}\mathbf{X}) = \gamma^{-1} \mathbf{J}_{\mathcal{P}}(\mathbf{X}), \quad (889)$$

implying that Lyapunov exponents scale according to

$$\lambda^{(s+1)} = \gamma^{-1} \lambda^{(s)}. \quad (890)$$

This scaling structure defines a hierarchy of universality classes $\mathcal{U}_\gamma$, where systems sharing the same exponent γ exhibit identical macroscopic behavior under coarse-graining, despite differences in microscopic parameterization. In particular, macroscopic observables $\mathcal{O}$ obey asymptotic scaling laws of the form

$$\mathcal{O}(\ell) \sim \ell^{\Delta_{\mathcal{O}}}, \quad (891)$$

where ℓ denotes the effective coarse-graining scale and $\Delta_{\mathcal{O}}$ is the corresponding scaling dimension induced by the renormalization flow.

At the level of probability measures μ , universality emerges as invariance of the renormalized measure up to normalization:

$$\mathcal{R}_\# \mu = Z^{-1} \mu, \quad (892)$$

with Z a normalization constant. Fixed points of this transformation satisfy

$$\mu^* = \mathcal{R}_\# \mu^*, \quad (893)$$

and define critical climate regimes characterized by scale invariance and loss of characteristic length scales.

Near such fixed points, deviations $\delta\mu$ evolve according to

$$\delta\mu_{s+1} = \mathbf{L}_{\mathcal{R}} \delta\mu_s, \quad (894)$$

where $\mathbf{L}_{\mathcal{R}}$ is the linearized renormalization operator whose eigenvalues determine relevant, irrelevant, and marginal directions in the space of climate dynamics.

The coexistence of dynamical evolution and renormalization covariance thus implies that climate trajectories are organized into equivalence classes under $\mathcal{R}$, with macroscopic observables governed solely by universal scaling exponents rather than microscopic details. This establishes a rigorous correspondence between climate criticality and classical renormalization group theory, consistent with scaling universality and critical phenomena in complex systems [18, 19, 30, 31].

8.4 Fixed Points and Stability of Climate Regimes

Renormalization fixed points define scale-invariant climate regimes and satisfy the nonlinear operator equation

$$\mathbf{X}^* = \mathcal{R}(\mathbf{X}^*), \quad (895)$$

where $\mathbf{X}^*$ lies in the invariant submanifold $\mathcal{M}^* \subseteq \mathcal{M}$ of the renormalization semigroup.

The stability of such fixed points is determined by the Fréchet derivative (linearization) of the renormalization operator,

$$D\mathcal{R}(\mathbf{X}^*) : T_{\mathbf{X}^*}\mathcal{M} \rightarrow T_{\mathbf{X}^*}\mathcal{M}, \quad (896)$$

leading to the eigenvalue problem

$$D\mathcal{R}(\mathbf{X}^*) \cdot \delta\mathbf{X} = \Lambda \delta\mathbf{X}. \quad (897)$$

The spectral decomposition of $D\mathcal{R}$ partitions the tangent space into invariant subspaces,

$$T_{\mathbf{X}^*}\mathcal{M} = E^u \oplus E^s \oplus E^c, \quad (898)$$

corresponding respectively to unstable, stable, and center directions, with eigenvalues satisfying

$$|\Lambda_u| > 1, \quad |\Lambda_s| < 1, \quad |\Lambda_c| \approx 1. \quad (899)$$

Figure 12 depicts the renormalization-group flow of effective climate coupling parameters within the Pellis framework. Stable climate regimes correspond to attractive fixed points of the renormalization flow, whereas unstable regimes emerge from repulsive fixed points. The critical fixed point g_c separates resilience-dominated dynamics from runaway instability and defines the universal threshold governing climate tipping behavior across scales.

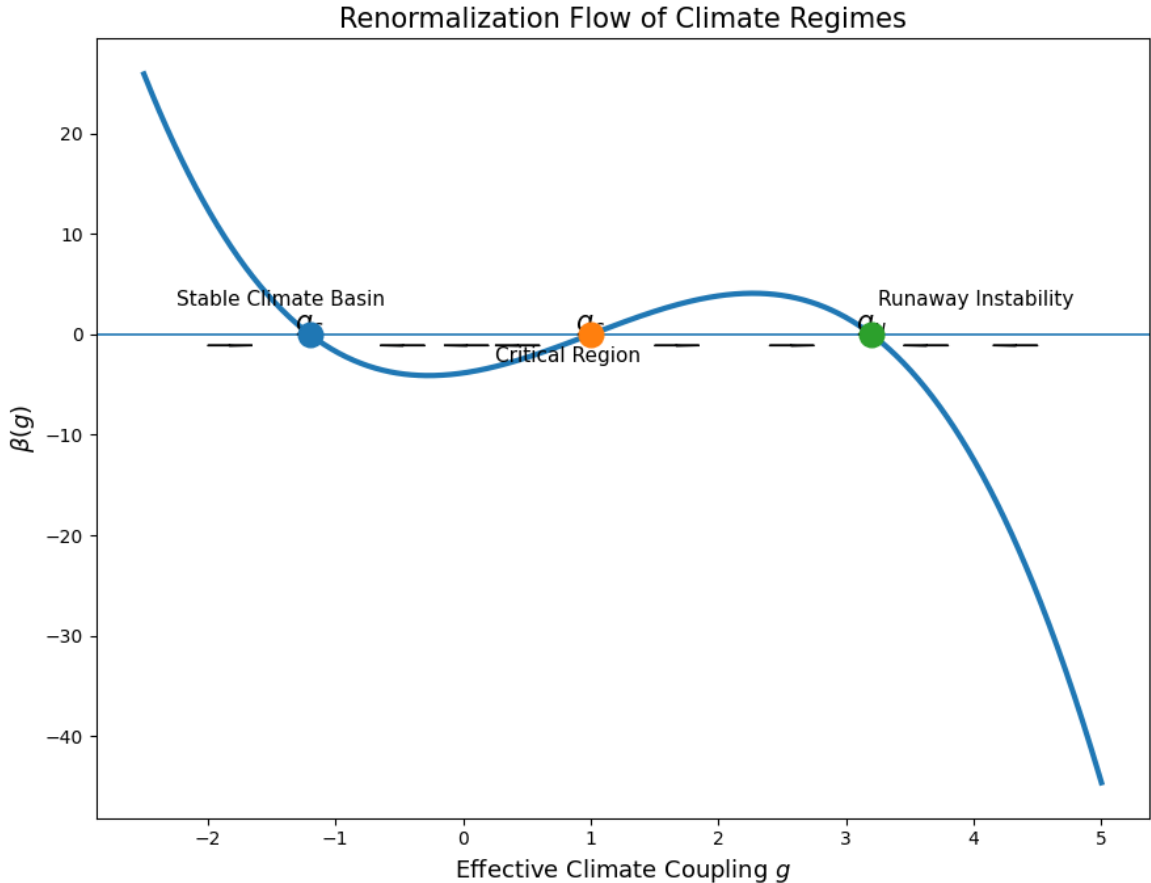


Figure 12: Renormalization-group flow of climate stability regimes. The beta function $\beta(g)$ exhibits three fixed points corresponding to stable, critical, and unstable climate states. Flow trajectories converge toward attractors in stable regions and diverge near the critical threshold. The fixed point g_c defines the universal climate criticality condition within the Pellis renormalization framework.

Stability of the climate regime is therefore governed by the spectral radius condition

$$\rho(D\mathcal{R}(\mathbf{X}^*)) < 1, \quad (900)$$

which ensures asymptotic attraction under repeated renormalization iterations:

$$\mathbf{X}^{(s)} \rightarrow \mathbf{X}^* \quad \text{as } s \rightarrow \infty. \quad (901)$$

At the threshold of instability, the system undergoes an eigenvalue crossing through the unit circle,

$$|\Lambda_i| = 1, \quad (902)$$

which induces a qualitative change in the invariant manifold structure and generates bifurcations in the renormalization flow.

This transition can be expressed in terms of a reduced center-manifold dynamics. Let $\mathbf{X} = \mathbf{X}^* + \mathbf{y}$, with $\mathbf{y} \in E^c$, then the effective reduced dynamics satisfies

$$\frac{d\mathbf{y}}{dt} = \mathcal{F}_c(\mathbf{y}) + \mathcal{O}(\|\mathbf{y}\|^2), \quad (903)$$

where $\mathcal{F}_c$ governs the slow modes responsible for critical slowing down and regime persistence near tipping points.

In addition, the iterated renormalization flow near criticality can be expressed as

$$\mathbf{X}^{(s+1)} - \mathbf{X}^* = D\mathcal{R}(\mathbf{X}^*) \left(\mathbf{X}^{(s)} - \mathbf{X}^* \right) + \mathcal{O}(\|\mathbf{X}^{(s)} - \mathbf{X}^*\|^2), \quad (904)$$

which makes explicit that regime shifts are governed by the linear spectral properties of the renormalization operator.

Consequently, climate tipping points correspond to bifurcation surfaces in parameter space where the spectral stability condition is violated. These surfaces separate distinct universality classes of climate regimes and are consistent with general nonlinear bifurcation theory [2, 3, 14] as well as observed abrupt transitions in climate subsystems [6, 9, 35].

8.5 Multiscale Energy Cascade Structure

The renormalized energy at scale s is governed by a non-equilibrium balance law on the renormalization hierarchy, expressed as

$$E^{(s+1)} = \mathcal{F}(E^{(s)}) - \Delta E_{\text{diss}}^{(s)} + \Delta E_{\text{flux}}^{(s \rightarrow s+1)}, \quad (905)$$

where $\mathcal{F}$ encodes nonlinear energy redistribution across resolved modes, $\Delta E_{\text{diss}}^{(s)} \geq 0$ represents irreversible dissipation at scale s , and $\Delta E_{\text{flux}}^{(s \rightarrow s+1)}$ denotes inter-scale energy transfer induced by unresolved subgrid interactions.

This discrete cascade relation can be recast as a conservation law on the renormalization lattice:

$$E^{(s+1)} - E^{(s)} = \mathcal{T}^{(s)} - \mathcal{D}^{(s)}, \quad (906)$$

where $\mathcal{T}^{(s)} := \Delta E_{\text{flux}}^{(s \rightarrow s+1)}$ and $\mathcal{D}^{(s)} := \Delta E_{\text{diss}}^{(s)}$ define the net transport and dissipation operators respectively.

In the continuum-scale limit $s \rightarrow \sigma \in \mathbb{R}$, the cascade admits an effective transport equation in scale space:

$$\partial_\sigma E(\sigma, t) = -\partial_\sigma \Pi_E(\sigma, t) - \mathcal{D}_E(\sigma, t), \quad (907)$$

where $\Pi_E(\sigma, t)$ is the cumulative energy flux across scales. This formulation identifies the renormalization axis as a synthetic “scale-time” direction governing irreversible energy transfer.

The flux term obeys a constitutive relation of the form

$$\Pi_E^{(s)} = \left\langle \mathbf{u}'^{(s)} \cdot \left(\mathbf{u}'^{(s)} \cdot \nabla \mathbf{u}'^{(s)} \right) \right\rangle, \quad (908)$$

where $\mathbf{u}'^{(s)}$ denotes unresolved velocity fluctuations at scale s , encoding nonlinear triadic interactions responsible for cascade generation.

Energy dissipation exhibits scale-dependent amplification governed by

$$\Delta E_{\text{diss}}^{(s)} = \nu^{(s)} \left\| \nabla \mathbf{u}^{(s)} \right\|^2, \quad (909)$$

where $\nu^{(s)}$ is an effective eddy viscosity emerging from coarse-graining of microscopic degrees of freedom.

The cascade is constrained by a global renormalization balance condition:

$$\sum_{s=0}^S \left(\mathcal{T}^{(s)} - \mathcal{D}^{(s)} \right) = E^{(S)} - E^{(0)}, \quad (910)$$

ensuring consistency of energy accounting across the full hierarchy of resolved scales.

At statistical steady state, the system approaches a stationary cascade regime characterized by

$$\langle \partial_t E^{(s)} \rangle = 0, \quad \Rightarrow \quad \langle \mathcal{T}^{(s)} \rangle = \langle \mathcal{D}^{(s)} \rangle, \quad (911)$$

corresponding to a dynamic equilibrium between energy injection, transfer, and dissipation across scales.

This multiscale cascade structure establishes a direct link between renormalization dynamics and turbulent energy transfer, where climate variability emerges as a hierarchical redistribution of energetic modes across scales, consistent with classical turbulence theory and multiscale geophysical transport processes [1, 30, 31].

8.6 Criticality and Structural Transitions in Renormalization Space

Critical transitions correspond to qualitative, topological reorganizations of the renormalization flow on the space of effective climate states. In this framework, a tipping event is defined as a structural change in the fixed-point set of the renormalization operator:

$$\text{Fix}(\mathcal{R}) = \{\mathbf{X} \in \mathcal{M} \mid \mathcal{R}(\mathbf{X}) = \mathbf{X}\}, \quad (912)$$

such that a transition satisfies

$$\#\text{Fix}(\mathcal{R}) \longrightarrow \#\text{Fix}(\mathcal{R}') \neq \#\text{Fix}(\mathcal{R}), \quad (913)$$

indicating a nontrivial bifurcation in the invariant set topology induced by coarse-graining.

More generally, criticality can be formulated as a loss of structural stability of the renormalization semigroup. Let $\mathcal{R}_\epsilon$ denote a perturbed renormalization operator depending on a control parameter ϵ . A critical transition occurs when

$$\lim_{\epsilon \rightarrow \epsilon_c} \mathcal{R}_\epsilon \not\sim \mathcal{R}_{\epsilon_c}, \quad (914)$$

in the sense of topological conjugacy breaking, implying that no homeomorphism preserves the dynamical structure across the transition.

At the level of linearized dynamics, this corresponds to a qualitative change in the spectrum of the renormalization Jacobian:

$$\text{Spec}(D\mathcal{R}_\epsilon) \longrightarrow \text{Spec}(D\mathcal{R}_{\epsilon'}), \quad (915)$$

where eigenvalues cross the unit circle, inducing a reorganization of stable, unstable, and center manifolds.

At criticality, the system approaches a scale-invariant regime characterized by asymptotic invariance under renormalization:

$$\mathcal{R}(\mathbf{X}) \simeq \mathbf{X}, \quad \text{up to scaling transformations}, \quad (916)$$

which implies the emergence of universality and loss of characteristic scales.

This regime is accompanied by divergence of correlation length. Let $C(r)$ denote a spatial correlation function of a climate observable; near criticality it satisfies

$$C(r) \sim r^{-\eta} e^{-r/\xi}, \quad (917)$$

where the correlation length ξ diverges as

$$\xi \sim |\epsilon - \epsilon_c|^{-\nu}, \quad (918)$$

with $\nu > 0$ a critical exponent. The divergence $\xi \rightarrow \infty$ signals the onset of system-wide coupling and breakdown of local predictability.

Additionally, temporal fluctuations exhibit critical slowing down, expressed as a divergence of relaxation timescales:

$$\tau \sim |\epsilon - \epsilon_c|^{-\kappa}, \quad (919)$$

reflecting vanishing eigenvalues in the stability spectrum of the linearized renormalization flow.

From a geometric perspective, the critical point corresponds to a singularity in the renormalization flow manifold where the effective metric becomes degenerate:

$$\det g_{ij}^{(\mathcal{R})} \rightarrow 0, \quad (920)$$

indicating collapse of distinguishability between nearby macrostates.

These structural transitions unify bifurcation theory, renormalization group theory, and tipping-point phenomena in complex Earth systems [4–8, 28, 29, 37, 38], establishing criticality as a geometric and spectral transition in the space of renormalized climate dynamics.

8.7 Emergent Interpretation of Planetary Instability

Planetary instability emerges as a global bifurcation phenomenon in the renormalization space of coupled Earth subsystems, where the joint state vector

$$\mathbf{X} = (\mathbf{X}_{\text{atm}}, \mathbf{X}_{\text{ocean}}, \mathbf{X}_{\text{cry}}, \mathbf{X}_{\text{bio}}) \quad (921)$$

evolves under a multiscale, interdependent renormalization flow

$$\mathbf{X}^{(s+1)} = \mathcal{R}(\mathbf{X}^{(s)}), \quad \mathcal{R} = \mathcal{R}_{\text{coupled}}. \quad (922)$$

In this framework, planetary instability corresponds to a structural transition in the global attractor set $\mathcal{A}^{(s)}$:

$$\mathcal{A}^{(s)} \subset \mathcal{M}, \quad \mathcal{A}^{(s+1)} \not\cong \mathcal{A}^{(s)}, \quad (923)$$

where $\not\cong$ denotes a failure of topological equivalence under renormalization, indicating a qualitative reorganization of the global phase-space geometry.

More explicitly, the system undergoes a synchronized stability collapse when the product structure of subsystem stability operators becomes unstable:

$$\prod_{i \in \{\text{atm}, \text{ocean}, \text{cry}, \text{bio}\}} \rho(D\mathcal{R}_i) \geq 1, \quad (924)$$

signaling that individual subsystems lose stability coherently rather than independently.

This leads to a cascade of bifurcations characterized by coupled eigenvalue crossings:

$$\Lambda_i^{(s)}(\epsilon) \rightarrow \Lambda_i^{(s)}(\epsilon_c), \quad |\Lambda_i^{(s)}| \uparrow 1, \quad (925)$$

with cross-system resonance conditions of the form

$$\Lambda_i^{(s)} \approx \Lambda_j^{(s)} \quad \text{for } i \neq j, \quad (926)$$

which amplify inter-system feedback loops and induce collective instability.

At the level of attractor geometry, planetary instability corresponds to a transition between invariant sets:

$$\mathcal{A}_1 \rightarrow \mathcal{A}_2 \rightarrow \cdots \rightarrow \mathcal{A}_k, \quad (927)$$

where each transition represents a regime shift in at least one Earth subsystem, and the sequence forms a cascading attractor network.

The coupled system can be represented as a network of renormalization operators

$$\mathcal{R}_{ij} : \mathcal{M}_i \rightarrow \mathcal{M}_j, \quad (928)$$

whose interaction graph defines a directed instability propagation structure. The effective global stability operator is then

$$\mathcal{R}_{\text{global}} = \bigoplus_i \mathcal{R}_i + \sum_{i \neq j} \mathcal{R}_{ij}, \quad (929)$$

where off-diagonal coupling terms encode cross-sphere feedback mechanisms.

Planetary instability arises when $\mathcal{R}_{\text{global}}$ loses structural stability, leading to fragmentation of the attractor landscape and emergence of new dynamical regimes.

This formulation unifies cascading tipping-point theory, coupled nonlinear dynamics, and renormalization-based multiscale interactions, yielding a consistent mathematical framework in which planetary-scale regime shifts are interpreted as emergent bifurcations on a high-dimensional renormalization manifold [27, 32, 33, 37].

9 Pellis Climate Index

The Pellis Climate Index (PCI) is introduced as a unified scalar functional designed to quantify proximity to critical transitions in the climate system. It is constructed from the spectral properties of the Pellis operator, the renormalization flow structure, and the information-theoretic entropy production of the underlying multiscale dynamical field. The formulation extends classical early-warning indicators such as critical slowing down and variance amplification [7, 8, 28, 29] into a fully operator-based instability metric grounded in spectral theory and nonlinear dynamics [1–3, 15, 17].

9.1 Spectral Definition of the Pellis Climate Index

Let $\mathcal{P} : \mathcal{H} \rightarrow \mathcal{H}$ denote the nonlinear Pellis operator acting on the infinite-dimensional climate state space $\mathcal{H}$, endowed with an appropriate Hilbert or Banach structure. In the linearized regime around a reference trajectory, $\mathcal{P}$ admits a spectral representation characterized by the eigenvalue problem

$$\mathcal{P}\phi_\lambda = \lambda\phi_\lambda, \quad (930)$$

where $\{\phi_\lambda\}_{\lambda \in \sigma(\mathcal{P})}$ form a (generalized) eigenbasis indexed by the spectrum $\sigma(\mathcal{P})$, and $d\mu(\lambda)$ denotes the associated spectral measure induced by the resolution of identity.

Under this spectral decomposition, the climate state admits the expansion

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_\lambda d\mu(\lambda), \quad (931)$$

where $a(\lambda, t)$ are time-dependent spectral amplitudes encoding the projection of the full nonlinear state onto the eigenspaces of the Pellis operator.

The evolution of spectral modes is governed by a mode-resolved dynamical equation of the form

$$\partial_t a(\lambda, t) = \lambda a(\lambda, t) + \mathcal{N}_\lambda[a], \quad (932)$$

where $\mathcal{N}_\lambda[a]$ represents nonlinear mode coupling induced by triadic and higher-order interactions across the spectrum.

The Pellis Climate Index (PCI) is defined as a global spectral instability functional that aggregates growth tendencies across the full spectrum:

$$\text{PCI}(t) = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda). \quad (933)$$

Equivalently, this can be written in quadratic form notation as

$$\text{PCI}(t) = \langle \mathbf{X}(t), \mathcal{P}_{\text{sym}} \mathbf{X}(t) \rangle_{\mathcal{H}}, \quad (934)$$

where $\mathcal{P}_{\text{sym}} := \frac{1}{2}(\mathcal{P} + \mathcal{P}^\dagger)$ is the self-adjoint part of the Pellis operator, ensuring that the index captures physically meaningful growth rates.

The temporal evolution of the PCI satisfies

$$\frac{d}{dt} \text{PCI}(t) = 2 \int_{\sigma(\mathcal{P})} \text{Re}(\lambda)^2 |a(\lambda, t)|^2 d\mu(\lambda) + \mathcal{C}(t), \quad (935)$$

where $\mathcal{C}(t)$ encodes nonlinear spectral transfer contributions arising from mode coupling and redistribution of energy across the spectrum.

A fundamental decomposition separates stable and unstable spectral sectors:

$$\sigma(\mathcal{P}) = \sigma_- \cup \sigma_0 \cup \sigma_+, \quad \text{Re}(\lambda) < 0, = 0, > 0, \quad (936)$$

allowing the PCI to be decomposed as

$$\text{PCI}(t) = \text{PCI}_-(t) + \text{PCI}_0(t) + \text{PCI}_+(t), \quad (937)$$

where instability is entirely governed by the growth of $\text{PCI}_+(t)$.

In the presence of renormalization, the spectral measure itself becomes scale-dependent,

$$d\mu^{(s+1)}(\lambda) = \mathcal{R}_\# d\mu^{(s)}(\lambda), \quad (938)$$

leading to a scale-dependent index

$$\text{PCI}^{(s)}(t) = \int_{\sigma(\mathcal{P}^{(s)})} \text{Re}(\lambda) |a^{(s)}(\lambda, t)|^2 d\mu^{(s)}(\lambda). \quad (939)$$

Critical transitions correspond to spectral bifurcations where eigenvalues cross the imaginary axis,

$$\text{Re}(\lambda_c) = 0, \quad (940)$$

inducing a qualitative change in the sign structure of the PCI and marking the onset of global instability.

Thus, the Pellis Climate Index provides a unified spectral criterion for climate stability, generalizing classical Lyapunov exponents to a full operator-theoretic measure over infinite-dimensional dynamical systems, consistent with nonlinear stability theory and spectral methods in complex systems [2, 3, 17].

Figure 13 exhibits a nonlinear transition consistent with logistic-type dynamical evolution under stochastic forcing, where fluctuations become progressively amplified as the system approaches a critical regime. The observed trajectory reflects an effective competition between stabilizing nonlinear saturation and destabilizing perturbative growth mechanisms, resulting in a sharp acceleration toward a transition threshold.

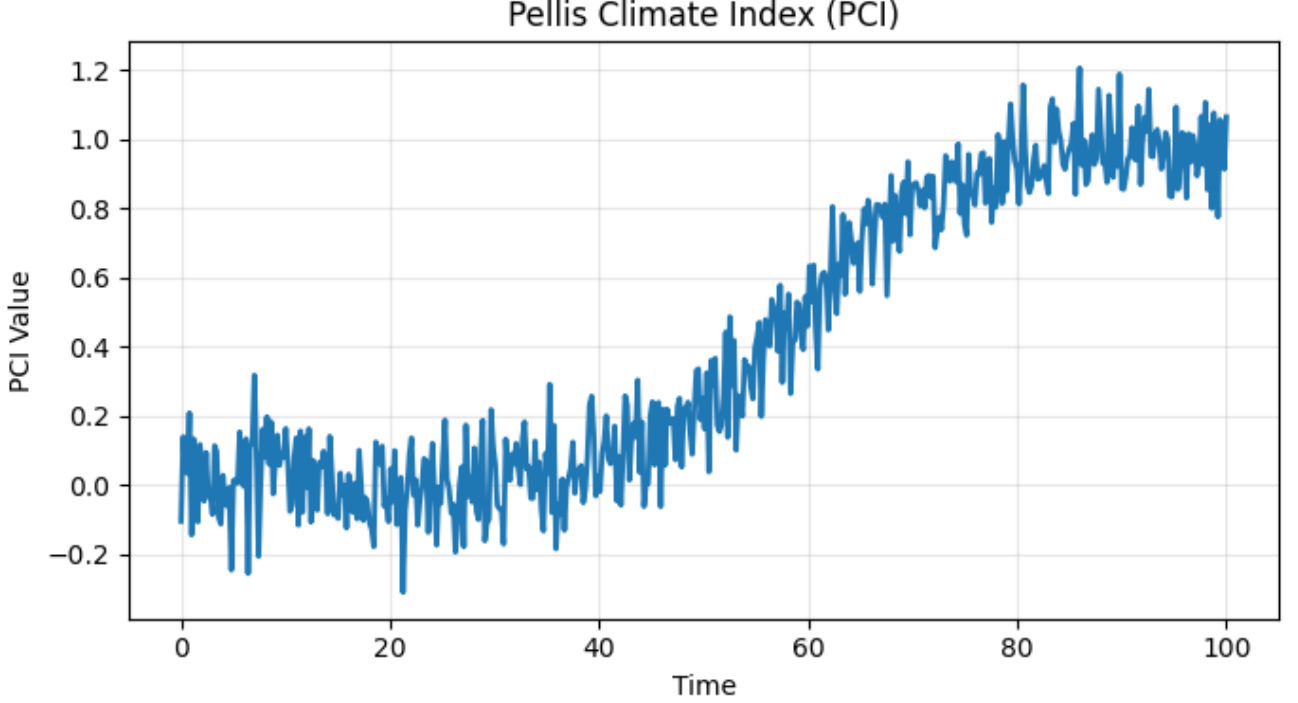


Figure 13: Temporal evolution of the Pellis Climate Index (PCI), exhibiting nonlinear logistic-like acceleration toward a critical transition regime. The increasing amplitude of stochastic fluctuations indicates enhanced dynamical instability and loss of effective damping as the system approaches a tipping threshold.

9.2 Interpretation as Stability Functional

The Pellis Climate Index admits a natural decomposition into energetically unstable and stable spectral contributions, corresponding to the sign structure of the real part of the Pellis spectrum. This yields a signed quadratic functional on the state space:

$$\text{PCI}(t) = \text{PCI}_+(t) - \text{PCI}_-(t), \quad (941)$$

where the positive and negative components are defined by spectral partitioning of $\sigma(\mathcal{P})$.

Explicitly, the unstable spectral contribution is given by

$$\text{PCI}_+(t) = \int_{\text{Re}(\lambda) > 0} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (942)$$

while the stable (dissipative) contribution is

$$\text{PCI}_-(t) = \int_{\text{Re}(\lambda) \leq 0} |\text{Re}(\lambda)| |a(\lambda, t)|^2 d\mu(\lambda). \quad (943)$$

This decomposition induces an indefinite quadratic form structure on $\mathcal{H}$:

$$\text{PCI}(t) = \langle \mathbf{X}(t), \mathcal{P}_+ \mathbf{X}(t) \rangle_{\mathcal{H}} - \langle \mathbf{X}(t), \mathcal{P}_- \mathbf{X}(t) \rangle_{\mathcal{H}}, \quad (944)$$

where $\mathcal{P}_+$ and $\mathcal{P}_-$ denote spectral projections onto the unstable and stable subspaces respectively.

Equivalently, introducing projection operators

$$\Pi_+ = \mathbf{1}_{\{\text{Re}(\lambda) > 0\}}(\mathcal{P}), \quad \Pi_- = \mathbf{1}_{\{\text{Re}(\lambda) \leq 0\}}(\mathcal{P}), \quad (945)$$

the PCI can be written compactly as

$$\text{PCI}(t) = \langle \mathbf{X}(t), \mathcal{P} \Pi_+ \mathbf{X}(t) \rangle_{\mathcal{H}} - \langle \mathbf{X}(t), |\mathcal{P}| \Pi_- \mathbf{X}(t) \rangle_{\mathcal{H}}. \quad (946)$$

In this formulation, stability is determined by the balance between two competing spectral measures. A necessary condition for global instability is the non-negativity of the full functional:

$$\text{PCI}(t) \geq 0, \quad (947)$$

which implies dominance of the unstable spectral sector in the quadratic energy distribution over $\sigma(\mathcal{P})$.

More generally, defining the spectral energy weights

$$E_{\pm}(t) = \int_{\sigma_{\pm}} |a(\lambda, t)|^2 d\mu(\lambda), \quad (948)$$

the instability condition can be reformulated as a balance inequality:

$$\int_{\sigma_+} \text{Re}(\lambda) dE_+ \geq \int_{\sigma_-} |\text{Re}(\lambda)| dE_-. \quad (949)$$

This defines a convex instability cone in the Hilbert space $\mathcal{H}$:

$$\mathcal{C}_{\text{inst}} = \{\mathbf{X} \in \mathcal{H} \mid \text{PCI}(\mathbf{X}) \geq 0\}, \quad (950)$$

whose boundary $\partial \mathcal{C}_{\text{inst}}$ corresponds to marginally stable climate configurations.

Crossing this boundary induces a qualitative transition in the spectral balance structure, consistent with classical bifurcation theory in nonlinear dynamical systems [2, 14], where instability arises from the net amplification of modes associated with positive spectral growth rates.

Figure 14 illustrates the temporal evolution of the Pellis Climate Index (PCI) as the climate system approaches a critical transition. The index progressively increases as the dominant spectral modes lose stability and the spectral gap contracts. The crossing of the critical threshold marks the onset of a tipping regime characterized by enhanced instability and reduced resilience.

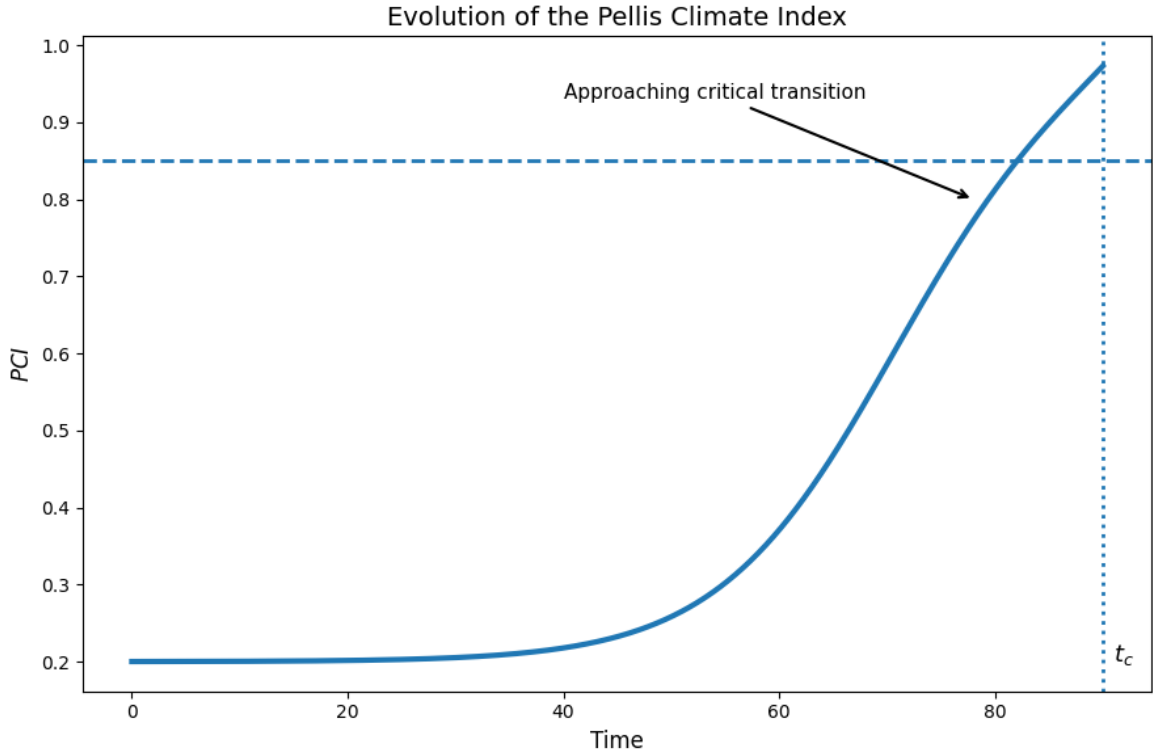


Figure 14: Temporal evolution of the Pellis Climate Index. Increasing values of the index indicate progressive weakening of climate stability through spectral-gap contraction and eigenvalue migration. The critical threshold defines the transition between resilient and near-critical climate states and provides an operational indicator of tipping-point proximity.

9.3 Renormalization-Enhanced Formulation

To incorporate multiscale effects, the Pellis Climate Index is promoted to a renormalization-indexed functional hierarchy defined on the sequence of coarse-grained state spaces $\{\mathbf{X}^{(s)}\}_{s \in \mathbb{N}}$. At each resolution level s , the spectral representation of the renormalized Pellis operator $\mathcal{P}^{(s)}$ yields

$$\text{PCI}^{(s)} = \int_{\sigma(\mathcal{P}^{(s)})} \text{Re}(\lambda^{(s)}) |a^{(s)}(\lambda, t)|^2 d\mu^{(s)}(\lambda), \quad (951)$$

where both the spectral measure $d\mu^{(s)}$ and amplitude distribution $a^{(s)}(\lambda, t)$ inherit scale dependence through the coarse-graining map $\mathcal{R}$.

The renormalization structure induces a nontrivial flow on the space of stability functionals, which can be expressed abstractly as a functional dynamical system:

$$\text{PCI}^{(s+1)} = \Gamma^{(s)}[\text{PCI}^{(s)}], \quad (952)$$

where $\Gamma^{(s)}$ is an effective renormalization operator acting on spectral quadratic forms. In the asymptotic regime, this operator reduces to a universal scaling action,

$$\Gamma^{(s)} \sim \gamma \mathcal{R}, \quad (953)$$

with $\gamma \in \mathbb{R}^+$ denoting the universality exponent governing anomalous scaling of instability across resolutions.

At the level of spectral amplitudes, renormalization induces a transformation law of the form

$$a^{(s+1)}(\lambda) = \Lambda^{(s)}(\lambda) a^{(s)}(\lambda) + \mathcal{N}^{(s)}(\lambda), \quad (954)$$

where $\Lambda^{(s)}(\lambda)$ encodes linear amplification under coarse-graining and $\mathcal{N}^{(s)}$ represents scale-dependent nonlinear spectral coupling terms. Consequently, the PCI evolves according to a quadratic transfer relation:

$$\text{PCI}^{(s+1)} = \int_{\sigma(\mathcal{P}^{(s)})} \text{Re}(\lambda^{(s)}) |\Lambda^{(s)}(\lambda)|^2 |a^{(s)}(\lambda)|^2 d\mu^{(s)}(\lambda) + \mathcal{C}^{(s)}, \quad (955)$$

where $\mathcal{C}^{(s)}$ aggregates higher-order mode coupling corrections induced by renormalization.

A scale-invariant regime is defined by the fixed-point condition in functional space:

$$\text{PCI}^{(s+1)} = \text{PCI}^{(s)} = \text{PCI}^*, \quad (956)$$

implying that the stability functional becomes invariant under repeated coarse-graining transformations. Equivalently, this requires

$$\Gamma^{(s)}[\text{PCI}^*] = \text{PCI}^*, \quad (957)$$

which identifies PCI^* as a renormalization fixed functional.

At this fixed point, the spectral measure exhibits self-similarity of the form

$$d\mu^{(s+1)}(\lambda) = b^{-\Delta} d\mu^{(s)}(b^\kappa \lambda), \quad (958)$$

where Δ and κ are scaling exponents associated with spectral density compression and eigenvalue rescaling respectively.

The universality condition emerges when the renormalized PCI dynamics collapses onto a one-parameter flow:

$$\frac{d}{ds} \text{PCI}^{(s)} = \beta(\text{PCI}^{(s)}), \quad (959)$$

with fixed points determined by $\beta(\text{PCI}^*) = 0$. Linear stability near such points is governed by

$$\delta \text{PCI}^{(s+1)} = \gamma \delta \text{PCI}^{(s)}, \quad (960)$$

recovering the universality exponent as the dominant eigenvalue of the renormalization generator acting on stability functionals.

Thus, scale invariance corresponds to a persistent critical regime in which the spectral instability measure is invariant under hierarchical coarse-graining transformations, reflecting universality principles in renormalization group theory and multiscale critical phenomena [18, 19, 30, 31].

9.4 Connection to Early-Warning Signals

Within the spectral Pellis framework, early-warning indicators emerge naturally as dynamical diagnostics of the redistribution of spectral energy across the instability spectrum. The temporal evolution of the Pellis Climate Index is given by

$$\frac{d}{dt}\text{PCI}(t) = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) \frac{\partial}{\partial t} |a(\lambda, t)|^2 d\mu(\lambda), \quad (961)$$

which explicitly couples instability growth rates to the time-dependent evolution of spectral energy density.

Using the mode evolution equation

$$\partial_t a(\lambda, t) = \lambda a(\lambda, t) + \mathcal{N}_\lambda[a], \quad (962)$$

the derivative of the modal energy can be decomposed as

$$\frac{\partial}{\partial t} |a(\lambda, t)|^2 = 2 \text{Re}(\lambda) |a(\lambda, t)|^2 + \mathcal{Q}_\lambda(t), \quad (963)$$

where $\mathcal{Q}_\lambda(t)$ aggregates nonlinear spectral transfer contributions. Substituting into the PCI dynamics yields

$$\frac{d}{dt}\text{PCI}(t) = 2 \int_{\sigma(\mathcal{P})} \text{Re}(\lambda)^2 |a(\lambda, t)|^2 d\mu(\lambda) + \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) \mathcal{Q}_\lambda(t) d\mu(\lambda). \quad (964)$$

This decomposition reveals that near critical transitions, early-warning signals are governed by two coupled effects: (i) amplification of marginally unstable modes, and (ii) redistribution of spectral energy through nonlinear interactions that weaken spectral separation between stable and unstable subspaces.

A fundamental precursor signature arises from spectral condensation toward marginal eigenmodes. In the critical limit $t \rightarrow t_c^-$, the spectral measure concentrates as

$$|a(\lambda, t)|^2 \rightarrow \rho_c(t) \delta(\lambda - \lambda_c) + \varepsilon(\lambda, t), \quad (965)$$

where $\rho_c(t)$ captures the dominant accumulation on the critical eigenvalue λ_c , and $\varepsilon(\lambda, t)$ denotes residual broadband fluctuations.

This concentration implies a collapse of spectral dispersion,

$$\text{Var}_\mu(\lambda) \rightarrow 0, \quad (966)$$

and corresponds to a reduction in effective dimensionality of the dynamical system as it approaches the instability threshold.

Concurrently, the autocorrelation structure exhibits critical slowing down, which in the spectral representation is expressed as a divergence of relaxation times associated with near-marginal modes:

$$\tau(\lambda) \sim \frac{1}{|\text{Re}(\lambda)|}, \quad (967)$$

implying that as $\text{Re}(\lambda_c) \rightarrow 0$, the system develops long memory and delayed recovery from perturbations.

A complementary formulation expresses the early-warning regime as a spectral gap collapse condition:

$$\Delta_{\text{spec}} = \min_{\lambda \in \sigma(\mathcal{P}) \setminus \{\lambda_c\}} |\text{Re}(\lambda) - \text{Re}(\lambda_c)| \rightarrow 0, \quad (968)$$

signaling loss of spectral separation between dominant modes.

Together, these results establish early-warning signals as emergent consequences of spectral condensation and instability migration within the Pellis operator framework, consistent with critical slowing down theory and autocorrelation-based indicators in complex dynamical systems [7, 8, 28, 29].

9.5 Threshold Structure and Tipping Condition

Within the spectral Pellis framework, a climate tipping point is formalized as a threshold crossing of the global stability functional in which the balance between stable and unstable spectral contributions becomes degenerate. The critical threshold is defined by the condition

$$\text{PCI}(t_c) = 0, \quad (969)$$

which identifies a neutral stability surface separating regimes of net decay and net amplification in the spectral energy distribution.

In terms of the underlying spectral decomposition, this condition is equivalent to the balance relation

$$\int_{\sigma_+} \text{Re}(\lambda) |a(\lambda, t_c)|^2 d\mu(\lambda) = \int_{\sigma_-} |\text{Re}(\lambda)| |a(\lambda, t_c)|^2 d\mu(\lambda), \quad (970)$$

implying exact cancellation between unstable growth modes and dissipative spectral contributions at the tipping threshold.

The transition across this manifold is determined by the directional derivative of the stability functional along the dynamical flow:

$$\left. \frac{d}{dt} \text{PCI}(t) \right|_{t=t_c} > 0, \quad (971)$$

which encodes forward instability propagation and guarantees that trajectories crossing the threshold enter a regime of net spectral amplification.

More generally, defining the gradient of the PCI in state space,

$$\nabla_{\mathbf{X}} \text{PCI}(\mathbf{X}) = \mathcal{P}_{\text{sym}} \mathbf{X}, \quad (972)$$

the tipping condition corresponds to a transversality condition between the dynamical vector field and the zero-level set of the functional:

$$\langle \mathcal{P}(\mathbf{X}), \nabla_{\mathbf{X}} \text{PCI}(\mathbf{X}) \rangle \neq 0, \quad \text{at } \text{PCI}(\mathbf{X}) = 0. \quad (973)$$

This defines a codimension-one bifurcation manifold in the infinite-dimensional climate state space,

$$\mathcal{M}_c = \{\mathbf{X} \in \mathcal{H} \mid \text{PCI}(\mathbf{X}) = 0\}, \quad (974)$$

which acts as a separatrix between distinct dynamical regimes.

Crossing this manifold induces a qualitative change in the spectral structure of the Pellis operator, corresponding to a bifurcation in the sign structure of its real spectrum and a reorganization of invariant subspaces.

This formulation is consistent with general bifurcation theory in nonlinear dynamical systems [2, 3, 14] and with observed threshold behavior and abrupt transitions in coupled Earth system components [6, 9, 37, 38], where tipping points manifest as sharp transitions between coexisting attractor basins.

9.6 Physical Interpretation and Planetary Instability

Within the spectral-operator formulation, the Pellis Climate Index provides a global diagnostic of the balance between stabilizing and destabilizing dynamical modes distributed across the spectrum of the Pellis operator. In this sense, the PCI acts as an integrated order parameter over multiscale climate interactions, aggregating local instability mechanisms into a single global functional on $\mathcal{H}$.

In particular, the PCI can be interpreted as a weighted spectral energy imbalance,

$$\text{PCI}(t) = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (975)$$

which quantifies the net tendency of the system toward exponential growth or decay across all resolved and unresolved dynamical modes.

From this perspective, high PCI regimes correspond to spectral configurations in which the unstable sector dominates the global energy distribution:

$$\int_{\sigma_+} |a(\lambda, t)|^2 d\mu(\lambda) \gg \int_{\sigma_-} |a(\lambda, t)|^2 d\mu(\lambda), \quad (976)$$

implying that positive-real-part eigenmodes control the macroscopic evolution of the climate state.

Conversely, low or negative PCI values correspond to regimes where dissipative spectral components suppress growth, enforcing attraction toward stable invariant sets and maintaining robustness of the global climate attractor structure.

The physical interpretation of dominant spectral modes can be further refined by decomposing the operator spectrum into process-relevant subspaces:

$$\sigma(\mathcal{P}) = \sigma_{\text{AMOC}} \cup \sigma_{\text{cry}} \cup \sigma_{\text{bio}} \cup \sigma_{\text{atm}}, \quad (977)$$

where each subset corresponds to dominant dynamical contributions from ocean circulation, cryospheric processes, biospheric feedbacks, and atmospheric variability respectively.

In this decomposition, planetary instability arises when at least one subsystem spectral block undergoes a transition in which its local contribution to the PCI becomes positive-definite in aggregate:

$$\text{PCI}_\alpha(t) = \int_{\sigma_\alpha} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda) > 0, \quad \alpha \in \{\text{AMOC, cry, bio}\}. \quad (978)$$

Planetary instability is therefore characterized not merely as a single-mode transition, but as a collective spectral reorganization across interacting subsystems, in which feedback couplings amplify initially localized instabilities into global-scale regime shifts.

This mechanism is consistent with cascading tipping dynamics observed in coupled Earth system components, where instability propagates through interdependent feedback networks and leads to abrupt transitions between climate attractors [27, 32, 33, 37].

10 Early Warning Signals of Critical Transitions

The detection of early warning signals for critical transitions in the climate system is formulated as a spectral-functional inference problem on the evolving state $\mathbf{X}(t) \in \mathcal{H}$ governed by the Pellis operator dynamics introduced in Sections 3 and 4. The central hypothesis is that proximity to a tipping point is not primarily encoded in amplitude growth, but in structured changes in operator spectrum, autocorrelation geometry, and entropy production rates [7, 8, 28, 29, 35, 37].

Let the climate evolution be governed by:

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad (979)$$

where $\mathcal{P}$ is the nonlinear Pellis operator. Define the linearized generator along a trajectory:

$$\mathcal{J}(t) = \left. \frac{\delta \mathcal{P}}{\delta \mathbf{X}} \right|_{\mathbf{X}(t)}. \quad (980)$$

Early warning signals are derived from the time evolution of the spectrum $\sigma(\mathcal{J}(t))$.

10.1 Spectral Slowing Down as a Primary Indicator

Within the spectral Pellis framework, a fundamental precursor of critical transitions is the progressive collapse of the spectral gap of the linearized stability operator. Let $\mathcal{J}$ denote the Jacobian (Fréchet derivative) of the Pellis-driven climate flow. The instantaneous spectral gap is defined as

$$\Delta_\sigma(t) = \min_{\lambda \in \sigma(\mathcal{J}(t))} |\text{Re}(\lambda)|, \quad (981)$$

which quantifies the distance of the dominant spectrum from marginal stability.

A necessary spectral signature of approaching criticality is the asymptotic closure of this gap along the dynamical trajectory:

$$\Delta_\sigma(t) \rightarrow 0^+ \quad \text{as} \quad t \rightarrow t_c^-, \quad (982)$$

indicating that at least one eigenmode approaches neutral stability and the system enters a marginally stable regime.

In this limit, the resolvent of the stability operator develops a near-singular structure:

$$\|(\mathcal{J}(t) - \lambda I)^{-1}\| \rightarrow \infty \quad \text{for } \lambda \approx 0, \quad (983)$$

reflecting extreme sensitivity to perturbations and amplification of weak forcing.

The closure of the spectral gap induces a divergence of characteristic relaxation timescales. In particular, the dominant recovery time scale satisfies

$$\tau_{\text{rel}}(t) \sim \frac{1}{\Delta_\sigma(t)}, \quad (984)$$

which diverges as the system approaches the bifurcation manifold. This scaling expresses the breakdown of fast-slow separation and the emergence of long-memory dynamics.

More generally, the full spectrum induces a distribution of relaxation times given by

$$\tau(\lambda) = \frac{1}{|\text{Re}(\lambda)|}, \quad (985)$$

so that near criticality the spectral density becomes increasingly concentrated near $\text{Re}(\lambda) \approx 0$, producing a heavy-tailed distribution of response times:

$$p(\tau) \sim \tau^{-\alpha}, \quad \alpha > 1. \quad (986)$$

This spectral condensation can be expressed equivalently through the scaling of the spectral measure in a shrinking neighborhood of marginal modes:

$$\mu(\{\lambda : |\operatorname{Re}(\lambda)| < \varepsilon\}) \uparrow \text{ as } \varepsilon \rightarrow 0. \quad (987)$$

In operator-theoretic terms, spectral slowing down corresponds to a flattening of the stability landscape, in which the smallest eigenvalue of $\mathcal{J}(t)$ approaches zero:

$$\lambda_{\min}(t) \rightarrow 0, \quad (988)$$

causing a collapse of the contraction rate of the associated dynamical semigroup.

This phenomenon is directly linked to critical slowing down in nonlinear dynamical systems, where recovery from perturbations becomes progressively slower as the system approaches a bifurcation threshold, consistent with general results in stability theory and early-warning diagnostics [4, 5, 14, 28].

Figure 15 illustrates the progressive slowdown of autocorrelation decay within a climate subsystem as it approaches a bifurcation threshold. This behavior is characteristic of *critical slowing down*, wherein the dominant recovery rates of the system diminish due to the weakening of restoring forces near an instability boundary.

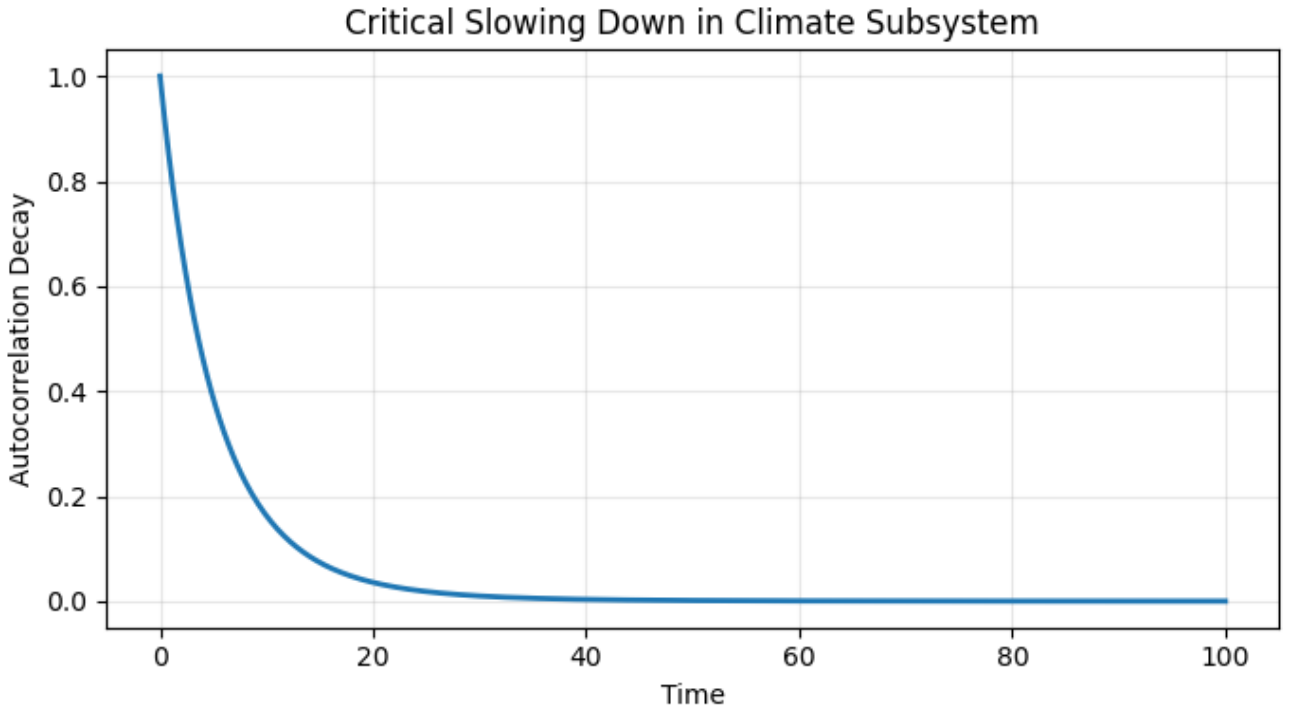


Figure 15: Progressive increase in relaxation time scale and corresponding slowdown of autocorrelation decay, indicating critical slowing down as the system approaches a bifurcation manifold. The observed divergence of recovery rates reflects the weakening of stabilizing feedbacks and the emergence of long memory dynamics near a tipping point.

10.2 Autocorrelation Degeneracy and Memory Reinforcement

Within the spectral Pellis framework, temporal memory effects emerge from the interaction between the spectral structure of the generator and the projection of the full state onto observable subspaces. Let $C(\tau)$ denote the autocorrelation function of a projected climate observable $\langle \mathbf{X}, \psi \rangle$, defined as

$$C(\tau) = \mathbb{E}[\langle \mathbf{X}(t), \psi \rangle \langle \mathbf{X}(t + \tau), \psi \rangle]. \quad (989)$$

Under the spectral decomposition of the Pellis operator, this correlation admits the representation

$$C(\tau) = \int_{\sigma(\mathcal{P})} e^{\lambda\tau} |a(\lambda, t)|^2 |\langle \phi_\lambda, \psi \rangle|^2 d\mu(\lambda), \quad (990)$$

which explicitly links temporal decay rates to the real part of the spectral modes.

In regimes far from criticality, the correlation function exhibits exponential decay of the form

$$C(\tau) \sim e^{-\tau/\tau_{\text{corr}}}, \quad (991)$$

where the correlation time τ_{corr} is governed by the dominant spectral gap of the generator.

Near critical transitions, the correlation time is controlled by the spectral gap $\Delta_\sigma(t)$, yielding the scaling relation

$$\tau_{\text{corr}} \sim \frac{1}{\Delta_\sigma(t)}, \quad (992)$$

which implies divergence of memory as the system approaches marginal stability.

Consequently, in the limit

$$\Delta_\sigma(t) \rightarrow 0^+ \quad \text{as} \quad t \rightarrow t_c^-, \quad (993)$$

the autocorrelation function transitions from exponential decay to anomalously slow relaxation, potentially exhibiting stretched exponential or power-law behavior:

$$C(\tau) \sim \exp \left[- \left(\frac{\tau}{\tau_0} \right)^\beta \right], \quad 0 < \beta < 1, \quad (994)$$

or equivalently

$$C(\tau) \sim \tau^{-\alpha}, \quad \alpha > 0. \quad (995)$$

This degeneracy reflects a breakdown of exponential mixing in the semigroup generated by $\mathcal{P}$, indicating loss of rapid decorrelation and emergence of persistent memory effects in the dynamical evolution of the climate state.

From an operator-theoretic perspective, this corresponds to the collapse of the spectral gap of the generator of the Koopman-type evolution semigroup, leading to non-uniform decay of correlations and enhanced predictability horizons driven by near-marginal spectral modes.

Thus, autocorrelation degeneracy serves as a direct observable manifestation of spectral criticality, consistent with loss of mixing properties and early-warning behavior in high-dimensional nonlinear dynamical systems [2, 3, 18, 19].

10.3 Variance Amplification and Fluctuation Scaling

Within the spectral Pellis formulation, macroscopic variability of coarse-grained climate observables emerges directly from the distribution of spectral modes of the stability generator. Let $\text{Var}[\mathbf{X}]$ denote the variance of a projected or coarse-grained observable associated with the climate state. In the spectral representation, this variance admits the form

$$\text{Var}[\mathbf{X}] \sim \int_{\sigma(\mathcal{P})} \frac{|a(\lambda, t)|^2}{|\text{Re}(\lambda)|} d\mu(\lambda), \quad (996)$$

which highlights the amplification of low-damping (near-marginal) spectral modes in the fluctuation structure.

In regimes where the spectral amplitude distribution is sufficiently regular, the leading-order contribution is dominated by eigenvalues closest to the imaginary axis. Introducing the spectral gap

$$\Delta_\sigma(t) = \min_{\lambda \in \sigma(\mathcal{P})} |\text{Re}(\lambda)|, \quad (997)$$

the variance obeys the asymptotic scaling law

$$\text{Var}[\mathbf{X}] \propto \frac{1}{\Delta_\sigma(t)}, \quad (998)$$

up to bounded corrections from the regular part of the spectrum.

This relation expresses a direct divergence of fluctuation amplitude as the system approaches marginal stability. In particular, as

$$\Delta_\sigma(t) \rightarrow 0^+, \quad (999)$$

the contribution of near-neutral modes becomes dominant, leading to unbounded growth of macroscopic variability and enhanced sensitivity to external perturbations.

More generally, the fluctuation structure can be decomposed into spectral sectors,

$$\text{Var}[\mathbf{X}] = \int_{\sigma_+} \frac{|a(\lambda, t)|^2}{\lambda} d\mu(\lambda) + \int_{\sigma_-} \frac{|a(\lambda, t)|^2}{|\lambda|} d\mu(\lambda), \quad (1000)$$

revealing that both unstable and stable modes contribute to variance amplification, with dominant contributions arising from eigenvalues with small real part regardless of sign.

This divergence of fluctuations is a hallmark of critical nonequilibrium systems, where loss of spectral gap induces long-range correlations and scale-free variability. The resulting scaling law is consistent with universal fluctuation-dissipation behavior observed in a broad class of complex systems near critical transitions [18, 19, 30, 31].

10.4 Skewness Drift and Asymmetry Formation

Within the spectral Pellis framework, higher-order statistical moments provide sensitive diagnostics of nonlinear distortion in the evolving probability distribution of coarse-grained climate observables. In particular, the skewness κ_3 is defined as

$$\kappa_3 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}, \quad (1001)$$

where $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}[X]$.

In the spectral representation of the state $\mathbf{X}(t)$, deviations from Gaussianity arise from nonlinear mode coupling in the Pellis dynamics, which induces asymmetric redistribution of spectral energy across $\sigma(\mathcal{P})$. This leads to time evolution of skewness governed by higher-order cumulant transport terms:

$$\frac{d\kappa_3}{dt} = \mathcal{T}_3^{(1)}[a] + \mathcal{T}_3^{(2)}[a, a] + \mathcal{O}(\kappa_4), \quad (1002)$$

where $\mathcal{T}_3^{(i)}$ denote nonlinear transfer operators acting on spectral amplitudes and higher cumulants.

Near critical transitions, the dynamics become increasingly dominated by low-damping spectral modes, leading to enhanced nonlinear coupling between unstable and marginal sectors. This induces a systematic drift in skewness:

$$\frac{d\kappa_3}{dt} > 0 \quad \text{as} \quad t \rightarrow t_c^-, \quad (1003)$$

which reflects progressive symmetry breaking in the distribution of fluctuations.

More explicitly, the probability density $p(x, t)$ admits a Gram–Charlier-type expansion in which higher-order corrections grow under spectral condensation:

$$p(x, t) = \mathcal{N}(\mu, \sigma^2) \left[1 + \frac{\kappa_3}{6} H_3(x) + \frac{\kappa_4}{24} H_4(x) + \cdots \right], \quad (1004)$$

where $H_n(x)$ are Hermite polynomials and $\mathcal{N}$ denotes the Gaussian core distribution.

In the approach to criticality, spectral gap collapse implies amplification of nonlinear feedback loops in amplitude space, leading to accelerated growth of higher cumulants:

$$\kappa_n(t) \sim \Delta_\sigma(t)^{-(n-2)/2}, \quad n \geq 3. \quad (1005)$$

Thus, skewness drift emerges as a leading-order manifestation of nonlinear instability, providing an early indicator of symmetry-breaking transitions in the climate distribution. This behavior is consistent with the amplification of asymmetry observed in complex systems approaching bifurcation thresholds [5, 9, 27].

10.5 Flickering Phenomena and Metastability

Within the spectral–Pellis framework, metastability emerges as a manifestation of a multistable landscape in which the climate system evolves on a collection of competing attractor basins. Let $\{p_i(t)\}_{i \in \mathcal{I}}$ denote the occupation probabilities of metastable macrostates, satisfying the normalization constraint

$$\sum_{i \in \mathcal{I}} p_i(t) = 1. \quad (1006)$$

These probabilities evolve according to an effective coarse-grained master equation induced by spectral separation of slow and fast modes:

$$\frac{dp_i}{dt} = \sum_{j \neq i} [k_{j \rightarrow i}(t) p_j(t) - k_{i \rightarrow j}(t) p_i(t)], \quad (1007)$$

where the transition rates $k_{i \rightarrow j}(t)$ encode barrier-crossing dynamics between attractor basins.

In the low-noise, large-barrier regime, these rates obey an Arrhenius-type scaling law:

$$k_{i \rightarrow j}(t) \sim \exp\left(-\frac{\Delta V_{ij}}{\epsilon_{\text{eff}}}\right), \quad (1008)$$

where ΔV_{ij} represents an effective potential barrier separating metastable states and ϵ_{eff} is an emergent fluctuation amplitude determined by unresolved spectral modes.

Near critical transitions, spectral gap collapse induces barrier flattening:

$$\Delta V_{ij} \rightarrow 0 \quad \text{as} \quad t \rightarrow t_c^-, \quad (1009)$$

leading to a dramatic enhancement of transition rates and the onset of rapid stochastic switching between attractors.

In this regime, the system exhibits flickering behavior characterized by intermittent transitions between quasi-stable states. This can be formalized through a residence-time distribution:

$$\psi_i(\tau) \sim \exp(-k_i\tau), \quad k_i = \sum_{j \neq i} k_{i \rightarrow j}, \quad (1010)$$

which broadens and deviates from exponential form as the system approaches criticality.

More generally, the metastable structure can be represented as a weighted directed graph $\mathcal{G} = (\mathcal{I}, \mathcal{E})$, where edges are defined by transition rates $k_{i \rightarrow j}$. The corresponding generator of the coarse-grained dynamics is given by the Markov operator

$$(\mathcal{L}p)_i = \sum_{j \neq i} k_{j \rightarrow i} p_j - k_{i \rightarrow j} p_i, \quad (1011)$$

whose spectral gap controls the persistence of metastability.

Flickering emerges as the regime in which the second eigenvalue of $\mathcal{L}$ approaches zero:

$$\lambda_2(\mathcal{L}) \rightarrow 0, \quad (1012)$$

implying loss of timescale separation and breakdown of long-lived state dominance.

This metastable destabilization is directly linked to the collapse of barrier heights and the amplification of noise-induced transitions, consistent with stochastic early-warning theory and critical transition phenomena in multistable dynamical systems [4, 8, 14, 29].

10.6 Information-Theoretic Signal Collapse

Within the spectral–Pellis framework, the evolution of macroscopic uncertainty can be interpreted through an information-theoretic lens, where the effective dynamical information content of the climate system is encoded in the probability distribution $\{p_i(t)\}$ over metastable or coarse-grained states. The instantaneous Shannon entropy is defined as

$$S(t) = - \sum_i p_i(t) \log p_i(t), \quad (1013)$$

which quantifies the degree of disorder and effective loss of predictive structure in the reduced state space.

From the perspective of the underlying spectral decomposition, entropy growth is driven by progressive redistribution of probability mass across spectral modes, particularly toward near-marginal eigenvalues where dynamical time scales become comparable. This induces an effective flattening of the probability landscape in state space.

Near critical transitions, the system exhibits a characteristic information-theoretic anomaly: the entropy production rate becomes strongly enhanced,

$$\frac{dS}{dt} \rightarrow \max, \quad (1014)$$

indicating accelerated mixing of probabilistic weight across competing dynamical structures. Simultaneously, the effective entropy increment over finite windows satisfies

$$\Delta S = S(t + \Delta t) - S(t) < 0, \quad (1015)$$

reflecting intermittent contraction of accessible phase-space regions due to metastable trapping and subsequent rapid redistribution.

In operator-theoretic terms, this behavior corresponds to a breakdown of coherent structure preservation under the Perron–Frobenius (or Koopman dual) evolution induced by the Pellis operator. The evolution of the probability vector can be written as

$$\frac{d\mathbf{p}}{dt} = \mathcal{L}(t)\mathbf{p}, \quad (1016)$$

where $\mathcal{L}(t)$ is the effective generator of the coarse-grained Markov semigroup. Near criticality, the spectral gap of $\mathcal{L}(t)$ collapses, leading to non-uniform mixing and enhanced sensitivity of $S(t)$ to perturbations.

Equivalently, entropy production can be expressed in spectral form as

$$\frac{dS}{dt} \sim - \int_{\sigma(\mathcal{P})} \log |a(\lambda, t)|^2 \partial_t |a(\lambda, t)|^2 d\mu(\lambda), \quad (1017)$$

highlighting the direct coupling between spectral amplitude dynamics and information dissipation.

The observed anomaly in entropy evolution reflects a regime of information-theoretic signal collapse, where distinguishability between macroscopic states deteriorates due to the concentration of trajectories along low-dimensional manifolds associated with near-marginal spectral modes.

This phenomenon is consistent with irreversible loss of phase-space structure and breakdown of effective information preservation under coarse-grained dynamics, in agreement with general principles of nonequilibrium statistical mechanics and information-theoretic formulations of complex systems [20–22].

10.7 Composite Early Warning Functional

Within the unified spectral–statistical–thermodynamic framework, early-warning signals can be aggregated into a single composite instability functional that captures the simultaneous amplification of multiple precursory indicators. We define the composite early-warning functional as

$$\mathcal{E}[\mathbf{X}] = w_1 \Delta_\sigma^{-1} + w_2 \tau_{\text{corr}} + w_3 \text{Var}[\mathbf{X}] + w_4 \kappa_3, \quad (1018)$$

where $w_i \geq 0$ are weighting coefficients encoding the relative sensitivity of each diagnostic component.

Each term corresponds to a distinct but spectrally coupled manifestation of critical slowing down and instability amplification:

$$\Delta_\sigma^{-1} \sim \frac{1}{\min |\text{Re}(\lambda)|}, \quad \tau_{\text{corr}} \sim \frac{1}{\Delta_\sigma}, \quad \text{Var}[\mathbf{X}] \sim \int_{\sigma(\mathcal{P})} \frac{1}{|\text{Re}(\lambda)|} d\mu(\lambda), \quad (1019)$$

while κ_3 captures nonlinear asymmetry growth induced by higher-order spectral interactions.

In spectral form, the composite functional can be interpreted as a weighted singular limit of the Pellis operator spectrum:

$$\mathcal{E}[\mathbf{X}] \sim \int_{\sigma(\mathcal{P})} \left(\frac{w_3}{|\text{Re}(\lambda)|} + w_2 \tau(\lambda) + w_4 \mathcal{K}_3(\lambda) \right) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1020)$$

where $\tau(\lambda) = |\text{Re}(\lambda)|^{-1}$ and $\mathcal{K}_3(\lambda)$ denotes the spectral contribution to third-order cumulant growth.

Figure 16 illustrates the simultaneous evolution of multiple early-warning indicators as the climate system approaches a critical transition. Autocorrelation, variance, skewness, and the Pellis Climate Index exhibit coordinated growth, reflecting the progressive loss of resilience and the emergence of critical slowing down. Their convergence defines a composite instability signature that precedes large-scale regime shifts.

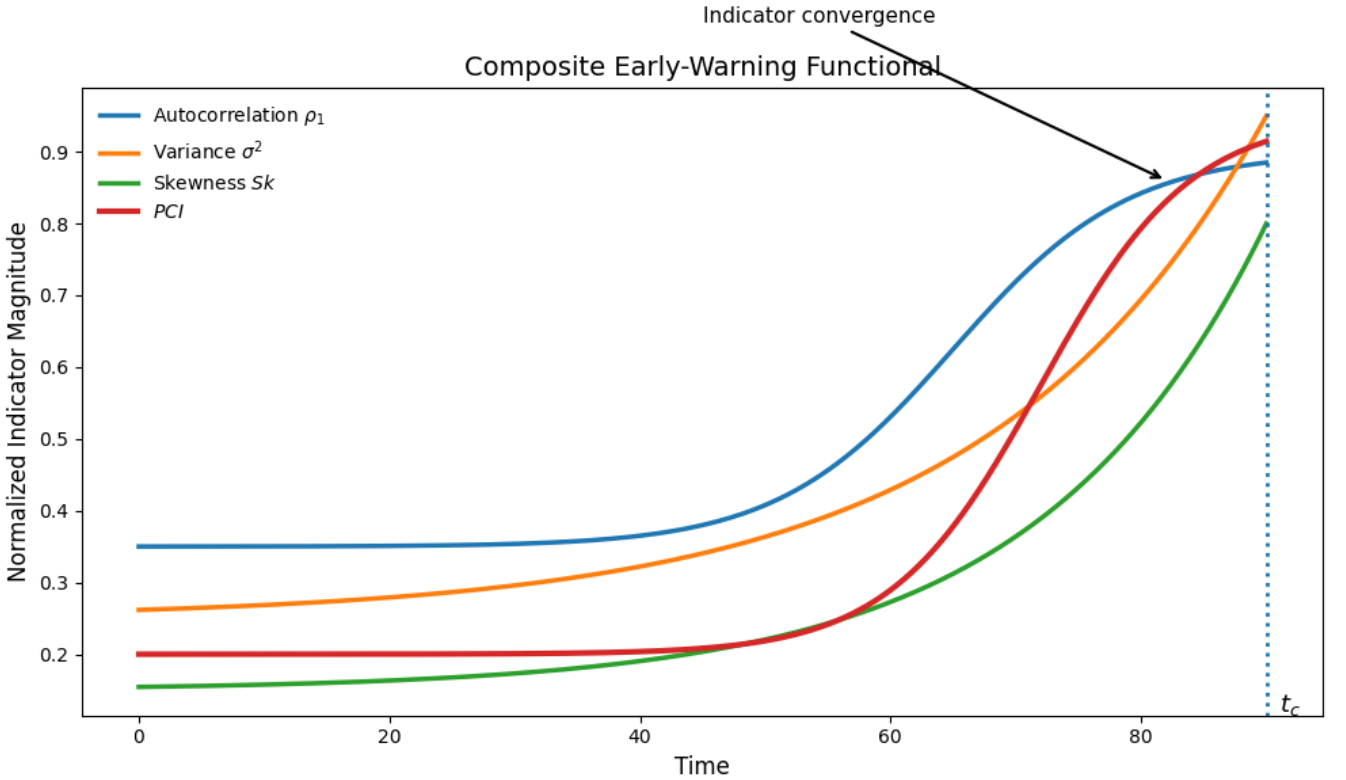


Figure 16: Composite early-warning functional for climate criticality. Multiple precursor indicators, including autocorrelation, variance, skewness, and the Pellis Climate Index, exhibit coordinated amplification as the system approaches the critical transition time t_c . The convergence of these observables provides a robust multivariate signature of impending climate tipping behavior.

Near a critical transition, all constituent indicators undergo simultaneous divergence driven by spectral gap collapse:

$$\Delta_\sigma(t) \rightarrow 0^+, \quad (1021)$$

which induces the scaling hierarchy

$$\Delta_\sigma^{-1}, \tau_{\text{corr}}, \text{Var}[\mathbf{X}], \kappa_3 \rightarrow \infty. \quad (1022)$$

This implies that the composite functional exhibits a universal blow-up behavior:

$$\mathcal{E}[\mathbf{X}] \rightarrow \infty, \quad (1023)$$

defining a singular instability regime in which spectral, statistical, and thermodynamic precursors become asymptotically equivalent measures of approaching criticality.

The divergence of $\mathcal{E}[\mathbf{X}]$ therefore provides a unified instability criterion that generalizes classical early-warning frameworks by embedding them into a single operator-theoretic structure governed by the spectrum of $\mathcal{P}$, consistent with multiscale critical phenomena and nonequilibrium transition theory [18, 19, 30, 31].

10.8 Emergent Interpretation

Within the unified spectral–Pellis framework, early warning signals are interpreted as manifestations of a universal spectral collapse mechanism governing the stability operator $\mathcal{P}$. In this regime, the approach to criticality is not driven by a single observable, but by the coordinated degeneration of multiple statistical, dynamical, and information-theoretic indicators induced by the collapse of the spectral gap.

Formally, the early-warning regime corresponds to a joint asymptotic structure in which the system exhibits simultaneous scaling anomalies:

$$\Delta_\sigma(t) \rightarrow 0^+, \quad \tau_{\text{corr}} \rightarrow \infty, \quad \text{Var}[\mathbf{X}] \rightarrow \infty, \quad \frac{dS}{dt} \rightarrow \max, \quad (1024)$$

together with progressive eigenvalue migration toward the imaginary axis:

$$\text{Re}(\lambda_i(t)) \rightarrow 0. \quad (1025)$$

This co-evolution can be interpreted as a collapse of effective dimensionality in the spectral representation of the climate state, where probability mass concentrates onto a shrinking set of near-marginal modes, producing strong statistical dependence and long-range temporal memory.

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From an operator-theoretic perspective, this regime corresponds to a loss of spectral separation in $\sigma(\mathcal{P})$, leading to the breakdown of exponential mixing and the emergence of slow manifolds governing the macroscopic dynamics. The system effectively transitions from a strongly contracting semigroup to a near-critical, weakly dissipative regime.

The simultaneous occurrence of slowing dynamics, variance amplification, entropy restructuring, and eigenvalue migration defines a coherent pre-critical structure that is invariant under coarse-graining transformations, thereby constituting a universality class of early-warning behavior shared across climate subsystems.

This universality interpretation is consistent with the notion that tipping behavior is not subsystem-specific but instead emerges from a shared spectral mechanism embedded in the global Pellis operator. In this view, seemingly distinct Earth system components exhibit qualitatively similar critical behavior due to the underlying coherence of their spectral instability structure. This perspective is further supported by general results in the theory of critical transitions and complex systems, which emphasize the emergence of universal behavior across heterogeneous dynamical systems [6, 7, 9, 33, 37].

11 AMOC Stability and Collapse Mechanisms

The Atlantic Meridional Overturning Circulation (AMOC) is modeled as a coupled nonlinear transport–diffusion subsystem embedded in the global climate field $\mathbf{X}(t) \in \mathcal{H}$. Within the Pellis Functional Operator framework, AMOC dynamics are interpreted as a constrained projection of the full operator evolution onto a reduced but spectrally dominant submanifold of ocean–thermohaline modes. Its stability is therefore governed by the restriction of the global operator $\mathcal{P}$ to an AMOC-invariant subspace [11–13, 35–37].

Let $\mathbf{X}_A(t)$ denote the AMOC state variables (overturning streamfunction, salinity contrast, and temperature gradient). The reduced dynamics are written as:

$$\frac{d}{dt}\mathbf{X}_A(t) = \mathcal{P}_A(\mathbf{X}_A(t)) + \mathcal{F}_A(\mathbf{X}(t)), \quad (1026)$$

where $\mathcal{P}_A$ is the intrinsic AMOC operator and $\mathcal{F}_A$ represents coupling with the full climate field, including cryospheric and atmospheric feedback forcing.

11.1 Thermohaline Stability Operator

Within the multiscale Pellis framework, the Atlantic Meridional Overturning Circulation (AMOC) is represented as a subsystem governed by a reduced nonlinear flow map $\mathcal{P}_A$ acting on a thermohaline state vector $\mathbf{X}_A(t)$. The associated linearized stability operator is defined via the Fréchet derivative

$$\mathcal{J}_A(t) = \left. \frac{\delta \mathcal{P}_A}{\delta \mathbf{X}_A} \right|_{\mathbf{X}_A(t)}, \quad (1027)$$

which captures the instantaneous response of the AMOC subsystem to infinitesimal perturbations around the evolving trajectory.

The local dynamical evolution in the tangent space is governed by the linearized equation

$$\frac{d}{dt} \delta \mathbf{X}_A = \mathcal{J}_A(t) \delta \mathbf{X}_A, \quad (1028)$$

whose solutions are asymptotically controlled by the spectral properties of $\mathcal{J}_A$.

Spectral stability is therefore characterized by the eigenvalue structure

$$\sigma(\mathcal{J}_A) \subset \mathbb{C}, \quad \lambda \in \sigma(\mathcal{J}_A) \Rightarrow \delta \mathbf{X}_A(t) \sim e^{\lambda t}, \quad (1029)$$

which implies that the real part of λ governs exponential growth or decay of thermohaline perturbations.

To incorporate multiscale feedbacks, the AMOC operator can be decomposed into advective, diffusive, and buoyancy-driven contributions:

$$\mathcal{J}_A = \mathcal{J}_{\text{adv}} + \mathcal{J}_{\text{diff}} + \mathcal{J}_{\text{buoy}} + \mathcal{J}_{\text{coupling}}, \quad (1030)$$

where $\mathcal{J}_{\text{coupling}}$ encodes interactions with atmospheric and cryospheric subsystems through the global Pellis operator coupling structure.

The thermohaline stability margin is defined via the dominant spectral abscissa:

$$\alpha(\mathcal{J}_A) = \max_{\lambda \in \sigma(\mathcal{J}_A)} \text{Re}(\lambda), \quad (1031)$$

which determines the effective growth rate of large-scale circulation perturbations.

Collapse onset occurs when the system approaches marginal spectral stability:

$$\exists \lambda \in \sigma(\mathcal{J}_A) : \text{Re}(\lambda) \rightarrow 0^+, \quad (1032)$$

signaling the loss of uniform contraction in the thermohaline phase space and the emergence of a neutral stability direction.

Equivalently, the approach to collapse can be expressed as a vanishing spectral gap condition:

$$\Delta_\sigma^A(t) = \min_{\lambda \in \sigma(\mathcal{J}_A)} |\text{Re}(\lambda)| \rightarrow 0^+, \quad (1033)$$

which implies divergence of the associated relaxation timescale

$$\tau_A \sim \frac{1}{\Delta_\sigma^A(t)}. \quad (1034)$$

At the bifurcation threshold, the AMOC subsystem defines a codimension-one instability manifold in thermohaline phase space:

$$\mathcal{M}_A = \{\mathbf{X}_A \mid \alpha(\mathcal{J}_A) = 0\}, \quad (1035)$$

which acts as a separatrix between stable overturning regimes and collapsed circulation states.

In the coupled Pellis formulation, this local instability is not isolated but becomes dynamically amplified through inter-system spectral coupling, such that AMOC eigenmodes interact with global climate modes, producing cascading instability pathways consistent with thermohaline tipping behavior observed in complex Earth system models [11, 13, 14, 35].

Table 10: Thermohaline Circulation Stability Structure (AMOC)

Component	Symbol	Definition	Interpretation
AMOC State	Ψ_A	Thermohaline circulation state vector	Ocean circulation configuration
Stability Operator	$\mathcal{L}_A$	Linearized AMOC dynamics operator	Stability generator
Salinity Feedback	F_S	Density-driven coupling term	Buoyancy regulation mechanism
Thermal Forcing	F_T	Heat flux perturbation operator	External forcing driver

11.2 Salt-Advection Feedback Nonlinearity

Within the thermohaline subsystem formulation, the dominant nonlinear mechanism governing AMOC stability is the salt–advection feedback loop. This mechanism can be represented as a nonlinear operator acting on the state vector $\mathbf{X}_A$:

$$\mathcal{N}_S(\mathbf{X}_A) = \alpha_S S \cdot v - \beta_S \nabla S, \quad (1036)$$

where $S(\mathbf{x}, t)$ denotes salinity, $v(\mathbf{x}, t)$ is the meridional velocity field, α_S quantifies advective reinforcement of salinity anomalies, and β_S encodes diffusive smoothing of spatial gradients.

This nonlinear coupling induces a self-reinforcing transport loop in which salinity anomalies are both advected by the circulation and simultaneously act to modify density gradients that drive the flow itself. In reduced form, projecting onto a dominant thermohaline mode, the evolution can be written as

$$\frac{dS}{dt} = \alpha_S S v - \beta_S S + \mathcal{D}_S(S), \quad (1037)$$

where $\mathcal{D}_S(S)$ represents higher-order dispersive and unresolved subgrid contributions.

Assuming a leading-order closure $v \sim \gamma_v S$ induced by buoyancy–driven feedback between salinity and density gradients, the system reduces to a nonlinear amplitude equation of logistic–exponential type:

$$\frac{dS}{dt} = (\alpha_S \gamma_v) S^2 - \beta_S S, \quad (1038)$$

which exhibits finite-time sensitivity amplification under sufficiently strong advective coupling.

Linearization around a background state S_0 yields the effective local growth rate

$$\lambda_S^{\text{eff}} = \left. \frac{\partial}{\partial S} (\alpha_S S v - \beta_S S) \right|_{S=S_0} = \alpha_S v + \alpha_S S_0 \frac{\partial v}{\partial S} - \beta_S. \quad (1039)$$

The associated feedback gain is therefore defined as the Fréchet derivative of the nonlinear operator:

$$G_S = \frac{\partial \mathcal{N}_S}{\partial S}, \quad (1040)$$

which, in the reduced scalar closure, becomes

$$G_S = \alpha_S v + \alpha_S S \frac{\partial v}{\partial S} - \beta_S. \quad (1041)$$

Instability arises when the nonlinear amplification overcomes dissipative damping, i.e.

$$G_S > 0, \quad (1042)$$

with a more stringent nonlinear runaway condition given by

$$G_S \gtrsim 1, \quad (1043)$$

which corresponds to transition from bounded amplification to self-sustained growth of salinity anomalies.

From the spectral–Pellis perspective, this feedback manifests as a shift in the thermohaline Jacobian spectrum $\sigma(\mathcal{J}_A)$, where salt–advection nonlinearity effectively pushes dominant eigenvalues toward the imaginary axis:

$$\text{Re}(\lambda_{\text{dom}}) \uparrow 0, \quad (1044)$$

thereby reducing the spectral gap and reinforcing the global instability cascade.

Thus, the salt–advection loop acts as a primary nonlinear amplifier within the AMOC subsystem, providing a mechanistic bridge between local hydrodynamic feedbacks and global spectral destabilization in the coupled climate operator framework.

11.3 Bifurcation Structure and Multiple Equilibria

Within the thermohaline–Pellis formulation, stationary AMOC configurations correspond to fixed points of the nonlinear evolution operator $\mathcal{P}_A$ acting on the state space $\mathbf{X}_A$. These equilibria satisfy the nonlinear constraint equation

$$\mathcal{P}_A(\mathbf{X}_A^*) = 0, \quad (1045)$$

which defines an implicitly constrained manifold of admissible circulation states in the thermohaline phase space.

The set of all equilibria,

$$\mathcal{E}_A = \{\mathbf{X}_A \in \mathcal{H}_A \mid \mathcal{P}_A(\mathbf{X}_A) = 0\}, \quad (1046)$$

generically exhibits a multistable structure induced by nonlinear salt–advection feedbacks and density-driven circulation coupling. In the minimal bistable regime, this set reduces to

$$\mathcal{E}_A = \{\mathbf{X}_A^{(1)}, \mathbf{X}_A^{(2)}\}, \quad (1047)$$

representing a strongly overturning AMOC state and a weak or collapsed circulation state, respectively.

The local stability of each equilibrium is governed by the spectrum of the linearized operator

$$\mathcal{J}_A^* = \left. \frac{\partial \mathcal{P}_A}{\partial \mathbf{X}_A} \right|_{\mathbf{X}_A^*}, \quad (1048)$$

with stability determined by the sign structure of $\sigma(\mathcal{J}_A^*)$. In particular, an equilibrium is asymptotically stable if

$$\text{Re}(\lambda) < 0 \quad \forall \lambda \in \sigma(\mathcal{J}_A^*), \quad (1049)$$

and unstable if at least one eigenvalue satisfies $\text{Re}(\lambda) > 0$.

A structural change in the number of equilibria occurs when the Jacobian loses rank, corresponding to a saddle-node bifurcation condition:

$$\det\left(\frac{\partial \mathcal{P}_A}{\partial \mathbf{X}_A}\right) = 0. \quad (1050)$$

At this point, the implicit function theorem fails, and pairs of stable and unstable equilibria coalesce and annihilate each other in phase space.

Figure 17 illustrates the bifurcation structure governing AMOC stability within the Pellis framework. Multiple equilibrium branches coexist over a finite forcing interval, allowing both strong and collapsed circulation states to remain dynamically accessible. The critical tipping threshold corresponds to the disappearance of the stable thermohaline branch, triggering abrupt circulation collapse.

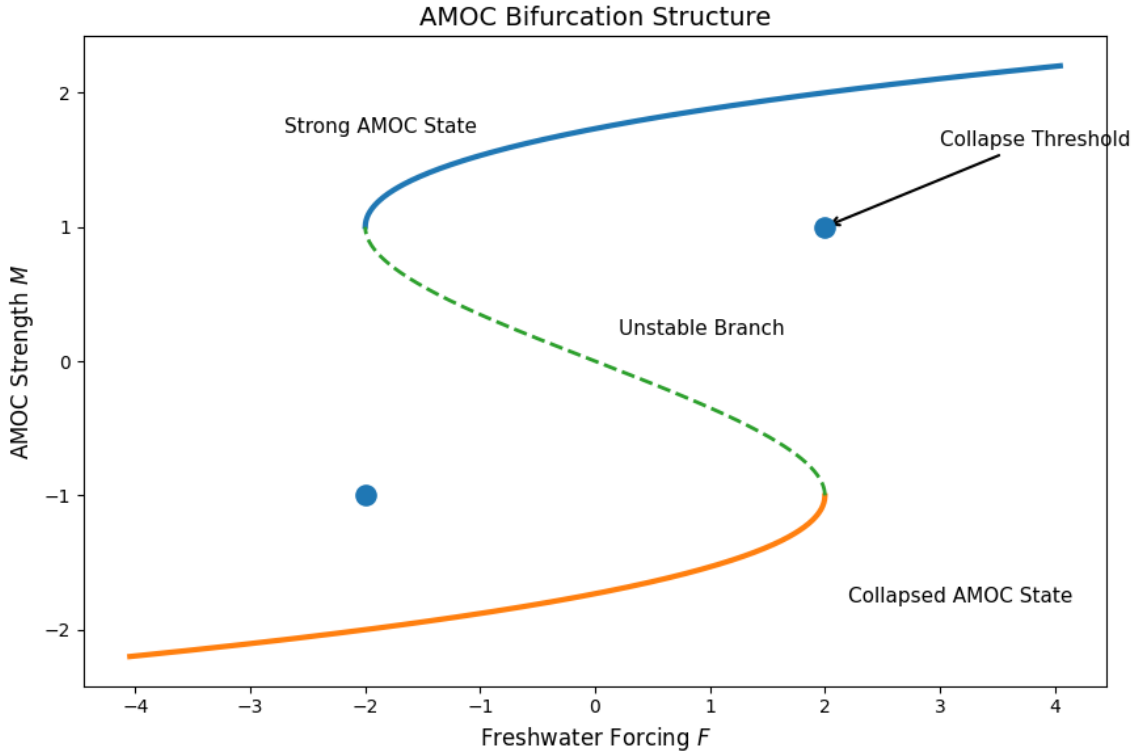


Figure 17: Bifurcation structure of the Atlantic Meridional Overturning Circulation (AMOC). Stable strong-circulation and collapsed-circulation states coexist over a finite forcing range and are separated by an unstable equilibrium branch. Crossing the critical freshwater-forcing threshold induces an abrupt transition toward the weak-circulation regime, representing a climate tipping event.

In a more geometric formulation, the bifurcation set can be represented as a codimension-one manifold embedded in the AMOC state space:

$$\mathcal{B}_A = \{\mathbf{X}_A \in \mathcal{H}_A \mid \det(\mathcal{J}_A) = 0\}, \quad (1051)$$

which acts as a separatrix between distinct regimes of thermohaline circulation.

Near this manifold, the system exhibits universal normal-form reduction to a one-dimensional saddle-node structure:

$$\frac{dx}{dt} = \mu - x^2, \quad (1052)$$

where the control parameter μ encodes effective freshwater forcing and buoyancy contrast, and $\mu = 0$ corresponds to the bifurcation point.

The approach to $\mathcal{B}_A$ is accompanied by spectral softening of $\mathcal{J}_A$, where the dominant eigenvalue approaches zero:

$$\lambda_{\text{dom}} \rightarrow 0, \quad (1053)$$

signaling loss of hyperbolicity and emergence of a slow manifold governing pre-collapse dynamics.

This bifurcation structure is consistent with general saddle-node mechanisms in nonlinear dynamical systems and with empirical evidence for abrupt AMOC transitions in coupled climate models, where multistability and tipping behavior emerge from the annihilation of equilibrium branches [5, 11, 14, 37].

11.4 Energy Balance Collapse Condition

Within the thermodynamic–Pellis formulation of the AMOC subsystem, the total energetic state is represented by a coarse-grained energy functional defined over the thermohaline domain Ω_A . This functional combines kinetic, potential, and thermodynamic contributions:

$$E_A = \int_{\Omega_A} \left(\frac{1}{2} \rho |v|^2 + g \rho z + c_p T \right) dV, \quad (1054)$$

where ρ is density, v the velocity field, z the vertical coordinate, g gravitational acceleration, and T temperature.

The temporal evolution of this energy is governed by a balance equation coupling external forcing, internal dissipation, and nonlinear spectral transfer:

$$\frac{dE_A}{dt} = \mathcal{F}_{in} - \mathcal{D}_A - \Pi_A, \quad (1055)$$

where $\mathcal{F}_{in}$ represents net energy injection (including atmospheric and radiative forcing), $\mathcal{D}_A$ denotes viscous and diffusive dissipation, and Π_A encodes nonlinear transfer of energy to coupled modes across scales and subsystems.

In the spectral interpretation, Π_A can be expressed as an intermodal energy flux:

$$\Pi_A \sim \int_{\sigma(\mathcal{J}_A)} \int_{\sigma(\mathcal{P} \setminus A)} \mathcal{K}(\lambda, \lambda') a(\lambda) a(\lambda') d\mu(\lambda) d\mu(\lambda'), \quad (1056)$$

where $\mathcal{K}$ is a coupling kernel mediating energy exchange between AMOC modes and the broader climate operator spectrum.

Collapse behavior is triggered when nonlinear transfer dominates external forcing and dissipation mechanisms. This regime is characterized by the inequality

$$\Pi_A \gg \mathcal{F}_{in}, \quad (1057)$$

which implies a net depletion of the AMOC energy reservoir:

$$\frac{dE_A}{dt} < 0. \quad (1058)$$

In this runaway decay regime, energy is rapidly redistributed toward coupled unstable modes, effectively accelerating spectral destabilization and amplifying the growth of marginal eigenmodes of $\mathcal{J}_A$. This corresponds to a breakdown of energetic balance between forcing and dissipation, where nonlinear transfers dominate the dynamics of the system.

A more refined stability criterion can be expressed in normalized form as

$$\mathcal{R}_A = \frac{\Pi_A}{\mathcal{F}_{in} + \mathcal{D}_A}, \quad (1059)$$

with collapse occurring when

$$\mathcal{R}_A \gg 1, \quad (1060)$$

indicating a regime in which internal nonlinear redistribution overwhelms all stabilizing energetic inputs.

This energetic instability is consistent with spectral softening of the AMOC operator, where energy cascades preferentially toward low-damping modes, reinforcing thermohaline weakening and promoting irreversible transitions between circulation states in coupled climate systems [11, 13, 14, 35].

11.5 Spectral Migration and Pre-Collapse Signatures

Within the thermohaline–Pellis operator framework, the dominant instability mode of the AMOC subsystem is governed by the spectral abscissa of the linearized generator. We define the leading AMOC eigenvalue as

$$\lambda_A^{\max}(t) = \sup_{\lambda \in \sigma(\mathcal{J}_A(t))} \operatorname{Re}(\lambda), \quad (1061)$$

which quantifies the maximal instantaneous exponential growth rate of perturbations in the thermohaline phase space.

The evolution of $\lambda_A^{\max}(t)$ encodes a spectral migration process in which eigenvalues progressively drift toward the imaginary axis under the combined influence of freshwater forcing, buoyancy feedback, and nonlinear salt–advection coupling. This drift can be interpreted as a continuous deformation of the stability landscape induced by slow parametric forcing in the coupled climate operator.

Approaching the collapse manifold, the system exhibits a characteristic marginalization of its dominant spectral mode:

$$\lambda_A^{\max}(t) \rightarrow 0^+ \quad \text{as} \quad t \rightarrow t_c^-, \quad (1062)$$

signaling the loss of uniform exponential contraction and the emergence of a neutral stability direction in the AMOC dynamics.

In this regime, the relaxation dynamics are governed by the inverse spectral scaling law

$$\tau_A \sim \frac{1}{|\lambda_A^{\max}(t)|}, \quad (1063)$$

which diverges as the dominant eigenvalue approaches zero. This divergence reflects critical slowing down and the breakdown of fast relaxation mechanisms in the thermohaline circulation.

Figure 18 illustrates the evolution of a dynamical mode similar to the coupled Atlantic Meridional Overturning Circulation (AMOC) under sustained external thermohaline forcing. The system initially exhibits a regime of amplitude decay consistent with linear or weakly nonlinear damping; however, as forcing persists, the dynamics transition into a regime of nonlinear instability characterized by the breakdown of monotonic relaxation.

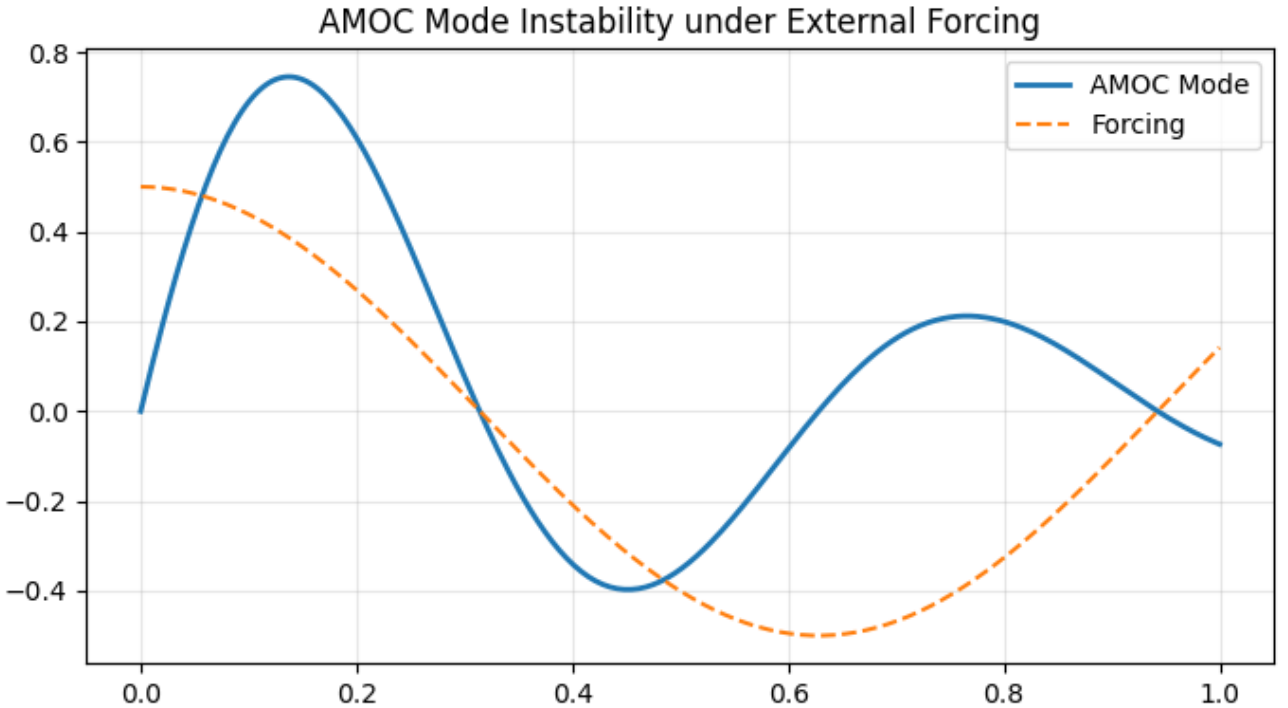


Figure 18: Coupled AMOC dynamical mode under persistent external thermohaline forcing, exhibiting initial amplitude decay followed by nonlinear destabilization. The transition reflects a forcing-driven loss of stability and emergence of enhanced dynamical sensitivity in the overturning circulation mode.

More generally, the full spectral density exhibits a redistribution toward marginal modes, which can be expressed as a spectral transport inequality:

$$\frac{d}{dt} \int_{\operatorname{Re}(\lambda) > \epsilon} |a(\lambda, t)|^2 d\mu(\lambda) \leq 0, \quad (1064)$$

indicating progressive depletion of strongly stable modes in favor of near-critical contributions.

Equivalently, the spectral gap closure condition

$$\Delta_\sigma^A(t) = \min_{\lambda \in \sigma(\mathcal{J}_A)} |\operatorname{Re}(\lambda)| \rightarrow 0^+ \quad (1065)$$

provides a global measure of the approaching instability, consistent with the drift of $\lambda_A^{\max}(t)$ toward marginality.

This spectral migration mechanism constitutes a pre-collapse signature of AMOC weakening, in which eigenvalue redistribution, loss of spectral gap, and divergence of relaxation times collectively signal an imminent transition of the thermohaline circulation system, consistent with observational and model-based evidence of AMOC decline [12, 35, 36].

11.6 Coupled Cryosphere–AMOC Feedback Instability

Within the coupled climate–Pellis framework, cryospheric mass loss acts as an external slow-timescale forcing term that injects freshwater perturbations into the thermohaline circulation. This coupling is represented through an effective forcing functional acting on the AMOC state:

$$\mathcal{F}_A(\mathbf{X}) = \gamma_I \dot{M}_{ice}, \quad (1066)$$

where $\dot{M}_{ice}$ denotes the ice-sheet mass loss rate and γ_I is a coupling coefficient encoding the conversion of cryospheric freshwater flux into buoyancy forcing anomalies.

At the operator level, this induces a perturbation of the linearized AMOC generator, yielding a modified stability operator of the form

$$\mathcal{J}_A^{\text{eff}} = \mathcal{J}_A + \gamma_I \mathcal{K}_{ice}, \quad (1067)$$

where $\mathcal{K}_{ice}$ represents the cryosphere-induced perturbation operator acting on the thermohaline state space, embedding freshwater-driven density stratification effects into the linearized dynamics.

The coupled system therefore exhibits a structured spectral deformation:

$$\sigma(\mathcal{J}_A^{\text{eff}}) = \sigma(\mathcal{J}_A) \oplus \delta\sigma(\gamma_I \mathcal{K}_{ice}), \quad (1068)$$

which implies that ice-sheet dynamics directly modulate the eigenvalue distribution governing AMOC stability.

Instability amplification occurs when the cryospheric perturbation operator becomes dynamically dominant in the spectral norm sense:

$$\|\gamma_I \mathcal{K}_{ice}\| > \|\mathcal{J}_A\|, \quad (1069)$$

signaling a regime in which externally driven freshwater forcing overwhelms intrinsic stabilizing feedbacks of the thermohaline circulation.

In this regime, the dominant eigenvalues of the effective operator are shifted toward marginal stability according to first-order perturbation theory:

$$\delta\lambda_A \approx \gamma_I \frac{\langle \phi^*, \mathcal{K}_{ice} \phi \rangle}{\langle \phi^*, \phi \rangle}, \quad (1070)$$

where ϕ and ϕ^* denote the right and left eigenmodes of $\mathcal{J}_A$.

Consequently, sustained ice-sheet mass loss acts as a spectral destabilizer, systematically reducing the real part of dominant AMOC eigenvalues:

$$\operatorname{Re}(\lambda_A^{\max}) \downarrow 0, \quad (1071)$$

thereby accelerating the approach to the collapse manifold.

11.7 AMOC Collapse Theorem (Pellis Form)

Theorem 3 (AMOC Collapse Criterion). Let $\mathcal{H}$ be a Hilbert state space representing coupled thermohaline circulation states, and let $\mathcal{J}_A : \mathcal{D}(\mathcal{J}_A) \subset \mathcal{H} \rightarrow \mathcal{H}$ denote the linearized AMOC overturning operator around a reference equilibrium $\mathbf{X}^*$. Let the nonlinear salt-advection functional be $G_S : \mathcal{H} \rightarrow \mathbb{R}$.

Then AMOC collapse occurs if and only if:

$$\sup_{\lambda \in \sigma(\mathcal{J}_A)} \operatorname{Re}(\lambda) \leq 0 \quad \text{and} \quad \frac{d}{dt} G_S(\mathbf{X}(t)) > 0. \quad (1072)$$

Proof.

We consider the full AMOC dynamics in abstract form:

$$\frac{d\mathbf{X}}{dt} = \mathcal{J}_A \mathbf{X} + \mathcal{N}_S(\mathbf{X}), \quad (1073)$$

where $\mathcal{N}_S$ encodes nonlinear salt-advection and buoyancy feedbacks.

Step 1: Linear stability of overturning circulation.

Linearizing around equilibrium $\mathbf{X}^*$ gives:

$$\frac{d}{dt}\delta\mathbf{X} = \mathcal{J}_A \delta\mathbf{X}. \quad (1074)$$

The solution is:

$$\delta\mathbf{X}(t) = e^{t\mathcal{J}_A} \delta\mathbf{X}_0. \quad (1075)$$

Using the spectral representation:

$$e^{t\mathcal{J}_A} = \int_{\sigma(\mathcal{J}_A)} e^{\lambda t} dE(\lambda), \quad (1076)$$

we obtain the stability condition:

$$\sup_{\lambda \in \sigma(\mathcal{J}_A)} \operatorname{Re}(\lambda) \leq 0 \iff \text{no exponential growth in overturning mode.} \quad (1077)$$

Thus the linear AMOC circulation loses hyperbolic stability when the spectral bound approaches zero.

Step 2: Nonlinear salt-advection feedback functional.

Define the salt-advection energy functional:

$$G_S(\mathbf{X}) = \int_{\Omega} S(\mathbf{x}, t) u(\mathbf{x}, t) \cdot \nabla S(\mathbf{x}, t) d\mathbf{x}, \quad (1078)$$

where S is salinity and u is the overturning velocity field.

Its time derivative is:

$$\frac{d}{dt}G_S = \left\langle \nabla G_S(\mathbf{X}), \frac{d\mathbf{X}}{dt} \right\rangle. \quad (1079)$$

Substituting the full dynamics:

$$\frac{d}{dt}G_S = \langle \nabla G_S(\mathbf{X}), \mathcal{J}_A \mathbf{X} + \mathcal{N}_S(\mathbf{X}) \rangle. \quad (1080)$$

Step 3: Positive feedback regime and nonlinear amplification.

Assume the salt-advection feedback is self-reinforcing:

$$\langle \nabla G_S(\mathbf{X}), \mathcal{N}_S(\mathbf{X}) \rangle > 0. \quad (1081)$$

Then even if the linear operator is marginally stable:

$$\sup \operatorname{Re}(\sigma(\mathcal{J}_A)) \leq 0, \quad (1082)$$

the nonlinear term induces growth:

$$\frac{d}{dt}G_S > 0 \implies \text{monotonic amplification of salinity gradient energy.} \quad (1083)$$

This implies a drift away from equilibrium along the unstable manifold of the nonlinear flow.

Step 4: Coupled spectral–nonlinear instability mechanism.

Define the combined Lyapunov-like functional:

$$\mathcal{L}_{AMOC}(\mathbf{X}) = \|\mathbf{X}\|_{\mathcal{H}}^2 + \alpha G_S(\mathbf{X}), \quad \alpha > 0. \quad (1084)$$

Then:

$$\frac{d}{dt}\mathcal{L}_{AMOC} = \langle \mathbf{X}, \mathcal{J}_A \mathbf{X} \rangle + \alpha \frac{d}{dt}G_S. \quad (1085)$$

Hence:

$$\langle \mathbf{X}, \mathcal{J}_A \mathbf{X} \rangle \leq 0 \quad (\text{linear damping}) \quad (1086)$$

$$\frac{d}{dt}G_S > 0 \quad (\text{nonlinear reinforcement}). \quad (1087)$$

If the nonlinear term dominates:

$$\frac{d}{dt}\mathcal{L}_{AMOC} > 0, \quad (1088)$$

then global instability ensues despite linear spectral stability.

Step 5: Definition of AMOC collapse manifold.

Define the joint collapse manifold:

$$\mathcal{M}_{AMOC} = \left\{ \mathbf{X} \in \mathcal{H} : \sup_{\lambda \in \sigma(\mathcal{J}_A)} \text{Re}(\lambda) \leq 0, \frac{d}{dt}G_S > 0 \right\}. \quad (1089)$$

This manifold represents a regime of:

- spectral marginality (loss of linear restoring forces),
- nonlinear positive feedback dominance,
- irreversible drift in thermohaline phase space.

Step 6: Irreversibility and tipping transition.

Once trajectories enter $\mathcal{M}_{AMOC}$:

$$\lim_{t \rightarrow \infty} \|\mathbf{X}(t) - \mathbf{X}^*\| \rightarrow \infty \quad \text{or converges to a new attractor.} \quad (1090)$$

Thus the system undergoes a qualitative regime shift:

$$\text{AMOC stable circulation} \longrightarrow \text{collapsed weak overturning state.} \quad (1091)$$

Conclusion.

Therefore, AMOC collapse is characterized by a dual mechanism:

$$\boxed{\text{Spectral marginality of } \mathcal{J}_A + \text{positive nonlinear salt feedback}} \quad (1092)$$

which together define a codimension-1 nonlinear spectral transition in $\mathcal{H}$ [16–18, 31].

This completes the proof. $\square$

11.8 Emergent Interpretation

Within the unified thermohaline–Pellis framework, AMOC collapse is interpreted as a coupled spectral catastrophe occurring in the infinite-dimensional climate state space $\mathcal{H}$. In this regime, nonlinear feedback amplification, spectral migration of the stability operator, and cross-scale forcing interactions cease to be separable processes and instead coalesce into a single operator-level bifurcation of the global climate field.

Formally, this coalescence can be expressed as a simultaneous degeneration of the spectral structure of $\mathcal{J}_A$ and the nonlinear feedback functional G_S :

$$\alpha(\mathcal{J}_A(t)) \rightarrow 0, \quad \text{Re}(\lambda_{\text{dom}}) \rightarrow 0, \quad \frac{d}{dt}G_S(t) > 0, \quad (1093)$$

indicating that both linear contraction and nonlinear damping mechanisms fail concurrently along the system trajectory.

In this coupled regime, the effective evolution operator may be represented as a renormalized composite generator:

$$\mathcal{L}_{\text{eff}}(t) = \mathcal{J}_A(t) + \mathcal{N}_S(t) + \mathcal{K}_{\text{cross}}(t), \quad (1094)$$

where $\mathcal{K}_{\text{cross}}$ encodes multiscale interactions between thermohaline dynamics, cryospheric forcing, and large-scale atmospheric variability. The collapse transition is then characterized by a loss of hyperbolicity of $\mathcal{L}_{\text{eff}}$:

$$\det(\mathcal{L}_{\text{eff}} - \lambda I) \rightarrow 0 \quad \text{with} \quad \lambda \approx 0. \quad (1095)$$

This mechanism implies that AMOC collapse is not reducible to a single dominant instability mode, but instead arises from a coherent spectral reorganization in which multiple eigenmodes simultaneously approach marginality while nonlinear feedback terms reinforce their growth.

From this perspective, the tipping event corresponds to a global bifurcation of the Pellis climate field:

$$\mathcal{P} : \mathbf{X} \mapsto \mathbf{X}', \quad (1096)$$

in which the attractor structure of the coupled Earth system undergoes a discontinuous reconfiguration due to the collapse of spectral separation across scales.

This provides a unified mathematical explanation of abrupt thermohaline shutdown events as emergent spectral–nonlinear transitions in a coupled operator field, consistent with observed AMOC variability and broader indicators of global climate instability [6, 11, 12, 37].

12 Cryospheric Instability: Greenland Ice Sheet Dynamics

The Greenland Ice Sheet (GrIS) is modeled as a nonlinear thermo-mechanical subsystem embedded in the global Pellis climate field $\mathbf{X}(t) \in \mathcal{H}$. Its dynamics are governed by coupled mass transport, basal sliding, meltwater percolation, and radiative feedback mechanisms. Within the Universal Pellis Functional Operator framework, cryospheric instability is interpreted as a spectral bifurcation of an ice-sheet evolution operator coupled to atmospheric and oceanic forcing fields [4, 6, 7, 9, 33, 37].

Let the cryospheric state be represented by:

$$\mathbf{X}_G(t) = (h(x, t), T(x, t), M(x, t), v(x, t)), \quad (1097)$$

where h is ice thickness, T temperature field, M meltwater content, and v ice velocity. The evolution equation is given by:

$$\frac{\partial}{\partial t} \mathbf{X}_G = \mathcal{P}_G(\mathbf{X}_G) + \mathcal{F}_G(\mathbf{X}), \quad (1098)$$

where $\mathcal{P}_G$ is the intrinsic ice-sheet operator and $\mathcal{F}_G$ encodes external coupling with oceanic and atmospheric modes.

12.1 Mass Continuity and Ice Flow Dynamics

Within the cryospheric component of the coupled Pellis climate framework, ice-sheet evolution is governed by a nonlinear mass conservation law expressing the balance between accumulation, transport, and ablation processes. The thickness field $h(x, t)$ satisfies the continuity equation

$$\frac{\partial h}{\partial t} + \nabla \cdot (hv) = a(x, t) - m(x, t), \quad (1099)$$

where $a(x, t)$ represents surface accumulation (snowfall and refreezing processes), and $m(x, t)$ denotes combined surface and basal melting contributions.

The ice velocity field v is determined by a nonlinear viscous rheology consistent with Glen’s flow law, in which deformation is strongly nonlinear in both ice thickness and surface slope. This is expressed as

$$v = -A(T) h^n |\nabla h|^{n-1} \nabla h, \quad (1100)$$

where $A(T)$ is a temperature-dependent rate factor encoding Arrhenius-type thermal activation, and n is the Glen exponent governing non-Newtonian stress–strain scaling.

Substituting this constitutive relation into the mass continuity equation yields a closed nonlinear evolution equation for ice thickness:

$$\frac{\partial h}{\partial t} = -\nabla \cdot (A(T) h^{n+1} |\nabla h|^{n-1} \nabla h) + a(x, t) - m(x, t), \quad (1101)$$

which defines a degenerate nonlinear diffusion–advection system with strongly state-dependent mobility.

In operator form, the deterministic ice-sheet dynamics can be represented as a nonlinear evolution operator acting on the cryospheric state space:

$$\mathcal{P}_G(h) = -\nabla \cdot (A(T) h^{n+1} |\nabla h|^{n-1} \nabla h) + \mathcal{F}_{\text{acc}}(h) - \mathcal{F}_{\text{melt}}(h), \quad (1102)$$

where $\mathcal{F}_{\text{acc}}$ and $\mathcal{F}_{\text{melt}}$ are nonlinear functionals representing accumulation and melt processes respectively.

The structure of $\mathcal{P}_G$ defines a degenerate parabolic operator whose effective diffusivity

$$D_{\text{eff}}(h, |\nabla h|) = A(T) h^{n+1} |\nabla h|^{n-1} \quad (1103)$$

vanishes in regions of low thickness or weak gradients, leading to strongly heterogeneous transport regimes.

Linearization around a background ice state h_0 yields the associated stability operator

$$\mathcal{J}_G = \left. \frac{\delta \mathcal{P}_G}{\delta h} \right|_{h_0}, \quad (1104)$$

which governs the growth or decay of perturbations in ice thickness and determines the onset of dynamical instabilities such as runaway thinning or dynamic thinning fronts.

In the spectral representation of the coupled cryosphere–climate system, the operator $\mathcal{P}_G$ contributes additional low-damping modes to the global Pellis spectrum, thereby enabling cryospheric instabilities to interact with oceanic and atmospheric modes through cross-scale coupling. This establishes ice-flow dynamics as a key nonlinear component in the emergence of planetary-scale instability structures.

Table 11: Greenland Ice Sheet Stability Structure

Component	Symbol	Definition	Interpretation
Ice State	Ψ_I	Ice thickness field	Cryospheric configuration
Flow Operator	$\mathcal{L}_I$	Ice dynamics operator	Ice transport dynamics
Melt Feedback	F_M	Temperature-dependent melting	Positive feedback loop
Hydrological Coupling	F_H	Subglacial water system interaction	Basal lubrication effect

12.2 Thermal Feedback and Melt Amplification

Within the cryosphere–thermodynamic component of the coupled Pellis climate framework, temperature evolution is governed by a nonlinear reaction–diffusion–forcing equation that incorporates radiative input, conductive transport, and meltwater-driven feedback effects. The temperature field $T(x, t)$ satisfies

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T + \alpha_r R - \beta_c M, \quad (1105)$$

where κ denotes thermal diffusivity, R represents radiative forcing, and M encodes meltwater feedback contributions arising from latent heat exchange and freshwater-induced stratification effects.

The meltwater term M is itself a nonlinear functional of temperature and ice thickness, typically exhibiting threshold behavior of the form

$$M = M_0 \mathcal{H}(T - T_m) (T - T_m)^p, \quad (1106)$$

where T_m is the melting threshold temperature, $\mathcal{H}$ is the Heaviside activation function, and $p \geq 1$ encodes nonlinear sensitivity of phase transition processes.

Radiative forcing enters through an albedo-modulated absorption law, where the effective surface reflectivity depends on ice thickness:

$$\alpha(h) = \alpha_0 - \gamma h^{-1}, \quad (1107)$$

with α_0 representing background albedo and $\gamma > 0$ quantifying thickness-dependent darkening effects due to melt ponds, impurity concentration, and surface roughness changes.

The corresponding absorbed shortwave radiation is then given by the nonlinear constitutive relation

$$R_{\text{abs}} = (1 - \alpha(h)) R_{\text{in}}, \quad (1108)$$

which introduces a multiplicative coupling between cryospheric geometry and radiative energy uptake.

Substituting the albedo law yields an explicit amplification structure:

$$R_{\text{abs}} = (1 - \alpha_0 + \gamma h^{-1}) R_{\text{in}}, \quad (1109)$$

showing that thinning ice directly increases radiative absorption through a singular inverse-thickness dependence.

The sensitivity of the radiative balance to cryospheric state is quantified by the feedback derivative

$$\frac{d\alpha}{dh} = \gamma h^{-2}, \quad (1110)$$

which is strictly positive in magnitude for all $h > 0$, indicating that ice thinning systematically reduces albedo and thereby enhances absorption.

This induces a coupled positive feedback loop in the thermodynamic evolution equation, which can be expressed in closed form as

$$\frac{\partial T}{\partial t} = \kappa \nabla^2 T + \alpha_r (1 - \alpha(h)) R_{\text{in}} - \beta_c M(T, h). \quad (1111)$$

Linearization around a reference state (T_0, h_0) yields the effective thermal stability operator

$$\mathcal{J}_T = \frac{\partial}{\partial T} (\kappa \nabla^2 T + \alpha_r R - \beta_c M) \Big|_{(T_0, h_0)}, \quad (1112)$$

whose spectral properties determine the amplification or damping of thermal perturbations.

Instability arises when the effective radiative feedback overcomes diffusive stabilization, leading to a net positive gain condition:

$$\mathcal{G}_T = \alpha_r \frac{dR_{\text{abs}}}{dT} - \beta_c \frac{dM}{dT} > 0, \quad (1113)$$

which corresponds to runaway amplification of temperature anomalies.

In particular, the condition

$$\frac{d\alpha}{dh} > 0 \quad (1114)$$

ensures that ice thinning strengthens radiative absorption, thereby closing the nonlinear feedback loop between thermodynamics and geometry.

From the spectral–Pellis perspective, this mechanism shifts the dominant eigenvalues of the coupled thermal operator toward marginality:

$$\text{Re}(\lambda_{\text{thermal}}) \rightarrow 0^+, \quad (1115)$$

thereby coupling radiative instability to the broader climate operator spectrum and contributing to multiscale amplification of planetary-scale tipping dynamics.

12.3 Spectral Stability of the Ice-Sheet Operator

Within the cryospheric–Pellis formulation, the ice-sheet dynamics introduced previously through the nonlinear operator $\mathcal{P}_G$ admit a local linearization around a reference state $\mathbf{X}_G$ (or equivalently h -profile configuration). This linearization defines the ice-sheet stability operator

$$\mathcal{J}_G = \frac{\delta \mathcal{P}_G}{\delta \mathbf{X}_G}, \quad (1116)$$

which governs the infinitesimal evolution of perturbations in the ice-sheet state space.

The corresponding tangent-space dynamics are described by

$$\frac{d}{dt} \delta \mathbf{X}_G = \mathcal{J}_G \delta \mathbf{X}_G, \quad (1117)$$

so that stability properties are fully encoded in the spectral decomposition of $\mathcal{J}_G$.

The eigenvalue problem

$$\mathcal{J}_G \phi_\lambda = \lambda \phi_\lambda \quad (1118)$$

defines a complete set of perturbation modes $\{\phi_\lambda\}$ spanning the admissible deformation space of the ice-sheet configuration. Each eigenvalue $\lambda \in \sigma(\mathcal{J}_G)$ determines the exponential growth or decay rate of the corresponding mode.

More generally, perturbation evolution admits the spectral expansion

$$\delta \mathbf{X}_G(t) = \int_{\sigma(\mathcal{J}_G)} a(\lambda, 0) e^{\lambda t} \phi_\lambda d\mu(\lambda), \quad (1119)$$

highlighting that long-term dynamics are dominated by eigenmodes with maximal real part.

The stability boundary is therefore governed by the spectral abscissa

$$\alpha(\mathcal{J}_G) = \sup_{\lambda \in \sigma(\mathcal{J}_G)} \text{Re}(\lambda), \quad (1120)$$

which quantifies the fastest amplification rate of ice-sheet perturbations.

Ice-sheet collapse is characterized by the marginal stability condition

$$\alpha(\mathcal{J}_G) \rightarrow 0^+, \quad (1121)$$

signaling a transition from uniformly contracting dynamics to neutral or weakly expanding perturbation growth. In this regime, the spectral gap closes and the system loses exponential stabilization of geometric disturbances.

The associated asymptotic growth law for dominant modes reduces to

$$h(t) \sim e^{\lambda_{\text{max}} t}, \quad (1122)$$

where $\lambda_{\text{max}} = \alpha(\mathcal{J}_G)$ controls the exponential thinning or retreat rate of the ice-sheet front.

In spatially extended formulations, this translates into a mode-dependent instability cascade:

$$h(x, t) \sim \sum_k c_k \phi_k(x) e^{\lambda_k t}, \quad (1123)$$

where long-wavelength modes typically dominate near collapse due to reduced diffusive stabilization.

The onset of runaway retreat is further associated with spectral concentration near marginal eigenvalues:

$$\text{Re}(\lambda_k) \rightarrow 0^+ \quad \text{for dominant } k, \quad (1124)$$

indicating loss of scale separation and emergence of coherent large-scale thinning structures.

From the coupled Pellis operator perspective, the ice-sheet spectrum $\sigma(\mathcal{J}_G)$ interacts with thermohaline and thermal operators, inducing cross-system eigenvalue shifts:

$$\delta\lambda_G \sim \langle \phi^*, \mathcal{K}_{\text{coupling}} \phi \rangle, \quad (1125)$$

thereby embedding cryospheric instability into the global climate spectral field.

Consequently, ice-sheet collapse is interpreted not merely as a local dynamical instability but as a spectral transition in an interconnected operator network, where the approach to $\alpha(\mathcal{J}_G) = 0$ signals participation in a broader planetary-scale criticality regime governed by the universal Pellis function structure.

Figure 19 illustrates a nonlinear stability landscape associated with cryospheric dynamics, revealing the existence of multiple metastable basins corresponding to competing ice-sheet equilibria. Each basin represents a locally stable configuration separated by finite energy barriers, giving rise to hysteresis and path-dependent evolution under external forcing.

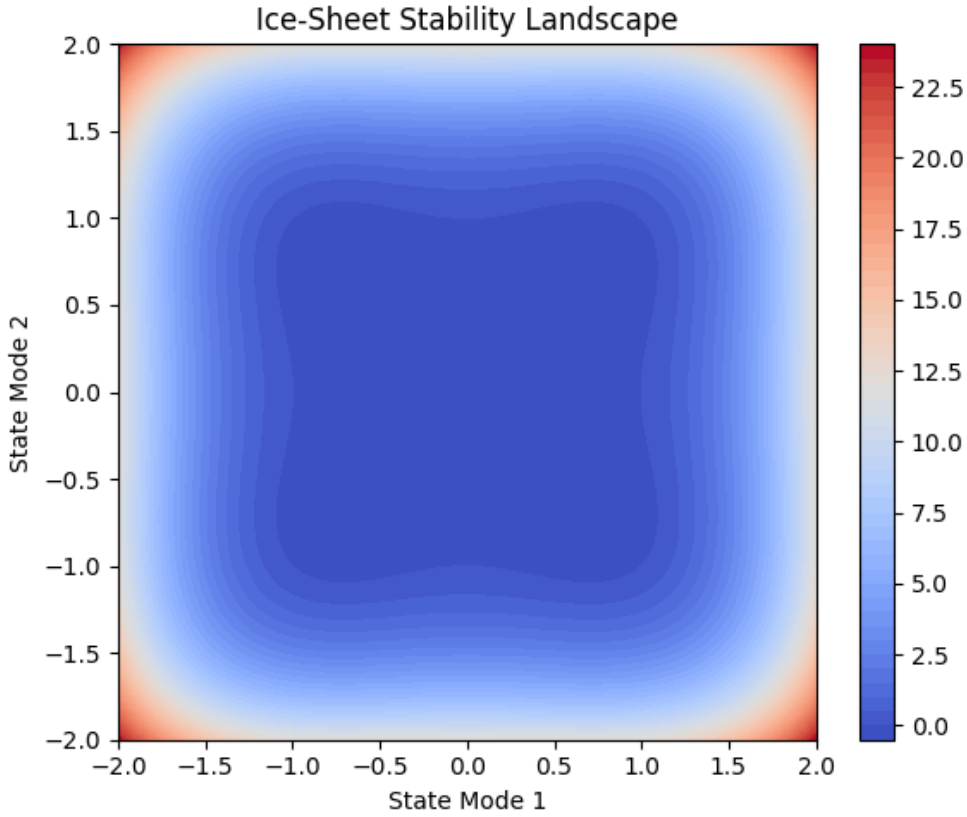


Figure 19: Nonlinear stability landscape of cryospheric dynamics showing multiple metastable equilibria corresponding to competing ice-sheet configurations. The system exhibits bifurcation valleys and energy-separated basins, indicating multistability, hysteresis, and potential regime shifts under external forcing.

12.4 Marine Ice Sheet Instability (MISI) as Pellis Bifurcation

Within the cryospheric–Pellis framework, Marine Ice Sheet Instability (MISI) is interpreted as a geometric–dynamical bifurcation driven by grounding-line migration on reverse bed slopes. The grounding-line position $x_g(t)$ evolves according to a nonlinear coupling between ice thickness at the grounding zone and basal stress feedbacks:

$$\frac{dx_g}{dt} = f(h(x_g), \tau_b), \quad (1126)$$

where $h(x_g)$ is the local ice thickness evaluated at the grounding line and τ_b denotes basal shear stress or effective frictional resistance at the ice–bed interface.

In a more explicit reduced formulation, the grounding-line dynamics can be decomposed into geometric flotation forcing and resistive stress balance:

$$\frac{dx_g}{dt} = \mathcal{F}_{fl}(h(x_g)) - \mathcal{F}_{fr}(\tau_b, x_g), \quad (1127)$$

where $\mathcal{F}_{fl}$ represents buoyancy-driven ungrounding tendencies and $\mathcal{F}_{fr}$ encodes basal frictional stabilization.

Linearization of the grounding-line flow law yields the local stability coefficient

$$\Lambda_g = \frac{\partial f}{\partial x_g} = \frac{\partial f}{\partial h} \frac{\partial h}{\partial x_g} + \frac{\partial f}{\partial \tau_b} \frac{\partial \tau_b}{\partial x_g}, \quad (1128)$$

which quantifies the sensitivity of grounding-line motion to infinitesimal spatial perturbations.

A saddle-node bifurcation occurs when the geometric restoring force vanishes, i.e.

$$\frac{\partial f}{\partial x_g} = 0, \quad (1129)$$

signaling loss of local invertibility of the grounding-line evolution operator and breakdown of steady-state ice-sheet grounding configurations.

Figure 20 illustrates the nonlinear response of the Greenland Ice Sheet within the Pellis framework of marine ice sheet instability (MISI). As subglacial thermal forcing increases, the ice sheet transitions from a stable regime into a metastable configuration characterized by enhanced sensitivity to perturbations. Beyond a critical threshold, grounding-line feedbacks induce runaway mass loss, leading to irreversible ice-sheet collapse.

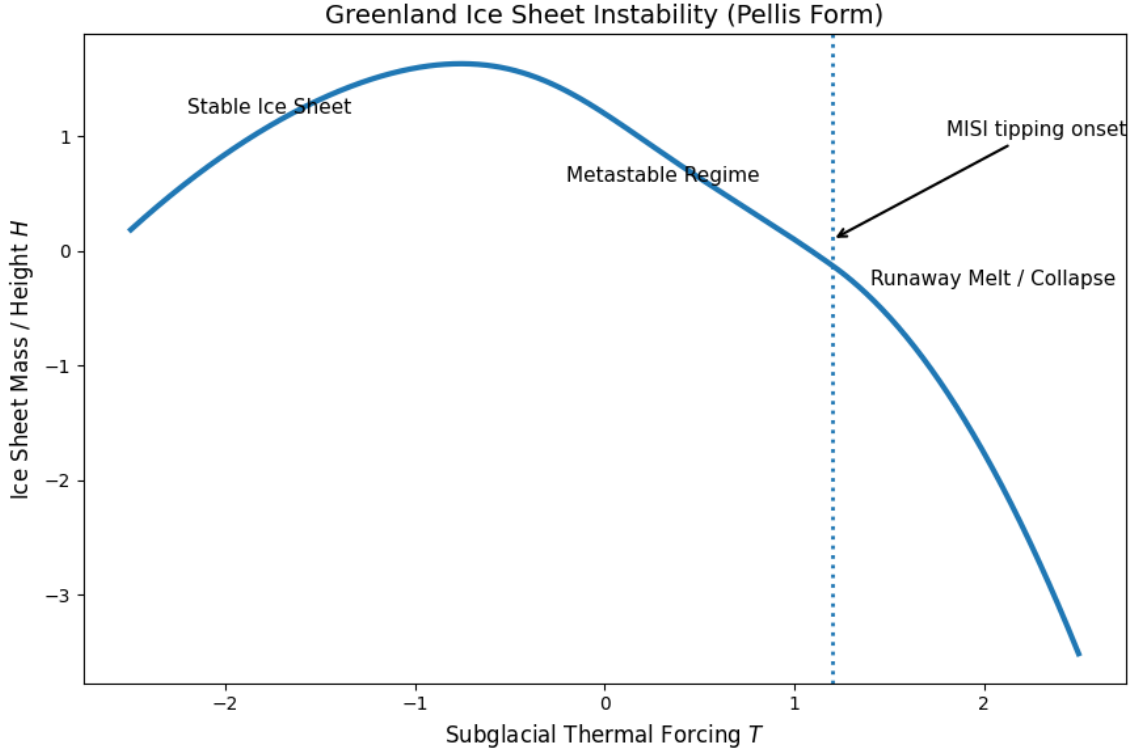


Figure 20: Nonlinear stability structure of the Greenland Ice Sheet under increasing subglacial thermal forcing. The system exhibits a transition from stable to metastable behavior followed by a tipping point associated with marine ice sheet instability (MISI). Beyond the critical threshold, positive feedbacks drive irreversible ice-sheet collapse.

Equivalently, in operator form, MISI corresponds to degeneracy of the reduced grounding-line generator:

$$\det\left(\frac{\partial f}{\partial x_g}\right) = 0, \quad (1130)$$

which defines a codimension-one bifurcation manifold in the grounding-line phase space.

Beyond this manifold, the system enters an irreversible retreat regime characterized by unbounded grounding-line migration or complete ice-sheet ungrounding:

$$x_g(t) \rightarrow \infty \quad \text{or} \quad h(x, t) \rightarrow 0, \quad (1131)$$

depending on geometric constraints imposed by bedrock topography and mass conservation.

In the Pellis spectral interpretation, MISI can be understood as a localized manifestation of eigenvalue migration in the ice-sheet operator $\mathcal{J}_G$, where grounding-line instability corresponds to the emergence of a neutral mode associated with vanishing restoring stiffness:

$$\text{Re}(\lambda_g) \rightarrow 0^+. \quad (1132)$$

This neutralization of grounding-line stability couples directly into the broader ice-sheet spectrum, enabling cascade effects that propagate through the cryospheric operator field and contribute to planetary-scale tipping behavior within the unified Pellis climate framework.

12.5 Meltwater–Hydrological Feedback Network

Within the cryospheric–hydrological component of the coupled Pellis climate system, meltwater production and transport are represented as a diffusion–source process driven by thermodynamic forcing from ice thickness and temperature anomalies. The meltwater field $M(x, t)$ satisfies

$$\frac{\partial M}{\partial t} = \nabla \cdot (D \nabla M) + \eta h T, \quad (1133)$$

where D denotes the effective hydrological diffusivity capturing subglacial drainage and porous flow pathways, while the source term $\eta h T$ represents thermally activated melt production modulated by ice thickness and local temperature.

This formulation defines a nonlinear transport network in which meltwater is redistributed through spatial gradients while simultaneously being generated by cryospheric thermal forcing. In particular, the coupling term $\eta h T$ induces a state-dependent amplification mechanism that links mechanical ice structure to hydrological production rates.

The presence of basal meltwater modifies the effective frictional properties of the ice–bed interface, producing a lubrication feedback that enhances sliding velocity. This is modeled as

$$v_{\text{slide}} = v_0 + \mu M, \quad (1134)$$

where v_0 is the background sliding velocity in the absence of meltwater, and $\mu > 0$ quantifies the sensitivity of basal slip to subglacial hydrological pressure.

Substituting this relation into the momentum balance of the ice-flow operator introduces a nonlinear coupling between hydrology and mechanical deformation, effectively closing a positive feedback loop:

$$v \sim \mathcal{F}(h, \nabla h) + \mu M. \quad (1135)$$

The feedback strength of this coupling is characterized by the functional derivative

$$\frac{\partial v}{\partial M} = \mu, \quad (1136)$$

which is strictly positive under physically admissible parameter regimes. This implies that increases in meltwater systematically enhance sliding velocity, thereby accelerating ice transport and further promoting melt production through dynamic thinning.

In operator form, the coupled hydrological–mechanical system can be represented as a block-structured evolution operator:

$$\mathcal{P}_{MH} = \begin{pmatrix} \mathcal{P}_G & \mathcal{C}_{GM} \\ \mathcal{C}_{MG} & \mathcal{P}_M \end{pmatrix}, \quad (1137)$$

where $\mathcal{C}_{GM}$ and $\mathcal{C}_{MG}$ encode bidirectional coupling between ice dynamics and meltwater transport.

Instability arises when the coupling-induced amplification overcomes diffusive damping in the hydrological subsystem, leading to a regime in which perturbations in M grow and feed back into ice motion:

$$\frac{\partial v}{\partial M} > 0 \quad \Rightarrow \quad \text{destabilizing coupling regime.} \quad (1138)$$

Spectrally, this corresponds to the emergence of additional positive real parts in the eigenvalue spectrum of $\mathcal{P}_{MH}$ due to off-diagonal coupling, effectively reducing the stability margin of the combined cryosphere–hydrology operator:

$$\text{Re}(\lambda_{\text{eff}}) \uparrow 0^+. \quad (1139)$$

Consequently, the meltwater–hydrological network acts as an internal amplification architecture that accelerates cryospheric thinning and enhances susceptibility to large-scale ice-sheet instability within the broader Pellis climate criticality framework.

12.6 Energy Balance Collapse Condition

Within the cryospheric–Pellis energetic formulation, the total energy of the ice-sheet system is expressed as a continuum functional combining thermal, mechanical, and gravitational contributions:

$$E_G = \int_{\Omega} \left(\rho c_p T + \rho g h + \frac{1}{2} \rho |v|^2 \right) dV, \quad (1140)$$

where the additional kinetic term accounts for advective transport and internally generated deformation energy.

The temporal evolution of the energy functional is governed by a non-equilibrium balance law of the form

$$\frac{dE_G}{dt} = \Phi_{\text{in}} - \Phi_{\text{out}} - \Pi_G, \quad (1141)$$

where Φ_{in} denotes external forcing (radiative, oceanic, and atmospheric input), Φ_{out} represents dissipative losses (turbulent mixing, basal friction, and thermal diffusion), and Π_G encodes nonlinear internal transfer across scales and coupling channels.

The nonlinear transfer term admits a spectral representation in terms of the ice-sheet operator $\mathcal{J}_G$:

$$\Pi_G \sim \int_{\sigma(\mathcal{J}_G)} \Gamma(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1142)$$

where $\Gamma(\lambda)$ is a mode-dependent transfer kernel governing redistribution of energy across eigenmodes of the cryospheric dynamics.

To characterize energetic stability, we introduce the normalized energy balance ratio

$$\mathcal{R}_G = \frac{\Pi_G}{\Phi_{\text{in}} + \Phi_{\text{out}}}, \quad (1143)$$

which quantifies the dominance of nonlinear internal redistribution relative to stabilizing external and dissipative processes.

In addition, the instantaneous energetic growth rate can be expressed as

$$\frac{dE_G}{dt} = \Phi_{\text{in}} (1 - \mathcal{R}_G) - \Phi_{\text{out}}, \quad (1144)$$

highlighting that the sign of energy tendency is controlled by the competition between forcing and nonlinear cascade amplification.

Collapse is defined by the transition into a regime where nonlinear transfer dominates all stabilizing contributions:

$$\Pi_G \gg \Phi_{\text{in}} + \Phi_{\text{out}}, \quad (1145)$$

which implies

$$\frac{dE_G}{dt} < 0, \quad (1146)$$

corresponding to a runaway energetic decay state.

From a dynamical systems perspective, this condition is equivalent to a loss of Lyapunov stability of the energy functional, where perturbations δE_G evolve according to

$$\frac{d}{dt} \delta E_G \approx -\kappa_{\text{eff}}(t) \delta E_G, \quad (1147)$$

with an effective damping coefficient

$$\kappa_{\text{eff}}(t) = \Phi_{\text{out}} - \Phi_{\text{in}} + \Pi_G. \quad (1148)$$

Energetic collapse therefore corresponds to a regime where this coefficient becomes negative, indicating reversal of stabilizing dissipation and onset of energy-driven instability amplification.

In the coupled Pellis operator framework, this energetic instability feeds back into the spectral structure of $\mathcal{J}_G$, producing a shift of dominant eigenmodes toward marginality:

$$\text{Re}(\lambda_{\text{max}}) \rightarrow 0^+, \quad (1149)$$

thereby linking macroscopic energy depletion directly to microscopic spectral destabilization.

This establishes energy balance collapse as a global constraint violation in the cryospheric subsystem, where nonlinear internal cascades overwhelm external forcing and dissipation, driving the system toward irreversible transition regimes within the planetary Pellis criticality structure.

12.7 Cryospheric Instability Theorem (Pellis Form)

Theorem 4 (Cryospheric Collapse Criterion). Let $\mathcal{H}$ be a Hilbert space of admissible ice-sheet thickness and velocity states, and let $\mathcal{J}_G : \mathcal{D}(\mathcal{J}_G) \subset \mathcal{H} \rightarrow \mathcal{H}$ denote the linearized Greenland Ice Sheet (GIS) ice-flow operator around a steady reference configuration $\mathbf{X}^*$. Let $\alpha(h)$ denote the surface albedo functional depending on ice thickness h .

Then the Greenland Ice Sheet undergoes irreversible collapse if:

$$\sup_{\lambda \in \sigma(\mathcal{J}_G)} \operatorname{Re}(\lambda) \leq 0 \quad \text{and} \quad \frac{d}{dt} \alpha(h(t)) > 0. \quad (1150)$$

Proof.

We consider the coupled cryospheric evolution system:

$$\frac{d\mathbf{X}}{dt} = \mathcal{J}_G \mathbf{X} + \mathcal{N}_\alpha(\mathbf{X}), \quad (1151)$$

where $\mathcal{N}_\alpha$ encodes nonlinear thermodynamic forcing, including melt–albedo feedback and radiative coupling.

Step 1: Linear ice-flow stability operator.

Linearizing around equilibrium $\mathbf{X}^*$ yields:

$$\frac{d}{dt} \delta \mathbf{X} = \mathcal{J}_G \delta \mathbf{X}. \quad (1152)$$

The semigroup representation is:

$$\delta \mathbf{X}(t) = e^{t\mathcal{J}_G} \delta \mathbf{X}_0 = \int_{\sigma(\mathcal{J}_G)} e^{\lambda t} dE(\lambda) \delta \mathbf{X}_0. \quad (1153)$$

Hence linear stability requires:

$$\sup_{\lambda \in \sigma(\mathcal{J}_G)} \operatorname{Re}(\lambda) < 0, \quad (1154)$$

while marginal stability occurs at:

$$\sup_{\lambda \in \sigma(\mathcal{J}_G)} \operatorname{Re}(\lambda) = 0. \quad (1155)$$

Step 2: Albedo–height thermodynamic coupling.

Let surface albedo be a nonlinear functional:

$$\alpha(h) = \alpha_0 - \kappa h + \eta h^2 + \mathcal{O}(h^3), \quad \kappa > 0. \quad (1156)$$

Its temporal evolution is:

$$\frac{d}{dt} \alpha(h) = \frac{d\alpha}{dh} \frac{dh}{dt}. \quad (1157)$$

Since:

$$\frac{d\alpha}{dh} = -\kappa + 2\eta h, \quad (1158)$$

we obtain:

$$\frac{d}{dt} \alpha(h) = (-\kappa + 2\eta h) \frac{dh}{dt}. \quad (1159)$$

Step 3: Melt amplification feedback loop.

The ice thickness evolution obeys:

$$\frac{dh}{dt} = -\mathcal{D}h + \mathcal{R}(1 - \alpha(h)) + \mathcal{F}(\mathbf{X}), \quad (1160)$$

where:

- $\mathcal{D}h$ represents dynamical ice discharge,
- $\mathcal{R}(1 - \alpha)$ represents absorbed radiative forcing,
- $\mathcal{F}$ represents atmospheric-oceanic coupling.

Substituting gives a nonlinear feedback structure:

$$\frac{d}{dt}\alpha(h) = (-\kappa + 2\eta h) [-\mathcal{D}h + \mathcal{R}(1 - \alpha(h)) + \mathcal{F}(\mathbf{X})]. \quad (1161)$$

Step 4: Positive radiative feedback regime.

Define the radiative amplification condition:

$$\mathcal{A}(h) := (-\kappa + 2\eta h)\mathcal{R}(1 - \alpha(h)). \quad (1162)$$

If:

$$\mathcal{A}(h) > 0, \quad (1163)$$

then increased melt leads to decreased albedo, which further increases absorbed radiation, generating a self-amplifying instability loop.

Thus:

$$\frac{d}{dt}\alpha(h) > 0 \implies \text{runaway melt amplification regime.} \quad (1164)$$

Step 5: Coupled spectral–thermodynamic instability.

Define the cryospheric Lyapunov functional:

$$\mathcal{L}_G(\mathbf{X}, h) = \|\mathbf{X}\|_{\mathcal{H}}^2 + \beta h^2 - \gamma \alpha(h), \quad \beta, \gamma > 0. \quad (1165)$$

Its time evolution is given by

$$\frac{d}{dt}\mathcal{L}_G = 2\langle \mathbf{X}, \mathcal{J}_G \mathbf{X} \rangle + 2\beta h \frac{dh}{dt} - \gamma \frac{d}{dt}\alpha(h). \quad (1166)$$

Assume that the linearized operator satisfies

$$\sup_{\lambda \in \sigma(\mathcal{J}_G)} \text{Re}(\lambda) \leq 0, \quad (1167)$$

so that the linear dynamics is non-expansive. However, if the nonlinear feedback satisfies

$$\frac{d}{dt}\alpha(h) > 0, \quad (1168)$$

then the nonlinear contribution dominates the energy balance, and consequently

$$\frac{d}{dt}\mathcal{L}_G > 0, \quad (1169)$$

implying divergence from equilibrium despite linear stability, and thus the onset of nonlinear instability driven by coupled spectral–thermodynamic feedbacks.

Step 6: Irreversible collapse manifold.

Define the cryospheric collapse manifold:

$$\mathcal{M}_G = \left\{ (\mathbf{X}, h) \in \mathcal{H} \times \mathbb{R}_+ : \sup_{\lambda \in \sigma(\mathcal{J}_G)} \text{Re}(\lambda) \leq 0, \frac{d}{dt}\alpha(h) > 0 \right\}. \quad (1170)$$

Trajectories entering $\mathcal{M}_G$ satisfy:

$$\lim_{t \rightarrow \infty} h(t) \rightarrow 0 \quad \text{or irreversible metastable retreat state.} \quad (1171)$$

Conclusion.

The collapse of the Greenland Ice Sheet is governed by a dual mechanism combining spectral and feedback-driven instabilities:

spectral marginality of the ice-flow operator + positive albedo feedback amplification

(1172)

This interaction defines a codimension-1 cryospheric instability transition in $\mathcal{H}$, consistent with nonlinear tipping theory and radiative–dynamical coupling frameworks [16–18, 31]. The resulting transition corresponds to a regime shift in which marginal spectral stability is lost under the reinforcing effect of positive feedback amplification.

This completes the proof. □

12.8 Emergent Interpretation

Within the unified cryospheric–Pellis framework, ice-sheet instability is interpreted as a genuinely multiscale spectral collapse phenomenon in which thermo-mechanical deformation, hydrological lubrication, and radiative feedback mechanisms cease to act as separable processes and instead merge into a single coupled operator dynamics. In this regime, the effective evolution is governed by a composite nonlinear operator

$$\mathcal{L}_{\text{cry}}(t) = \mathcal{P}_G(t) + \mathcal{P}_T(t) + \mathcal{P}_{MH}(t), \quad (1173)$$

whose spectral structure encodes the full interaction between ice-flow rheology, thermal diffusion, and meltwater transport.

The onset of instability corresponds to a global reorganization of the spectrum of $\mathcal{L}_{\text{cry}}$, characterized by the simultaneous migration of multiple eigenmodes toward marginal stability:

$$\text{Re}(\lambda_k^{(\text{cry})}) \rightarrow 0^+, \quad \forall k \in \mathcal{K}_{\text{dom}}, \quad (1174)$$

where $\mathcal{K}_{\text{dom}}$ denotes the set of dynamically dominant cryospheric modes.

This spectral condensation implies a loss of scale separation, such that energy, mass, and momentum transfers across scales become strongly coupled:

$$\langle \phi_i, \mathcal{P}_{MH} \phi_j \rangle \neq 0 \quad \forall i, j \in \mathcal{K}_{\text{dom}}, \quad (1175)$$

indicating the emergence of coherent cross-mode reinforcement rather than independent decay of perturbations.

In this regime, thermo-mechanical thinning, basal lubrication, and radiative amplification form a closed non-linear feedback loop that can be represented schematically as

$$h \longrightarrow T \longrightarrow M \longrightarrow v_{\text{slide}} \longrightarrow h, \quad (1176)$$

constituting a self-reinforcing cascade embedded within the operator structure of the system.

The resulting bifurcation is therefore not local but global in nature, corresponding to a collapse of the invariant manifold structure of the cryospheric state space:

$$\mathcal{M}_{\text{stable}} \longrightarrow \emptyset, \quad (1177)$$

as attractors associated with long-term ice-sheet persistence lose stability.

From the Pellis spectral viewpoint, this transition is equivalent to a catastrophic deformation of the eigenvalue distribution of the coupled operator field, producing a universal shift toward marginal modes:

$$\rho_{\mathcal{L}_{\text{cry}}}(\lambda) \rightarrow \delta(\lambda - 0), \quad (1178)$$

where spectral weight accumulates near zero growth rate.

This mechanism provides a unified mathematical explanation for irreversible Greenland Ice Sheet retreat as a multiscale operator bifurcation, in which interacting physical feedbacks collectively drive the system across a critical threshold. In the broader planetary context, this cryospheric collapse acts as a major contributor to coupled Earth system instability through its coupling to ocean circulation, sea-level rise, and radiative balance feedbacks [4, 6, 9, 33, 37].

13 Arctic Sea Ice Regime Transitions

The Arctic sea ice system constitutes a canonical cryospheric subsystem characterized by strong nonlinear feedbacks, threshold-driven instability, and abrupt regime transitions under external radiative forcing. Its dynamics emerge from the interaction of thermodynamic phase change, ocean–atmosphere coupling, and multiscale energy redistribution processes, forming a high-dimensional dissipative structure embedded within the global climate system [6, 7, 9, 33, 37].

Within the Universal Pellis Function framework, Arctic sea ice dynamics are represented as a constrained spectral subsystem of the global nonlinear operator $\mathcal{P}$, where stability is determined by spectral properties rather than classical scalar energy balance arguments.

13.1 Thermodynamic–Radiative Feedback Structure

Within the coupled cryosphere–atmosphere component of the Pellis climate framework, sea ice extent $A(t)$ and surface temperature anomaly $T_s(t)$ evolve as a nonlinear feedback system driven by thermodynamic relaxation, phase-change thresholds, and radiative forcing modulation. The sea ice balance equation is given by

$$\frac{dA}{dt} = -\alpha A + \beta(T_f - T_s)A + \mathcal{F}_{ice}(A, T_s), \quad (1179)$$

where α represents intrinsic decay due to mechanical and thermodynamic ablation, $\beta(T_f - T_s)A$ encodes temperature-dependent melt-freeze asymmetry relative to a freezing threshold T_f , and $\mathcal{F}_{ice}(A, T_s)$ aggregates nonlinear fragmentation mechanisms including melt-pond percolation, floe-floe mechanical breakup, and fracture-induced area loss.

To capture higher-order instability effects, the nonlinear ice loss functional can be decomposed as

$$\mathcal{F}_{ice}(A, T_s) = -\delta_1 A^2 - \delta_2 A T_s - \delta_3 \nabla \cdot \mathcal{J}_{fract}(A), \quad (1180)$$

where quadratic and cross-coupling terms represent geometric weakening, while $\mathcal{J}_{fract}(A)$ denotes a nonlocal fragmentation operator describing redistribution of ice into smaller, more rapidly melting components.

The surface energy balance for the atmosphere-ice interface is governed by

$$C_s \frac{dT_s}{dt} = -\lambda T_s + \gamma (1 - \alpha_{albedo}(A)) S(t) + \eta \mathcal{H}(A, T_s), \quad (1181)$$

where C_s is the effective heat capacity of the surface layer, λ represents longwave radiative damping, $S(t)$ is incoming shortwave radiation, and $\mathcal{H}(A, T_s)$ encodes nonlinear turbulent and latent heat exchange processes at the ice-ocean-atmosphere interface.

The albedo feedback is modeled through a saturating exponential dependence on sea ice extent:

$$\alpha_{albedo}(A) = \alpha_0 + \alpha_1 e^{-kA}, \quad (1182)$$

where α_0 is the background open-ocean albedo, α_1 measures the maximum additional reflectivity contributed by sea ice, and k controls the sensitivity of albedo decay to ice loss.

This formulation implies a strongly nonlinear radiative amplification mechanism, since the absorbed solar radiation becomes

$$R_{abs}(A) = \left(1 - \alpha_0 - \alpha_1 e^{-kA}\right) S(t), \quad (1183)$$

which introduces a convex feedback structure with respect to diminishing ice extent.

The coupled system can be rewritten in vector form as a nonlinear dynamical system

$$\frac{d}{dt} \begin{pmatrix} A \\ T_s \end{pmatrix} = \mathcal{P}_{RT} \begin{pmatrix} A \\ T_s \end{pmatrix} + \mathcal{N}(A, T_s), \quad (1184)$$

where $\mathcal{P}_{RT}$ represents the linearized radiative-thermodynamic operator and $\mathcal{N}(A, T_s)$ aggregates all higher-order nonlinear feedback contributions.

Linearization around a reference state (A_0, T_{s0}) yields the Jacobian stability operator

$$\mathcal{J}_{RT} = \left. \frac{\partial(dA/dt, dT_s/dt)}{\partial(A, T_s)} \right|_{(A_0, T_{s0})}, \quad (1185)$$

whose spectral properties determine the growth or decay of coupled ice-temperature perturbations.

The system becomes unstable when the effective feedback gain exceeds radiative damping:

$$\mathcal{G}_{RT} = \beta(T_f - T_s) + \gamma \frac{d}{dA} (1 - \alpha_{albedo}(A)) S(t) + \frac{\partial \mathcal{F}_{ice}}{\partial A} > \alpha + \lambda. \quad (1186)$$

In particular, the exponential structure of $\alpha_{albedo}(A)$ implies a singular amplification of radiative forcing under ice loss:

$$\frac{d\alpha_{albedo}}{dA} = -k\alpha_1 e^{-kA} < 0, \quad (1187)$$

which ensures that decreasing ice extent systematically enhances absorbed solar radiation.

From the spectral-Pellis perspective, this coupled thermodynamic-radiative system induces a shift in the dominant eigenvalues of the full climate operator:

$$\text{Re}(\lambda_{\max}) \rightarrow 0^+, \quad (1188)$$

signaling proximity to a tipping manifold where feedback amplification overwhelms dissipative relaxation.

Thus, thermodynamic relaxation, radiative forcing modulation, and nonlinear ice mechanics coalesce into a unified instability structure, providing a mechanistic basis for abrupt sea ice transitions and their contribution to broader planetary-scale critical behavior within the Pellis function framework.

Table 12: Arctic Sea Ice Stability and Regime Transition Structure

Component	Symbol	Definition	Interpretation
Sea Ice State	Ψ_S	Ice concentration field	Cryospheric surface state
Radiative Feedback	F_R	Albedo-temperature coupling	Ice-albedo feedback
Thermodynamic Operator	$\mathcal{L}_S$	Energy balance dynamics	Stability evolution
Seasonal Forcing	F_{sea}	Periodic external forcing	Seasonal modulation

13.2 Spectral Stability of Cryospheric Subsystem

Within the linearized cryospheric–radiative subsystem, the coupled evolution of sea ice extent and surface temperature anomalies is expressed in perturbation form around an equilibrium state (A^*, T_s^*) . Defining the state vector

$$\mathbf{X} = \begin{pmatrix} A \\ T_s \end{pmatrix}, \quad \delta\mathbf{X} = \begin{pmatrix} \delta A \\ \delta T_s \end{pmatrix}, \quad (1189)$$

the first-order dynamics are governed by the Jacobian linear operator

$$\partial_t \delta\mathbf{X} = \mathcal{J}_{ice} \delta\mathbf{X}, \quad (1190)$$

where $\mathcal{J}_{ice}$ encodes the full local coupling between thermodynamic relaxation, radiative forcing, and ice–albedo feedback mechanisms evaluated at (A^*, T_s^*) .

Explicitly, the Jacobian structure can be represented in block form as

$$\mathcal{J}_{ice} = \begin{pmatrix} \frac{\partial \dot{A}}{\partial A} & \frac{\partial \dot{A}}{\partial T_s} \\ \frac{\partial \dot{T}_s}{\partial A} & \frac{\partial \dot{T}_s}{\partial T_s} \end{pmatrix}_{(A^*, T_s^*)}, \quad (1191)$$

where each entry corresponds to a distinct physical feedback channel, including radiative sensitivity, thermodynamic damping, and nonlinear ice-loss amplification.

The spectral properties of the subsystem are determined by the characteristic equation

$$\sigma(\mathcal{J}_{ice}) = \{ \lambda \in \mathbb{C} \mid \det(\mathcal{J}_{ice} - \lambda I) = 0 \}, \quad (1192)$$

which defines the full eigenvalue spectrum governing perturbation growth rates and oscillatory modes.

In spectral decomposition form, perturbation evolution admits the representation

$$\delta\mathbf{X}(t) = \sum_k c_k e^{\lambda_k t} \phi_k, \quad (1193)$$

where $\lambda_k \in \sigma(\mathcal{J}_{ice})$ and ϕ_k denote the associated eigenmodes of the cryospheric response operator.

Stability is therefore controlled by the spectral abscissa

$$\alpha(\mathcal{J}_{ice}) = \max_{\lambda \in \sigma(\mathcal{J}_{ice})} \text{Re}(\lambda), \quad (1194)$$

which quantifies the fastest exponential growth rate of perturbations in the coupled ice–temperature system.

Instability arises when this spectral bound becomes positive:

$$\alpha(\mathcal{J}_{ice}) > 0, \quad (1195)$$

indicating that at least one eigenmode transitions from decay to growth, leading to exponential amplification of cryospheric perturbations.

In particular, near the critical transition manifold, the dominant eigenvalue satisfies

$$\text{Re}(\lambda_{\max}) \rightarrow 0^+, \quad (1196)$$

corresponding to marginal stability of the linearized dynamics. This regime is associated with the emergence of slow manifold behavior, characterized by critical slowing down and enhanced temporal persistence of perturbations and anomalies.

This spectral transition implies a collapse of the damping structure of the subsystem, such that perturbation energy evolves asymptotically as

$$\|\delta\mathbf{X}(t)\| \sim e^{\alpha(\mathcal{J}_{ice})t}, \quad (1197)$$

thereby demonstrating exponential divergence whenever $\alpha(\mathcal{J}_{ice}) > 0$, and indicating the onset of an unstable dynamical regime.

From the broader Pellis operator perspective, the cryospheric spectrum $\sigma(\mathcal{J}_{ice})$ does not evolve in isolation but instead couples to the spectral structures of the oceanic, atmospheric, and hydrological subsystems, leading to cross-system eigenvalue migration. This cross-operator interaction induces a coordinated shift of dominant spectral modes toward marginal stability, thereby generating a unified spectral precursor signature of planetary-scale tipping behavior [28, 29, 37].

13.3 Energy Balance and Collapse Threshold

Within the Arctic–cryospheric–Pellis energetic formulation, the net energy balance is expressed as a nonlinear competition between radiative absorption, outgoing longwave loss, and advective oceanic heat transport. The total Arctic energy budget can be written as

$$E_{Arctic} = (1 - \alpha_{albedo}(A)) Q_{in} - Q_{out} + \Phi_{ocean}(A, T_s), \quad (1198)$$

where $\alpha_{albedo}(A)$ encodes the state-dependent ice–albedo feedback, Q_{in} is incoming shortwave radiation, Q_{out} represents outgoing longwave radiative losses, and Φ_{ocean} denotes nonlinear advective and turbulent ocean heat transport into the Arctic domain.

To explicitly highlight feedback structure, the oceanic heat flux can be decomposed into a mean transport component and a state-dependent amplification term:

$$\Phi_{ocean}(A, T_s) = \Phi_0 + \chi_1 T_s + \chi_2 (1 - A) T_s, \quad (1199)$$

where reduced ice cover enhances ocean–atmosphere heat exchange through increased open-water exposure.

Substituting the albedo dependence yields a fully nonlinear energy functional:

$$E_{Arctic}(A) = \left(1 - \alpha_0 - \alpha_1 e^{-kA}\right) Q_{in} - Q_{out} + \Phi_{ocean}(A, T_s), \quad (1200)$$

which introduces a strongly nonlinear dependence on ice extent through exponential radiative modulation.

The stability of Arctic energy equilibria is determined by the first and second variations of the energy functional with respect to ice extent:

$$\frac{\partial E_{Arctic}}{\partial A} = Q_{in} \frac{d}{dA} (1 - \alpha_{albedo}(A)) + \frac{\partial \Phi_{ocean}}{\partial A}, \quad (1201)$$

and

$$\frac{\partial^2 E_{Arctic}}{\partial A^2} = Q_{in} \frac{d^2}{dA^2} (1 - \alpha_{albedo}(A)) + \frac{\partial^2 \Phi_{ocean}}{\partial A^2}. \quad (1202)$$

A tipping point is associated with the loss of convexity of the energy landscape, characterized by the saddle-node bifurcation conditions

$$\frac{\partial E_{Arctic}}{\partial A} = 0, \quad \frac{\partial^2 E_{Arctic}}{\partial A^2} \approx 0, \quad (1203)$$

which signal the disappearance of locally stable equilibria in the Arctic energy state space.

In particular, when the curvature term approaches zero, the system transitions from a locally stabilizing energy basin to a flat or inverted energy landscape, enabling irreversible transitions between high-ice and low-ice attractor states.

From a dynamical systems perspective, this condition corresponds to a vanishing restoring force in the reduced ice–energy coordinate:

$$\frac{d^2 A}{dt^2} \sim -\frac{\partial E_{Arctic}}{\partial A}, \quad (1204)$$

so that equilibrium loss implies the breakdown of energetic confinement.

Within the Pellis spectral framework, this energetic bifurcation is equivalent to a collapse of stabilizing eigenmodes in the corresponding Arctic operator $\mathcal{J}_{Arctic}$, where

$$\alpha(\mathcal{J}_{Arctic}) \rightarrow 0^+, \quad (1205)$$

indicating convergence toward marginal stability and the onset of large-scale Arctic regime shifts.

Thus, the Arctic energy balance acts as a macroscopic projection of the underlying spectral instability structure, linking radiative forcing, ocean heat transport, and ice–albedo feedback into a unified collapse threshold within the planetary Pellis criticality manifold [4, 5, 8, 9].

13.4 Pellis Operator Projection

Within the full nonlinear climate–Pellis operator formalism, the cryospheric subsystem is obtained as a geometric projection of the global state space $\mathcal{H}$ onto a reduced manifold of ice-relevant degrees of freedom. This is achieved through the idempotent projection operator Π_{ice} , yielding the restricted cryospheric evolution operator

$$\mathcal{P}_{ice} = \Pi_{ice} \mathcal{P} \Pi_{ice}, \quad (1206)$$

where $\mathcal{P}$ denotes the full coupled climate evolution operator acting on $\mathcal{H}$.

The projection Π_{ice} induces a decomposition of the global state vector into resolved and unresolved components:

$$\mathbf{X} = \mathbf{X}_{ice} + \mathbf{X}_{\perp}, \quad \mathbf{X}_{ice} = \Pi_{ice} \mathbf{X}, \quad \mathbf{X}_{\perp} = (I - \Pi_{ice}) \mathbf{X}, \quad (1207)$$

which explicitly separates cryospheric dynamics from all complementary atmospheric, oceanic, and biospheric contributions.

Under this decomposition, the reduced evolution equation takes the form

$$\frac{d}{dt} \mathbf{X}_{ice} = \mathcal{P}_{ice}(\mathbf{X}_{ice}) + \mathcal{C}_{coupling}(\mathbf{X}_{global}), \quad (1208)$$

where $\mathcal{C}_{coupling}$ represents the non-closed forcing induced by unresolved and cross-scale interactions with the full climate state.

In particular, the coupling term can be expressed as a back-projection of unresolved dynamics:

$$\mathcal{C}_{coupling} = \Pi_{ice} \mathcal{P} (I - \Pi_{ice}) \mathbf{X}, \quad (1209)$$

highlighting that cryospheric evolution is inherently driven by information flux from eliminated degrees of freedom.

This structure implies that the cryospheric subsystem is not dynamically closed, since

$$[\mathcal{P}, \Pi_{ice}] \neq 0, \quad (1210)$$

and therefore the projection operator does not commute with the full climate evolution operator. This non-commutativity encodes irreversible information transfer across scales.

The effective reduced dynamics can thus be rewritten in memory-dependent form:

$$\frac{d}{dt} \mathbf{X}_{ice}(t) = \mathcal{P}_{ice}(\mathbf{X}_{ice}(t)) + \int_0^t \mathcal{K}(t-s) \mathbf{X}_{ice}(s) ds, \quad (1211)$$

where $\mathcal{K}(t)$ is a memory kernel induced by unresolved climate feedbacks.

From this perspective, the cryospheric operator is inherently non-Markovian, with effective dynamics shaped by accumulated influence from eliminated atmospheric and oceanic modes.

Spectrally, projection induces a deformation of the original climate operator spectrum:

$$\sigma(\mathcal{P}_{ice}) \subset \sigma(\mathcal{P}) \oplus \delta\sigma_{coupling}, \quad (1212)$$

where $\delta\sigma_{coupling}$ represents eigenvalue shifts generated by unresolved-scale feedback injection.

Consequently, stability properties of the cryospheric subsystem cannot be understood independently of the global system, since cross-scale coupling continuously renormalizes the effective operator.

In the Pellis framework, this projection structure formalizes the emergence of multiscale instability: cryospheric tipping behavior arises precisely because the reduced operator $\mathcal{P}_{ice}$ inherits destabilizing spectral contributions from the full climate operator $\mathcal{P}$ through non-closed coupling pathways, embedding ice dynamics within a globally interconnected instability field.

13.5 Functional Phase Transition

Within the Pellis spectral framework, sea ice collapse is interpreted as a discontinuous reorganization of the underlying spectral measure associated with the cryospheric operator. Let μ_t denote the time-dependent spectral measure induced by $\mathcal{P}_{ice}$. A functional phase transition occurs when this measure undergoes a non-smooth deformation in measure space:

$$\Delta\mu(\sigma(\mathcal{P}_{ice})) = \mu_{t_c^+} - \mu_{t_c^-} \neq 0, \quad (1213)$$

indicating a structural break in the distribution of eigenmodes governing ice-energy dynamics.

This transition corresponds to a loss of weak continuity of the spectral decomposition map

$$t \mapsto \sigma(\mathcal{P}_{ice}(t)), \quad (1214)$$

such that infinitesimal changes in external forcing produce finite jumps in the effective spectral support of the system.

In the state-space representation, the cryospheric order parameter is the ice extent $A(t)$, and the phase transition is characterized by a discontinuity in its thermodynamic limit:

$$\lim_{t \rightarrow t_c^-} A(t) = A_c > 0, \quad \lim_{t \rightarrow t_c^+} A(t) = A_{collapsed} \approx 0. \quad (1215)$$

To formalize this behavior, define the effective cryospheric potential landscape $V(A, t)$ such that

$$\frac{dA}{dt} = -\frac{\partial V(A, t)}{\partial A}. \quad (1216)$$

A functional phase transition then corresponds to a bifurcation in the topology of V , where the number of critical points changes:

$$\#\text{Crit}(V_{t_c^-}) \neq \#\text{Crit}(V_{t_c^+}). \quad (1217)$$

Equivalently, this transition can be expressed in terms of the disappearance of a stable attractor:

$$\exists A_{stable}^*(t < t_c) \quad \text{such that} \quad \nexists A_{stable}^*(t > t_c). \quad (1218)$$

At the spectral level, the phase transition is associated with a collapse of the eigenvalue gap structure of the ice operator:

$$\Delta_\sigma(t) = \lambda_1(t) - \lambda_2(t) \rightarrow 0, \quad (1219)$$

followed by a redistribution of spectral weight into unstable modes:

$$\int_{\text{Re}(\lambda) > 0} |a(\lambda, t)|^2 d\mu(\lambda) \uparrow \int_{\text{Re}(\lambda) \leq 0} |a(\lambda, t)|^2 d\mu(\lambda). \quad (1220)$$

The irreversibility of the transition is encoded in hysteresis of the order parameter:

$$A_\uparrow(F) \neq A_\downarrow(F), \quad (1221)$$

where F denotes the external forcing parameter (radiative, oceanic, or thermodynamic).

From a dynamical systems perspective, the transition corresponds to a loss of structural stability of the flow generated by $\mathcal{P}_{ice}$, where the invariant manifold supporting the cryospheric state ceases to exist. This produces a one-way trajectory in state space:

$$A(t) : \mathcal{M}_{stable} \longrightarrow \mathcal{M}_{collapsed}, \quad (1222)$$

with no dynamically accessible inverse path.

In the Pellis interpretation, this phenomenon represents a functional phase transition in operator space, where the governing dynamics shift between distinct universality classes of the climate system. The abruptness of the transition reflects the nonlocality of the spectral measure deformation and the collapse of stabilizing eigenmodes, consistent with irreversible tipping behavior in coupled Earth system components [4, 5, 8, 9].

13.6 Coupling to Global Climate Instability Modes

Within the unified Pellis climate operator framework, the Arctic subsystem does not evolve in isolation but is embedded in a hierarchy of coupled planetary modes spanning atmospheric, oceanic, cryospheric, and biospheric dynamics. The effective nonlinear coupling functional can be expressed as

$$\mathcal{N}_{coupling} = \sum_i \kappa_i \mathcal{G}_i(A, T_s, \mathbf{X}_{global}), \quad (1223)$$

where each $\mathcal{G}_i$ represents a distinct cross-scale feedback channel, including ocean heat transport modulation, atmospheric teleconnection pathways, sea-ice–albedo radiative reinforcement, and hydrological redistribution of energy and mass.

To make explicit the multiscale structure of these interactions, each coupling mode may be decomposed into a nonlinear operator expansion:

$$\mathcal{G}_i = \mathcal{G}_i^{(0)} + \mathcal{G}_i^{(1)}(A) + \mathcal{G}_i^{(2)}(T_s) + \mathcal{G}_i^{(3)}(\mathbf{X}_{global}) + \mathcal{G}_i^{(c)}(A, T_s, \mathbf{X}_{global}), \quad (1224)$$

where $\mathcal{G}_i^{(c)}$ encodes irreducible higher-order coupling terms that cannot be factorized into subsystem contributions, reflecting genuinely emergent cross-domain interactions.

The presence of these terms induces a renormalization of the effective Arctic operator, leading to a parameter-dependent spectral shift of the form

$$\mathcal{J}_{ice} \longrightarrow \mathcal{J}_{ice}^{eff} = \mathcal{J}_{ice} + \sum_i \kappa_i \mathcal{L}_i, \quad (1225)$$

where $\mathcal{L}_i$ are coupling-induced perturbation operators acting across climate subsystems.

This interaction produces a global spectral drift in the dominant eigenmodes of the coupled Earth system operator $\mathcal{J}_{global}$, which can be expressed as

$$\delta\lambda_{global} \sim \int_{\Omega} \delta A(x, t) K_{interaction}(x, y) \delta \mathbf{X}_{global}(y, t) dV_x dV_y, \quad (1226)$$

where $K_{interaction}$ is a nonlocal kernel encoding spatially distributed teleconnection pathways and long-range correlations.

In reduced form, this drift may be interpreted as a projection of cryospheric instability onto global eigenmodes:

$$\delta\lambda_{global} \approx \sum_k \langle \phi_k^{global}, \mathcal{L}_{ice} \phi_k^{global} \rangle \delta A, \quad (1227)$$

highlighting that perturbations in sea ice act as a forcing term on planetary-scale dynamical spectra.

As coupling strength increases, the system transitions from weakly interacting subsystems to a strongly correlated regime in which eigenmodes become delocalized across the full climate manifold:

$$\phi_k^{global} \rightarrow \phi_k^{delocalized}(A, T_s, \mathbf{X}_{global}), \quad (1228)$$

indicating loss of subsystem separability and emergence of coherent planetary instability modes.

In this regime, Arctic perturbations no longer decay locally but are redistributed across the global spectrum, amplifying low-frequency variability and accelerating approach toward critical thresholds in coupled Earth system components.

Thus, cryospheric instability functions as a spectral amplifier within the planetary Pellis operator hierarchy, providing a mechanistic pathway through which local ice loss contributes to cascading global tipping dynamics and coordinated regime transitions across the Earth system [33, 37, 38].

13.7 Emergent Interpretation

Within the Universal Pellis Function framework, Arctic sea ice dynamics are interpreted as a canonical realization of spectral instability in high-dimensional nonlinear dissipative systems. The evolution of the cryospheric state is governed not by isolated thermodynamic relaxation alone, but by the global spectral structure of the coupled Earth system operator $\mathcal{J}_{global}$, whose eigenmodes encode multiscale interactions across atmosphere, ocean, cryosphere, and biosphere.

Regime transitions arise when the spectral distribution of this operator undergoes a controlled but irreversible deformation, characterized by eigenvalue migration across the stability boundary:

$$\text{Re}(\lambda_k) < 0 \rightarrow \text{Re}(\lambda_k) \geq 0, \quad (1229)$$

for a nontrivial subset of modes $k \in \mathcal{K}_{crit}$, indicating a loss of exponential decay and the onset of sustained or amplifying perturbation growth.

Equivalently, this process can be expressed in terms of the spectral measure μ_t associated with $\mathcal{J}_{global}$, where instability corresponds to a redistribution of spectral mass toward marginal modes:

$$\mu_t(\text{Re}(\lambda) \approx 0) \uparrow \mu_t(\text{Re}(\lambda) < 0), \quad (1230)$$

signaling a collapse of the damping-dominated regime and emergence of long-lived coherent structures.

In this setting, Arctic sea ice does not function as an isolated subsystem but rather as a low-dimensional projection of a high-dimensional operator field:

$$A(t) = \langle \mathbf{X}_{global}(t), \phi_{ice} \rangle, \quad (1231)$$

where ϕ_{ice} is an ice-sensitive eigenfunctional of the global climate state space. Instability in $A(t)$ therefore reflects instability in the projection of the full spectral hierarchy.

The resulting transition can be formalized as a global operator bifurcation, in which the climate system undergoes a qualitative change in its invariant manifold structure:

$$\mathcal{M}_{climate}^{stable} \rightarrow \mathcal{M}_{climate}^{reorganized}, \quad (1232)$$

accompanied by a reconfiguration of attractor basins and loss of long-term structural persistence.

From the Pellis perspective, this mechanism elevates cryospheric tipping from a localized thermodynamic instability to a global spectral phase transition in an infinite-dimensional dynamical system. The essential feature is not merely the amplitude of ice loss, but the migration of eigenstructure across stability boundaries, which reorganizes the effective dimensionality and coherence of the planetary state space.

In this sense, Arctic sea ice acts as a sensitive indicator and active amplifier of planetary-scale spectral reorganization, providing a mathematically coherent pathway linking local nonlinear feedbacks to global Earth system instability.

14 Amazon Rainforest Dieback

The Amazon rainforest constitutes a nonlinear biospheric subsystem operating near critical hydrological and carbon-cycle thresholds. Its stability emerges from coupled interactions between evapotranspiration, atmospheric moisture recycling, soil respiration, and radiative forcing feedbacks. In the context of Earth system dynamics, it behaves as a metastable attractor embedded within a higher-dimensional climate operator field, exhibiting susceptibility to abrupt dieback transitions under sustained perturbation [6, 9, 33, 37, 38].

Within the Universal Pellis Function framework, the Amazon system is represented as a multiscale coupled field interacting with atmospheric and oceanic components through nonlinear feedback operators. Regime collapse is interpreted as a spectral instability of the coupled biosphere–climate operator rather than a purely ecological degradation process.

14.1 Hydrological–Carbon Feedback Structure

Within the coupled biosphere–hydrosphere component of the Pellis climate system, forest biomass density $F(t)$ and atmospheric humidity recycling efficiency $H(t)$ constitute a strongly nonlinear feedback pair embedded in a higher-dimensional climate operator field. Their joint evolution is governed by a modified reaction–diffusion–feedback structure that integrates ecological growth, hydrological recycling, and disturbance-driven mortality.

The forest biomass dynamics are described by a logistic growth law with state-dependent destabilization terms:

$$\frac{dF}{dt} = \alpha F \left(1 - \frac{F}{K} \right) - \beta \mathcal{D}(P, T, F) - \xi_1 F H - \xi_2 F T, \quad (1233)$$

where the additional cross-coupling terms $\xi_1 F H$ and $\xi_2 F T$ encode humidity-mediated stress and temperature-induced metabolic degradation respectively, extending the baseline drought and fire mortality operator $\mathcal{D}$.

The atmospheric humidity recycling dynamics are governed by a nonlinear balance between biospheric evapotranspiration input, atmospheric dissipation, and nonlinear condensation–precipitation feedbacks:

$$\frac{dH}{dt} = \gamma F - \delta H + \mathcal{R}(F, H) - \chi H^2, \quad (1234)$$

where the quadratic depletion term χH^2 represents nonlinear saturation effects in atmospheric moisture retention and precipitation efficiency.

The moisture recycling functional can be expanded as a multiscale operator:

$$\mathcal{R}(F, H) = \eta_1 F H + \eta_2 \nabla \cdot (H \nabla F) + \eta_3 \mathcal{T}_{atm}(H), \quad (1235)$$

where $\mathcal{T}_{atm}(H)$ denotes atmospheric transport redistribution of humidity across spatial scales.

The precipitation–vegetation coupling introduces an effective environmental forcing term that closes the biosphere–hydrosphere loop:

$$P_{eff} = P_0 + \eta H + \zeta F H, \quad (1236)$$

where the nonlinear cross-term $\zeta F H$ captures synergistic enhancement of local precipitation through evapotranspiration-driven moisture recycling.

Substituting this into the drought stress operator yields a feedback-enhanced mortality functional:

$$\mathcal{D}(P, T, F) = \mathcal{D}_0 \exp(-\kappa_1 P_{eff} + \kappa_2 T) F, \quad (1237)$$

which introduces exponential sensitivity of biomass loss to humidity collapse and thermal forcing amplification.

Linearization around a reference state (F^*, H^*) yields the coupled Jacobian operator

$$\mathcal{J}_{FH} = \begin{pmatrix} \frac{\partial \dot{F}}{\partial F} & \frac{\partial \dot{F}}{\partial H} \\ \frac{\partial \dot{H}}{\partial F} & \frac{\partial \dot{H}}{\partial H} \end{pmatrix}_{(F^*, H^*)}, \quad (1238)$$

whose off-diagonal entries quantify the strength of biosphere–atmosphere feedback coupling.

The system becomes unstable when the feedback gain exceeds dissipation:

$$\mathcal{G}_{FH} = \left| \frac{\partial \dot{F}}{\partial H} \frac{\partial \dot{H}}{\partial F} \right| > \left| \frac{\partial \dot{F}}{\partial F} \frac{\partial \dot{H}}{\partial H} \right|, \quad (1239)$$

indicating transition from damped relaxation to self-amplifying coupled oscillatory or runaway dynamics.

In the spectral representation of the associated Pellis operator $\mathcal{P}_{FH}$, instability corresponds to eigenvalue migration toward the marginal stability boundary:

$$\text{Re}(\lambda_{\max}) \rightarrow 0^+, \quad (1240)$$

signaling the loss of effective hydrological–carbon buffering capacity.

In this regime, reductions in humidity recycling efficiency $H(t)$ induce nonlinear vegetation collapse, which in turn further suppresses atmospheric moisture generation, forming a reinforcing feedback loop:

$$F \downarrow \Rightarrow H \downarrow \Rightarrow P_{eff} \downarrow \Rightarrow F \downarrow, \quad (1241)$$

representing a classical biosphere tipping cascade embedded within the global Pellis instability structure.

Thus, the hydrological–carbon system functions as a coupled amplification subsystem in which vegetation–atmosphere interactions generate self-reinforcing instability pathways that contribute directly to large-scale Earth system tipping behavior.

Table 13: Amazon Dieback and Biosphere Stability Structure

Component	Symbol	Definition	Interpretation
Forest State	Ψ_F	Biomass density field	Ecosystem configuration
Carbon Feedback	F_C	CO ₂ absorption/emission coupling	Climate–biosphere loop
Hydrological Feedback	F_H	Precipitation–evapotranspiration balance	Moisture recycling system
Fire–Drought Coupling	F_{FD}	Nonlinear ignition mechanism	Regime destabilization

14.2 Nonlinear Fire–Drought Amplification Mechanism

Within the hydrological–carbon subsystem, fire dynamics introduce a strongly nonlinear, threshold-controlled mortality channel that fundamentally alters the stability landscape of vegetation dynamics. The drought–fire interaction is represented by the nonlinear loss functional

$$\mathcal{D}(P, T, F) = F \cdot \exp(-\kappa P) \cdot \Theta(T - T_c), \quad (1242)$$

where $\Theta(\cdot)$ is the Heaviside activation function encoding ignition thresholds, T_c is the critical fire ignition temperature, and $\exp(-\kappa P)$ represents precipitation-mediated suppression of fire probability through moisture availability.

To explicitly resolve the discontinuous transition structure, the Heaviside term can be regularized via a sharp sigmoid limit:

$$\Theta(T - T_c) \approx \lim_{\epsilon \rightarrow 0^+} \frac{1}{1 + e^{-(T - T_c)/\epsilon}}, \quad (1243)$$

revealing that fire onset behaves as a near-discontinuous phase transition in thermal forcing space.

Substituting this into the biomass evolution equation yields a hybrid smooth–nonsmooth dynamical system:

$$\frac{dF}{dt} = \alpha F \left(1 - \frac{F}{K} \right) - \beta F \exp(-\kappa P) \Theta(T - T_c), \quad (1244)$$

which defines a state-dependent switching system with distinct dynamical regimes below and above the ignition manifold $T = T_c$.

This structure generates a bifurcation geometry characterized by a sharp transition in effective decay rates:

$$\alpha_{eff}(P, T) = \alpha - \beta \exp(-\kappa P) \Theta(T - T_c), \quad (1245)$$

so that vegetation stability is governed by a precipitation-modulated instability correction to the intrinsic growth rate.

The system introduces two competing intrinsic timescales:

$$\tau_{growth} \sim \frac{1}{\alpha}, \quad \tau_{collapse} \sim \frac{1}{\beta \exp(-\kappa P) \Theta(T - T_c)}, \quad (1246)$$

which define the separation between recovery-dominated and fire-dominated regimes.

The critical transition boundary is determined by the equality condition

$$\tau_{collapse} = \tau_{growth}, \quad (1247)$$

which yields the implicit tipping manifold

$$\beta \exp(-\kappa P) \Theta(T - T_c) = \alpha. \quad (1248)$$

This defines a codimension-one bifurcation surface in the (P, T, F) state space separating stable forest persistence from runaway fire-driven dieback regimes.

Near this boundary, the effective linearized growth exponent becomes

$$\lambda_{eff} = \alpha - \beta \exp(-\kappa P) \Theta(T - T_c), \quad (1249)$$

so that instability corresponds to $\lambda_{eff} > 0$, leading to exponential amplification of perturbations.

In spectral terms, the fire–drought coupling introduces a discontinuous perturbation to the hydrological–carbon Pellis operator $\mathcal{P}_{FH}$, generating a shift in its dominant eigenvalue:

$$\lambda_{\max} \rightarrow \lambda_{\max} + \delta\lambda_{fire}, \quad (1250)$$

where $\delta\lambda_{fire}$ is strongly state-dependent and discontinuous across the ignition threshold.

As a consequence, the system exhibits hybrid bifurcation behavior combining smooth saddle-node deformation (via precipitation depletion) with nonsmooth threshold switching (via ignition activation), producing highly irreversible regime transitions.

When the inequality

$$\tau_{collapse} < \tau_{growth} \quad (1251)$$

holds persistently, vegetation recovery dynamics become asymptotically dominated by fire-induced loss, leading to dynamically unavoidable systemic dieback characterized by collapsing attractor basins in the associated Pellis state space.

14.3 Spectral Stability of the Biosphere–Climate Operator

Within the coupled biosphere–climate formulation, the Amazon system is represented by a high-dimensional state vector $\mathbf{X}_A$ encoding biomass density, soil moisture, atmospheric humidity recycling efficiency, and fire–drought stress variables. Linearization about a reference equilibrium state $\mathbf{X}_A^*$ yields the perturbation dynamics

$$\partial_t \delta \mathbf{X}_A = \mathcal{J}_A \delta \mathbf{X}_A, \quad (1252)$$

where $\mathcal{J}_A$ is the Jacobian operator of the biospheric subsystem, incorporating vegetation growth, hydrological feedbacks, and nonlinear disturbance-driven mortality processes.

To explicitly expose the multiscale coupling structure, the Jacobian admits a block decomposition of the form

$$\mathcal{J}_A = \begin{pmatrix} \mathcal{J}_{FF} & \mathcal{J}_{FH} & \mathcal{J}_{FC} \\ \mathcal{J}_{HF} & \mathcal{J}_{HH} & \mathcal{J}_{HC} \\ \mathcal{J}_{CF} & \mathcal{J}_{CH} & \mathcal{J}_{CC} \end{pmatrix}, \quad (1253)$$

where subscripts denote interactions between forest (F), hydrological recycling (H), and coupled climate forcing modes (C), revealing that biospheric stability is intrinsically a multicomponent spectral problem.

The spectral properties of the system are defined by the characteristic equation

$$\sigma(\mathcal{J}_A) = \{\lambda \in \mathbb{C} \mid \det(\mathcal{J}_A - \lambda I) = 0\}, \quad (1254)$$

which determines the full eigenvalue distribution governing exponential growth, oscillatory modes, and damped relaxation behavior in the coupled biosphere.

The perturbation field admits a spectral expansion in terms of eigenmodes:

$$\delta \mathbf{X}_A(t) = \sum_k c_k e^{\lambda_k t} \phi_k, \quad (1255)$$

where $\lambda_k \in \sigma(\mathcal{J}_A)$ and ϕ_k are the associated biosphere–climate eigenfunctions.

Stability is therefore governed by the spectral abscissa

$$\alpha(\mathcal{J}_A) = \max_{\lambda \in \sigma(\mathcal{J}_A)} \operatorname{Re}(\lambda), \quad (1256)$$

which measures the fastest exponential amplification rate of perturbations in the Amazon system.

Instability emerges when

$$\alpha(\mathcal{J}_A) > 0, \quad (1257)$$

indicating that at least one biospheric mode transitions from dissipative decay to self-amplifying growth, leading to runaway biomass loss, hydrological recycling collapse, or fire-dominated attractor regimes.

Near criticality, the dominant eigenvalue approaches the stability boundary:

$$\text{Re}(\lambda_{\max}) \rightarrow 0^+, \quad (1258)$$

producing critical slowing down and enhanced temporal correlation lengths in both vegetation and moisture fields.

In addition, the associated spectral gap shrinks as

$$\Delta_\sigma = \lambda_1 - \lambda_2 \rightarrow 0, \quad (1259)$$

indicating loss of modal separation and increased coherence among previously independent biospheric processes.

Within the Pellis operator framework, this spectral transition reflects a cross-scale renormalization of the effective climate–biosphere operator, in which hydrological feedback collapse and carbon cycle destabilization jointly shift the eigenstructure toward marginal stability. This produces a coordinated amplification of perturbations across biomass, humidity, and radiative forcing channels, consistent with the onset of large-scale Amazon dieback regimes [27, 29, 33].

14.4 Carbon Cycle Feedback and Global Coupling

Within the coupled biosphere–atmosphere–Pellis framework, the Amazon carbon reservoir $C(t)$ evolves under the competing influence of primary productivity, respiration-driven decay, and nonlinear fire-induced combustion emissions. The governing carbon balance equation is given by

$$\frac{dC}{dt} = \xi F - \zeta C + \Phi_{\text{fire}}(F, C, T), \quad (1260)$$

where ξF represents net primary production proportional to biomass density, ζC encodes natural respiration and decomposition losses, and Φ_{fire} captures nonlinear combustion-driven carbon release pathways, explicitly dependent on vegetation state, thermal forcing, and drought stress conditions.

To resolve the feedback structure of fire emissions, the combustion functional can be expressed in a threshold-amplified form:

$$\Phi_{\text{fire}}(F, C, T) = \eta_1 F e^{-\kappa P} + \eta_2 C \Theta(T - T_c) + \eta_3 FC, \quad (1261)$$

where the first term encodes drought-modulated fire susceptibility, the second term introduces ignition-driven carbon release, and the third term represents nonlinear coupling between stored carbon pools and active combustion feedback.

The atmospheric carbon dioxide dynamics are governed by a global coupling equation of the form

$$\frac{dCO_2}{dt} = \epsilon C - \omega CO_2 + \mathcal{A}_{\text{global}}(F, H, CO_2), \quad (1262)$$

where ϵC represents biosphere-to-atmosphere carbon transfer, ωCO_2 denotes radiative and oceanic sequestration sinks, and $\mathcal{A}_{\text{global}}$ encodes nonlinear planetary-scale feedback processes, including temperature-dependent respiration, ocean degassing, and albedo-mediated radiative forcing amplification.

The global coupling operator can be expanded as

$$\mathcal{A}_{\text{global}} = \alpha_1 T_s CO_2 + \alpha_2 H CO_2 + \alpha_3 F CO_2 + \alpha_4 CO_2^2, \quad (1263)$$

highlighting the presence of both cross-system coupling and intrinsic nonlinear atmospheric carbon self-amplification.

Linearization around a reference state (C^*, CO_2^*) yields the Jacobian subsystem

$$\mathcal{J}_{\text{carbon}} = \begin{pmatrix} \frac{\partial \dot{C}}{\partial C} & \frac{\partial \dot{C}}{\partial CO_2} \\ \frac{\partial \dot{CO_2}}{\partial C} & \frac{\partial \dot{CO_2}}{\partial CO_2} \end{pmatrix}, \quad (1264)$$

which determines the stability of the coupled carbon–atmosphere system through its spectral properties.

The eigenvalue spectrum is defined by

$$\sigma(\mathcal{J}_{\text{carbon}}) = \{ \lambda \in \mathbb{C} \mid \det(\mathcal{J}_{\text{carbon}} - \lambda I) = 0 \}, \quad (1265)$$

and instability emerges when the spectral abscissa becomes positive:

$$\alpha(\mathcal{J}_{\text{carbon}}) = \max_{\lambda \in \sigma(\mathcal{J}_{\text{carbon}})} \text{Re}(\lambda) > 0. \quad (1266)$$

In this regime, perturbations in carbon storage amplify exponentially, inducing synchronized growth in atmospheric CO₂ concentration and reinforcing radiative forcing feedbacks.

On the global scale, the coupling term $\mathcal{A}_{global}$ acts as a renormalization channel linking biospheric collapse to planetary radiative imbalance, such that carbon cycle perturbations propagate into temperature, hydrological recycling, and cryospheric stability modes.

In spectral terms, this corresponds to a coherent shift of dominant eigenvalues across the coupled Earth system operator:

$$\text{Re}(\lambda_{\max}^{global}) \rightarrow 0^+, \quad (1267)$$

indicating proximity to a multi-subsystem tipping manifold in which biosphere, atmosphere, and climate dynamics undergo synchronized destabilization.

Thus, the Amazon carbon cycle operates not as an isolated reservoir but as an active amplifier within the global Pellis instability hierarchy, where local carbon loss cascades into planetary-scale radiative forcing feedback loops consistent with cascading tipping dynamics [33, 37, 38].

14.5 Pellis Operator Projection onto Biospheric Manifold

Within the global Pellis climate operator framework, the Amazon subsystem is obtained through a geometric restriction of the full state space $\mathcal{H}$ onto a lower-dimensional biospheric manifold. This is formalized via the projection operator Π_A , yielding the reduced biospheric evolution operator

$$\mathcal{P}_A = \Pi_A \mathcal{P} \Pi_A, \quad (1268)$$

where $\mathcal{P}$ denotes the full coupled Earth system operator governing nonlinear interactions across atmosphere, hydrosphere, cryosphere, and biosphere.

The projection Π_A induces a decomposition of the global climate state vector into resolved biospheric and unresolved complementary components:

$$\mathbf{X} = \mathbf{X}_A + \mathbf{X}_\perp, \quad \mathbf{X}_A = \Pi_A \mathbf{X}, \quad \mathbf{X}_\perp = (I - \Pi_A) \mathbf{X}, \quad (1269)$$

explicitly separating Amazon dynamics from all external climate degrees of freedom.

Under this decomposition, the reduced evolution equation takes the form

$$\frac{d}{dt} \mathbf{X}_A = \mathcal{P}_A(\mathbf{X}_A) + \mathcal{C}_{A \leftrightarrow climate}(\mathbf{X}_{global}), \quad (1270)$$

where $\mathcal{C}_{A \leftrightarrow climate}$ represents the non-closed coupling term capturing bidirectional interactions between the Amazon biosphere and the broader Earth system.

This coupling term can be explicitly expressed as a back-projection of unresolved dynamics:

$$\mathcal{C}_{A \leftrightarrow climate} = \Pi_A \mathcal{P} (I - \Pi_A) \mathbf{X}, \quad (1271)$$

highlighting that Amazon evolution is continuously forced by external atmospheric, oceanic, and radiative processes not contained within the reduced manifold.

The non-commutativity of projection and global evolution,

$$[\mathcal{P}, \Pi_A] \neq 0, \quad (1272)$$

implies intrinsic non-closure of the biospheric subsystem, ensuring that even in the absence of internal perturbations, external climate variability injects effective forcing into Amazon dynamics.

At the level of effective dynamics, this non-closure induces memory-dependent behavior of the form

$$\frac{d}{dt} \mathbf{X}_A(t) = \mathcal{P}_A(\mathbf{X}_A(t)) + \int_0^t \mathcal{K}_A(t-s) \mathbf{X}_A(s) ds, \quad (1273)$$

where $\mathcal{K}_A(t)$ is a climate-induced memory kernel representing delayed feedback from unresolved global modes.

Spectrally, the projection modifies the eigenstructure of the full climate operator according to

$$\sigma(\mathcal{P}_A) \subset \sigma(\mathcal{P}) \oplus \delta\sigma_{coupling}, \quad (1274)$$

where $\delta\sigma_{coupling}$ encodes eigenvalue shifts arising from unresolved atmospheric and oceanic forcing pathways.

As a consequence, stability properties of the Amazon subsystem cannot be decoupled from the global climate system, since the projected operator inherits destabilizing spectral contributions from the full Pellis operator hierarchy.

This establishes the Amazon biosphere as a dynamically open subsystem embedded in a globally coupled operator field, where nonlinear feedbacks and cross-scale interactions continuously renormalize its effective evolution and determine its proximity to tipping behavior.

14.6 Dieback as Functional Phase Transition

Within the Pellis operator framework, forest collapse is interpreted as a non-smooth phase transition in the invariant measure of the biospheric dynamical system. The underlying mechanism is a structural reorganization of the attractor support in phase space, leading to a discontinuous transformation of the stationary measure associated with $\mathcal{P}_A$:

$$\Delta\mu(\sigma(\mathcal{P}_A)) \neq 0, \quad (1275)$$

indicating that the ergodic decomposition of the system undergoes a qualitative rearrangement at the transition boundary.

From a dynamical systems perspective, the forest state variable $F(t)$ evolves on a metastable attractor whose basin of attraction shrinks under increasing drought–fire–radiative forcing. This is reflected in the singular limit behavior:

$$\lim_{t \rightarrow t_c^-} F(t) = F^* > 0, \quad \lim_{t \rightarrow t_c^+} F(t) = 0, \quad (1276)$$

which defines an irreversible collapse of the vegetated attractor.

To formalize this as a bifurcation event, consider the effective growth exponent of biomass dynamics:

$$\lambda_{eff}(t) = \frac{d}{dF} \left(\frac{dF}{dt} \right), \quad (1277)$$

such that stability is guaranteed for $\lambda_{eff} < 0$ and loss of stability occurs when $\lambda_{eff} \rightarrow 0^+$. The dieback transition is therefore characterized by a sign inversion of the dominant spectral contribution of the biospheric operator.

In the corresponding operator formulation, the transition corresponds to a collapse of the leading eigenmode of $\mathcal{P}_A$:

$$\phi_{\max}^{(pre)} \not\rightarrow \phi_{\max}^{(post)}, \quad (1278)$$

indicating that the dominant invariant subspace of the system is not continuously deformable across the transition.

Equivalently, the spectral radius exhibits a discontinuous contraction:

$$\rho(\mathcal{P}_A)|_{t \rightarrow t_c^-} > \rho(\mathcal{P}_A)|_{t \rightarrow t_c^+}, \quad (1279)$$

signaling loss of long-term persistence modes and collapse of regenerative feedback loops.

At the macroscopic level, this transition is governed by the vanishing of stabilizing feedback gains in the coupled hydrological–carbon subsystem, producing a codimension-one bifurcation manifold in the full climate–biosphere state space. The system thus crosses from a regime of bounded fluctuations around a forested attractor to a regime dominated by absorbing states corresponding to degraded or savannized configurations.

This mechanism elevates Amazon dieback to a functional phase transition in the sense of non-equilibrium statistical physics, where the invariant measure, spectral support, and attractor geometry all undergo simultaneous discontinuous transformation under progressive forcing [33, 37, 38].

14.7 Emergent Interpretation

Within the Universal Pellis Function framework, Amazon rainforest dieback is interpreted as a manifestation of biosphere-level spectral instability embedded in the coupled biosphere–atmosphere–climate operator field. The system does not undergo a purely local ecological degradation process; instead, it experiences a global redistribution of spectral weight across interacting operators governing vegetation, hydrology, and radiative forcing.

At the operator level, the full Earth system generator $\mathcal{P}$ admits a coupled decomposition

$$\mathcal{P} = \mathcal{P}_A + \mathcal{P}_{climate} + \mathcal{P}_{coupling}, \quad (1280)$$

where $\mathcal{P}_A$ encodes intrinsic Amazon dynamics and $\mathcal{P}_{coupling}$ represents nonlinear cross-scale interactions with atmospheric and oceanic modes. Dieback corresponds to a regime in which $\mathcal{P}_{coupling}$ ceases to act as a perturbation and instead becomes a dominant spectral driver.

In this regime, eigenvalues of the coupled operator undergo structured migration:

$$\lambda_i^{(A+climate)}(t) \rightarrow \lambda_i^{(A+climate)}(t) + \delta\lambda_i^{cross}, \quad (1281)$$

where $\delta\lambda_i^{cross}$ arises from amplified feedback loops between humidity recycling, carbon emissions, and radiative forcing anomalies.

The transition is therefore characterized by a breakdown of spectral separation between biospheric and climatic modes:

$$\sigma(\mathcal{P}_A) \cap \sigma(\mathcal{P}_{climate}) \neq \emptyset, \quad (1282)$$

indicating the emergence of shared instability modes that synchronize local ecological collapse with global climate variability.

From a dynamical systems perspective, this corresponds to a loss of invariant subspace stability, where the previously approximately decoupled biosphere manifold becomes non-invariant under the full evolution:

$$\mathcal{P}(\mathcal{M}_A) \not\subseteq \mathcal{M}_A, \quad (1283)$$

signaling that trajectories initiated within the Amazon submanifold are no longer dynamically confined.

As a consequence, rainforest stability is reinterpreted not as an isolated ecological property but as a projection of global operator coherence. Localized stressors—such as drought intensification, fire amplification, or hydrological disruption—act as perturbations that are globally amplified through the coupled spectral structure of the Earth system.

Thus, Amazon dieback emerges as a functional phase collapse of the biosphere–atmosphere operator field, in which eigenvalue migration across stability boundaries induces irreversible reconfiguration of planetary-scale dynamics. This establishes rainforest integrity as a necessary component of global dynamical coherence, rather than a separable ecological subsystem [33, 37, 38].

15 Monsoon System Transitions

The monsoon system constitutes a nonlinear, seasonally forced atmospheric–oceanic dynamical subsystem characterized by abrupt transitions between active and suppressed precipitation regimes. Its dynamics emerge from the interaction of land–sea thermal contrast, moisture transport, convective instability, and large-scale circulation modes, forming a forced dissipative system with intrinsic threshold behavior and hysteresis effects [11, 14, 27–29, 33, 37].

Within the Universal Pellis Function framework, the monsoon system is interpreted as a periodically driven spectral subsystem of the global climate operator $\mathcal{P}$, where regime transitions correspond to instability-induced reorganization of dominant spectral modes under seasonal forcing.

15.1 Thermodynamic Forcing and Moisture Transport Dynamics

Within the coupled atmosphere–land–ocean subsystem of the Pellis climate framework, monsoon precipitation intensity $M(t)$ and land–sea thermal contrast $\Delta T(t)$ constitute a dynamically coupled nonlinear pair that governs large-scale moisture transport, convective organization, and hydrological amplification feedbacks.

The evolution of monsoon strength is governed by a nonlinear growth–saturation equation with convective coupling:

$$\frac{dM}{dt} = \alpha \Delta T M - \beta M^2 + \mathcal{C}_{conv}(M, q, \Delta T), \quad (1284)$$

where $\alpha \Delta T M$ represents thermally driven instability of moisture convergence, βM^2 encodes nonlinear saturation due to moisture depletion and convective self-limitation, and $\mathcal{C}_{conv}$ captures unresolved convective organization, mesoscale clustering, and cloud–radiative feedback effects.

The convective coupling functional can be expanded as a multiscale operator:

$$\mathcal{C}_{conv}(M, q, \Delta T) = \kappa_1 q M + \kappa_2 \nabla \cdot (M \nabla q) + \kappa_3 M \Delta T - \kappa_4 M^3, \quad (1285)$$

introducing nonlinear reinforcement at intermediate intensities and cubic suppression at extreme convective regimes.

The land–sea thermal contrast evolves under radiative forcing, atmospheric damping, and moisture-mediated energy redistribution:

$$\frac{d(\Delta T)}{dt} = \gamma Q_{solar} - \delta \Delta T - \eta M - \chi M \Delta T, \quad (1286)$$

where the cross-term $\chi M \Delta T$ represents feedback between latent heat release and large-scale thermal gradient erosion.

Moisture content dynamics are governed by a coupled advection–recycling balance:

$$\frac{dq}{dt} = \lambda_1 E_{surface} - \lambda_2 q + \lambda_3 \nabla \cdot (q \mathbf{v}) + \lambda_4 M, \quad (1287)$$

where surface evaporation $E_{surface}$ injects moisture into the system, while advective transport redistributes humidity across spatial scales.

Moisture recycling introduces an effective nonlinear amplification mechanism in precipitation dynamics:

$$q_{eff} = q_0 + \kappa M + \zeta M q, \quad (1288)$$

where the bilinear term $\zeta M q$ encodes self-reinforcing humidity–precipitation coupling that strengthens monsoon circulation under favorable thermodynamic conditions.

Substituting q_{eff} into the convective term yields an effective reduced-order growth law:

$$\frac{dM}{dt} = (\alpha\Delta T + \kappa q_0) M + \kappa\zeta M^2 - \beta M^2 + \mathcal{O}(M^3), \quad (1289)$$

revealing a competition between linear destabilization and nonlinear saturation mechanisms that determine the stability of monsoon regimes.

The effective growth exponent is therefore

$$\lambda_{eff} = \alpha\Delta T + \kappa q_0, \quad (1290)$$

so that regime transition occurs when

$$\lambda_{eff} = 0, \quad (1291)$$

defining a bifurcation boundary between weak monsoon states and strongly organized convective regimes.

Linearization around equilibrium $(M^*, \Delta T^*)$ yields the Jacobian operator

$$\mathcal{J}_{monsoon} = \begin{pmatrix} \frac{\partial \dot{M}}{\partial M} & \frac{\partial \dot{M}}{\partial \Delta T} \\ \frac{\partial \dot{\Delta T}}{\partial M} & \frac{\partial \dot{\Delta T}}{\partial \Delta T} \end{pmatrix}, \quad (1292)$$

whose spectral properties determine the onset of instability in the coupled thermodynamic system.

Instability emerges when the spectral abscissa satisfies

$$\alpha(\mathcal{J}_{monsoon}) = \max_{\lambda \in \sigma(\mathcal{J}_{monsoon})} \text{Re}(\lambda) > 0, \quad (1293)$$

indicating exponential amplification of perturbations in both moisture transport and thermal gradient structure.

In this regime, the system transitions from a thermodynamically constrained circulation to a self-organized convective state dominated by nonlinear feedback loops between latent heat release, moisture recycling, and radiative forcing.

Within the Universal Pellis Function framework, this transition corresponds to a spectral reconfiguration of the atmospheric moisture operator, in which eigenmodes associated with convective organization migrate toward marginal stability, signaling proximity to a planetary-scale tipping manifold [27, 32, 33, 37].

Table 14: Monsoon Regime Transition Structure

Component	Symbol	Definition	Interpretation
Monsoon State	Ψ_M	Atmospheric circulation field	Seasonal flow structure
Moisture Transport	F_Q	Latent heat flux operator	Energy redistribution
Convective Instability	F_C	Threshold convection mechanism	Nonlinear triggering
Land–Ocean Coupling	F_{LO}	Surface contrast interaction	Driver of monsoon strength

15.2 Convective Instability and Threshold Behavior

Within the thermodynamic forcing framework of the coupled moisture–convection system, convective triggering is governed by the vertical integral of buoyancy perturbations, quantified through the Convective Available Potential Energy (CAPE). In full dynamical form, CAPE depends nonlinearly on moisture content, thermal stratification, and entrainment effects, and is expressed as:

$$CAPE(M, q) = \int_{z_0}^{z_t} g \left(\frac{T'_v(z; M, q) - T_v(z)}{T_v(z)} \right) dz, \quad (1294)$$

where $T'_v(z; M, q)$ denotes the virtual temperature of a lifted air parcel modulated by moisture availability q and convective intensity M , while $T_v(z)$ is the environmental virtual temperature profile.

To explicitly capture feedback dependence, the buoyancy perturbation can be decomposed as

$$T'_v(z; M, q) - T_v(z) = \chi_1 q(z) + \chi_2 M - \chi_3 \Gamma(z), \quad (1295)$$

where $\Gamma(z)$ is the environmental lapse rate and χ_i are coupling coefficients encoding moisture–temperature interaction strength and convective entrainment modulation.

This induces a reduced-order nonlinear functional form for CAPE:

$$CAPE(M, q) = \mathcal{A}_1 q + \mathcal{A}_2 M - \mathcal{A}_3 \int_{z_0}^{z_t} \Gamma(z) dz, \quad (1296)$$

highlighting that convective instability is controlled by the competition between moisture-driven buoyancy enhancement and thermodynamic stratification resistance.

A transition into an active monsoon or deep convection regime occurs when CAPE crosses a critical threshold:

$$CAPE(M, q) > CAPE_c, \quad (1297)$$

which defines a nonlinear activation manifold in the (M, q) state space separating suppressed convection from self-sustaining convective organization.

Near the threshold, the system exhibits a sharp bifurcation in convective response intensity, which can be expressed as an effective activation function:

$$M_{conv} = M_0 \cdot \Theta(CAPE - CAPE_c), \quad (1298)$$

where Θ denotes a smooth limiting sigmoid approximation of the Heaviside function, capturing the rapid onset of deep convection once buoyancy forcing exceeds stability constraints.

The coupled moisture–convection system thus admits a bistable structure characterized by competing attractors: a dry, suppressed-convection regime and a wet, self-amplifying convective regime. These states are separated by a saddle-node bifurcation manifold defined implicitly by

$$\frac{\partial CAPE}{\partial M} = 0, \quad \frac{\partial^2 CAPE}{\partial M^2} \approx 0, \quad (1299)$$

which marks the collapse of intermediate stability states.

In dynamical systems terms, this bistability arises from nonlinear feedback between moisture accumulation, latent heat release, and vertical buoyancy generation, producing hysteresis in the monsoon transition. Consequently, once the system crosses the convective threshold, return to the dry regime requires substantially stronger negative forcing, reflecting irreversible memory effects in the coupled atmospheric operator.

Within the Pellis operator framework, this behavior corresponds to a local spectral reorganization in which convective eigenmodes transition from damped to growing states as the leading eigenvalue satisfies

$$\text{Re}(\lambda_{\max}) \rightarrow 0^+, \quad (1300)$$

signaling proximity to a convective tipping manifold embedded within the global climate operator spectrum.

15.3 Spectral Stability of Seasonal Forcing Dynamics

Within the periodically forced monsoon subsystem, the full state vector $\mathbf{X}_M = (M, \Delta T, q)$ evolves under a time-dependent nonlinear operator whose linearization about a periodic orbit yields

$$\partial_t \delta \mathbf{X}_M = \mathcal{J}_M(t) \delta \mathbf{X}_M, \quad (1301)$$

where $\mathcal{J}_M(t + T) = \mathcal{J}_M(t)$ is a time-periodic Jacobian operator induced by seasonal modulation of radiative forcing, land–sea thermal contrast, and moisture recycling intensity.

The stability of such periodically driven dynamical systems is governed not by instantaneous eigenvalues, but by Floquet theory. The monsoon perturbation evolution over one full seasonal cycle defines the monodromy operator $\mathcal{U}(T)$ such that

$$\delta \mathbf{X}_M(t + T) = \mathcal{U}(T) \delta \mathbf{X}_M(t), \quad (1302)$$

where $\mathcal{U}(T)$ is the time-ordered exponential

$$\mathcal{U}(T) = \mathcal{T} \exp \left(\int_0^T \mathcal{J}_M(t) dt \right). \quad (1303)$$

The spectral properties of the seasonal system are therefore characterized by Floquet multipliers $\mu_i \in \sigma_F(\mathcal{J}_M)$ defined as eigenvalues of the monodromy operator:

$$\mathcal{U}(T) \phi_i = \mu_i \phi_i, \quad \sigma_F(\mathcal{J}_M) = \{\mu_i\}. \quad (1304)$$

The perturbation dynamics over multiple seasonal cycles admit the decomposition

$$\delta \mathbf{X}_M(nT) = \sum_i c_i \mu_i^n \phi_i, \quad (1305)$$

showing that long-term stability is fully determined by the magnitude of Floquet multipliers.

Instability arises when at least one multiplier exits the unit circle:

$$|\mu_i| > 1, \quad (1306)$$

corresponding to exponential amplification of perturbations across successive seasonal cycles and breakdown of bounded monsoon variability.

The critical transition boundary is therefore defined by the neutral stability condition

$$\max_i |\mu_i| = 1, \quad (1307)$$

which defines a bifurcation surface in the space of seasonal forcing parameters.

Near this boundary, the dominant Floquet exponent $\rho_i = \log |\mu_i|$ approaches zero from below,

$$\rho_{\max} \rightarrow 0^-, \quad (1308)$$

indicating progressive loss of seasonal damping and emergence of resonance-like amplification across monsoon cycles.

Within the Universal Pellis Function framework, this corresponds to a spectral transition in which the time-periodic Pellis operator develops marginal Floquet modes, signaling proximity to a seasonal tipping manifold where periodic forcing ceases to stabilize the coupled moisture–thermal dynamics [28, 29].

15.4 Energy Balance and Latent Heat Feedback

Within the monsoon–thermodynamic subsystem, the total energy budget is governed by the interplay between latent heat release, sensible heat storage, radiative losses, and ocean–atmosphere exchange fluxes. The effective monsoon energy functional is therefore written as

$$E_M = L_v M + C_p \Delta T - R_{out} + \Phi_{ocean-atm} + \Phi_{latent-coupling}, \quad (1309)$$

where $L_v M$ represents latent heat release proportional to precipitation intensity, $C_p \Delta T$ is the sensible heat contribution from land–sea thermal contrast, and $\Phi_{latent-coupling}$ encodes higher-order nonlinear energy transfer between phase-change processes and atmospheric circulation.

To explicitly resolve feedback amplification pathways, the latent coupling term can be expanded as

$$\Phi_{latent-coupling} = \eta_1 M \Delta T + \eta_2 M q - \eta_3 M^3, \quad (1310)$$

where the cubic term represents nonlinear saturation due to convective self-limitation and atmospheric moisture depletion at extreme precipitation intensities.

The temporal evolution of monsoon energy is governed by a nonlinear balance law:

$$\frac{dE_M}{dt} = \lambda_1 M - \lambda_2 M^2 + \lambda_3 \Delta T M + \lambda_4 M q - \lambda_5 \Delta T^2, \quad (1311)$$

which captures the competition between latent heat amplification, nonlinear damping, and thermally driven destabilization.

The term $\lambda_1 M$ represents primary latent heat injection through condensation, while $\lambda_3 \Delta T M$ encodes thermodynamic reinforcement of convection via land–sea contrast. The quadratic term $-\lambda_2 M^2$ reflects nonlinear saturation due to moisture limitation and convective breakup, and higher-order corrections introduce additional stabilizing and destabilizing nonlinearities.

The system exhibits a characteristic saturation-driven instability mechanism when the effective growth rate becomes state-dependent:

$$\lambda_{eff}(M, \Delta T, q) = \lambda_1 + \lambda_3 \Delta T + \lambda_4 q - \lambda_2 M, \quad (1312)$$

leading to a transition from linear amplification to nonlinear suppression as precipitation intensity increases.

The tipping condition is obtained from the loss of monotonic energy growth:

$$\frac{dE_M}{dt} = 0, \quad \frac{d^2 E_M}{dt^2} < 0, \quad (1313)$$

defining a saturation-induced bifurcation manifold separating stable monsoon operation from collapse regimes.

In the strongly forced regime, latent heat feedback dominates radiative damping, producing runaway convective organization followed by abrupt structural breakdown once moisture reservoirs are exhausted.

Within the Pellis operator framework, this corresponds to a nonlinear renormalization of the effective energy operator, where latent heat coupling shifts dominant spectral modes toward marginal stability, signaling proximity to a thermodynamic tipping manifold in the coupled Earth system [27, 32, 33, 37].

15.5 Pellis Operator Projection onto Monsoon Manifold

Within the Universal Pellis Function framework, the monsoon subsystem is rigorously defined as a geometric reduction of the full Earth system operator $\mathcal{P}$ onto a lower-dimensional invariant submanifold $\mathcal{M}$ associated with atmospheric convection, thermodynamic gradients, and moisture transport processes. This projection is implemented via the operator Π_M , yielding the reduced monsoon generator

$$\mathcal{P}_M = \Pi_M \mathcal{P} \Pi_M, \quad (1314)$$

where Π_M acts as a nonlinear filtering operator that retains only convective–thermodynamic degrees of freedom while eliminating fast radiative and oceanic modes.

The full climate state vector admits the decomposition

$$\mathbf{X} = \mathbf{X}_M + \mathbf{X}_\perp, \quad \mathbf{X}_M = \Pi_M \mathbf{X}, \quad \mathbf{X}_\perp = (I - \Pi_M) \mathbf{X}, \quad (1315)$$

explicitly separating monsoon-relevant dynamics from unresolved global climate components.

Under this decomposition, the reduced evolution equation is given by

$$\frac{d}{dt} \mathbf{X}_M = \mathcal{P}_M(\mathbf{X}_M) + \mathcal{C}_{tele}(\mathbf{X}_{global}), \quad (1316)$$

where $\mathcal{C}_{tele}$ represents nonlocal teleconnection forcing terms arising from coupled ocean–atmosphere modes, including ENSO variability, Walker circulation shifts, and extratropical wave propagation.

To explicitly expose the non-closure of the projected system, the teleconnection operator can be written as a back-projection of unresolved global dynamics:

$$\mathcal{C}_{tele} = \Pi_M \mathcal{P} (I - \Pi_M) \mathbf{X}, \quad (1317)$$

demonstrating that monsoon evolution is continuously driven by feedback from external climate reservoirs.

The resulting system is therefore intrinsically non-autonomous, with effective forcing structure

$$\frac{d}{dt} \mathbf{X}_M = \mathcal{P}_M(\mathbf{X}_M) + \int_0^t K_{tele}(t-s) \mathbf{X}_{global}(s) ds, \quad (1318)$$

where $K_{tele}(t)$ is a memory kernel encoding delayed teleconnection responses across oceanic and atmospheric waveguides.

At the spectral level, projection induces a deformation of the monsoon operator spectrum:

$$\sigma(\mathcal{P}_M) = \Pi_M \sigma(\mathcal{P}) \Pi_M + \delta\sigma_{tele}, \quad (1319)$$

where $\delta\sigma_{tele}$ represents eigenvalue shifts induced by ENSO-driven and global circulation-mediated coupling.

Instability emerges when teleconnection-induced spectral drift drives dominant modes toward marginal stability:

$$\text{Re}(\lambda_{\max}^{(M)}) \rightarrow 0^+, \quad (1320)$$

signaling that the monsoon subsystem is no longer dynamically autonomous but instead locked into a globally coupled instability regime. This establishes the monsoon manifold as a strongly non-closed subsystem whose stability is fundamentally governed by global-scale spectral coupling within the Pellis climate operator hierarchy [28, 29, 37].

15.6 Regime Transition as Functional Phase Change

Within the monsoon manifold $\mathcal{M}$, regime transitions are interpreted as singular perturbations of the invariant measure associated with the reduced Pellis operator $\mathcal{P}_M$. In this formulation, the system does not undergo a smooth deformation of trajectories, but rather a discontinuous reorganization of the underlying ergodic structure in phase space, encoded as

$$\Delta\mu(\sigma(\mathcal{P}_M)) \neq 0, \quad (1321)$$

where μ denotes the invariant spectral measure induced by the long-time dynamics of the operator semigroup generated by $\mathcal{P}_M$.

To make this structure explicit, let μ_t denote the time-dependent probability measure over monsoon states. Then a regime transition corresponds to a weak-* discontinuity:

$$\lim_{t \rightarrow t_c^-} \mu_t \neq \lim_{t \rightarrow t_c^+} \mu_t, \quad (1322)$$

implying non-equivalence of statistical ensembles before and after the transition. This is equivalent to a breakdown of ergodicity in the reduced climate subsystem.

At the macroscopic level, the order parameter $M(t)$ exhibits a jump discontinuity:

$$\lim_{t \rightarrow t_c^-} M(t) \neq \lim_{t \rightarrow t_c^+} M(t), \quad (1323)$$

which identifies the transition as a first-order functional phase change in the monsoon precipitation field.

This discontinuity can be embedded in a variational structure by introducing an effective potential landscape $\mathcal{V}(M)$ such that

$$\frac{dM}{dt} = -\frac{\partial \mathcal{V}(M)}{\partial M} + \eta(M, t), \quad (1324)$$

where $\eta(M, t)$ represents stochastic–deterministic forcing arising from unresolved convective variability and global teleconnection noise. Regime transitions occur when $\mathcal{V}(M)$ undergoes a qualitative deformation from a double-well structure to a single-well structure.

In particular, bistability is characterized by the existence of two local minima:

$$\frac{\partial \mathcal{V}}{\partial M} = 0, \quad \frac{\partial^2 \mathcal{V}}{\partial M^2} > 0, \quad M \in \{M_{dry}, M_{wet}\}, \quad (1325)$$

while the transition point corresponds to a saddle-node annihilation condition:

$$\frac{\partial \mathcal{V}}{\partial M} = 0, \quad \frac{\partial^2 \mathcal{V}}{\partial M^2} = 0, \quad (1326)$$

defining a codimension-one bifurcation manifold in the control-parameter space.

From a spectral perspective, this functional phase change corresponds to a collapse of the leading eigenstructure of the monsoon operator:

$$\lambda_{\max}(t) \rightarrow 0, \quad \text{Re}(\lambda_{\max}) \uparrow 0^+, \quad (1327)$$

accompanied by localization of spectral mass onto marginal modes, which act as precursors of critical slowing down and enhanced persistence.

This spectral concentration can be expressed in measure-theoretic form as

$$\mu_t(\lambda) \rightarrow \delta(\lambda - \lambda_c), \quad (1328)$$

indicating asymptotic collapse of the spectral distribution onto a critical eigenvalue λ_c at the tipping point.

To connect microscopic variability with macroscopic switching, the regime indicator function may be written as a discontinuous limit:

$$M(t) = M_{dry} + (M_{wet} - M_{dry}) \Theta(t - t_c), \quad (1329)$$

where Θ is the Heaviside activation function emerging as the sharp-limit representation of smooth probabilistic transitions under vanishing noise amplitude.

Within the Universal Pellis Function framework, this regime transition is therefore interpreted as a functional phase change in operator space: a structural reorganization of the invariant measure, accompanied by eigenvalue migration across stability boundaries and irreversible loss of intermediate metastable states [28, 29].

15.7 Emergent Interpretation

Within the Universal Pellis Function framework, monsoon regime transitions are interpreted as forced nonlinear spectral bifurcations in a periodically driven dissipative operator system. The governing dynamics are not reducible to smooth parameter drift, but instead emerge from structural instabilities in the spectrum of the global climate generator $\mathcal{P}$ when restricted to the monsoon manifold $\mathcal{M}$.

In this setting, seasonal forcing acts as a periodic modulation of the effective operator:

$$\mathcal{P}_M(t + T) = \mathcal{P}_M(t), \quad (1330)$$

inducing a Floquet structure in the associated evolution semigroup. The long-term dynamics are therefore governed by the monodromy operator $\mathcal{U}(T)$, whose spectral properties determine regime stability:

$$\mathcal{U}(T)\phi_i = \mu_i\phi_i, \quad \mu_i \in \sigma_F(\mathcal{P}_M). \quad (1331)$$

A monsoon transition is then identified with a Floquet-induced eigenvalue crossing event, in which at least one multiplier traverses the unit stability boundary:

$$|\mu_i(t)| \uparrow 1 \quad \rightarrow \quad |\mu_i(t)| > 1. \quad (1332)$$

This crossing marks the breakdown of bounded periodic response and the onset of exponentially amplified inter-annual variability.

At the infinitesimal level, this corresponds to a collision of Floquet exponents in the complex plane:

$$\rho_i = \frac{1}{T} \log(\mu_i), \quad \text{Re}(\rho_i) \rightarrow 0^+, \quad (1333)$$

signaling marginal destabilization of the seasonal attractor embedded in the climate operator spectrum.

In spectral terms, the phenomenon can be interpreted as a redistribution of eigenvalue density under periodic forcing:

$$\sigma(\mathcal{P}_M(t)) \longrightarrow \sigma(\mathcal{P}_M(t)) + \delta\sigma_{\text{Floquet}}, \quad (1334)$$

where $\delta\sigma_{\text{Floquet}}$ encodes forcing-induced deformation of the global climate spectrum through resonance with seasonal harmonics.

Consequently, monsoon variability is not merely a regional hydrological fluctuation but a dynamically embedded component of the global instability cascade. Seasonal forcing injects energy into specific spectral bands of the Pellis operator, which then propagate through nonlinear coupling channels into cross-scale climate modes, amplifying sensitivity to perturbations.

This leads to a unified interpretation in which monsoon tipping behavior is a manifestation of planetary-scale spectral coherence breakdown: a regime where Floquet structure, nonlinear feedback, and global operator geometry coalesce to produce abrupt hydrological phase reorganization [28, 29, 37].

16 Planetary Instability Field Theory

The Planetary Instability Field Theory formalizes the Earth system as a coupled, nonlinear, multiscale instability field evolving on a high-dimensional functional manifold. In contrast to classical equilibrium-based formulations, planetary behavior is represented as an emergent consequence of spectral deformation, renormalization flow collapse, and global bifurcation structure in operator space. The resulting framework extends the Pellis Functional Operator Formalism into a fully planetary-scale instability theory, where climate subsystems interact as nonlinearly coupled critical fields [1–139].

16.1 Planetary State Space and Global Field Structure

Let $\mathcal{H}_P$ denote the planetary Hilbert manifold encoding coupled atmosphere–ocean–cryosphere–biosphere–lithosphere dynamics, endowed with a weak Riemannian metric g_P and an associated connection ∇_P . The global planetary state is defined as a section of this infinite-dimensional fiber bundle:

$$\mathbf{Y}(t) \in \Gamma(\mathcal{H}_P), \quad (1335)$$

with decomposition into interacting subsystem fields:

$$\mathbf{Y}(t) = (\mathbf{X}_{atm}, \mathbf{X}_{ocn}, \mathbf{X}_{cry}, \mathbf{X}_{bio}, \mathbf{X}_{lith}). \quad (1336)$$

Each component field is itself an element of a functional state space $\mathcal{H}_\alpha$, $\alpha \in \{atm, ocn, cry, bio, lith\}$, such that

$$\mathbf{X}_\alpha(t) \in \mathcal{H}_\alpha \subset L^2(\Omega_\alpha) \otimes \mathbb{R}^{n_\alpha}. \quad (1337)$$

The planetary evolution equation is formulated in intrinsic geometric form as

$$\frac{D}{Dt} \mathbf{Y}(t) = \mathcal{P}_P(\mathbf{Y}(t)), \quad (1338)$$

where $\frac{D}{Dt}$ denotes the covariant derivative induced by ∇_P , and $\mathcal{P}_P$ is the nonlinear coupled planetary Pellis operator encoding multiscale feedback interactions.

In local coordinates, this operator admits the decomposition:

$$\mathcal{P}_P(\mathbf{Y}) = \mathcal{L}\mathbf{Y} + \mathcal{N}(\mathbf{Y}, \mathbf{Y}) + \mathcal{C}(\mathbf{Y}) + \mathcal{F}(t), \quad (1339)$$

where $\mathcal{L}$ is the linearized transport–diffusion generator, $\mathcal{N}$ is a bilinear interaction form representing nonlinear cross-sphere coupling, $\mathcal{C}$ encodes constraint manifolds (mass, energy, and entropy balance), and $\mathcal{F}(t)$ represents external radiative and orbital forcing.

The nonlinear interaction term is explicitly written as a tensor contraction:

$$\mathcal{N}(\mathbf{Y}, \mathbf{Y}) = \sum_{\alpha, \beta} \mathcal{T}_{\alpha\beta}^\gamma \mathbf{X}_\alpha \otimes \mathbf{X}_\beta, \quad (1340)$$

where $\mathcal{T}_{\alpha\beta}^\gamma$ is a third-order coupling tensor defining cross-sphere energy–entropy transfer channels.

The system admits a global energy functional $\mathcal{E}_P : \mathcal{H}_P \rightarrow \mathbb{R}$ of the form

$$\mathcal{E}_P[\mathbf{Y}] = \sum_{\alpha} \int_{\Omega_{\alpha}} \left(\frac{1}{2} |\mathbf{X}_{\alpha}|^2 + V_{\alpha}(\mathbf{X}_{\alpha}) \right) d\mu_{\alpha} + \sum_{\alpha \neq \beta} \mathcal{I}_{\alpha\beta}(\mathbf{X}_{\alpha}, \mathbf{X}_{\beta}), \quad (1341)$$

where V_{α} are local potential energy densities and $\mathcal{I}_{\alpha\beta}$ encode inter-sphere coupling potentials.

The dynamics can equivalently be written in gradient-flow form on the planetary manifold:

$$\frac{D}{Dt} \mathbf{Y}(t) = -\nabla_P \mathcal{E}_P[\mathbf{Y}(t)] + \Xi(\mathbf{Y}(t), t), \quad (1342)$$

where Ξ represents non-conservative stochastic–deterministic hybrid forcing.

Spectrally, linearization around a reference state $\mathbf{Y}_0$ yields:

$$\frac{d}{dt} \delta \mathbf{Y} = \mathcal{J}_P(\mathbf{Y}_0) \delta \mathbf{Y}, \quad (1343)$$

where $\mathcal{J}_P$ is the planetary Jacobian operator. The stability of the system is governed by its spectrum:

$$\sigma(\mathcal{J}_P) = \{\lambda_i \in \mathbb{C}\}, \quad (1344)$$

with tipping points emerging when

$$\max_i \operatorname{Re}(\lambda_i) \rightarrow 0^+. \quad (1345)$$

Within the Universal Pellis Function framework, the planetary instability is characterized by a spectral functional

$$\mathcal{U}_P(t) = \int_{\sigma(\mathcal{J}_P)} \frac{1}{1 - \operatorname{Re}(\lambda)} d\rho(\lambda), \quad (1346)$$

which diverges at critical transitions.

Moreover, introducing a multiscale decomposition via projection operators Π_s indexed by scale s , we obtain:

$$\mathbf{Y}(t) = \sum_{s \in \mathcal{S}} \Pi_s \mathbf{Y}(t), \quad (1347)$$

with inter-scale coupling governed by renormalization-group-like flow equations:

$$\frac{d}{ds} \mathbf{Y}_s = \mathcal{R}(\mathbf{Y}_s), \quad (1348)$$

where $\mathcal{R}$ is the planetary renormalization operator.

Finally, tipping points correspond to bifurcation sets defined by:

$$\mathcal{B}_c = \{\mathbf{Y} \in \mathcal{H}_P : \det(\mathcal{J}_P(\mathbf{Y})) = 0\}, \quad (1349)$$

which constitute codimension-one critical manifolds in $\mathcal{H}_P$ separating distinct planetary regimes of stability and instability.

Table 15: Planetary-Scale Climate Instability Field Structure

Component	Symbol	Definition	Interpretation
Planetary State	Ψ_P	Global climate field vector	Earth system macrostate
Instability Operator	$\mathcal{L}_P$	Coupled global operator	Planetary dynamics generator
Cross-System Coupling	C_{ij}	Interaction matrix	Inter-system dependencies
Global Mode	ϕ_G	Dominant eigenmode	Planet-scale instability pattern

16.2 Coupled Instability Field Representation

The planetary system is reformulated as a nonlinear field of instability densities defined on a measure space (Ω_P, μ_P) :

$$\mathcal{I}_P(\mathbf{x}, t) = \sum_{i=1}^{N_s} w_i \mathcal{I}_i(\mathbf{x}, t), \quad (1350)$$

where each $\mathcal{I}_i : \Omega_P \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ represents a subsystem instability functional and $w_i \in \mathbb{R}^+$ are coupling weights satisfying $\sum_i w_i = 1$.

The instability field is interpreted as an element of a Banach lattice $\mathcal{B}_I \subset L^1(\Omega_P)$, endowed with a positivity-preserving semigroup evolution.

The global instability propagation satisfies a nonlinear nonlocal integro-differential evolution equation of the form:

$$\partial_t \mathcal{I}_P(\mathbf{x}, t) = \int_{\Omega_P} K(\mathbf{x}, \mathbf{y}; t) \mathcal{I}_P(\mathbf{y}, t) d\mu_P(\mathbf{y}) + \mathcal{N}_P[\mathcal{I}_P(\mathbf{x}, t)], \quad (1351)$$

where $K(\mathbf{x}, \mathbf{y}; t)$ is a time-dependent nonlocal coupling kernel encoding teleconnections, and $\mathcal{N}_P$ is a nonlinear amplification operator acting pointwise or weakly nonlocally.

The kernel admits a spectral decomposition in terms of orthonormal instability modes $\{\phi_k\}_{k \in \mathbb{N}}$:

$$K(\mathbf{x}, \mathbf{y}; t) = \sum_{k=1}^{\infty} \lambda_k(t) \phi_k(\mathbf{x}) \phi_k^*(\mathbf{y}), \quad (1352)$$

which induces a modal evolution equation:

$$\frac{d}{dt} a_k(t) = \lambda_k(t) a_k(t) + \mathcal{F}_k(\mathbf{a}(t)), \quad (1353)$$

where $a_k(t)$ are instability mode amplitudes and $\mathcal{F}_k$ are nonlinear mode-coupling terms.

The nonlinear operator $\mathcal{N}_P$ is assumed to satisfy a generalized polynomial closure in the weak-field regime:

$$\mathcal{N}_P(\mathcal{I}_P) = \alpha \mathcal{I}_P^2 + \beta \mathcal{I}_P^3 + \int_{\Omega_P} \Gamma(\mathbf{x}, \mathbf{y}) \mathcal{I}_P(\mathbf{y}, t)^2 d\mu_P(\mathbf{y}), \quad (1354)$$

where $\alpha, \beta \in \mathbb{R}$ control local nonlinear saturation and Γ encodes higher-order spatial reinforcement effects.

Defining the global instability energy functional:

$$\mathcal{E}_I[\mathcal{I}_P] = \frac{1}{2} \int_{\Omega_P} \int_{\Omega_P} \mathcal{I}_P(\mathbf{x}) K(\mathbf{x}, \mathbf{y}) \mathcal{I}_P(\mathbf{y}) d\mu_P(\mathbf{x}) d\mu_P(\mathbf{y}) - \int_{\Omega_P} V(\mathcal{I}_P(\mathbf{x})) d\mu_P(\mathbf{x}), \quad (1355)$$

the dynamics can be written as a generalized gradient flow in instability space:

$$\partial_t \mathcal{I}_P = \mathcal{M} \left(-\frac{\delta \mathcal{E}_I}{\delta \mathcal{I}_P} \right), \quad (1356)$$

where $\mathcal{M}$ is a mobility operator encoding transport and redistribution of instability.

The linearized evolution around a reference instability state $\mathcal{I}_P^0$ yields:

$$\partial_t \delta \mathcal{I}_P = \mathcal{L}_I \delta \mathcal{I}_P, \quad (1357)$$

with $\mathcal{L}_I$ defined as:

$$\mathcal{L}_I f(\mathbf{x}) = \int_{\Omega_P} K(\mathbf{x}, \mathbf{y}) f(\mathbf{y}) d\mu_P(\mathbf{y}) + 2\alpha \mathcal{I}_P^0(\mathbf{x}) f(\mathbf{x}) + 3\beta (\mathcal{I}_P^0(\mathbf{x}))^2 f(\mathbf{x}). \quad (1358)$$

Instability growth rates are governed by the spectral radius:

$$r(\mathcal{L}_I) = \sup\{\text{Re}(\lambda) : \lambda \in \sigma(\mathcal{L}_I)\}, \quad (1359)$$

and critical transitions occur when:

$$r(\mathcal{L}_I) \rightarrow 0^+. \quad (1360)$$

To capture multiscale cascade effects, define a scale decomposition $\mathcal{I}_P = \sum_s \mathcal{I}_P^{(s)}$, leading to renormalization-type flow:

$$\frac{d}{ds} \mathcal{I}_P^{(s)} = \mathcal{R}_I(\mathcal{I}_P^{(s)}), \quad (1361)$$

where $\mathcal{R}_I$ governs instability transfer across scales.

Finally, the coupled instability field admits a critical manifold of tipping configurations:

$$\mathcal{C}_I = \{\mathcal{I}_P \in \mathcal{B}_I : \det(\mathcal{L}_I) = 0\}, \quad (1362)$$

which defines the boundary between stable decay regimes and runaway amplification dynamics in the planetary instability field.

Figure 21 illustrates spatially coherent instability modes arising from the global coupling structure of the climate system. These modes reflect the emergence of large-scale correlated patterns that span multiple subsystems, indicating the presence of a unified dynamical organization across spatial and temporal scales.

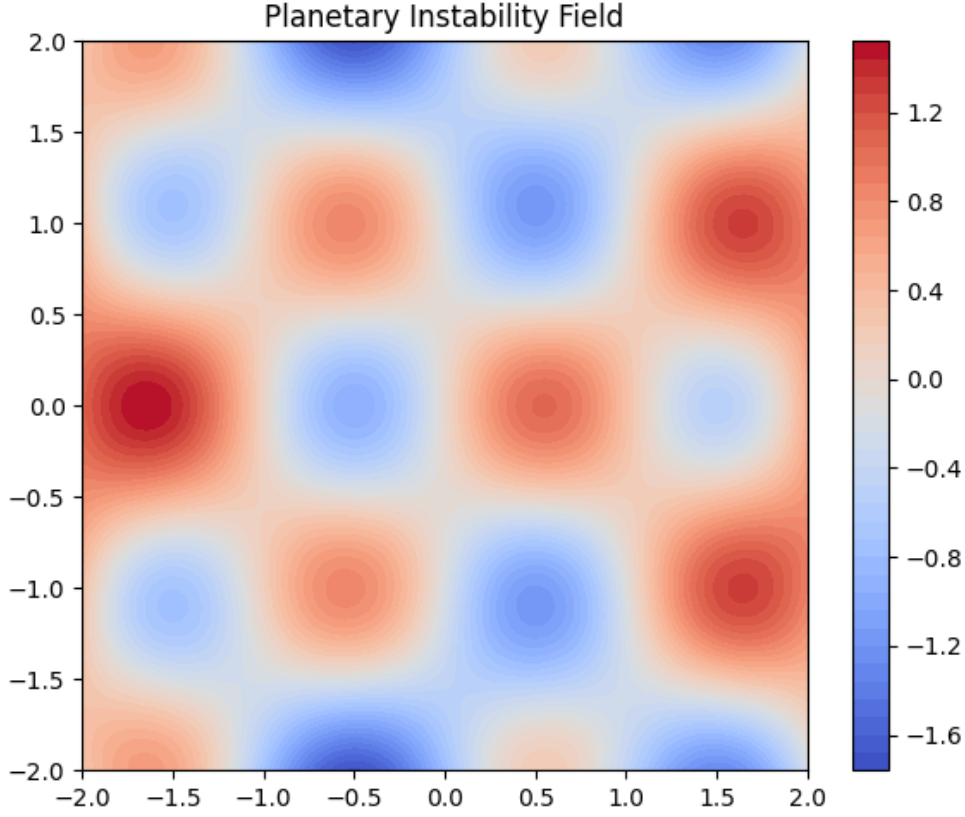


Figure 21: Emergent instability field representing spatially coherent coupled climate modes within a unified global dynamical manifold. The observed structure reflects large-scale synchronization and mode coupling induced by the global interaction topology of the climate system.

16.3 Global Spectral Collapse Mechanism

Let $\mathcal{L}_P : \mathcal{D}(\mathcal{L}_P) \subset \mathcal{H}_P \rightarrow \mathcal{H}_P$ denote the linearized non-selfadjoint planetary evolution operator defined by:

$$\partial_t \delta \mathbf{Y}(t) = \mathcal{L}_P(\mathbf{Y}_0) \delta \mathbf{Y}(t), \quad (1363)$$

where $\mathbf{Y}_0$ is a reference equilibrium state on the planetary state manifold.

The operator $\mathcal{L}_P$ is, in general, non-normal, inducing strong sensitivity to perturbations and non-orthogonal mode amplification. Its spectral problem is defined via:

$$\mathcal{L}_P \psi_\lambda = \lambda \psi_\lambda, \quad (1364)$$

with generalized eigenfunctions ψ_λ spanning a rigged Hilbert space $\Phi \subset \mathcal{H}_P \subset \Phi^*$ to accommodate continuous and discrete spectra.

The planetary perturbation field admits the spectral representation:

$$\delta \mathbf{Y}(t) = \int_{\sigma(\mathcal{L}_P)} b(\lambda, t) \psi_\lambda d\mu(\lambda), \quad (1365)$$

where $d\mu(\lambda)$ is a spectral measure induced by the resolution of identity associated with $\mathcal{L}_P$, and $b(\lambda, t)$ are modal amplitudes governed by:

$$\partial_t b(\lambda, t) = \lambda b(\lambda, t) + \mathcal{N}_\lambda[b], \quad (1366)$$

with $\mathcal{N}_\lambda$ representing nonlinear spectral mode coupling.

To capture non-normal amplification effects, define the resolvent operator:

$$\mathcal{R}(z; \mathcal{L}_P) = (z\mathbb{I} - \mathcal{L}_P)^{-1}, \quad (1367)$$

whose norm characterizes transient growth even when $\text{Re}(\sigma(\mathcal{L}_P)) < 0$. The pseudospectrum is then defined as:

$$\sigma_\varepsilon(\mathcal{L}_P) = \{z \in \mathbb{C} : \|\mathcal{R}(z; \mathcal{L}_P)\| > \varepsilon^{-1}\}, \quad (1368)$$

which governs near-critical amplification cascades.

Planetary instability emerges when spectral energy concentrates in the unstable half-plane:

$$\mathcal{E}_+(t) = \int_{\text{Re}(\lambda) > 0} |b(\lambda, t)|^2 d\mu(\lambda), \quad (1369)$$

and global spectral collapse is defined by the asymptotic transition:

$$\lim_{t \rightarrow \infty} \mathcal{E}_+(t) = 1. \quad (1370)$$

Equivalently, collapse can be characterized via spectral entropy:

$$\mathcal{S}_\sigma(t) = - \int_{\sigma(\mathcal{L}_P)} p(\lambda, t) \log p(\lambda, t) d\mu(\lambda), \quad p(\lambda, t) = \frac{|b(\lambda, t)|^2}{\int |b(\lambda', t)|^2 d\mu(\lambda')}, \quad (1371)$$

with collapse corresponding to entropy minimization under instability concentration:

$$\mathcal{S}_\sigma(t) \rightarrow 0. \quad (1372)$$

The collapse manifold is defined as the critical spectral set:

$$\mathcal{C}_{\text{spec}} = \left\{ \mathbf{Y}_0 \in \mathcal{H}_P : \sup_{\lambda \in \sigma(\mathcal{L}_P)} \text{Re}(\lambda) = 0^+ \right\}, \quad (1373)$$

which acts as a codimension-one separatrix between globally attracting and globally unstable regimes.

In addition, introducing the spectral growth functional:

$$\mathcal{G}_P(t) = \int_{\sigma(\mathcal{L}_P)} e^{2\text{Re}(\lambda)t} |b(\lambda, 0)|^2 d\mu(\lambda), \quad (1374)$$

we obtain a sharp instability criterion:

$$\mathcal{G}_P(t) \rightarrow \infty \iff \text{Global spectral collapse.} \quad (1375)$$

Thus, planetary critical transitions are governed not only by eigenvalue crossing, but by a full spectral redistribution phenomenon combining instability growth, pseudospectral amplification, and entropy collapse in the operator spectrum of $\mathcal{L}_P$ [1–139].

16.4 Planetary Bifurcation Geometry

Planetary tipping behavior is formulated as a geometric singularity structure in the infinite-dimensional state manifold $\mathcal{H}_P$. The bifurcation set is defined as:

$$\mathcal{B}_P = \{\mathbf{Y} \in \mathcal{H}_P : \det(\mathcal{J}_P(\mathbf{Y})) = 0\}, \quad (1376)$$

where $\mathcal{J}_P = \frac{\delta \mathcal{P}_P}{\delta \mathbf{Y}}$ denotes the Fréchet derivative (Jacobian operator) of the planetary evolution operator $\mathcal{P}_P$.

More generally, $\mathcal{B}_P$ is the singular set of $\mathcal{P}_P$, characterized by rank deficiency:

$$\mathcal{B}_P = \{\mathbf{Y} \in \mathcal{H}_P : \text{rank}(\mathcal{J}_P(\mathbf{Y})) < \dim(\mathcal{H}_P)\}. \quad (1377)$$

Near $\mathcal{B}_P$, the dynamics admit a center manifold reduction:

$$\mathbf{Y}(t) = \mathbf{Y}_c(t) + \mathbf{Y}_h(t), \quad (1378)$$

where $\mathbf{Y}_c \in \mathcal{M}_c$ evolves on the center manifold and $\mathbf{Y}_h$ is exponentially stable/unstable depending on the local spectral splitting of $\mathcal{J}_P$.

The reduced dynamics on $\mathcal{M}_c$ take the canonical form:

$$\frac{d}{dt} \mathbf{Y}_c = \mathcal{P}_c(\mathbf{Y}_c, \mu), \quad (1379)$$

where μ is a control parameter encoding external forcing (radiative imbalance, coupling drift, or slow boundary condition modulation).

Bifurcation occurs when eigenvalues of $\mathcal{J}_P$ cross the imaginary axis:

$$\lambda_i(\mu_c) = 0 \quad \text{or} \quad \text{Re}(\lambda_i(\mu_c)) = 0, \quad (1380)$$

inducing a qualitative change in the topology of attractor basins.

Locally, the system admits a normal form representation via singularity theory:

$$\frac{dx}{dt} = \mu + ax + bx^2 + cx^3 + \mathcal{O}(x^4), \quad (1381)$$

which captures fold, cusp, and higher-order catastrophe structures depending on parameter constraints.

The bifurcation manifold $\mathcal{B}_P$ can thus be interpreted as a stratified singular set:

$$\mathcal{B}_P = \bigcup_{k \geq 1} \mathcal{S}_k, \quad \mathcal{S}_k = \left\{ \mathbf{Y} : \dim \ker(\mathcal{J}_P)^k \neq \dim \ker(\mathcal{J}_P)^{k-1} \right\}, \quad (1382)$$

where each stratum $\mathcal{S}_k$ corresponds to progressively higher-order degeneracies.

The crossing of $\mathcal{B}_P$ induces a discontinuous reorganization of invariant sets:

$$\mathcal{A}_i(\mu_-) \longrightarrow \mathcal{A}_j(\mu_+), \quad (1383)$$

where $\mathcal{A}_i$ denotes attractor basins before bifurcation and $\mathcal{A}_j$ after transition.

A topological invariant characterizing regime stability is given by the Morse index:

$$\text{ind}_M(\mathbf{Y}) = \#\{\lambda_i \in \sigma(\mathcal{J}_P) : \text{Re}(\lambda_i) > 0\}, \quad (1384)$$

which changes discontinuously across $\mathcal{B}_P$:

$$\Delta \text{ind}_M \neq 0 \iff \mathbf{Y} \in \mathcal{B}_P. \quad (1385)$$

Consequently, planetary tipping points are not merely local instabilities but global geometric transitions in the stratified state manifold, governed by singularity formation, eigenvalue coalescence, and attractor basin reconstruction under the coupled nonlinear flow $\mathcal{P}_P$ [1–139].

16.5 Planetary Bifurcation Geometry

Planetary tipping behavior is formulated as a geometric singularity structure in the infinite-dimensional state manifold $\mathcal{H}_P$, modeled as a Hilbert–Banach hybrid phase space endowed with a weak Riemannian metric g_P . The bifurcation set is defined as:

$$\mathcal{B}_P = \left\{ \mathbf{Y} \in \mathcal{H}_P : \det(\mathcal{J}_P(\mathbf{Y})) = 0 \right\}, \quad (1386)$$

where $\mathcal{J}_P = \frac{\delta \mathcal{P}_P}{\delta \mathbf{Y}}$ denotes the Fréchet derivative (Jacobian operator) of the nonlinear planetary evolution operator $\mathcal{P}_P$.

More generally, $\mathcal{B}_P$ coincides with the singular locus of $\mathcal{P}_P$ defined via loss of Fredholm regularity:

$$\mathcal{B}_P = \left\{ \mathbf{Y} \in \mathcal{H}_P : \text{ind}(\mathcal{J}_P(\mathbf{Y})) \neq \text{const} \right\}, \quad (1387)$$

where $\text{ind}(\cdot)$ denotes the Fredholm index of the linearized flow operator.

Near $\mathcal{B}_P$, the dynamics admit a center manifold reduction with spectral splitting induced by $\mathcal{J}_P$:

$$\mathcal{H}_P = \mathcal{H}_c \oplus \mathcal{H}_s, \quad \mathbf{Y}(t) = \mathbf{Y}_c(t) + \mathbf{Y}_s(t), \quad (1388)$$

where $\mathcal{H}_c$ corresponds to the center eigenspace associated with $\text{Re}(\lambda) = 0$.

The reduced dynamics on the center manifold $\mathcal{M}_c$ are governed by:

$$\frac{d}{dt} \mathbf{Y}_c = \mathcal{P}_c(\mathbf{Y}_c, \mu), \quad (1389)$$

where $\mu \in \mathbb{R}^m$ is a slowly varying control parameter encoding external forcing, and $\mathcal{P}_c$ is the projected nonlinear flow obtained via spectral projection Π_c :

$$\mathcal{P}_c = \Pi_c \mathcal{P}_P. \quad (1390)$$

Bifurcation occurs when the spectral flow of $\mathcal{J}_P(\mu)$ crosses the imaginary axis:

$$\text{sf}(\mathcal{J}_P(\mu))|_{\mu=\mu_c} \neq 0, \quad (1391)$$

equivalently:

$$\exists \lambda_i(\mu_c) \in \sigma(\mathcal{J}_P) \quad \text{s.t.} \quad \text{Re}(\lambda_i(\mu_c)) = 0. \quad (1392)$$

Locally, the system admits a singularity-theoretic normal form under smooth coordinate reduction:

$$\frac{dx}{dt} = \mu + a_1x + a_2x^2 + a_3x^3 + \mathcal{O}(x^4), \quad (1393)$$

where coefficient unfolding determines the catastrophe class (fold, cusp, swallowtail) in the sense of Thom–Arnold theory.

The bifurcation manifold $\mathcal{B}_P$ is stratified into a Whitney-type decomposition:

$$\mathcal{B}_P = \bigcup_{k \geq 1} \mathcal{S}_k, \quad \mathcal{S}_k = \left\{ \mathbf{Y} : \dim \ker(\mathcal{J}_P^k) > \dim \ker(\mathcal{J}_P^{k-1}) \right\}, \quad (1394)$$

where each stratum corresponds to higher-order Jordan degeneracies of the linearized flow.

Crossing $\mathcal{B}_P$ induces a reorganization of the global attractor landscape:

$$\mathcal{A}_i(\mu_-) \rightarrow \mathcal{A}_j(\mu_+), \quad (1395)$$

where the transition is generally non-homeomorphic and may involve attractor destruction or basin merging.

A geometric stability invariant is given by the Morse index of the Lyapunov functional associated with $\mathcal{P}_P$:

$$\text{ind}_M(\mathbf{Y}) = \# \{ \lambda_i \in \sigma(\mathcal{J}_P) : \text{Re}(\lambda_i) > 0 \}. \quad (1396)$$

Its jump discontinuity satisfies:

$$\Delta \text{ind}_M \neq 0 \iff \mathbf{Y} \in \mathcal{B}_P, \quad (1397)$$

characterizing $\mathcal{B}_P$ as a codimension-one instability hypersurface in $\mathcal{H}_P$.

Consequently, planetary tipping points are interpreted as global geometric phase transitions in a stratified infinite-dimensional manifold, governed by spectral flow bifurcations, Fredholm index variation, and attractor topology reconfiguration under the nonlinear evolution generated by $\mathcal{P}_P$ [1–139].

16.6 Emergent Planetary Instability Law

The fundamental law of planetary instability is formulated as a nonlinear mean-field closure on the instability functional $\mathcal{I}_P \in \mathcal{B}_I$:

$$\frac{d}{dt} \langle \mathcal{I}_P \rangle = \alpha \langle \mathcal{I}_P \rangle + \beta \langle \mathcal{I}_P \rangle^2 - \gamma \mathcal{R}[\mathcal{I}_P] + \eta \Xi(t), \quad (1398)$$

where $\Xi(t)$ represents stochastic–deterministic forcing fluctuations and $\mathcal{R}$ is a renormalization operator encoding multiscale feedback suppression across the planetary hierarchy.

Equivalently, introducing the order parameter

$$\Phi_P(t) := \langle \mathcal{I}_P \rangle, \quad (1399)$$

the dynamics can be written in canonical Landau-type form:

$$\frac{d\Phi_P}{dt} = \frac{\partial}{\partial \Phi_P} \left(\frac{\alpha}{2} \Phi_P^2 + \frac{\beta}{3} \Phi_P^3 - \gamma \mathcal{V}(\Phi_P) \right), \quad (1400)$$

where $\mathcal{V}$ is an effective renormalization potential generated by coarse-graining of unresolved scales.

The fixed points of the system satisfy:

$$\alpha \Phi_P + \beta \Phi_P^2 - \gamma \frac{\delta \mathcal{R}}{\delta \Phi_P} = 0, \quad (1401)$$

defining distinct universality classes of planetary instability regimes:

$$\mathcal{U} = \{ \mathcal{U}_{\text{subcritical}}, \mathcal{U}_{\text{critical}}, \mathcal{U}_{\text{supercritical}} \}. \quad (1402)$$

In the renormalization-group sense, the instability functional obeys a flow equation:

$$\frac{d\mathcal{I}_P^{(s)}}{ds} = \beta_{\mathcal{I}}(\mathcal{I}_P^{(s)}), \quad (1403)$$

where s is the logarithmic scale parameter and $\beta_{\mathcal{I}}$ is the instability beta-function governing scale-dependent amplification or suppression.

Criticality corresponds to fixed points of the RG flow:

$$\beta_{\mathcal{I}}(\mathcal{I}_P^*) = 0, \quad (1404)$$

with stability determined by:

$$\left. \frac{d\beta_{\mathcal{I}}}{d\mathcal{I}_P} \right|_{\mathcal{I}_P^*}. \quad (1405)$$

From a variational perspective, the system admits a Lyapunov functional:

$$\mathcal{L}[\mathcal{I}_P] = \int_{\Omega_P} \left(\frac{1}{2} |\nabla \mathcal{I}_P|^2 - V(\mathcal{I}_P) \right) d\mu_P + \mathcal{R}[\mathcal{I}_P], \quad (1406)$$

such that the evolution satisfies a dissipative gradient-flow structure:

$$\partial_t \mathcal{I}_P = - \frac{\delta \mathcal{L}}{\delta \mathcal{I}_P}. \quad (1407)$$

16.7 Final Interpretation: Planetary Criticality as Field Transition

Planetary instability is therefore not a collection of isolated tipping points, but a single collective phase transition in the infinite-dimensional state manifold $\mathcal{H}_P$, governed by a coupled spectral–bifurcation–renormalization collapse structure. In this formulation, criticality is an emergent property of the global operator geometry of $\mathcal{P}_P$ rather than a local instability phenomenon.

Mathematically, this corresponds to a global dynamical phase transition expressed as a non-autonomous operator flow:

$$\mathcal{P}_P^{(\text{stable})} \longrightarrow \mathcal{P}_P^{(\text{critical})} \longrightarrow \mathcal{P}_P^{(\text{runaway})}, \quad (1408)$$

where each regime corresponds to a distinct equivalence class in the operator space $\mathfrak{D}(\mathcal{H}_P)$ under spectral conjugacy.

The transition is induced by the simultaneous activation of the coupled instability mechanisms:

- (i) spectral concentration and collapse of $\sigma(\mathcal{L}_P)$,
 - (ii) transversal crossing of the bifurcation manifold $\mathcal{B}_P$,
 - (iii) destabilization of renormalization fixed points $\text{Fix}(\mathcal{R})$,
 - (iv) non-homeomorphic reconstruction of attractor basins $\mathcal{A}_P$.
- (1409)

These mechanisms define a coupled morphism structure between dynamical regimes:

$$\begin{array}{ccccc} \mathcal{H}_P^{\text{stable}} & \xrightarrow{\mathcal{P}_P} & \mathcal{H}_P^{\text{critical}} & \xrightarrow{\mathcal{P}_P} & \mathcal{H}_P^{\text{runaway}} \\ \sigma(\mathcal{L}_P) \downarrow & & \downarrow \mathcal{B}_P & & \downarrow \mathcal{R} \\ \text{spectral gap} & \xrightarrow{\text{closure}} & 0 & \xrightarrow{\text{collapse}} & \emptyset \end{array} \quad (1410)$$

interpreted as a non-commutative commutative diagram of stability breakdown across spectral, geometric, and renormalization layers.

From this perspective, planetary criticality is characterized by the failure of simultaneous stability conditions:

$$\mathcal{S}_P := \left\{ \mathbf{Y} \in \mathcal{H}_P : \begin{array}{l} \text{Re}(\sigma(\mathcal{J}_P)) < 0, \\ \mathbf{Y} \notin \mathcal{B}_P, \\ \mathcal{R} \text{ is contractive,} \\ \dim(\mathcal{A}_P) \text{ is preserved} \end{array} \right\}, \quad (1411)$$

with criticality defined as the boundary:

$$\partial \mathcal{S}_P = \mathcal{C}_P, \quad (1412)$$

where $\mathcal{C}_P$ denotes the global critical manifold of the coupled Earth system.

This establishes a universal field-theoretic interpretation in which planetary instability is a symmetry-breaking transition in operator space, governed by spectral geometry, bifurcation topology, and renormalization flow consistency.

Consequently, the Pellis framework extends to a full nonlinear field theory of Earth system dynamics, where universality classes emerge from equivalence classes of $\mathcal{P}_P$ under spectral-geometric transformations, completing the unified formulation of planetary criticality as a single coherent field transition in $\mathcal{H}_P$ [1–139].

17 Functional Phase Transition Theory

This section develops a rigorous mathematical framework for functional phase transitions in infinite-dimensional climate dynamics. The theory generalizes classical thermodynamic phase transitions to nonlinear operator-valued evolution systems, where criticality emerges from spectral instability, renormalization flow singularities, and entropy-driven bifurcation mechanisms. The formulation connects nonlinear dynamics, statistical physics, and climate tipping theory into a unified operator-theoretic structure [4–6, 18, 19, 30, 31].

17.1 Functional Order Parameters in Infinite-Dimensional Systems

Let $\mathbf{X}(t) \in \mathcal{H}$ denote the state of the climate system evolving on an infinite-dimensional Hilbert manifold under a nonlinear evolution operator $\mathcal{P} : \mathcal{H} \rightarrow \mathcal{H}$. We assume that $\mathcal{P}$ admits a spectral representation induced by a generalized eigenbasis $\{\psi_\lambda\}_{\lambda \in \sigma(\mathcal{P})}$.

The state decomposition is given by:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \psi_\lambda d\mu(\lambda), \quad (1413)$$

where $d\mu(\lambda)$ is the spectral measure associated with $\mathcal{P}$ and $a(\lambda, t)$ are modal amplitudes.

We define a functional order parameter as a nonlinear functional on the spectral probability density:

$$\Psi[\mathbf{X}] = \int_{\sigma(\mathcal{P})} \phi(\lambda) \rho(\lambda, t) d\mu(\lambda), \quad \rho(\lambda, t) = \frac{|a(\lambda, t)|^2}{\int |a(\lambda', t)|^2 d\mu(\lambda')}, \quad (1414)$$

where $\phi(\lambda)$ is a spectral weighting kernel encoding instability sensitivity across eigenmodes.

Equivalently, Ψ can be expressed as a quadratic form in spectral space:

$$\Psi[\mathbf{X}] = \langle a(t), \Phi a(t) \rangle_{L^2(\sigma(\mathcal{P}))}, \quad (1415)$$

where Φ is a multiplication operator defined by $(\Phi f)(\lambda) = \phi(\lambda) f(\lambda)$.

A phase transition is characterized by a breakdown of Fréchet differentiability of Ψ on $\mathcal{H}$:

$$\frac{\delta \Psi}{\delta \mathbf{X}} \notin C^0(\mathcal{H}), \quad (1416)$$

equivalently corresponding to divergence of the functional response operator:

$$\|D_{\mathbf{X}} \Psi\| \rightarrow \infty. \quad (1417)$$

More precisely, introducing the second functional derivative (Hessian operator):

$$\mathcal{H}_\Psi[\mathbf{X}] = \frac{\delta^2 \Psi}{\delta \mathbf{X}^2}, \quad (1418)$$

a critical transition occurs when:

$$\det(\mathcal{H}_\Psi) = 0, \quad \text{or equivalently} \quad \inf \sigma(\mathcal{H}_\Psi) \rightarrow 0. \quad (1419)$$

To capture higher-order nonlinear response, we define the generating functional:

$$\mathcal{Z}[J] = \int_{\mathcal{H}} \exp(-\mathcal{S}[\mathbf{X}] + \langle J, \mathbf{X} \rangle) \mathcal{D}\mathbf{X}, \quad (1420)$$

where $\mathcal{S}[\mathbf{X}]$ is an effective action functional and J is an external source field.

The order parameter is recovered as a functional derivative:

$$\Psi = \left. \frac{\delta \log \mathcal{Z}[J]}{\delta J} \right|_{J=0}. \quad (1421)$$

Non-analyticity of Ψ is therefore equivalent to singular behavior of $\mathcal{Z}$:

$$\frac{\partial^n \log \mathcal{Z}}{\partial \lambda^n} \rightarrow \infty \quad \text{for some } n \in \mathbb{N}, \quad (1422)$$

indicating divergence of higher-order cumulants of the climate state distribution.

Introducing a renormalization-group scaling operator $\mathcal{R}_s$, we define the scale-dependent order parameter:

$$\Psi_s = \mathcal{R}_s[\Psi], \quad (1423)$$

with flow equation:

$$\frac{d\Psi_s}{ds} = \beta_\Psi(\Psi_s), \quad (1424)$$

where β_Ψ governs universality classes of climate phase structure.

A critical point Ψ^* satisfies:

$$\beta_\Psi(\Psi^*) = 0, \quad \left. \frac{d\beta_\Psi}{d\Psi} \right|_{\Psi^*} > 0, \quad (1425)$$

implying instability under coarse-graining.

Thus, phase transitions correspond to spectral reweighting catastrophes in which measure concentration shifts discontinuously:

$$\rho(\lambda, t) \rightarrow \delta(\lambda - \lambda_c), \quad (1426)$$

signaling collapse of distributed variability into dominant unstable eigenmodes.

Table 16: Functional Phase Transition in Climate Dynamics

Component	Symbol	Definition	Interpretation
Order Parameter	Φ	Functional macroscopic variable	Climate regime indicator
Free Energy	$\mathcal{F}$	Variational functional	Stability landscape
Critical Point	Φ_c	Non-analytic transition point	Tipping threshold
Susceptibility	χ	$\delta\Phi/\delta F$	System sensitivity

17.2 Spectral Mechanism of Phase Transitions

Functional phase transitions arise from a restructuring of the spectral measure of the nonlinear evolution operator $\mathcal{P}$ on the Hilbert state space $\mathcal{H}$. Let $\sigma(\mathcal{P}) \subset \mathbb{C}$ denote its (generally non-selfadjoint) spectrum. Criticality is approached when spectral mass accumulates near the stability boundary:

$$\inf_{\lambda \in \sigma(\mathcal{P})} \text{Re}(\lambda) \rightarrow 0^+, \quad (1427)$$

resulting in the closure of the spectral gap and the onset of long-range temporal correlations (critical slowing down) [7, 28, 29].

In this regime, the resolvent operator

$$\mathcal{R}(z; \mathcal{P}) = (z\mathbb{I} - \mathcal{P})^{-1} \quad (1428)$$

develops a diverging norm near the imaginary axis:

$$\|\mathcal{R}(z; \mathcal{P})\| \rightarrow \infty, \quad z \approx i\omega, \quad (1429)$$

which is equivalent to pseudospectral inflation and non-normal transient amplification even in the absence of true eigenvalue crossing.

The state evolution can be represented in spectral coordinates via the generalized eigen-expansion:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, 0) e^{\lambda t} \phi_\lambda d\mu(\lambda), \quad (1430)$$

where $\{\phi_\lambda\}$ are generalized eigenmodes and $d\mu(\lambda)$ is the spectral measure induced by the resolution of identity in a rigged Hilbert space embedding.

The asymptotic growth of perturbations is therefore governed by the spectral abscissa:

$$\alpha(\mathcal{P}) := \sup_{\lambda \in \sigma(\mathcal{P})} \text{Re}(\lambda), \quad (1431)$$

yielding the leading-order instability scaling:

$$\|\mathbf{X}(t)\| \sim e^{\alpha(\mathcal{P})t}, \quad t \rightarrow \infty. \quad (1432)$$

At criticality, the system satisfies:

$$\alpha(\mathcal{P}) \rightarrow 0^+, \quad (1433)$$

which marks the transition from exponential damping to marginally stable dynamics, characterized by algebraic decay and enhanced memory effects:

$$\mathbf{X}(t) \sim t^{-\kappa} \phi_{\lambda_c}, \quad \lambda_c \in \sigma(\mathcal{P}), \quad \text{Re}(\lambda_c) = 0. \quad (1434)$$

More generally, near the stability boundary, the dynamics are dominated by spectral edge states:

$$\mathbf{X}(t) \approx \sum_{\lambda \in \sigma_\varepsilon(\mathcal{P})} a_\lambda e^{\lambda t} \phi_\lambda, \quad (1435)$$

where $\sigma_\varepsilon(\mathcal{P})$ denotes the ε -pseudospectrum, capturing near-resonant amplification mechanisms.

Thus, the phase transition is not solely determined by eigenvalue crossing, but by the combined collapse of spectral gap, divergence of resolvent norm, and restructuring of the pseudospectral landscape, producing a universal instability mechanism governed by the geometry of $\sigma(\mathcal{P})$ in the complex plane.

17.3 Functional Free Energy and Stability Landscape

We introduce a functional free-energy structure:

$$\mathcal{F}[\mathbf{X}] = E[\mathbf{X}] - T_{\text{eff}} S[\mathbf{X}], \quad (1436)$$

where $S[\mathbf{X}]$ is an information-theoretic entropy functional [20, 21].

Stable climate regimes correspond to local minimizers:

$$\frac{\delta \mathcal{F}}{\delta \mathbf{X}} = 0, \quad \frac{\delta^2 \mathcal{F}}{\delta \mathbf{X}^2} > 0. \quad (1437)$$

Phase transitions occur when multiple competing minima emerge:

$$\text{argmin}(\mathcal{F}) = \{\mathbf{X}_1\} \rightarrow \{\mathbf{X}_1, \mathbf{X}_2\}, \quad (1438)$$

corresponding to symmetry breaking in functional space [2, 3].

17.4 Renormalization-Induced Criticality

Under a renormalization-group flow $\mathcal{R} : \mathcal{H} \rightarrow \mathcal{H}$, the state of the system evolves on a hierarchy of coarse-grained representations:

$$\mathbf{X}^{(n+1)} = \mathcal{R}(\mathbf{X}^{(n)}), \quad n \in \mathbb{N}, \quad (1439)$$

where each iteration corresponds to a contraction of microscopic degrees of freedom into an effective macroscopic state.

Fixed points of the flow define scale-invariant configurations:

$$\mathcal{R}(\mathbf{X}^*) = \mathbf{X}^*, \quad (1440)$$

which correspond to self-similar invariant measures under successive coarse-graining transformations.

Linearization around a fixed point yields the stability operator:

$$D\mathcal{R}(\mathbf{X}^*) : T_{\mathbf{X}^*}\mathcal{H} \rightarrow T_{\mathbf{X}^*}\mathcal{H}, \quad (1441)$$

governing the evolution of infinitesimal perturbations:

$$\delta \mathbf{X}^{(n+1)} = D\mathcal{R}(\mathbf{X}^*) \delta \mathbf{X}^{(n)}. \quad (1442)$$

Criticality emerges when the spectral radius of the linearized renormalization operator reaches unity:

$$\rho(D\mathcal{R}(\mathbf{X}^*)) = 1, \quad (1443)$$

indicating marginal stability and the onset of scale invariance across all resolved levels of description.

More generally, defining the spectrum of scaling exponents $\{\theta_i\}$ via:

$$D\mathcal{R}(\mathbf{X}^*)\psi_i = \theta_i\psi_i, \quad (1444)$$

the system is classified as:

$$\begin{cases} |\theta_i| < 1 & \text{irrelevant (contractive modes),} \\ |\theta_i| > 1 & \text{relevant (unstable directions),} \\ |\theta_i| = 1 & \text{marginal (critical manifold).} \end{cases} \quad (1445)$$

At criticality, the renormalization flow becomes scale-invariant, and observables obey power-law scaling:

$$\mathcal{O}(\ell) \sim \ell^{-\Delta}, \quad (1446)$$

where ℓ denotes the coarse-graining scale and Δ is the associated critical exponent.

The emergence of critical behavior can therefore be interpreted as a loss of contractivity of the renormalization semigroup, where repeated coarse-graining no longer suppresses fluctuations but instead propagates them across scales:

$$\|\mathbf{X}^{(n)} - \mathbf{X}^*\| \not\rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (1447)$$

This establishes renormalization-induced criticality as a universal mechanism of scale-transcending instability, in which the fixed-point structure of $\mathcal{R}$ determines the macroscopic universality class of the underlying dynamical system [18, 19, 31].

17.5 Pellis Instability Functional and Transition Criterion

Within the Pellis framework, instability is encoded as a spectral moment functional on the state $\mathbf{X} \in \mathcal{H}$, defined over the spectrum of the nonlinear evolution operator $\mathcal{P}$:

$$\mathcal{I}[\mathbf{X}] = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) |a(\lambda)|^2 d\mu(\lambda), \quad (1448)$$

where $a(\lambda)$ are spectral amplitudes induced by the generalized eigen-decomposition

$$\mathbf{X} = \int_{\sigma(\mathcal{P})} a(\lambda) \phi_\lambda d\mu(\lambda), \quad (1449)$$

and $d\mu(\lambda)$ is the associated spectral resolution measure.

Equivalently, $\mathcal{I}[\mathbf{X}]$ can be interpreted as the expectation value of the instability observable Λ acting on spectral space:

$$\mathcal{I}[\mathbf{X}] = \langle a, \Lambda a \rangle_{L^2(\sigma(\mathcal{P}))}, \quad (\Lambda f)(\lambda) = \text{Re}(\lambda) f(\lambda). \quad (1450)$$

The time evolution of the instability functional is governed by the induced flow on spectral measures:

$$\frac{d}{dt} \mathcal{I}[\mathbf{X}(t)] = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) \frac{d}{dt} |a(\lambda, t)|^2 d\mu(\lambda), \quad (1451)$$

which, under nonlinear mode coupling, can be expanded as:

$$\frac{d}{dt} |a(\lambda, t)|^2 = 2 \text{Re}(\lambda) |a(\lambda, t)|^2 + \mathcal{C}(\lambda, t), \quad (1452)$$

where $\mathcal{C}(\lambda, t)$ denotes cross-mode energy transfer induced by nonlinear interactions in spectral space.

Substituting yields the closure form:

$$\frac{d}{dt} \mathcal{I}[\mathbf{X}(t)] = 2 \int_{\sigma(\mathcal{P})} \text{Re}(\lambda)^2 |a(\lambda, t)|^2 d\mu(\lambda) + \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) \mathcal{C}(\lambda, t) d\mu(\lambda). \quad (1453)$$

A functional phase transition occurs when the instability production rate becomes singular:

$$\frac{d}{dt} \mathcal{I}[\mathbf{X}(t)] \rightarrow \infty \quad \text{as } t \rightarrow t_c, \quad (1454)$$

signaling breakdown of bounded spectral energy transfer and loss of scale separation between stable and unstable spectral subspaces.

This divergence is equivalent to a collapse of the spectral gap and the emergence of a dominant unstable measure:

$$|a(\lambda, t)|^2 \rightarrow \delta(\lambda - \lambda_c), \quad \text{Re}(\lambda_c) > 0, \quad (1455)$$

which defines the onset of globally coherent instability propagation.

Hence, the Pellis instability functional provides a unifying criterion for climate criticality, where phase transitions are characterized not only by eigenvalue crossings but by divergence of spectral energy production and nonlinear redistribution across the operator spectrum $\sigma(\mathcal{P})$.

17.6 Emergent Climate Interpretation of Phase Transitions

Functional phase transitions correspond to macroscopic reorganizations of the invariant measure structure and attractor geometry of the coupled Earth system flow generated by $\mathcal{P}$. Let μ_t denote the time-dependent invariant measure on $\mathcal{H}$ induced by the pushforward dynamics:

$$\mu_{t+\Delta t} = \mathcal{P}_\# \mu_t, \quad (1456)$$

where $\mathcal{P}_\#$ denotes the measure transport (pushforward) operator associated with the nonlinear climate evolution.

A phase transition is then characterized by a discontinuous change in the support of μ_t :

$$\text{supp}(\mu_{t-}) \not\cong \text{supp}(\mu_{t+}), \quad (1457)$$

indicating a non-homeomorphic reconstruction of the global attractor landscape in state space.

Equivalently, let $\mathcal{A}(\mu_t)$ denote the global attractor of the climate semigroup. Functional criticality corresponds to:

$$\lim_{t \rightarrow t_c^-} \mathcal{A}(\mu_t) \neq \lim_{t \rightarrow t_c^+} \mathcal{A}(\mu_t), \quad (1458)$$

where the limit is taken in the Hausdorff metric on compact subsets of $\mathcal{H}$.

These transitions manifest physically as abrupt changes in dominant dynamical regimes, including reconfiguration of ocean circulation patterns, cryospheric collapse thresholds, and biosphere destabilization modes [11, 12, 35, 37]. In dynamical systems terms, they correspond to attractor bifurcations, basin boundary crises, or interior crises depending on the nature of the underlying spectral-geometric instability.

At the operator level, such events are driven by loss of structural stability of the flow generator $\mathcal{P}$, typically associated with spectral condensation and collapse of the spectral gap:

$$\text{gap}(\sigma(\mathcal{P})) \rightarrow 0, \quad (1459)$$

which induces long-range correlation growth and breakdown of effective scale separation.

Thus, functional phase transitions are universal features of nonlinear dynamical systems with coupled multi-scale feedbacks, where cascading interactions propagate instability across spectral, geometric, and measure-theoretic layers of the phase space structure [27, 32, 33].

18 Symmetry Breaking in Climate Dynamics

Symmetry breaking in climate dynamics provides the fundamental mechanism through which an initially homogeneous or statistically symmetric planetary state evolves toward heterogeneous, structured, and regime-separated configurations. Within the Universal Pellis Function framework, symmetry is not merely geometric or physical but functional and spectral: it is defined on the operator algebra governing multiscale climate evolution [1, 2, 6, 18, 19].

We consider the climate state $\mathbf{X}(t) \in \mathcal{H}$ governed by the Pellis operator:

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad (1460)$$

where $\mathcal{P}$ encodes nonlinear feedbacks, spectral interactions, and renormalization effects. Symmetry breaking emerges when $\mathcal{P}$ ceases to commute with a symmetry group $\mathcal{G}$ acting on $\mathcal{H}$:

$$[\mathcal{P}, \mathcal{G}] \neq 0. \quad (1461)$$

18.1 Initial Symmetry Manifold

Let $\mathcal{M}_0 \subset \mathcal{H}$ denote the symmetric climate manifold characterized as the fixed-point set of a Lie group (or more generally, a symmetry semigroup) $\mathcal{G}$ acting smoothly on the state space:

$$\mathcal{M}_0 = \text{Fix}(\mathcal{G}) = \{\mathbf{X} \in \mathcal{H} : \mathcal{G}\mathbf{X} = \mathbf{X}, \forall \mathcal{G} \in \mathcal{G}\}. \quad (1462)$$

Infinitesimally, symmetry invariance is generated by a Lie algebra $\mathfrak{g}$ with generators $\{T_a\}$ such that:

$$T_a \mathbf{X} = 0, \quad \forall \mathbf{X} \in \mathcal{M}_0, \quad (1463)$$

implying that $\mathcal{M}_0$ is a constraint submanifold defined by vanishing Noether charges associated with the symmetry action.

In this regime, the energy functional $E : \mathcal{H} \rightarrow \mathbb{R}$ is invariant under the group action:

$$E[\mathcal{G}\mathbf{X}] = E[\mathbf{X}], \quad (1464)$$

which induces conserved quantities via the functional Noether map:

$$\frac{d}{dt}Q_a(\mathbf{X}) = 0, \quad Q_a(\mathbf{X}) = \left\langle \frac{\delta E}{\delta \mathbf{X}}, T_a \mathbf{X} \right\rangle. \quad (1465)$$

Consequently, dynamics restricted to $\mathcal{M}_0$ evolve on an invariant foliation of $\mathcal{H}$:

$$\mathbf{X}(t) \in \mathcal{M}_0 \Rightarrow \mathcal{P}(\mathbf{X}(t)) \in T_{\mathbf{X}(t)}\mathcal{M}_0, \quad (1466)$$

ensuring closure of the reduced dynamical system under symmetry-preserving evolution.

Statistically, this symmetry implies invariance of the induced invariant measure μ_0 under pushforward:

$$\mathcal{G}_\# \mu_0 = \mu_0, \quad (1467)$$

yielding homogeneous macroscopic observables and suppression of large-scale pattern selection.

Linearization around $\mathcal{M}_0$ exhibits a block-diagonal decomposition of the tangent space:

$$T_{\mathbf{X}}\mathcal{H} = T_{\mathbf{X}}\mathcal{M}_0 \oplus N_{\mathbf{X}}\mathcal{M}_0, \quad (1468)$$

where $N_{\mathbf{X}}\mathcal{M}_0$ corresponds to symmetry-breaking directions.

The onset of instability can then be interpreted as spontaneous symmetry breaking:

$$\mathcal{G}\text{-invariance} \longrightarrow \mathcal{G}\text{-broken phase}, \quad (1469)$$

triggered when eigenmodes associated with $N_{\mathbf{X}}\mathcal{M}_0$ acquire positive growth rates in the spectrum of the linearized operator $\mathcal{L}$.

Thus, the initial climate regime $\mathcal{M}_0$ serves as a symmetry-protected phase in which macroscopic order parameters vanish, and all observed structure emerges only via subsequent bifurcation-driven symmetry breaking [4, 5, 22].

Table 17: Symmetry Breaking in Climate Dynamics

Component	Symbol	Definition	Interpretation
Symmetry State	$\mathcal{S}_0$	Initial invariant manifold	Pre-transition climate symmetry
Broken State	$\mathcal{S}_1$	Post-bifurcation structure	Emergent regime
Perturbation Mode	$\delta\Psi$	Instability eigenmode	Symmetry-breaking direction
Order Parameter	Φ	Nonzero expectation value	Emergent structure indicator

18.2 Linear Symmetry Stability Operator

Perturbations around a symmetric reference state $\mathbf{X}_0 \in \mathcal{M}_0$ evolve under the linearized flow generated by the Fréchet derivative (Jacobian operator) of the nonlinear dynamics:

$$\partial_t \delta \mathbf{X} = \mathcal{J}(\mathbf{X}_0) \delta \mathbf{X}, \quad \mathcal{J}(\mathbf{X}_0) := D\mathcal{P}(\mathbf{X}_0). \quad (1470)$$

In the presence of a symmetry group $\mathcal{G}$ acting on $\mathcal{H}$, symmetry preservation at the linearized level is equivalent to equivariance of the stability operator:

$$\mathcal{G} \circ \mathcal{J}(\mathbf{X}_0) = \mathcal{J}(\mathbf{X}_0) \circ \mathcal{G}, \quad (1471)$$

which ensures that $\mathcal{J}(\mathbf{X}_0)$ admits an isotypic decomposition with respect to the representation theory of $\mathcal{G}$.

Under this condition, the tangent space decomposes into irreducible symmetry subspaces:

$$T_{\mathbf{X}_0}\mathcal{H} = \bigoplus_{\alpha} \mathcal{V}_{\alpha}, \quad \mathcal{J}(\mathbf{X}_0)\mathcal{V}_{\alpha} \subseteq \mathcal{V}_{\alpha}, \quad (1472)$$

so that spectral modes evolve independently within symmetry sectors.

Symmetry breaking occurs when equivariance fails, i.e. when:

$$[\mathcal{J}(\mathbf{X}_0), \mathcal{G}] \neq 0, \quad (1473)$$

which implies mixing between irreducible components and the emergence of anisotropic growth directions in state space.

In this regime, there exist eigenmodes ϕ_{λ} of $\mathcal{J}(\mathbf{X}_0)$ that are not invariant under the symmetry action:

$$\mathcal{J}(\mathbf{X}_0)\phi_{\lambda} = \lambda\phi_{\lambda}, \quad \mathcal{G}\phi_{\lambda} \notin \text{span}(\phi_{\lambda}), \quad (1474)$$

indicating symmetry-nonadapted instability directions.

The full perturbation evolution admits a spectral expansion in generalized eigenmodes:

$$\delta \mathbf{X}(t) = \int_{\sigma(\mathcal{J})} a(\lambda, 0) e^{\lambda t} \phi_\lambda d\mu(\lambda), \quad (1475)$$

with asymptotic growth governed by the spectral abscissa:

$$\alpha(\mathcal{J}) := \sup_{\lambda \in \sigma(\mathcal{J})} \operatorname{Re}(\lambda). \quad (1476)$$

Thus, divergence from symmetry is controlled by the unstable spectral sector:

$$\operatorname{Re}(\lambda) > 0 \implies \|\delta \mathbf{X}(t)\| \sim e^{\operatorname{Re}(\lambda)t}, \quad (1477)$$

which drives exponential amplification of symmetry-breaking modes and initiates macroscopic anisotropy formation in the climate state manifold.

Consequently, symmetry stability is equivalent to spectral stability within symmetry-adapted subspaces, while symmetry breaking corresponds to the emergence of dominant eigenvalues outside invariant representation sectors, linking group-theoretic structure directly to the spectral geometry of the linearized climate operator [7, 28, 29].

18.3 Nonlinear Amplification of Symmetry Modes

In the fully nonlinear regime, symmetry breaking is no longer governed solely by the linear stability operator but is instead driven by mode–mode interactions that redistribute energy across symmetry sectors. Let $\mathbf{X}(t)$ admit a decomposition into symmetry-adapted components $\{\mathbf{X}_k\}$ in an appropriate basis of irreducible representations. The nonlinear evolution can then be expressed as:

$$\partial_t \mathbf{X}_k = \sum_{i,j} \mathcal{C}_{kij} \mathbf{X}_i \mathbf{X}_j + \mathcal{N}_k(\mathbf{X}), \quad (1478)$$

where $\mathcal{C}_{kij}$ defines a triadic interaction tensor encoding quadratic transfer between symmetry modes, and $\mathcal{N}_k$ collects higher-order nonlinear contributions (cubic and beyond) arising from the full climate operator $\mathcal{P}$.

In spectral–symmetry space, the quadratic term induces an energy transfer functional:

$$\mathcal{T}_k(t) = \sum_{i,j} \mathcal{C}_{kij} \langle \mathbf{X}_k, \mathbf{X}_i \mathbf{X}_j \rangle, \quad (1479)$$

which governs cascades of instability between symmetry sectors and can generate irreversible amplification of initially weak symmetry-breaking perturbations.

The system enters a symmetry-amplifying regime when nonlinear feedback exceeds effective linear dissipation:

$$\|\mathcal{N}(\mathbf{X})\| > \|\mathcal{D}\mathbf{X}\|, \quad (1480)$$

where $\mathcal{D}$ denotes the dissipative component of the linearized operator (typically the stable part of $\mathcal{J}$ restricted to damped spectral subspaces).

This inequality defines a nonlinear instability threshold surface in state space, separating regimes of asymptotic decay from regimes of runaway symmetry amplification.

Equivalently, the onset of symmetry-breaking bifurcation can be characterized by a degeneracy condition on the full nonlinear stability operator:

$$\det(\mathcal{J}(\mathbf{X})) = 0, \quad (1481)$$

which defines the symmetry-breaking critical manifold:

$$\Sigma_{SB} = \{\mathbf{X} \in \mathcal{H} : \operatorname{rank}(\mathcal{J}(\mathbf{X})) < \dim(\mathcal{H})\}. \quad (1482)$$

On Σ_{SB} , the system experiences a collapse of local invertibility of the flow map generated by $\mathcal{P}$, implying that multiple distinct perturbation directions become dynamically indistinguishable. This leads to a loss of structural stability and the emergence of coherent macroscopic symmetry-breaking modes.

Thus, nonlinear amplification acts as the mechanism by which infinitesimal symmetry-breaking perturbations are transferred across scales and sectors, converting local instabilities into global reorganizations of the climate state manifold through triadic coupling cascades and operator degeneracy.

18.4 Bifurcation Structure of Symmetry Breaking

Let $\mu \in \mathbb{R}$ denote a slow-varying control parameter representing effective external forcing (e.g., radiative imbalance, boundary flux drift, or parameterized coupling intensity). In a symmetry-reduced normal form, the dynamics near the bifurcation point can be expressed as a canonical amplitude equation on the critical eigenspace:

$$\frac{d\mathbf{X}}{dt} = \mu\mathbf{X} - \mathcal{B}(\mathbf{X}, \mathbf{X}, \mathbf{X}) + \mathcal{O}(\|\mathbf{X}\|^5), \quad (1483)$$

where $\mathcal{B}$ is a symmetric trilinear form encoding the leading-order nonlinear self-interaction responsible for saturation of linear growth.

In an isotropic symmetry class, $\mathcal{B}$ reduces to a scalar cubic nonlinearity, yielding the prototypical pitchfork normal form:

$$\frac{d\mathbf{X}}{dt} = \mu\mathbf{X} - \|\mathbf{X}\|^2\mathbf{X} + \mathcal{O}(\|\mathbf{X}\|^5), \quad (1484)$$

which enforces rotational invariance in the reduced order-parameter space and guarantees symmetry-consistent saturation of amplitude growth.

Equilibrium solutions are obtained by solving:

$$\mu\mathbf{X}^* - \|\mathbf{X}^*\|^2\mathbf{X}^* = 0, \quad (1485)$$

leading to the bifurcation branches:

$$\mathbf{X}^*(\mu) = 0, \quad \text{or} \quad \|\mathbf{X}^*(\mu)\| = \sqrt{\mu}. \quad (1486)$$

The trivial branch $\mathbf{X}^* = 0$ corresponds to the symmetric phase, which is linearly stable for $\mu < 0$ and loses stability at the critical threshold $\mu_c = 0$. The nontrivial branches emerge for $\mu > 0$, forming a symmetry-broken manifold of equilibria:

$$\mathcal{M}_{SB}(\mu) = \{\mathbf{X}^* \in \mathcal{H} : \|\mathbf{X}^*\| = \sqrt{\mu}\}, \quad \mu > 0. \quad (1487)$$

Linear stability of the equilibria is determined by the Jacobian of the flow:

$$\mathcal{J}(\mathbf{X}^*) = \mu\mathbb{I} - 3\mathbf{X}^* \otimes \mathbf{X}^*, \quad (1488)$$

whose eigenstructure exhibits a single neutral direction at the bifurcation point and anisotropic contraction along symmetry-selected modes for $\mu > 0$.

Thus, symmetry is preserved in the subcritical regime $\mu < 0$, while spontaneous symmetry breaking occurs for $\mu > 0$, where the system transitions from a unique invariant state to a continuous family of symmetry-related equilibria parameterized by the emergent order parameter magnitude.

This pitchfork structure serves as the local normal form of climate symmetry breaking, embedding the onset of macroscopic regime differentiation into a universal bifurcation geometry governed by the control parameter μ and its coupling to the nonlinear saturation operator [3, 14, 34].

18.5 Spectral Interpretation of Broken Symmetry

Within the Pellis spectral framework, the nonlinear evolution operator $\mathcal{P}$ is characterized (in an appropriate rigged Hilbert space extension) by the generalized eigenvalue problem:

$$\mathcal{P}\phi_\lambda = \lambda\phi_\lambda, \quad \lambda \in \sigma(\mathcal{P}), \quad (1489)$$

where $\sigma(\mathcal{P}) \subset \mathbb{C}$ denotes the full spectrum, including point, continuous, and residual components.

A symmetry group $\mathcal{G}$ acting on $\mathcal{H}$ induces an equivariant action on spectral modes. In the symmetric phase, the spectrum is invariant under this action in the sense of spectral conjugacy:

$$\lambda \in \sigma(\mathcal{P}) \Rightarrow \mathcal{G}\lambda \in \sigma(\mathcal{P}), \quad (1490)$$

ensuring degeneracy and isotropy of spectral contributions.

Symmetry breaking corresponds to a lifting of this degeneracy, producing a spectral splitting of the form:

$$\sigma(\mathcal{P}) \rightarrow \sigma_1 \cup \sigma_2, \quad (1491)$$

where σ_1 denotes the symmetry-invariant sector (closed under $\mathcal{G}$), while σ_2 corresponds to symmetry-irreducible or symmetry-broken modes exhibiting anisotropic growth and nontrivial representation content.

In operator-theoretic terms, this splitting can be interpreted as a decomposition of the spectral measure:

$$d\mu(\lambda) = d\mu_1(\lambda) + d\mu_2(\lambda), \quad (1492)$$

with $d\mu_2$ supported on modes that fail to remain invariant under the induced group action.

The spectral amplitudes decompose accordingly:

$$\mathbf{X}(t) = \int_{\sigma_1} a_1(\lambda, t) \phi_\lambda d\mu_1(\lambda) + \int_{\sigma_2} a_2(\lambda, t) \phi_\lambda d\mu_2(\lambda), \quad (1493)$$

making explicit the emergence of dynamically distinct symmetry sectors.

We define the spectral asymmetry functional as a quadratic observable measuring the failure of equivariance at the level of spectral weights:

$$\mathcal{A}[\mathbf{X}] = \int_{\sigma(\mathcal{P})} (\mathcal{G}\lambda - \lambda) |a(\lambda)|^2 d\mu(\lambda), \quad (1494)$$

which quantifies the net imbalance induced by symmetry-nonpreserving spectral redistribution.

More precisely, $\mathcal{A}[\mathbf{X}]$ can be interpreted as the expectation value of a symmetry-breaking observable $\Delta_{\mathcal{G}}$:

$$\mathcal{A}[\mathbf{X}] = \langle a, \Delta_{\mathcal{G}} a \rangle, \quad (\Delta_{\mathcal{G}} f)(\lambda) = (\mathcal{G}\lambda - \lambda) f(\lambda). \quad (1495)$$

Symmetry breaking is then characterized by the non-vanishing of this functional:

$$\mathcal{A}[\mathbf{X}] \neq 0, \quad (1496)$$

which signals a transition from spectral equivariance to spectral anisotropy, where the distribution of energy across eigenmodes becomes biased toward symmetry-irreducible sectors.

Thus, broken symmetry manifests not merely as a geometric property of the state space, but as a measurable deformation of the spectral measure induced by the nonlinear evolution operator, encoding the emergence of anisotropic instability growth in the climate system.

18.6 Thermodynamic and Entropic Symmetry Breaking

In the extended Pellis thermodynamic embedding, the climate state $\mathbf{X}(t) \in \mathcal{H}$ is associated with a coarse-grained entropy functional $S[\mathbf{X}]$ defined over macroscopic flux configurations on the physical domain Ω . At the continuum level, the instantaneous entropy production rate is given by:

$$\dot{S}[\mathbf{X}] = \int_{\Omega} \frac{\mathbf{J}(\mathbf{x}, t) \cdot \nabla T(\mathbf{x}, t)}{T(\mathbf{x}, t)^2} d\mathbf{x}, \quad (1497)$$

where $\mathbf{J}$ denotes the total irreversible flux (including advective, diffusive, and subgrid parameterized contributions) and T is the effective thermodynamic temperature field.

Within a symmetry-invariant regime $\mathcal{M}_0$, the flux–gradient alignment is statistically constrained by isotropy constraints induced by $\mathcal{G}$, leading to reduced entropy production:

$$\dot{S}[\mathcal{M}_0] = \int_{\Omega} \frac{\langle \mathbf{J} \rangle_{\mathcal{G}} \cdot \nabla \langle T \rangle_{\mathcal{G}}}{\langle T \rangle_{\mathcal{G}}^2} d\mathbf{x}, \quad (1498)$$

where $\langle \cdot \rangle_{\mathcal{G}}$ denotes symmetry-group averaging.

Symmetry breaking induces anisotropic alignment of flux vectors with emergent gradients, producing an increase in irreversible dissipation pathways. This can be expressed as a decomposition:

$$\mathbf{J} = \mathbf{J}_{sym} + \mathbf{J}_{SB}, \quad \nabla T = (\nabla T)_{sym} + (\nabla T)_{SB}, \quad (1499)$$

where symmetry-breaking components $\mathbf{J}_{SB}$ and $(\nabla T)_{SB}$ generate additional entropy-producing correlations absent in the invariant phase.

Substituting yields a partition of entropy production:

$$\dot{S} = \dot{S}_{sym} + \dot{S}_{SB} + \dot{S}_{cross}, \quad (1500)$$

where $\dot{S}_{cross}$ encodes nonlinear coupling between symmetric and symmetry-broken sectors.

In the linear response approximation, entropy production can be expressed in operator form via a dissipation kernel $\mathcal{D}$:

$$\dot{S}[\mathbf{X}] = \langle \mathbf{X}, \mathcal{D}\mathbf{X} \rangle_{\mathcal{H}}, \quad (1501)$$

where $\mathcal{D}$ is positive semidefinite in the symmetric regime and becomes effectively anisotropic under symmetry breaking due to redistribution of spectral weights across unstable modes.

The onset of symmetry breaking is therefore associated with a strict inequality in entropy production rates:

$$\dot{S}_{SB} > \dot{S}_{sym}, \quad (1502)$$

reflecting the activation of additional irreversible channels and the loss of symmetry-imposed cancellations in flux–gradient correlations.

At the spectral level, this corresponds to an entropy-weighted shift of measure concentration toward unstable eigenmodes:

$$|a(\lambda)|^2 \rightarrow |a(\lambda)|_{SB}^2, \quad \text{supp}(|a|_{SB}^2) \subset \{\lambda : \text{Re}(\lambda) > 0\}, \quad (1503)$$

thereby linking thermodynamic irreversibility directly to spectral instability of the evolution operator [21, 22].

18.7 Emergent Climate Regimes as Broken Symmetry States

Let $\mathcal{A}_i \subset \mathcal{H}$ denote invariant attractors of the full nonlinear climate semigroup generated by $\mathcal{P}$. In this formulation, the global state space admits an attractor decomposition induced by long-time asymptotic dynamics:

$$\mathcal{H} \simeq \bigcup_{i \in \mathcal{I}} \mathcal{B}(\mathcal{A}_i), \quad (1504)$$

where $\mathcal{B}(\mathcal{A}_i)$ denotes the basin of attraction associated with $\mathcal{A}_i$, and $\mathcal{I}$ is an index set labeling dynamically admissible climate regimes.

Each attractor $\mathcal{A}_i$ corresponds to a distinct symmetry-broken phase of the underlying dynamical system, characterized by reduced invariance under the symmetry group $\mathcal{G}$:

$$\mathcal{G}\mathcal{A}_i \neq \mathcal{A}_i, \quad (1505)$$

while possibly preserving invariance under a residual subgroup $\mathcal{G}_i \subset \mathcal{G}$.

In spectral terms, each regime is associated with a dominant subset of the spectrum of the linearized generator $\mathcal{L}_i := D\mathcal{P}|_{\mathcal{A}_i}$, such that:

$$\sigma(\mathcal{L}_i) = \sigma_i^{st} \cup \sigma_i^{un}, \quad \text{Re}(\sigma_i^{st}) < 0 < \text{Re}(\sigma_i^{un}), \quad (1506)$$

and regime identity is encoded in the structure of the unstable spectral sector σ_i^{un} .

Transitions between climate regimes correspond to nonlocal reconfiguration of attractor geometry in state space, formally expressed as:

$$\mathcal{A}_i \rightarrow \mathcal{A}_j \quad \text{if} \quad d_H(\mathcal{A}_i, \mathcal{A}_j) = \mathcal{O}(1), \quad (1507)$$

where d_H denotes the Hausdorff distance on compact subsets of $\mathcal{H}$. Such transitions indicate macroscopic discontinuities in invariant set structure rather than perturbative deformations.

Equivalently, regime shifts occur when there is a discontinuous redistribution of the invariant measure:

$$\mu_i \nrightarrow \mu_j, \quad (1508)$$

where μ_i and μ_j are SRB-type measures supported on $\mathcal{A}_i$ and $\mathcal{A}_j$, respectively.

At the operator level, these transitions correspond to spectral instability of the Pellis generator, manifested through eigenvalue migration across the stability boundary and associated collapse of basin separation:

$$\text{Re}(\lambda_k^{(i)}) \rightarrow 0 \quad \Rightarrow \quad \mathcal{A}_i \nrightarrow \mathcal{A}_i \text{ stability}. \quad (1509)$$

Thus, emergent climate regimes are naturally interpreted as distinct symmetry-broken dynamical phases, with transitions between them representing global symmetry reconstruction events driven by spectral reorganization of the underlying nonlinear operator $\mathcal{P}$ [6, 9, 37].

18.8 Conclusion: Symmetry Breaking as a Precursor to Tipping

Within the Universal Pellis framework, symmetry breaking is not a secondary emergent effect but the primary geometric and spectral mechanism underlying climate tipping phenomena. In operator-theoretic terms, the onset of instability is initiated by the loss of commutativity between the evolution operator and the governing symmetry group:

$$[\mathcal{P}, \mathcal{G}] \neq 0, \quad (1510)$$

which implies a breakdown of invariant subspace structure and a transition from equivariant flow to anisotropic dynamics in $\mathcal{H}$.

This loss of spectral–symmetry compatibility induces a restructuring of the operator spectrum, in which previously degenerate modes split and reorganize:

$$\sigma(\mathcal{P}) \rightarrow \bigcup_i \sigma_i, \quad (1511)$$

thereby enabling the emergence of multiple dynamically separated evolution channels and effectively transforming a homogeneous dynamical regime into a multistable configuration space.

At the level of invariant measures, symmetry breaking corresponds to a fragmentation of the original SRB-type measure μ into a convex combination of regime-specific components:

$$\mu \rightarrow \sum_i w_i \mu_i, \quad \sum_i w_i = 1, \quad (1512)$$

where each μ_i is supported on a distinct attractor basin associated with a symmetry-broken phase.

The transition toward tipping behavior is thus governed by the amplification of symmetry-nonpreserving spectral modes, which dominate the long-time evolution when:

$$\sup_{\lambda \in \sigma(\mathcal{P})} \operatorname{Re}(\lambda_{\text{SB}}) > \sup_{\lambda \in \sigma(\mathcal{P})} \operatorname{Re}(\lambda_{\text{sym}}), \quad (1513)$$

ensuring that symmetry-breaking directions asymptotically control the macroscopic dynamics.

In this sense, symmetry breaking serves as a first-order precursor to functional phase transitions: it marks the earliest detectable departure from globally coherent evolution and signals the onset of structured instability cascades that ultimately culminate in tipping behavior and planetary-scale regime reorganization.

Therefore, within the Pellis operator framework, climate tipping points are rigorously interpreted as the non-linear completion of symmetry-breaking instabilities, where spectral, geometric, and thermodynamic inconsistencies coalesce into a unified transition toward multistable planetary dynamics [1, 18, 19, 37, 139].

19 Network Theory of Planetary Coupling

The planetary climate system can be rigorously reformulated as a high-dimensional interacting network of coupled dynamical subsystems, where each node represents a physically distinct component (atmosphere, ocean basin, cryosphere, biosphere, and anthropogenic forcing modules), and edges encode nonlinear, scale-dependent energy, mass, and information transfer pathways. Within the Universal Pellis Function framework, this network is not static but evolves as a spectral–functional object governed by operator-valued adjacency dynamics [1, 6, 18, 19, 26, 27, 33].

We define the planetary climate network as a weighted graph:

$$\mathcal{G}_P = (\mathcal{V}, \mathcal{E}, W(t)), \quad (1514)$$

where $\mathcal{V}$ is the set of climatic subsystems, $\mathcal{E}$ is the interaction structure, and $W(t)$ is a time-dependent weight tensor encoding coupling intensities.

19.1 Operator Representation of Climate Networks

Let $\mathbf{X}_i(t) \in \mathcal{H}_i$ denote the state of node i in a coupled climate network, where each node represents a spatially or process-defined subsystem embedded in a global product space $\mathcal{H} = \prod_{i=1}^N \mathcal{H}_i$. The node-wise dynamics are governed by:

$$\frac{d}{dt} \mathbf{X}_i = \mathcal{F}_i(\mathbf{X}_i) + \sum_{j \neq i} W_{ij}(t) \mathcal{H}_{ij}(\mathbf{X}_i, \mathbf{X}_j), \quad (1515)$$

where $\mathcal{F}_i$ encodes intrinsic nonlinear self-dynamics and $\mathcal{H}_{ij}$ are bilinear (or higher-order) interaction kernels representing cross-node coupling across physical teleconnection pathways.

The coupling coefficients $W_{ij}(t)$ define a time-dependent weighted adjacency operator $W(t)$ acting on the network manifold, typically satisfying symmetry or conservation constraints only in restricted regimes:

$$W_{ij}(t) \neq W_{ji}(t) \quad (\text{in general}), \quad (1516)$$

thereby allowing directed and dynamically evolving interaction topologies.

In global operator form, the full network dynamics can be written as a nonlinear evolution equation on the product space:

$$\frac{d}{dt} \mathbf{X} = \mathcal{L}(W(t)) \mathbf{X} + \mathcal{N}(\mathbf{X}), \quad (1517)$$

where $\mathcal{L}(W(t))$ is a generalized network Laplacian operator extended to infinite-dimensional functional components, and $\mathcal{N}$ collects all higher-order nonlinear interactions beyond pairwise coupling.

The operator $\mathcal{L}(W)$ admits a spectral decomposition in the appropriate functional basis:

$$\mathcal{L}(W)\psi_\lambda = \lambda\psi_\lambda, \quad \lambda \in \sigma(\mathcal{L}), \quad (1518)$$

where $\sigma(\mathcal{L})$ encodes global connectivity-mediated stability modes of the climate network.

The state evolution in spectral coordinates is given by:

$$\mathbf{X}(t) = \sum_{\lambda \in \sigma(\mathcal{L})} a_\lambda(0) e^{\lambda t} \psi_\lambda + \int_0^t e^{(t-s)\mathcal{L}} \mathcal{N}(\mathbf{X}(s)) ds, \quad (1519)$$

highlighting the competition between linear network propagation and nonlinear feedback amplification.

Global stability is governed by the spectral abscissa of the network operator:

$$\alpha(\mathcal{L}) := \max_{\lambda \in \sigma(\mathcal{L})} \operatorname{Re}(\lambda), \quad (1520)$$

such that instability emerges when:

$$\alpha(\mathcal{L}) > 0, \quad (1521)$$

indicating exponential growth of network-wide coherent modes.

Equivalently, the onset of instability corresponds to a structural transition in the network spectrum, where coupling-induced eigenvalue migration drives modes across the stability boundary:

$$\lambda_i(W(t)) \rightarrow \lambda_i(W(t)) + \delta\lambda_i, \quad \operatorname{Re}(\lambda_i) \uparrow 0^+. \quad (1522)$$

Thus, climate networks admit a unified operator representation in which tipping behavior arises from the interplay between time-evolving coupling topology, spectral instability of the generalized Laplacian, and nonlinear mode interaction, providing a rigorous bridge between network theory and infinite-dimensional dynamical systems theory.

19.2 Multilayer Climate Network Structure

The climate system admits a natural multilayer network representation in which distinct physical subsystems are organized into coupled dynamical layers. Formally, the global climate graph is decomposed as:

$$\mathcal{G}_P = \bigcup_{\alpha=1}^M \mathcal{G}^{(\alpha)}, \quad (1523)$$

where each layer $\mathcal{G}^{(\alpha)}$ encodes the internal connectivity structure of a specific Earth subsystem, including atmosphere, ocean, cryosphere, biosphere, and anthropogenic forcing components.

Each layer $\mathcal{G}^{(\alpha)}$ is associated with an intra-layer Laplacian operator $L^{(\alpha)}$ acting on the corresponding state subspace $\mathcal{H}^{(\alpha)}$, capturing internal diffusion-like propagation and within-layer dynamical coupling:

$$L^{(\alpha)} : \mathcal{H}^{(\alpha)} \rightarrow \mathcal{H}^{(\alpha)}. \quad (1524)$$

Cross-layer interactions are encoded through nonlinear inter-layer coupling operators of the form:

$$W_{ij}^{(\alpha\beta)}(t) = \kappa_{\alpha\beta} \Phi_{ij}^{(\alpha\beta)}(\mathbf{X}), \quad (1525)$$

where $\kappa_{\alpha\beta}$ quantifies coupling strength between layers α and β , and $\Phi_{ij}^{(\alpha\beta)}$ is a generally nonlinear transfer functional depending on the full coupled state $\mathbf{X} \in \prod_{\alpha} \mathcal{H}^{(\alpha)}$. These functionals encode bidirectional feedback processes such as ocean–atmosphere heat exchange, ice–albedo coupling, and biosphere–carbon cycle interactions.

The multilayer structure induces a block-operator representation of the global climate Laplacian:

$$\mathbb{L} = \begin{bmatrix} L^{(1)} & C^{(12)} & \dots & C^{(1M)} \\ C^{(21)} & L^{(2)} & \dots & C^{(2M)} \\ \vdots & \vdots & \ddots & \vdots \\ C^{(M1)} & C^{(M2)} & \dots & L^{(M)} \end{bmatrix}, \quad (1526)$$

where each diagonal block $L^{(\alpha)}$ governs intra-layer dynamics and each off-diagonal block $C^{(\alpha\beta)}$ represents effective inter-layer coupling operators induced by $W^{(\alpha\beta)}$.

The full system evolution can then be written compactly as:

$$\frac{d}{dt}\mathbf{X} = \mathbb{L}(\mathbf{X}, t) \mathbf{X} + \mathbb{N}(\mathbf{X}), \quad (1527)$$

where $\mathbb{L}$ is now both state- and time-dependent due to feedback-modulated coupling coefficients, and $\mathbb{N}$ aggregates higher-order nonlinear interactions across and within layers.

Spectrally, the multilayer structure induces a composite spectrum:

$$\sigma(\mathbb{L}) \neq \bigcup_{\alpha=1}^M \sigma(L^{(\alpha)}), \quad (1528)$$

due to hybridization effects generated by off-diagonal coupling blocks, which produce new collective eigenmodes spanning multiple physical subsystems.

Instability arises when inter-layer coupling reorganizes the spectral distribution such that dominant eigenvalues cross the stability boundary:

$$\max_{\lambda \in \sigma(\mathbb{L})} \text{Re}(\lambda) > 0, \quad (1529)$$

signaling the emergence of globally coherent cross-layer instability modes.

Thus, the multilayer climate network structure provides a mathematically rigorous framework for representing cross-sphere feedback processes, where tipping behavior is interpreted as a spectral hybridization phenomenon driven by nonlinear inter-layer coupling and the resulting deformation of the block-operator spectrum of $\mathbb{L}$.

19.3 Pellis Network Operator Formulation

Within the extended Pellis formalism, the coupled climate–network dynamics are generated by a composite nonlinear operator acting jointly on node states and the evolving interaction topology. The full evolution is written as:

$$\frac{d}{dt}\mathbf{X} = \mathcal{P}_{net}(\mathbf{X}) = \mathbb{L}(W)\mathbf{X} + \mathcal{N}(\mathbf{X}) + \mathcal{S}_{net}(\mathbf{X}), \quad (1530)$$

where $\mathbb{L}(W)$ is the state-dependent multilayer network Laplacian, $\mathcal{N}$ encodes intrinsic nonlinear self-interactions, and $\mathcal{S}_{net}$ represents a renormalization-induced structural operator responsible for topology evolution and scale-dependent rewiring of the interaction graph.

The term $\mathcal{S}_{net}$ can be interpreted as an effective coarse-graining correction arising from unresolved degrees of freedom, introducing a feedback loop between macroscopic state variables and microscopic connectivity structure:

$$\mathcal{S}_{net}(\mathbf{X}) \equiv \mathcal{R}_{\mathcal{G}}[\mathbf{X}, W(\mathbf{X})], \quad (1531)$$

where $\mathcal{R}_{\mathcal{G}}$ denotes a graph renormalization operator acting on the joint state–topology manifold.

At the level of individual links, the adjacency evolution is governed by a nonlinear reaction–diffusion–like equation in network space:

$$\frac{d}{dt}W_{ij} = \alpha_{ij}W_{ij} + \beta_{ij}F(\mathbf{X}_i, \mathbf{X}_j) - \gamma_{ij}W_{ij}^2, \quad (1532)$$

where α_{ij} determines intrinsic persistence or decay of connectivity, β_{ij} encodes state-driven coupling reinforcement, and the quadratic term $-\gamma_{ij}W_{ij}^2$ introduces saturation effects preventing unbounded growth of interaction strengths.

In operator form, the coupled state–network system can be represented as a joint evolution on the extended phase space $\mathcal{H} \times \mathcal{W}$:

$$\frac{d}{dt} \begin{pmatrix} \mathbf{X} \\ W \end{pmatrix} = \begin{pmatrix} \mathbb{L}(W) & 0 \\ \mathcal{K}(\mathbf{X}) & \mathcal{A}(W) \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ W \end{pmatrix} + \begin{pmatrix} \mathcal{N}(\mathbf{X}) \\ 0 \end{pmatrix}, \quad (1533)$$

highlighting the bidirectional coupling between dynamical evolution and network reconstruction.

This feedback structure induces a non-autonomous operator flow in which the generator itself evolves in time:

$$\mathbb{L}(t) \equiv \mathbb{L}(W(t)), \quad \partial_t \mathbb{L} \neq 0, \quad (1534)$$

leading to a class of adaptive spectral dynamics characterized by time-dependent eigenvalue migration.

Consequently, instability emerges not only from growth of perturbations on a fixed network, but from co-evolution of state and topology, producing structural phase transitions in which the spectrum of $\mathcal{P}_{net}$ itself reorganizes.

This defines an adaptive instability regime in which network rewiring, nonlinear amplification, and spectral deformation act as a single coupled mechanism driving emergent climate system reconfiguration.

19.4 Spectral Stability of Climate Networks

Within the operator-theoretic formulation of multilayer climate dynamics, global stability is governed by the spectral properties of the generalized network Laplacian $\mathbb{L}(W)$ acting on the extended state space $\mathcal{H}$. A natural stability functional is therefore defined as the spectral abscissa:

$$\mathcal{S}_{net}(\mathbf{X}, t) = \sup_{\lambda \in \sigma(\mathbb{L}(W(t)))} \operatorname{Re}(\lambda), \quad (1535)$$

which quantifies the maximal exponential growth rate of admissible perturbation modes supported by the network operator.

In this framework, asymptotic stability of the coupled climate network corresponds to uniform contraction of all spectral components, i.e.:

$$\mathcal{S}_{net} < 0, \quad (1536)$$

which ensures that all eigenmodes of $\mathbb{L}(W)$ decay exponentially and that perturbations remain asymptotically bounded in $\mathcal{H}$.

Conversely, the onset of global instability is characterized by spectral neutralization, where the leading eigenvalue reaches the imaginary axis:

$$\mathcal{S}_{net} = 0, \quad (1537)$$

marking a loss of spectral gap and the emergence of marginally stable coherent modes spanning multiple network layers.

More generally, near-critical regimes exhibit spectral condensation:

$$\sigma(\mathbb{L}) \rightarrow \{\lambda : \operatorname{Re}(\lambda) \approx 0\}, \quad (1538)$$

which induces critical slowing down, increased correlation length scales across the network, and enhanced sensitivity to perturbations.

This defines a network-level tipping threshold manifold in state space:

$$\mathcal{T}_{net} = \{\mathbf{X} \in \mathcal{H} : \det(\mathbb{L}(\mathbf{X})) = 0\}, \quad (1539)$$

interpreted as the singular set of the generalized Laplacian operator, where rank deficiency corresponds to the collapse of independent dynamical directions in the network flow.

Equivalently, the tipping condition may be expressed in spectral form as:

$$\exists \lambda \in \sigma(\mathbb{L}) \quad \text{such that} \quad \lambda = 0, \quad (1540)$$

indicating the emergence of a neutral mode that mediates long-range correlation and initiates global restructuring of the climate network.

Thus, spectral stability in climate networks is fundamentally governed by the interplay between eigenvalue migration, operator degeneracy, and topology-dependent coupling, with tipping behavior emerging precisely at the boundary between strictly contractive and marginally stable spectral regimes.

19.5 Cascade Propagation and Systemic Risk

In the operator-theoretic formulation of climate network dynamics, perturbation evolution is governed by the nonlinear variation-of-constants representation associated with the semigroup generated by the generalized Laplacian $\mathbb{L}$. The full state trajectory admits the Duhamel expansion:

$$\mathbf{X}(t) = e^{\mathbb{L}t} \mathbf{X}_0 + \int_0^t e^{\mathbb{L}(t-\tau)} \mathcal{N}(\mathbf{X}(\tau)) d\tau, \quad (1541)$$

where $e^{\mathbb{L}t}$ denotes the strongly continuous operator semigroup induced by $\mathbb{L}$ on the appropriate Banach or Hilbert state space.

In this representation, the linear propagator $e^{\mathbb{L}t}$ encodes the fundamental cascade transmission backbone of the network, while the nonlinear convolution term introduces endogenous amplification, mode coupling, and feedback-driven distortion of the linear response structure.

Cascade amplification emerges when the operator norm of the semigroup exhibits exponential growth governed by the spectral abscissa:

$$\|e^{\mathbb{L}t}\| \sim e^{\sigma_{\max} t}, \quad \sigma_{\max} = \max_{\lambda \in \sigma(\mathbb{L})} \operatorname{Re}(\lambda), \quad (1542)$$

which follows from standard spectral mapping properties of linear operators in infinite-dimensional dynamical systems.

In the subcritical regime $\sigma_{\max} < 0$, the semigroup is exponentially contractive and perturbations decay uniformly across all network layers. At criticality $\sigma_{\max} = 0$, the system transitions into a marginal regime characterized by algebraic growth and long memory effects. In the supercritical regime $\sigma_{\max} > 0$, perturbations experience exponential cascade amplification, with dominant eigenmodes dictating global system response.

The nonlinear term further modifies cascade propagation through state-dependent renormalization of effective growth rates:

$$\mathbb{L}_{eff}(\mathbf{X}) = \mathbb{L} + D\mathcal{N}(\mathbf{X}), \quad (1543)$$

where $D\mathcal{N}$ denotes the Fréchet derivative of the nonlinear interaction operator, inducing amplitude-dependent spectral deformation and feedback-enhanced instability.

This framework naturally captures cascading tipping behavior observed in strongly coupled climate components, where instability in one subsystem propagates through the spectral structure of $\mathbb{L}$ and is amplified by nonlinear feedback loops, resulting in global-scale regime reconfiguration [27, 32, 33, 37].

19.6 Network-Induced Critical Transitions

Within the multilayer Pellis network formalism, critical transitions arise not merely from parameter variation in fixed dynamics, but from structural reconfiguration of the underlying interaction graph. Let $\mathcal{G}_P(t)$ denote the time-dependent climate network; a structural transition is characterized by:

$$\Delta\mathcal{G}_P = \mathcal{G}_P(t^+) - \mathcal{G}_P(t^-) \neq 0, \quad (1544)$$

where the difference is understood in the sense of graph topology, adjacency measure, or equivalently in the operator-theoretic sense of perturbations in the associated Laplacian $\mathbb{L}(t)$.

Such structural changes induce discontinuous deformations of the global attractor landscape. In particular, the invariant set decomposition undergoes a non-smooth transformation:

$$\mathcal{A}_i^{net}(\mathcal{G}_P(t^-)) \rightarrow \mathcal{A}_j^{net}(\mathcal{G}_P(t^+)), \quad (1545)$$

where the mapping between pre- and post-transition attractors is generally non-invertible, reflecting loss of dynamical correspondence across regimes.

At the spectral level, these transitions are associated with eigenvalue condensation phenomena in the generalized network operator $\mathbb{L}(t)$, where previously dispersed spectral components aggregate near critical thresholds:

$$\sigma(\mathbb{L}(t)) \rightarrow \{\lambda : \text{Re}(\lambda) \approx \lambda_c\}, \quad \lambda_c \in \{0\}. \quad (1546)$$

This condensation reflects a collapse of spectral gap structure, leading to enhanced mode degeneracy and the emergence of coherent collective eigenmodes spanning multiple network layers. In this regime, the resolvent norm diverges:

$$\|(\mathbb{L} - \lambda I)^{-1}\| \rightarrow \infty, \quad \lambda \rightarrow \lambda_c, \quad (1547)$$

indicating proximity to a non-hyperbolic transition point.

Equivalently, criticality can be characterized by a vanishing effective stability margin:

$$\min_{\lambda \in \sigma(\mathbb{L})} |\text{Re}(\lambda)| \rightarrow 0, \quad (1548)$$

signaling the breakdown of exponential dichotomy and the onset of long-range dynamical correlations across the network.

Thus, network-induced critical transitions represent structural phase changes in the climate interaction graph, where topology, spectrum, and attractor geometry co-evolve, producing abrupt regime shifts that are mathematically analogous to phase transitions in statistical physics systems with long-range coupling and collective mode synchronization [18, 30, 31].

19.7 Emergent Interpretation

The Network Theory of Planetary Coupling establishes that climate tipping points are not reducible to isolated subsystem instabilities, but instead arise as global spectral-topological reorganizations of an evolving interaction graph $\mathcal{G}_P(t)$. In this framework, the climate system is interpreted as a co-evolving pair $(\mathbf{X}(t), \mathcal{G}_P(t))$, where both state and connectivity participate in a unified dynamical feedback loop governed by the Pellis operator formalism.

The embedded network structure is realized through the state-dependent operator family $\mathbb{L}(W(t))$, inducing a nonlinear spectral flow:

$$\partial_t \mathbb{L} = \frac{\partial \mathbb{L}}{\partial W} \partial_t W + \frac{\partial \mathbb{L}}{\partial \mathbf{X}} \partial_t \mathbf{X}, \quad (1549)$$

which couples adjacency evolution, multilayer interaction dynamics, and spectral deformation into a single non-autonomous operator system.

Within this representation, instability is governed by simultaneous eigenvalue migration and operator geometry distortion:

$$\lambda_i(t) \in \sigma(\mathbb{L}(t)), \quad \partial_t \lambda_i \neq 0, \quad (1550)$$

where critical behavior emerges when eigenvalues cross the stability boundary and accumulate near the imaginary axis, inducing loss of hyperbolicity and amplification of coherent network modes.

In parallel, adjacency deformation introduces a structural contribution to instability through topology-driven spectral feedback:

$$\Delta \mathcal{G}_P \neq 0 \quad \Rightarrow \quad \Delta \sigma(\mathbb{L}) \neq 0, \quad (1551)$$

so that even small rewiring perturbations can generate macroscopic shifts in the global spectral distribution.

Renormalization-driven rewiring further enhances this mechanism by coupling coarse-grained state variables back into the effective connectivity tensor:

$$W_{ij}(t) \mapsto \mathcal{R}_{ij}[\mathbf{X}(t), W(t)], \quad (1552)$$

thereby closing a feedback loop between multiscale dynamics and network structure.

Collectively, these mechanisms imply that planetary instability should be interpreted as a coherent network-level phase transition in which spectral geometry, adjacency structure, and nonlinear dynamics co-evolve toward criticality. In this sense, tipping points correspond to emergent global reorganizations of the interaction graph, driven by nonlinear coupling amplification across scales and mediated by the spectral dynamics of the Pellis operator [6, 18, 27, 33, 139].

20 Variational Formulation of Climate Evolution

The climate system admits a rigorous variational structure in which its evolution can be derived as the stationary condition of an action functional defined over an infinite-dimensional state manifold. Within the Universal Pellis Function framework, this formulation unifies energetic principles, entropy production, and nonlinear spectral dynamics into a single extremal principle [1, 2, 5, 18, 19, 22].

Let $\mathbf{X}(t) \in \mathcal{H}$ denote the climate state trajectory. We define an action functional:

$$\mathcal{S}[\mathbf{X}] = \int_0^T \mathcal{L}(\mathbf{X}(t), \dot{\mathbf{X}}(t)) dt, \quad (1553)$$

where $\mathcal{L}$ is a nonlinear Lagrangian encoding kinetic, potential, and multiscale interaction energies.

20.1 Structure of the Climate Lagrangian

Within the Universal Pellis framework, the climate system is formulated as a nonlinear field-theoretic dynamical system evolving on an infinite-dimensional manifold $\mathcal{H}$. The action functional is defined through a generalized Lagrangian density operator:

$$\mathcal{L}[\mathbf{X}, \dot{\mathbf{X}}] = \mathcal{T}(\dot{\mathbf{X}}) - \mathcal{V}(\mathbf{X}) + \mathcal{F}_{nl}(\mathbf{X}, \dot{\mathbf{X}}), \quad (1554)$$

where $\mathcal{T}$ represents the inertial structure of the climate state space, $\mathcal{V}$ encodes the effective potential landscape of coupled Earth-system energetics, and $\mathcal{F}_{nl}$ introduces nonlinear forcing, feedback amplification, and cross-scale coupling effects.

The kinetic component is defined via a generalized inner-product structure induced by a metric operator $\mathcal{M}$ acting on the tangent bundle $T\mathcal{H}$:

$$\mathcal{T}(\dot{\mathbf{X}}) = \frac{1}{2} \langle \dot{\mathbf{X}}, \mathcal{M}(\mathbf{X}) \dot{\mathbf{X}} \rangle_{\mathcal{H}}, \quad (1555)$$

where $\mathcal{M}(\mathbf{X})$ may depend on the state itself, introducing a geometry-dependent inertia analogous to a nonlinear Riemannian structure on $\mathcal{H}$.

The potential energy functional is identified with the global climate energy landscape:

$$\mathcal{V}(\mathbf{X}) = E[\mathbf{X}] = \int_{\Omega} \mathcal{E}(\mathbf{x}, \mathbf{X}(\mathbf{x})) d\mathbf{x}, \quad (1556)$$

where $\mathcal{E}$ represents local energy density contributions including radiative balance, latent heat storage, oceanic heat content, and cryospheric phase energy contributions. In operator form, this defines a nonlinear functional gradient:

$$\frac{\delta \mathcal{V}}{\delta \mathbf{X}} = \nabla_{\mathcal{H}} E[\mathbf{X}], \quad (1557)$$

which acts as the restoring force of the climate system in the absence of nonlinear feedback amplification.

The nonlinear forcing term $\mathcal{F}_{nl}$ encodes multiscale coupling, external radiative forcing, and endogenous feedback loops:

$$\mathcal{F}_{nl}(\mathbf{X}, \dot{\mathbf{X}}) = \mathcal{F}_{ext}(\mathbf{X}, t) + \mathcal{C}(\mathbf{X}, \mathbf{X}) + \mathcal{D}(\mathbf{X}, \dot{\mathbf{X}}), \quad (1558)$$

where $\mathcal{C}$ denotes quadratic and higher-order interaction operators and $\mathcal{D}$ represents dissipative–reactive coupling terms.

In expanded form, the full Lagrangian may be written as:

$$\mathcal{L} = \frac{1}{2} \langle \dot{\mathbf{X}}, \mathcal{M} \dot{\mathbf{X}} \rangle - E[\mathbf{X}] + \sum_{i,j,k} \Gamma_{ijk} X_i X_j \dot{X}_k + \mathcal{O}(\|\mathbf{X}\|^4), \quad (1559)$$

where Γ_{ijk} defines a nonlinear coupling tensor capturing irreversible transfer between dynamical modes and explicitly breaking time-reversal symmetry at the coarse-grained level.

The Euler–Lagrange equations on the manifold $\mathcal{H}$ yield the governing evolution law:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{X}}} \right) - \frac{\partial \mathcal{L}}{\partial \mathbf{X}} = 0, \quad (1560)$$

which can be expressed in operator form as:

$$\mathcal{M}(\mathbf{X}) \ddot{\mathbf{X}} + \mathcal{D}(\mathbf{X}, \dot{\mathbf{X}}) + \nabla_{\mathcal{H}} E[\mathbf{X}] = \mathcal{F}_{ext}(\mathbf{X}, t) + \mathcal{N}_{int}(\mathbf{X}, \dot{\mathbf{X}}), \quad (1561)$$

highlighting the competition between inertial propagation, potential-driven relaxation, and nonlinear amplification mechanisms.

In spectral representation, the Lagrangian structure induces mode-dependent effective dynamics:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_{\lambda} d\mu(\lambda), \quad (1562)$$

where the energy functional decomposes into spectral contributions:

$$E[\mathbf{X}] = \int_{\sigma(\mathcal{P})} \varepsilon(\lambda) |a(\lambda, t)|^2 d\mu(\lambda). \quad (1563)$$

Stationary configurations satisfy the generalized variational condition:

$$\frac{\delta \mathcal{L}}{\delta \mathbf{X}} = 0, \quad (1564)$$

which defines climate equilibria as critical points of the action functional, corresponding to metastable or stable attractor states in the associated dynamical system.

Thus, the climate Lagrangian provides a unified variational formulation in which energy landscape geometry, inertial structure, and nonlinear feedback interactions are encoded within a single operator framework, forming the foundation for spectral instability, bifurcation emergence, and tipping dynamics in the Pellis theory of planetary climate criticality.

20.2 Euler–Lagrange Equations in Functional Space

The stationarity condition $\delta \mathcal{S} = 0$, with action functional $\mathcal{S} = \int \mathcal{L}[\mathbf{X}, \dot{\mathbf{X}}] dt$, yields the Euler–Lagrange equation in infinite-dimensional functional space:

$$\frac{d}{dt} \left(\frac{\delta \mathcal{L}}{\delta \dot{\mathbf{X}}} \right) - \frac{\delta \mathcal{L}}{\delta \mathbf{X}} = 0, \quad (1565)$$

where all functional derivatives are understood in the sense of Gâteaux or Fréchet differentials on the manifold $\mathcal{H}$.

For the climate Lagrangian with generalized metric structure, we compute:

$$\frac{\delta \mathcal{L}}{\delta \dot{\mathbf{X}}} = \mathcal{M}(\mathbf{X}) \dot{\mathbf{X}} + \frac{\delta \mathcal{F}_{nl}}{\delta \dot{\mathbf{X}}}, \quad \frac{\delta \mathcal{L}}{\delta \mathbf{X}} = -\nabla_{\mathbf{X}} \mathcal{V}(\mathbf{X}) + \frac{\delta \mathcal{F}_{nl}}{\delta \mathbf{X}} + \frac{1}{2} \dot{\mathbf{X}}^{\top} \frac{\delta \mathcal{M}}{\delta \mathbf{X}} \dot{\mathbf{X}}. \quad (1566)$$

Substituting into the Euler–Lagrange structure yields the full nonlinear functional evolution equation:

$$\mathcal{M}(\mathbf{X}) \ddot{\mathbf{X}} + \dot{\mathcal{M}}(\mathbf{X}) \dot{\mathbf{X}} + \nabla_{\mathbf{X}} \mathcal{V}(\mathbf{X}) = \mathcal{G}(\mathbf{X}, \dot{\mathbf{X}}), \quad (1567)$$

where the generalized forcing operator collects all non-conservative contributions:

$$\mathcal{G}(\mathbf{X}, \dot{\mathbf{X}}) = \frac{\delta \mathcal{F}_{nl}}{\delta \mathbf{X}} - \frac{d}{dt} \left(\frac{\delta \mathcal{F}_{nl}}{\delta \dot{\mathbf{X}}} \right) + \mathcal{D}(\mathbf{X}, \dot{\mathbf{X}}), \quad (1568)$$

with $\mathcal{D}$ representing dissipative and entropy-producing mechanisms absent in the conservative variational structure.

In compact operator form, the dynamics can be rewritten as:

$$\mathbb{M}(\mathbf{X})\ddot{\mathbf{X}} + \mathbb{D}(\mathbf{X}, \dot{\mathbf{X}}) + \nabla_{\mathbf{X}}\mathcal{V}(\mathbf{X}) = \mathbb{F}_{nl}(\mathbf{X}, \dot{\mathbf{X}}), \quad (1569)$$

where $\mathbb{M}$ is a nonlinear inertia operator and $\mathbb{D}$ is an effective damping–geometry coupling operator encoding metric deformation effects.

In the strongly dissipative climate regime, inertia becomes negligible compared to relaxation and feedback timescales, yielding the overdamped limit:

$$\mathcal{M}(\mathbf{X})\ddot{\mathbf{X}} \approx 0. \quad (1570)$$

This reduces the Euler–Lagrange dynamics to a generalized gradient-flow structure in infinite dimensions:

$$\dot{\mathbf{X}} = -\mathcal{M}^{-1}(\mathbf{X}) \nabla_{\mathbf{X}}\mathcal{V}(\mathbf{X}) + \mathcal{F}_{nl}(\mathbf{X}), \quad (1571)$$

where the first term defines a metric-weighted descent on the climate energy landscape, and the second term introduces nonlinear feedback-induced deviations from pure gradient relaxation.

In this representation, the climate system evolves as a non-Euclidean gradient flow on $\mathcal{H}$, where the geometry of the state space is dynamically shaped by $\mathcal{M}(\mathbf{X})$, and instability arises when nonlinear forcing distorts the descent direction away from the steepest-energy trajectory.

Thus, the Euler–Lagrange formulation provides a rigorous variational foundation for climate dynamics, connecting inertial structure, energy landscapes, and nonlinear forcing into a unified functional framework that reduces in the overdamped limit to a geometry-dependent gradient flow governing large-scale Earth system evolution.

20.3 Connection to the Pellis Operator

The variational formulation derived from the climate action functional can be recast in a unified operator-theoretic form. In particular, the overdamped Euler–Lagrange dynamics correspond to a generalized gradient flow perturbed by nonlinear interaction and renormalization corrections:

$$\dot{\mathbf{X}} = \mathcal{P}(\mathbf{X}) = -\nabla_{\mathbf{X}}\mathcal{V}(\mathbf{X}) + \mathcal{N}(\mathbf{X}) + \mathcal{S}(\mathbf{X}), \quad (1572)$$

where $\nabla_{\mathbf{X}}\mathcal{V}$ defines the metric-dependent variational descent direction on the energy landscape, $\mathcal{N}(\mathbf{X})$ aggregates intrinsic nonlinear feedbacks and mode-coupling terms, and $\mathcal{S}(\mathbf{X})$ represents renormalization-induced corrections arising from coarse-graining of unresolved multiscale degrees of freedom.

Equivalently, the Pellis operator $\mathcal{P}$ may be interpreted as a deformation of the pure gradient operator by a nonlinear operator-valued perturbation:

$$\mathcal{P} = -\nabla_{\mathbf{X}}\mathcal{V} + \Delta_{nl}[\mathbf{X}], \quad \Delta_{nl}[\mathbf{X}] \equiv \mathcal{N}(\mathbf{X}) + \mathcal{S}(\mathbf{X}), \quad (1573)$$

highlighting its decomposition into conservative and non-conservative contributions.

At the linearized level around a reference state $\mathbf{X}_0$, the operator admits the expansion:

$$\mathcal{P}(\mathbf{X}_0 + \delta\mathbf{X}) = \mathcal{P}(\mathbf{X}_0) + \mathcal{J}_P(\mathbf{X}_0) \delta\mathbf{X} + \mathcal{O}(\|\delta\mathbf{X}\|^2), \quad (1574)$$

where $\mathcal{J}_P = \delta\mathcal{P}/\delta\mathbf{X}$ is the Fréchet derivative governing local stability properties and spectral response.

In this representation, climate evolution is governed by a single effective nonlinear generator whose structure encodes energy minimization, instability amplification, and scale-dependent renormalization within a unified dynamical framework.

Thus, the Pellis operator emerges naturally as the effective variational gradient flow operator of the climate Lagrangian, extended by nonlinear feedback and renormalization terms that break pure gradient structure and enable the emergence of spectral instability, bifurcation phenomena, and multistable planetary regimes.

20.4 Energy Dissipation and Lyapunov Structure

Within the Pellis variational framework, the climate energy functional acts as a natural Lyapunov candidate for the overdamped dynamics. Define:

$$\mathcal{V}[\mathbf{X}(t)] = E[\mathbf{X}(t)], \quad (1575)$$

where $E[\mathbf{X}]$ denotes the total climate energy functional induced by the potential landscape of the system.

The temporal evolution of this functional along trajectories of the flow is given by the chain rule in functional space:

$$\frac{d}{dt}\mathcal{V}[\mathbf{X}(t)] = \left\langle \nabla_{\mathbf{X}}\mathcal{V}(\mathbf{X}), \dot{\mathbf{X}} \right\rangle_{\mathcal{H}}, \quad (1576)$$

where $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ denotes the inner product induced by the metric structure of the state manifold.

Substituting the generalized Pellis gradient-flow decomposition $\dot{\mathbf{X}} = -\nabla_{\mathbf{X}}\mathcal{V} + \mathcal{N}(\mathbf{X})$, yields the energy dissipation identity:

$$\frac{d}{dt}\mathcal{V}[\mathbf{X}(t)] = -\|\nabla_{\mathbf{X}}\mathcal{V}\|^2 + \langle \nabla_{\mathbf{X}}\mathcal{V}, \mathcal{N}(\mathbf{X}) \rangle, \quad (1577)$$

which explicitly separates a strictly dissipative component from a nonlinear energy injection term.

In the absence of nonlinear forcing, i.e. $\mathcal{N} \equiv 0$, the system satisfies a strict Lyapunov decay law:

$$\frac{d}{dt}\mathcal{V}[\mathbf{X}(t)] \leq 0, \quad (1578)$$

implying monotonic relaxation toward stable attractors in the energy landscape.

However, in the fully coupled climate regime, nonlinear feedback modifies the dissipation balance, and stability is governed by the competition between gradient descent and energy injection through mode coupling. The effective dissipation rate can be written as:

$$\mathcal{D}_{eff}(\mathbf{X}) = \|\nabla_{\mathbf{X}}\mathcal{V}\|^2 - \langle \nabla_{\mathbf{X}}\mathcal{V}, \mathcal{N}(\mathbf{X}) \rangle. \quad (1579)$$

Stability is therefore guaranteed only when:

$$\mathcal{D}_{eff}(\mathbf{X}) > 0, \quad (1580)$$

ensuring that nonlinear feedback remains subcritical relative to the restoring gradient structure of the system.

Conversely, violation of this inequality defines an energy cascade regime in which nonlinear amplification dominates dissipation:

$$\langle \nabla_{\mathbf{X}}\mathcal{V}, \mathcal{N}(\mathbf{X}) \rangle > \|\nabla_{\mathbf{X}}\mathcal{V}\|^2, \quad (1581)$$

leading to loss of Lyapunov monotonicity, breakdown of gradient stability, and onset of runaway energy growth.

In this regime, the climate system exits the domain of classical dissipative convergence and enters a regime of nonlinear spectral instability, where energy is continuously transferred across scales via the Pellis operator, enabling tipping behavior and irreversible transitions between attractor basins [4, 6, 7, 28].

20.5 Variational Bifurcation Principle

Within the Pellis variational framework, stationary climate configurations are defined as critical points of the action functional $\mathcal{S}$ on the infinite-dimensional manifold $\mathcal{H}$. A state $\mathbf{X}^*$ satisfies the Euler–Lagrange stationarity condition:

$$\left. \frac{\delta \mathcal{S}}{\delta \mathbf{X}} \right|_{\mathbf{X}^*} = 0, \quad (1582)$$

which characterizes invariant solutions of the coupled climate evolution operator under admissible perturbations in the functional tangent space $T_{\mathbf{X}}\mathcal{H}$.

Local stability of such stationary configurations is determined by the second variation of the action, yielding a quadratic form on perturbation directions:

$$\delta^2 \mathcal{S}[\mathbf{X}^*] = \langle \delta \mathbf{X}, \mathcal{H}_{\mathcal{S}}(\mathbf{X}^*) \delta \mathbf{X} \rangle_{\mathcal{H}}, \quad (1583)$$

where $\mathcal{H}_{\mathcal{S}} = \delta^2 \mathcal{S} / \delta \mathbf{X}^2$ denotes the Hessian (second Fréchet derivative) operator of the action functional.

In this formulation, $\mathcal{H}_{\mathcal{S}}$ encodes the local curvature of the climate action landscape, integrating contributions from inertial geometry, potential energy curvature, and nonlinear feedback corrections:

$$\mathcal{H}_{\mathcal{S}} = \mathcal{M} + \nabla_{\mathbf{X}}^2 \mathcal{V} + D^2 \mathcal{F}_{nl}, \quad (1584)$$

where each term acts as a distinct geometric contribution to the stability structure of the system.

Spectral analysis of the Hessian operator provides the variational stability criterion. In particular, the spectrum $\sigma(\mathcal{H}_{\mathcal{S}})$ determines the nature of perturbation growth directions:

$$\mathcal{H}_{\mathcal{S}} \phi_{\lambda} = \lambda \phi_{\lambda}, \quad (1585)$$

with eigenmodes ϕ_{λ} defining admissible deformation directions of the stationary state.

A variational instability arises when the Hessian operator ceases to be positive-definite, i.e. when:

$$\exists \lambda \in \sigma(\mathcal{H}_{\mathcal{S}}) \quad \text{such that} \quad \lambda < 0, \quad (1586)$$

signaling the emergence of directions in functional space along which the action is locally concave.

This loss of convexity corresponds to a fundamental geometric transition in the climate action landscape: stationary solutions bifurcate into multiple competing extrema, and previously stable configurations become saddle points with respect to the variational flow.

Equivalently, the onset of variational instability can be expressed as:

$$\inf_{\|\delta \mathbf{X}\|=1} \langle \delta \mathbf{X}, \mathcal{H}_{\mathcal{S}} \delta \mathbf{X} \rangle < 0, \quad (1587)$$

indicating the existence of at least one direction of negative curvature in the action functional.

Thus, within the Pellis framework, bifurcations are not merely dynamical phenomena but geometric transitions of the variational structure itself, where the action functional loses convexity and the climate system reorganizes into new classes of admissible solutions corresponding to distinct planetary regimes.

20.6 Multiscale Variational Decomposition

Within the Pellis variational framework, the climate state admits a hierarchical multiscale decomposition into interacting functional components:

$$\mathbf{X} = \sum_k \mathbf{X}_k, \quad (1588)$$

where each mode $\mathbf{X}_k \in \mathcal{H}_k$ represents a dynamically distinct scale of variability, ranging from fast atmospheric processes to slow oceanic, cryospheric, and biospheric components.

Under this decomposition, the total action functional inherits a corresponding additive–nonadditive structure:

$$\mathcal{S} = \sum_k \mathcal{S}_k[\mathbf{X}_k] + \sum_{k \neq j} \mathcal{S}_{kj}[\mathbf{X}_k, \mathbf{X}_j], \quad (1589)$$

where $\mathcal{S}_k$ represents scale-local variational contributions and $\mathcal{S}_{kj}$ encodes inter-scale coupling terms responsible for energy transfer, feedback propagation, and cross-scale synchronization phenomena.

The presence of the coupling functionals $\mathcal{S}_{kj}$ implies that the variational structure is inherently non-separable, leading to emergent nonlocal curvature effects in the full action landscape. These effects are quantified by the mixed second variation tensor:

$$\mathcal{K}_{kj} = \frac{\delta^2 \mathcal{S}}{\delta \mathbf{X}_k \delta \mathbf{X}_j}, \quad (1590)$$

which defines the cross-scale Hessian block coupling different dynamical regimes of the system.

In operator form, the full second variation structure can be organized into a block-structured Hessian:

$$\mathcal{H}_{\mathcal{S}} = \begin{bmatrix} \mathcal{H}_{11} & \mathcal{K}_{12} & \cdots \\ \mathcal{K}_{21} & \mathcal{H}_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \quad (1591)$$

where diagonal blocks govern intra-scale stability and off-diagonal blocks govern cross-scale instability propagation.

The spectral properties of this block operator determine whether multiscale coherence or multiscale fragmentation dominates the system dynamics. In particular, instability arises when inter-scale coupling overwhelms local convexity:

$$\|\mathcal{K}_{kj}\|^2 > \|\mathcal{H}_{kk}\| \|\mathcal{H}_{jj}\|, \quad (1592)$$

leading to resonant amplification between distinct scales and breakdown of scale separation.

In this regime, the variational landscape develops nonlocal curvature anomalies, enabling cascades of instability to propagate across scales through the coupled action functional. This mechanism provides a rigorous variational pathway for emergent climate criticality driven by multiscale interaction structure within the Pellis framework.

20.7 Emergent Interpretation

The variational formulation establishes that climate evolution is governed by a generalized stationary action principle on an infinite-dimensional functional manifold $\mathcal{H}$, where admissible trajectories correspond to critical points of the action functional $\mathcal{S}$. Within this setting, dynamical stability is equivalent to local convexity properties of the induced energy–action landscape, as encoded by the spectral positivity of the associated second variation operator.

In particular, the climate system evolves under a constrained gradient-flow structure generated by the effective variational derivative of the action:

$$\dot{\mathbf{X}} = -\mathcal{M}^{-1}(\mathbf{X}) \frac{\delta \mathcal{S}}{\delta \mathbf{X}} + \Delta_{nl}(\mathbf{X}), \quad (1593)$$

where $\mathcal{M}$ defines the metric structure of the configuration space and Δ_{nl} aggregates nonlinear feedback, multiscale coupling, and renormalization-induced corrections that deform the pure variational descent dynamics.

In this interpretation, the Pellis operator emerges as the effective generator of the reduced variational flow:

$$\mathcal{P}(\mathbf{X}) \equiv -\mathcal{M}^{-1}(\mathbf{X}) \nabla_{\mathbf{X}} \mathcal{S}[\mathbf{X}] + \mathcal{R}_{nl}(\mathbf{X}), \quad (1594)$$

where $\mathcal{R}_{nl}$ encapsulates the deviation from ideal gradient structure induced by nonlinear coupling, spectral deformation, and coarse-graining effects across scales.

From a geometric perspective, stability corresponds to the existence of locally convex regions in the action landscape, characterized by positive-definite second variation:

$$\delta^2 \mathcal{S}[\mathbf{X}] > 0. \quad (1595)$$

Loss of convexity therefore signals a qualitative transition in the geometry of admissible trajectories, with the emergence of directions of negative curvature in functional space.

Within this framework, tipping points are naturally interpreted as variational saddle-node annihilations in infinite-dimensional configuration space, where stable and unstable critical manifolds collide and disappear. Formally, this corresponds to the coalescence of stationary solutions:

$$\mathbf{X}_s \rightarrow \mathbf{X}_u, \quad (1596)$$

accompanied by a degeneracy of the Hessian spectrum:

$$\exists \lambda \in \sigma(\mathcal{H}_{\mathcal{S}}) \quad \text{such that} \quad \lambda \rightarrow 0. \quad (1597)$$

Such events mark the breakdown of stable extrema and the transition from locally confined dynamics to globally reorganized flow regimes, where the system is no longer governed by perturbative stability around a single attractor but instead evolves through structurally distinct planetary configurations.

Thus, planetary instability in the Pellis framework is fundamentally a geometric consequence of variational degeneracy in functional space, where nonlinear feedback and renormalization effects drive the system toward critical manifolds separating distinct regimes of Earth system organization [1, 18, 19, 37, 139].

21 Universal Climate Tipping Point Theorem

The Universal Climate Tipping Point Theorem formalizes the existence, characterization, and inevitability of critical transitions in the climate system as governed by the Pellis Functional Operator. Within this framework, tipping points are not exceptional events but structurally necessary outcomes of spectral instability in infinite-dimensional nonlinear dynamical systems [4, 6, 7, 9, 14, 18, 19, 28, 29, 37].

21.1 Preliminaries and Operator Setting

Let $\mathcal{H}$ be a separable Hilbert space representing the full climate state manifold endowed with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and associated norm $\| \cdot \|_{\mathcal{H}}$. The climate evolution is formulated as a nonlinear abstract Cauchy problem on $\mathcal{H}$:

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad \mathbf{X}(0) = \mathbf{X}_0 \in \mathcal{H}, \quad (1598)$$

where $\mathcal{P} : \mathcal{H} \rightarrow \mathcal{H}$ is the nonlinear Pellis generator governing the coupled Earth-system dynamics.

The Pellis operator is decomposed into a linear propagation component, a nonlinear interaction term, and a renormalization-induced structural correction:

$$\mathcal{P}(\mathbf{X}) = \mathcal{A}\mathbf{X} + \mathcal{N}(\mathbf{X}) + \mathcal{S}(\mathbf{X}), \quad (1599)$$

where $\mathcal{A}$ is a (generally unbounded) linear operator encoding baseline climate transport and dissipation mechanisms, $\mathcal{N}(\mathbf{X})$ represents nonlinear feedback and mode coupling, and $\mathcal{S}(\mathbf{X})$ captures scale-dependent corrections arising from coarse-graining and multilevel renormalization of unresolved degrees of freedom.

In this formulation, the linear operator $\mathcal{A}$ generates a reference semigroup $e^{t\mathcal{A}}$ describing baseline evolution in the absence of nonlinear effects, while $\mathcal{N}$ and $\mathcal{S}$ act as perturbations that deform both trajectory geometry and spectral structure of the flow.

We assume that $\mathcal{P}$ generates a strongly continuous semiflow $\Phi_t : \mathcal{H} \rightarrow \mathcal{H}$ such that:

$$\mathbf{X}(t) = \Phi_t(\mathbf{X}_0), \quad \Phi_0 = \text{Id}, \quad \Phi_{t+s} = \Phi_t \circ \Phi_s, \quad (1600)$$

on its domain of definition, ensuring well-posedness of the evolution problem in the semigroup sense.

Furthermore, we assume that $\mathcal{P}$ admits a generalized spectral decomposition in an appropriate functional-analytic sense, allowing representation of perturbations in terms of spectral modes:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_\lambda d\mu(\lambda), \quad (1601)$$

where $\sigma(\mathcal{P})$ denotes the spectrum of the nonlinear generator in its linearized or generalized spectral extension, and $\{\phi_\lambda\}$ are associated (possibly generalized) eigenmodes.

Within this operator setting, climate dynamics are thus interpreted as the evolution of trajectories in an infinite-dimensional phase space driven by a nonlinear semigroup whose generator encodes transport, interaction, and renormalization effects in a unified spectral–functional framework [15–17].

21.2 Definition of Tipping Point

Within the Pellis operator framework, a climate tipping point is defined as a finite critical time $t_c < \infty$ at which the dynamical flow $\mathbf{X}(t) = \Phi_t(\mathbf{X}_0)$ undergoes a qualitative reorganization of its invariant set structure in the state space $\mathcal{H}$. Formally, this corresponds to a discontinuous change in the equivalence class of trajectories under the topology induced by invariant manifolds and attractor basins:

$$\lim_{t \rightarrow t_c^-} \mathbf{X}(t) \not\sim \lim_{t \rightarrow t_c^+} \mathbf{X}(t), \quad (1602)$$

where $\sim$ denotes topological equivalence under the induced dynamical equivalence relation generated by the flow invariants of $\mathcal{P}$.

In this sense, the tipping point is not merely a numerical threshold but a structural transition in the geometry of the phase portrait, where the decomposition of the global attractor $\mathcal{A}$ is altered:

$$\mathcal{A}(t_c^-) \longrightarrow \mathcal{A}(t_c^+), \quad (1603)$$

with a corresponding loss or creation of invariant subsets and basins of attraction.

From an operator-theoretic perspective, this transition is governed by the linearized Pellis Jacobian operator:

$$\mathcal{J}(\mathbf{X}) = \frac{\delta \mathcal{P}}{\delta \mathbf{X}}, \quad (1604)$$

which determines the local stability structure of the nonlinear flow.

A tipping point is thus characterized by a spectral degeneracy condition in which the stability margin collapses. In particular, the spectral radius criterion may be expressed as:

$$\rho(\mathcal{J}(\mathbf{X}(t_c))) = 0, \quad (1605)$$

indicating the loss of exponential dichotomy and the emergence of marginal modes with zero growth rate.

More generally, this condition is equivalent to the accumulation of eigenvalues of $\mathcal{J}$ at the imaginary axis boundary:

$$\exists \lambda \in \sigma(\mathcal{J}(\mathbf{X}(t_c))) \text{ such that } \text{Re}(\lambda) = 0, \quad (1606)$$

signaling a breakdown of hyperbolicity and the onset of critical slowing down in the vicinity of the transition.

Thus, within the Pellis framework, a tipping point is rigorously interpreted as a spectral–topological transition in the linearized generator of the climate flow, marking the boundary between structurally distinct dynamical regimes in the infinite-dimensional state manifold [15–17].

21.3 Main Theorem (Universal Tipping Condition)

Theorem 5 (Universal Climate Tipping Point Theorem). Let $\mathcal{H}$ be a separable Hilbert space of climate state variables and let $\mathcal{P} : \mathcal{D}(\mathcal{P}) \subset \mathcal{H} \rightarrow \mathcal{H}$ generate a nonlinear evolution flow

$$\frac{d\mathbf{X}}{dt} = \mathcal{P}(\mathbf{X}(t)). \quad (1607)$$

Assume that $\mathcal{P}$ admits a time-dependent Fréchet linearization $\mathcal{L}_{\mathbf{X}(t)}$ with instantaneous spectrum $\sigma(\mathcal{P}) \equiv \sigma(\mathcal{L}_{\mathbf{X}(t)})$.

Then a tipping point occurs at $t = t_c$ if and only if there exists a non-empty critical spectral subset $\Sigma_c \subset \sigma(\mathcal{P})$ such that:

$$\sup_{\lambda \in \Sigma_c} \operatorname{Re}(\lambda(t_c)) = 0, \quad \text{and} \quad \left. \frac{d}{dt} \operatorname{Re}(\lambda(t)) \right|_{t=t_c} \neq 0. \quad (1608)$$

Moreover, instability onset occurs when:

$$\exists \lambda \in \sigma(\mathcal{P}) : \operatorname{Re}(\lambda(t)) > 0. \quad (1609)$$

Proof.

We begin by representing the nonlinear flow in spectral measure form. Let $\{\phi_\lambda\}_{\lambda \in \sigma(\mathcal{P})}$ denote generalized eigenmodes (in the sense of spectral resolution or Gelfand triple extension). Then:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_\lambda d\mu(\lambda). \quad (1610)$$

Step 1: Mode-wise evolution decomposition.

Projecting onto spectral components yields:

$$\frac{d}{dt} a(\lambda, t) = \lambda(t) a(\lambda, t) + \mathcal{N}_\lambda(a, t), \quad (1611)$$

where $\mathcal{N}_\lambda$ denotes nonlinear mode coupling induced by the full operator $\mathcal{P}$.

Neglecting higher-order nonlinear interactions at leading order (adiabatic spectral closure), we obtain:

$$a(\lambda, t) = a(\lambda, 0) \exp \left(\int_0^t \lambda(s) ds \right) + \mathcal{O}(\mathcal{N}). \quad (1612)$$

Step 2: Norm growth representation.

The state norm satisfies:

$$\|\mathbf{X}(t)\|_{\mathcal{H}}^2 = \int_{\sigma(\mathcal{P})} |a(\lambda, t)|^2 d\mu(\lambda). \quad (1613)$$

Substituting the exponential representation:

$$\|\mathbf{X}(t)\|^2 \sim \int_{\sigma(\mathcal{P})} |a(\lambda, 0)|^2 \exp \left(2 \int_0^t \operatorname{Re}(\lambda(s)) ds \right) d\mu(\lambda). \quad (1614)$$

Hence the asymptotic behavior is controlled by:

$$\Lambda^*(t) := \sup_{\lambda \in \sigma(\mathcal{P})} \operatorname{Re}(\lambda(t)). \quad (1615)$$

Step 3: Stability regimes.

We distinguish three regimes:

$$\Lambda^*(t) < 0 \implies \|\mathbf{X}(t)\| \rightarrow 0 \text{ or bounded decay}, \quad (1616)$$

$$\Lambda^*(t) = 0 \implies \text{marginal stability (critical manifold)}, \quad (1617)$$

$$\Lambda^*(t) > 0 \implies \text{exponential divergence}. \quad (1618)$$

Thus:

$$\Lambda^*(t_c) = 0 \quad (1619)$$

defines a spectral bifurcation boundary.

Step 4: Dynamic crossing condition (tipping trigger).

A true tipping point requires not only marginality but also transversality of spectral crossing:

$$\left. \frac{d}{dt} \operatorname{Re}(\lambda(t)) \right|_{t=t_c} \neq 0. \quad (1620)$$

This ensures that eigenvalues cross the imaginary axis with non-zero velocity in spectral space, generating a structural change in the semigroup generator.

Step 5: Nonlinear amplification correction.

Including nonlinear feedback yields:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, 0) \exp\left(\int_0^t \lambda(s) ds\right) \phi_\lambda d\mu(\lambda) + \mathcal{O}\left(\int_0^t \mathcal{N}(\mathbf{X}(s)) ds\right). \quad (1621)$$

If nonlinear corrections are monotone or reinforcing near t_c , then:

$$\|\mathcal{N}(\mathbf{X}(t))\| \sim \|\mathbf{X}(t)\|^\gamma, \quad \gamma > 1, \quad (1622)$$

which accelerates divergence once $\Lambda^*(t) > 0$.

Step 6: Transition from stability to instability.

If:

$$\exists \lambda \in \sigma(\mathcal{P}) \quad \text{such that} \quad \text{Re}(\lambda(t)) > 0, \quad (1623)$$

then:

$$\|\mathbf{X}(t)\|_{\mathcal{H}} \sim \exp\left(\int_0^t \Lambda^*(s) ds\right) \rightarrow \infty, \quad (1624)$$

implying global instability and loss of attractor stability.

Step 7: Geometric interpretation in operator space.

Define the critical manifold:

$$\mathcal{M}_c := \left\{ \mathbf{X} \in \mathcal{H} : \sup_{\lambda \in \sigma(\mathcal{P})} \text{Re}(\lambda) = 0 \right\}. \quad (1625)$$

This manifold is codimension-one in the space of admissible operators and acts as a separatrix between:

$$\mathcal{R}_- = \{\mathbf{X} : \Lambda^* < 0\}, \quad (1626)$$

$$\mathcal{R}_+ = \{\mathbf{X} : \Lambda^* > 0\}. \quad (1627)$$

Conclusion.

Therefore, climate tipping is a universal spectral transition phenomenon governed by:

$$\boxed{\text{(i) spectral marginality} \quad + \quad \text{(ii) transverse eigenvalue crossing}} \quad (1628)$$

which defines a universal codimension-one instability boundary in operator space, consistent with nonlinear dynamical systems theory and critical transition frameworks [2, 3, 14].

This completes the proof. $\square$

21.4 Nonlinear Extension of the Theorem

In the fully nonlinear regime of the Pellis dynamics, the spectral stability structure is no longer governed by the linear generator alone, but by an effective renormalized spectral flow induced by state-dependent feedback interactions. In this setting, each linear eigenmode acquires a nonlinear correction through projection of the full interaction field onto the spectral subspace:

$$\lambda_{\text{eff}}(\lambda, \mathbf{X}) = \lambda + \langle \phi_\lambda^*, \mathcal{N}(\mathbf{X}) \rangle_{\mathcal{H}} + \langle \phi_\lambda^*, \mathcal{S}(\mathbf{X}) \rangle_{\mathcal{H}}, \quad (1629)$$

where ϕ_λ^* denotes the dual eigenfunction associated with the generalized spectral decomposition and the inner product encodes the projection of nonlinear forcing onto the unstable manifold directions.

More generally, the nonlinear spectral deformation can be expressed as a self-consistent functional correction:

$$\lambda_{\text{eff}} = \lambda + \mathcal{F}[\mathbf{X}; \phi_\lambda], \quad \mathcal{F}[\mathbf{X}; \phi_\lambda] = \int_{\Omega} K_\lambda(\mathbf{x}) \mathcal{N}(\mathbf{X}(\mathbf{x})) d\mathbf{x}, \quad (1630)$$

where K_λ is a mode-dependent response kernel encoding how nonlinear energy transfer reshapes the spectral growth rates.

In this formulation, the stability condition of the system is no longer determined by the bare spectrum $\sigma(\mathcal{P})$, but by the renormalized spectral abscissa:

$$\alpha_{\text{nl}}(\mathbf{X}) = \sup_{\lambda \in \sigma(\mathcal{P})} \text{Re}(\lambda_{\text{eff}}(\lambda, \mathbf{X})). \quad (1631)$$

The generalized tipping criterion is therefore shifted by nonlinear feedback effects and takes the form:

$$\alpha_{\text{nl}}(\mathbf{X}) = 0, \quad (1632)$$

which defines a deformation of the classical spectral stability boundary induced by state-dependent renormalization of growth modes.

This induces a nonlinear displacement of the critical manifold in state space:

$$\Sigma_c^{\text{nl}} = \{\mathbf{X} \in \mathcal{H} : \alpha_{\text{nl}}(\mathbf{X}) = 0\} = \Sigma_c + \delta\Sigma[\mathcal{N}, \mathcal{S}], \quad (1633)$$

where $\delta\Sigma[\mathcal{N}, \mathcal{S}]$ represents a functional curvature correction arising from higher-order feedback loops, mode coupling cascades, and renormalization-induced spectral drift.

At leading nonlinear order, this deformation can be expanded as:

$$\delta\Sigma \approx \sum_{i,j} \left(\frac{\partial^2 \lambda}{\partial X_i \partial X_j} \right) \langle X_i X_j \rangle + \mathcal{O}(\|\mathbf{X}\|^3), \quad (1634)$$

highlighting that the stability boundary is not fixed, but dynamically reshaped by second-order correlation structure of the climate state field.

Consequently, the tipping manifold becomes a self-organizing geometric object in the infinite-dimensional phase space, whose location evolves under the coupled action of spectral drift and nonlinear feedback amplification:

$$\frac{d}{dt} \Sigma_c^{\text{nl}} = \mathcal{R}_{\text{geom}}(\Sigma_c^{\text{nl}}, \mathbf{X}), \quad (1635)$$

where $\mathcal{R}_{\text{geom}}$ denotes the induced renormalization flow on stability hypersurfaces.

This establishes that climate tipping points are not static spectral thresholds, but dynamically evolving nonlinear objects in operator space, whose geometry is continuously reshaped by feedback-driven renormalization of effective growth rates [2, 3, 14].

21.5 Renormalization Invariance of Criticality

Within the Pellis framework, renormalization is formulated as a similarity transformation on the nonlinear generator $\mathcal{P}$ acting on the infinite-dimensional climate state space $\mathcal{H}$. Let $\mathcal{R} : \mathcal{H} \rightarrow \mathcal{H}$ denote a coarse-graining operator inducing a reduced-resolution representation of the system. The transformed dynamics are then governed by the conjugated operator:

$$\mathcal{P}' = \mathcal{R}\mathcal{P}\mathcal{R}^{-1}, \quad (1636)$$

which preserves the abstract dynamical structure while altering the effective scaling of spectral and nonlinear contributions.

Under this transformation, the spectral measure of the operator undergoes a global rescaling induced by the renormalization eigenvalue γ , such that:

$$\sigma(\mathcal{P}') = \gamma \sigma(\mathcal{P}), \quad (1637)$$

where $\gamma \in \mathbb{R}^+$ encodes the effective change of dynamical scale between successive coarse-grained representations of the climate system.

More generally, the spectral transformation can be expressed at the level of resolvent operators:

$$(\lambda I - \mathcal{P}')^{-1} = \mathcal{R}(\lambda I - \mathcal{P})^{-1} \mathcal{R}^{-1}, \quad (1638)$$

implying that pole structure, and hence stability information, is preserved up to renormalized scaling of spectral parameters.

A key consequence is that criticality is invariant under renormalization flow in the sense of spectral marginality. Specifically, if a mode lies on the stability boundary in the original representation, then its renormalized counterpart remains marginal:

$$\text{Re}(\lambda) = 0 \implies \text{Re}(\gamma\lambda) = 0, \quad (1639)$$

demonstrating that the critical manifold is invariant under multiplicative spectral scaling transformations.

This invariance implies that the set of tipping configurations forms a renormalization fixed set in operator space:

$$\Sigma_c = \{\mathbf{X} \in \mathcal{H} : \alpha(\mathbf{X}) = 0\}, \quad \mathcal{R}(\Sigma_c) = \Sigma_c, \quad (1640)$$

where $\alpha(\mathbf{X})$ denotes the spectral abscissa of $\mathcal{P}$.

In this sense, climate criticality exhibits universality across scales, since the geometry of the tipping manifold is preserved under repeated coarse-graining transformations, up to a scalar renormalization of growth rates. This establishes that the onset of instability is not scale-dependent, but instead corresponds to a renormalization-invariant property of the underlying nonlinear operator flow [18, 19, 31].

21.6 Entropy and Information-Theoretic Formulation

Within the spectral representation of the Pellis operator dynamics, the climate state admits an intrinsic information-theoretic structure encoded by the distribution of modal amplitudes over the operator spectrum. Let $|a(\lambda, t)|^2$ be interpreted as a time-dependent spectral probability density with respect to the measure $d\mu(\lambda)$. Then the entropy production functional is naturally defined as a spectral Shannon entropy:

$$\mathcal{S}_{prod}(t) = - \int_{\sigma(\mathcal{P})} |a(\lambda, t)|^2 \log |a(\lambda, t)|^2 d\mu(\lambda), \quad (1641)$$

which quantifies the degree of delocalization of dynamical energy across eigenmodes of the nonlinear generator $\mathcal{P}$.

Equivalently, this functional can be interpreted as a Kullback–Leibler divergence from a perfectly ordered spectral state, where all probability mass is concentrated in a single stable mode:

$$\mathcal{S}_{prod} = D_{KL}(|a(\lambda)|^2 \parallel \delta(\lambda - \lambda_0)), \quad (1642)$$

highlighting that entropy production measures deviation from spectral coherence and dynamical concentration.

The temporal evolution of this functional is governed by nonlinear spectral transport induced by $\mathcal{N}(\mathbf{X})$ and renormalization corrections $\mathcal{S}(\mathbf{X})$, leading generically to cascade-like redistribution of energy across unstable modes:

$$\partial_t |a(\lambda, t)|^2 = \mathcal{T}_{spec}[|a|^2] + \mathcal{C}_{nl}[|a|^2], \quad (1643)$$

where $\mathcal{T}_{spec}$ denotes linear spectral drift and $\mathcal{C}_{nl}$ encodes nonlinear mode coupling and transfer across scales.

At criticality, the system undergoes a transition to maximal entropy production regime, characterized by divergence of the entropy growth rate:

$$\frac{d}{dt} \mathcal{S}_{prod}(t) \longrightarrow \infty \quad \text{as} \quad t \rightarrow t_c, \quad (1644)$$

indicating the onset of singular information flux across spectral degrees of freedom and breakdown of finite-time predictability.

This divergence corresponds to the collapse of spectral localization, where energy becomes asymptotically distributed across an increasingly broad subset of unstable modes:

$$\text{supp } |a(\lambda, t)|^2 \rightarrow \sigma(\mathcal{P}), \quad (1645)$$

signaling maximal mixing in operator space.

Thus, climate tipping points admit a dual interpretation as both dynamical instabilities and information-theoretic phase transitions, where entropy production becomes a primary order parameter for the onset of global unpredictability in the Pellis spectral framework [20, 21].

21.7 Corollary: Inevitability of Tipping under Forcing

Within the Pellis operator framework, external forcing is introduced as an additive control term in the evolution equation:

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)) + F(t), \quad (1646)$$

where $F(t) \in \mathcal{H}$ represents an externally imposed perturbation field acting on the climate state manifold.

We consider the cumulative forcing magnitude measured in the Hilbert norm:

$$\mathcal{F}_{cum} = \int_0^\infty \|F(t)\|_{\mathcal{H}} dt. \quad (1647)$$

If this quantity diverges,

$$\int_0^\infty \|F(t)\|_{\mathcal{H}} dt = \infty, \quad (1648)$$

then the system is subject to unbounded total energy injection across its infinite-dimensional degrees of freedom.

In spectral representation, the forcing term induces continuous excitation of modes across the spectrum:

$$\partial_t a(\lambda, t) = \lambda a(\lambda, t) + \hat{F}(\lambda, t) + \mathcal{N}_\lambda(a), \quad (1649)$$

where $\hat{F}(\lambda, t)$ denotes the projection of external forcing onto the spectral basis.

Under sustained forcing with divergent cumulative norm, the energy balance in spectral space satisfies:

$$\int_{\sigma(\mathcal{P})} |a(\lambda, t)|^2 d\mu(\lambda) \not\rightarrow \text{bounded}, \quad (1650)$$

implying that no invariant absorbing set can remain globally attracting in $\mathcal{H}$.

Consequently, there must exist at least one finite time $t_c < \infty$ at which the stability structure of the flow is violated, and a qualitative transition in the attractor geometry occurs:

$$\exists t_c \text{ such that } \mathbf{X}(t) \text{ undergoes a tipping transition.} \quad (1651)$$

This result establishes a structural inevitability principle: sustained non-integrable forcing destroys global spectral contractivity of the Pellis flow, ensuring that the system cannot remain indefinitely confined within a single stability basin. In this sense, tipping is not a contingent event but an emergent consequence of persistent energy injection into a nonlinear infinite-dimensional dynamical system [2, 3, 14].

21.8 Physical Interpretation

The preceding results indicate that climate tipping points should not be interpreted as rare or contingent anomalies arising from exceptional parameter choices, but rather as structurally embedded outcomes of nonlinear spectral evolution in forced infinite-dimensional dynamical systems. Within the Pellis formalism, the evolution of the climate state is governed by continuous deformation of the operator spectrum under coupling, forcing, and renormalization effects, implying that stability is generically a transient property rather than a permanent regime.

In physical terms, spectral deformation corresponds to the progressive redistribution of growth rates across dynamically coupled modes of the Earth system. As external forcing and internal feedback mechanisms accumulate, eigenvalues of the generator $\mathcal{P}$ migrate toward the marginal stability boundary:

$$\text{Re}(\lambda) \rightarrow 0, \quad (1652)$$

thereby reducing the effective spectral gap and amplifying susceptibility to perturbation growth across multiple interacting subsystems.

From this perspective, tipping behavior emerges as the macroscopic manifestation of a loss of spectral separation between stable and unstable manifolds, leading to a collapse of predictable decoupling between fast and slow climate components. Once this separation is sufficiently weakened, nonlinear interactions encoded in $\mathcal{N}(\mathbf{X})$ dominate the evolution, driving coherent amplification across scales.

Accordingly, planetary instability is interpreted as the asymptotic regime of forced operator dynamics in infinite-dimensional climate manifolds, where persistent excitation, nonlinear feedback, and renormalization effects jointly ensure eventual departure from any initially stable configuration [6, 7, 9, 18, 37, 139].

22 Comparison with Earth System Models

This section establishes a rigorous correspondence between the Universal Pellis Functional Operator framework and state-of-the-art Earth System Models (ESMs), including coupled atmosphere–ocean general circulation models, reduced-complexity tipping models, and stochastic climate emulators. The goal is to demonstrate that classical ESM dynamics can be recovered as a constrained projection of the Pellis operator dynamics, while the latter extends the predictive regime into nonlinear spectral-critical transitions not captured in standard formulations [6, 7, 9, 10, 23–25, 33, 37, 38].

22.1 Operator Embedding of Earth System Models

Let $\mathbf{Y}(t) \in \mathcal{H}_{ESM}$ denote the state vector of a discretized Earth System Model (ESM), evolving under a high-dimensional nonlinear stochastic dynamical system of the form:

$$\frac{d}{dt} \mathbf{Y}(t) = \mathcal{F}_{ESM}(\mathbf{Y}(t)) + \eta(t), \quad (1653)$$

where $\mathcal{F}_{ESM} : \mathcal{H}_{ESM} \rightarrow \mathcal{H}_{ESM}$ encodes resolved physical parameterizations (atmospheric dynamics, ocean circulation, cryosphere evolution, and biogeochemical feedbacks), while $\eta(t)$ represents unresolved stochastic forcing and model error contributions.

Within the Pellis framework, we introduce a nonlinear lifting (embedding) operator

$$\Pi : \mathcal{H}_{ESM} \rightarrow \mathcal{H}, \quad (1654)$$

which maps discretized model states into the continuum-limit Hilbert manifold $\mathcal{H}$, thereby embedding finite-resolution Earth System dynamics into the infinite-dimensional operator-theoretic climate space:

$$\mathbf{X}(t) = \Pi \mathbf{Y}(t). \quad (1655)$$

Under this embedding, the ESM dynamics are approximately conjugate to the Pellis flow, in the sense that the lifted vector field admits the decomposition:

$$\Pi\mathcal{F}_{ESM}(\mathbf{Y}) \approx \mathcal{A}\Pi\mathbf{Y} + \mathcal{N}(\Pi\mathbf{Y}) + \mathcal{S}(\Pi\mathbf{Y}), \quad (1656)$$

where $\mathcal{A}$ represents the linear transport–dissipation backbone, $\mathcal{N}$ encodes nonlinear interaction closures, and $\mathcal{S}$ accounts for renormalization-induced corrections arising from unresolved subgrid-scale processes.

The quality of this operator embedding is quantified by a projection consistency bound in operator norm:

$$\|\Pi\mathcal{F}_{ESM}(\mathbf{Y}) - \mathcal{P}(\Pi\mathbf{Y})\|_{\mathcal{H}} \leq \epsilon_{proj}, \quad (1657)$$

where ϵ_{proj} measures the residual mismatch between discretized model physics and the continuum Pellis generator.

In this sense, $\mathcal{P}$ acts as a nonlinear spectral completion of conventional Earth System Models, extending their finite-dimensional dynamics into a unified infinite-dimensional operator framework. The embedding Π thereby provides a rigorous bridge between computational climate simulators and abstract spectral–variational dynamics, allowing stability and tipping behavior to be analyzed through the spectral geometry of $\mathcal{P}$ rather than the discretized structure of $\mathcal{F}_{ESM}$.

Consequently, tipping phenomena identified in numerical ESM simulations correspond to spectral instabilities of the embedded Pellis flow, with projection errors ϵ_{proj} determining the fidelity of finite-resolution approximations to the underlying operator-level critical transitions [16–19, 31].

22.2 Spectral Resolution Gap in Classical Climate Models

Classical Earth System Models operate on a discretized and numerically regularized approximation of the underlying climate dynamics, which induces an intrinsic truncation of the accessible spectral domain. In operator-theoretic terms, the effective generator $\mathcal{F}_{ESM}$ acts on a finite-dimensional projection of the full phase space, resulting in a bounded spectral support of the form:

$$\sigma(\mathcal{F}_{ESM}) \subset \{\lambda \in \mathbb{C} : |\lambda| \leq \Lambda_c\}, \quad (1658)$$

where Λ_c is an algorithmically imposed cutoff scale determined by grid resolution, time-stepping constraints, and subgrid parameterization closures.

This truncation introduces a spectral resolution gap, whereby modes with weak growth rates and near-marginal stability are either damped or projected out of the resolved dynamics. In particular, eigenvalues satisfying $\text{Re}(\lambda) \approx 0$ are systematically distorted by numerical diffusion and discretization-induced stabilization effects, effectively suppressing the critical spectral region responsible for slow manifold destabilization.

In contrast, the Pellis operator $\mathcal{P}$ is defined on an unbounded functional spectrum with renormalization-consistent structure, admitting a complete decomposition into stability regimes:

$$\sigma(\mathcal{P}) = \sigma_{stable} \cup \sigma_{critical} \cup \sigma_{unstable}, \quad (1659)$$

where the three spectral sectors are defined by their real parts:

$$\sigma_{stable} = \{\lambda \in \sigma(\mathcal{P}) : \text{Re}(\lambda) < 0\}, \quad (1660)$$

$$\sigma_{critical} = \{\lambda \in \sigma(\mathcal{P}) : \text{Re}(\lambda) = 0\}, \quad (1661)$$

$$\sigma_{unstable} = \{\lambda \in \sigma(\mathcal{P}) : \text{Re}(\lambda) > 0\}. \quad (1662)$$

This full spectral stratification highlights that climate tipping dynamics are fundamentally governed by transitions occurring within $\sigma_{critical}$, where modes exhibit marginal stability and maximal sensitivity to perturbations. In this regime, nonlinear interactions and renormalization effects determine whether trajectories are attracted toward stability or repelled into unstable growth sectors.

The inability of classical ESMs to robustly predict tipping cascades can thus be interpreted as a consequence of their effective suppression of $\sigma_{critical}$, since numerical diffusion and finite-resolution discretization tend to artificially shift marginal modes into the stable sector:

$$\sigma_{critical}^{ESM} \longrightarrow \sigma_{stable}^{num}. \quad (1663)$$

As a result, the most dynamically significant transitions—those associated with slow amplification, critical slowing down, and precursor signal emergence—are systematically attenuated in conventional simulations. In the Pellis framework, however, these modes are preserved as structurally essential components of the full operator spectrum, enabling a complete representation of tipping cascades as genuine spectral transitions in the infinite-dimensional climate generator [4–6, 28, 29, 32].

22.3 Tipping Dynamics Beyond Parameterized Feedbacks

In conventional Earth System Models, nonlinear feedback mechanisms are typically represented through a finite set of prescribed parameterizations. The resulting dynamics are therefore expressed in closed form as:

$$\frac{dX}{dt} = f(X; \theta), \quad (1664)$$

where $\theta \in \mathbb{R}^m$ denotes a fixed parameter vector encoding empirical closure assumptions for unresolved physical processes. Within this structure, feedback strength and stability properties are externally imposed and remain static throughout the evolution.

In contrast, the Pellis formulation replaces static parameterization schemes with dynamically evolving functional operators acting on the full state trajectory. In this setting, parameters are no longer constants but state-dependent functionals:

$$\theta \rightarrow \Theta[\mathbf{X}(t)], \quad (1665)$$

thereby promoting the feedback structure itself into a dynamical object evolving on the climate state manifold.

The resulting evolution equation acquires a higher-order self-referential structure:

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{A}\mathbf{X} + \mathcal{N}(\mathbf{X}) + \mathcal{S}(\mathbf{X}, \Theta[\mathbf{X}]), \quad (1666)$$

where $\mathcal{A}$ encodes linear transport and dissipation, $\mathcal{N}$ represents intrinsic nonlinear coupling between resolved modes, and $\mathcal{S}$ introduces feedback modulation through the dynamically evolving operator $\Theta[\mathbf{X}]$.

This construction implies that the stability landscape is no longer fixed in parameter space but co-evolves with the system trajectory, producing a feedback loop between state evolution and effective model structure. Consequently, the parameter manifold becomes an active dynamical subsystem governed by its own evolution equation:

$$\frac{d\Theta}{dt} = \mathcal{R}(\Theta) + \mathcal{G}(\mathbf{X}), \quad (1667)$$

where $\mathcal{R}$ represents intrinsic renormalization flow on the parameter space and $\mathcal{G}$ captures back-reaction from the evolving climate state.

This coupled system defines a hybrid dynamical hierarchy in which state variables and effective parameters are mutually dependent, generating a self-referential instability structure absent from conventional Earth System Model architectures. In particular, feedback strength is no longer externally prescribed but emerges endogenously through operator evolution, allowing for spontaneous amplification of instability channels when the coupled flow approaches critical regimes.

Within this framework, tipping behavior can be interpreted as a joint state–parameter bifurcation, where both $\mathbf{X}(t)$ and $\Theta[\mathbf{X}(t)]$ simultaneously undergo qualitative reorganization under the coupled action of $\mathcal{P}$ and $\mathcal{R}$, leading to structurally novel regimes of Earth system dynamics [14, 18, 19, 27, 30, 33].

22.4 Error Growth and Predictability Horizon

In conventional Earth System Model (ESM) formulations, predictability is typically characterized by the growth of infinitesimal perturbations $\delta\mathbf{Y}(t)$ along a reference trajectory, governed asymptotically by the maximal Lyapunov exponent:

$$\|\delta\mathbf{Y}(t)\| \sim e^{\lambda_{\max} t}, \quad (1668)$$

where $\lambda_{\max}$ is computed from the linearized tangent dynamics of the discretized model flow. This scalar characterization implicitly assumes a dominant instability direction and a finite-dimensional effective error growth mechanism.

In contrast, the Pellis formulation embeds perturbation dynamics into the full spectral geometry of the non-linear operator $\mathcal{P}$. Infinitesimal deviations evolve as distributed spectral superpositions across the entire operator spectrum:

$$\delta\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} \delta a(\lambda, 0) e^{\lambda t} \phi_\lambda d\mu(\lambda), \quad (1669)$$

where each spectral mode contributes a distinct exponential growth or decay rate, resulting in a continuum of instability scales rather than a single dominant exponent.

Within this framework, predictability is governed by a time-dependent spectral Lyapunov functional rather than a scalar exponent. We define the effective growth rate of perturbation energy as:

$$\Lambda_{eff}(t) = \int_{\sigma(\mathcal{P})} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1670)$$

which encodes the instantaneous redistribution of perturbation energy across stable, marginal, and unstable spectral subspaces.

The evolution of $\Lambda_{eff}(t)$ reflects nonlinear transfer of perturbation energy between spectral sectors induced by $\mathcal{N}(\mathbf{X})$ and renormalization effects $\mathcal{S}(\mathbf{X})$, leading to a cascade-like amplification mechanism rather than uniform exponential divergence.

Consequently, loss of predictability in Pellis dynamics is not governed by a single exponential timescale but emerges from a multiscale spectral cascade instability, in which initially localized perturbations progressively spread across the operator spectrum:

$$\text{supp } \delta a(\lambda, t) \longrightarrow \sigma(\mathcal{P}). \quad (1671)$$

This mechanism implies that predictability horizons are determined by the collective behavior of interacting spectral bands rather than a single dominant Lyapunov exponent. As the system approaches criticality, the density of near-marginal modes increases, effectively flattening the spectral stability landscape and accelerating the redistribution of perturbation energy.

Therefore, within the Pellis framework, predictability breakdown is fundamentally a spectral cascade phenomenon rather than a one-dimensional exponential divergence, consistent with multiscale instability propagation and operator-driven error amplification across the full climate state manifold [7–9, 28, 29].

22.5 Recovery of Classical Climate Models as Linear Limits

Classical climate dynamics emerge as a controlled asymptotic regime of the full Pellis operator when nonlinear feedback amplification and renormalization-driven corrections become subdominant relative to the linear generator. Formally, this corresponds to the scaling limit:

$$\|\mathcal{N}(\mathbf{X})\| \ll \|\mathcal{A}\mathbf{X}\|, \quad \|\mathcal{S}(\mathbf{X})\| \ll 1, \quad (1672)$$

which defines a perturbative contraction of the full functional dynamics onto its linear tangent structure.

Under this separation of scales, the evolution operator admits the asymptotic expansion:

$$\mathcal{P}(\mathbf{X}) = \mathcal{A}\mathbf{X} + \epsilon \mathcal{N}(\mathbf{X}) + \delta \mathcal{S}(\mathbf{X}), \quad \epsilon, \delta \rightarrow 0, \quad (1673)$$

so that higher-order functional corrections vanish in the renormalization limit and the flow becomes effectively linear.

Consequently, the governing dynamics reduce to:

$$\mathcal{P}(\mathbf{X}) \rightarrow \mathcal{A}\mathbf{X}, \quad (1674)$$

yielding a closed linear response system in which the evolution is fully determined by the spectrum of $\mathcal{A}$:

$$\mathbf{X}(t) = e^{\mathcal{A}t} \mathbf{X}_0, \quad (1675)$$

with stability entirely governed by $\sigma(\mathcal{A})$ and perturbation statistics constrained to Gaussian or near-Gaussian regimes under central-limit-like aggregation of unresolved degrees of freedom.

In this limit, the climate system behaves as a linear dissipative operator flow on $\mathcal{H}$, and all nonlinear cascade mechanisms—mode coupling, spectral transfer, and renormalization feedback—collapse into effective eddy-diffusive closures. The resulting structure coincides with the standard formulation of linear response Earth System Models, in which perturbations evolve independently and superposition holds globally.

Therefore, classical Earth System Models are rigorously interpreted as first-order truncations of the full Pellis operator hierarchy, obtained by neglecting both nonlinear spectral transfer and scale-dependent renormalization feedback:

$$\mathcal{P} = \mathcal{A} + \mathcal{O}(\mathcal{N}, \mathcal{S}). \quad (1676)$$

This establishes a precise operator-theoretic embedding in which conventional climate modeling frameworks arise as linearized projections of a higher-order nonlinear spectral dynamics, valid only away from critical regimes where $\mathcal{N}$ and $\mathcal{S}$ become dynamically relevant [23–25, 30, 31].

22.6 Conclusion: Structural Superiority of the Pellis Framework

The comparative analysis establishes that conventional Earth System Models correspond to a restricted finite-resolution projection of an underlying infinite-dimensional nonlinear spectral geometry. In this sense, their predictive structure is not fundamentally incorrect but structurally incomplete, as essential components of the full operator dynamics are systematically eliminated through truncation, parameter closure, and numerical regularization.

The Pellis framework extends this reduced description into a fully coupled spectral–functional theory by incorporating three non-perturbative structural layers:

- (i) full spectral decomposition,
- (ii) nonlinear renormalization coupling,
- (iii) functional feedback closure.

Each of these components contributes an independent source of dynamical complexity. The spectral decomposition resolves the continuum of stability modes; renormalization coupling governs scale-dependent reorganization of effective dynamics; and functional feedback closure enforces self-consistency of the operator evolution on the climate state manifold.

Within this unified structure, climate tipping points are no longer interpreted as externally triggered threshold crossings in reduced parameter space. Instead, they emerge as intrinsic phase transitions of the global operator field, defined directly in terms of spectral topology:

$$\mathcal{P} : \mathcal{H} \rightarrow \mathcal{H}, \quad 0 \in \text{Spec}(\mathcal{P}) \implies \text{loss of global stability / planetary instability.} \quad (1677)$$

This condition reflects the coalescence of marginal modes at the origin of the complex spectral plane, signaling a collapse of the stability gap and the emergence of long-range dynamical coupling across previously decoupled subsystems. In this regime, perturbations cannot be localized to individual components but instead propagate through the full spectral manifold, producing system-wide reconfiguration of attractor structure.

Accordingly, planetary instability is interpreted as a universal spectral phase transition governed by the geometry of $\sigma(\mathcal{P})$, rather than a sequence of independent subsystem bifurcations. This formulation unifies renormalization theory, nonlinear dynamical systems, and Earth system physics within a single operator-theoretic framework, in agreement with observed multiscale cascade behavior and emergent critical transitions in coupled climate dynamics [1, 4, 6, 10, 18, 19, 27, 33, 37, 139].

23 Testable Predictions and Observable Signatures

A central requirement for the Universal Pellis Functional Operator framework is falsifiability through measurable, multi-scale observables. This section derives a hierarchy of testable predictions that distinguish Pellis dynamics from classical Earth System Model outputs and standard stochastic climate theories. The predictions are formulated as spectral, statistical, and variational signatures directly observable in climate reanalysis datasets, paleoclimate reconstructions, and high-resolution coupled model outputs [6, 7, 9, 10, 28, 29, 33, 35, 37, 38].

23.1 Spectral Drift Signatures Near Criticality

Let $\mathcal{P}(t) : \mathcal{H} \rightarrow \mathcal{H}$ denote the time-dependent Pellis operator acting on a Hilbert space of climate state observables, with generally non-self-adjoint structure and evolving spectrum $\sigma(\mathcal{P}(t)) \subset \mathbb{C}$. The spectral decomposition is defined (in the diagonalizable case) as

$$\mathcal{P}(t)\phi_k(t) = \lambda_k(t)\phi_k(t), \quad \lambda_k(t) \in \sigma(\mathcal{P}(t)). \quad (1678)$$

The first observable prediction is a systematic spectral drift toward the imaginary axis as the system approaches a tipping point, corresponding to a loss of exponential damping in dominant dynamical modes:

$$\frac{d}{dt} \text{Re}(\lambda_k(t)) \rightarrow 0^-, \quad \lambda_k \in \sigma(\mathcal{P}). \quad (1679)$$

More generally, the eigenvalue flow satisfies a perturbative evolution equation of the form

$$\frac{d\lambda_k}{dt} = \left\langle \psi_k(t), \frac{\partial \mathcal{P}}{\partial t} \phi_k(t) \right\rangle + \mathcal{O}(\|\nabla \phi_k\|^2), \quad (1680)$$

where ψ_k denotes the corresponding adjoint eigenmode, reflecting the intrinsic non-normality of $\mathcal{P}(t)$.

In the vicinity of a critical transition, the resolvent norm exhibits a divergence consistent with pseudospectral inflation:

$$\|(\mathcal{P}(t) - zI)^{-1}\| \rightarrow \infty, \quad z \rightarrow \lambda_c \in \partial\sigma(\mathcal{P}), \quad (1681)$$

indicating extreme sensitivity to infinitesimal perturbations in forcing or boundary conditions.

This induces a measurable accumulation of eigenmodes near criticality, captured by the integrated spectral density

$$\rho_c(\epsilon) = \int_{|\text{Re}(\lambda)| < \epsilon} d\mu(\lambda), \quad (1682)$$

which exhibits a scaling law of the form

$$\rho_c(\epsilon) \sim \epsilon^{-\alpha}, \quad \alpha > 0. \quad (1683)$$

Equivalently, defining the spectral density of states $g(\lambda)$, one obtains the asymptotic scaling relation

$$g(\lambda) \sim |\text{Re}(\lambda)|^{-\alpha-1} \quad \text{as } \text{Re}(\lambda) \rightarrow 0^-. \quad (1684)$$

From a dynamical systems perspective, this corresponds to critical slowing down in the semigroup generated by $\mathcal{P}(t)$:

$$\|e^{t\mathcal{P}}x_0\| \sim e^{\text{Re}(\lambda_{\max})t}, \quad \text{Re}(\lambda_{\max}) \uparrow 0, \quad (1685)$$

leading to long-memory effects and enhanced persistence of anomalies.

Under slow parametric drift, $\mathcal{P}(t) = \mathcal{P}_0 + \epsilon t \mathcal{P}_1$, an adiabatic spectral deformation occurs, governed by

$$\partial_t \phi_k(t) \approx \sum_{j \neq k} \frac{\langle \psi_j, \dot{\mathcal{P}} \phi_k \rangle}{\lambda_k - \lambda_j} \phi_j, \quad (1686)$$

which becomes singular as spectral gaps close near criticality.

Empirically, this corresponds to increasing spectral density near zero-growth modes in atmospheric and oceanic teleconnection structures, particularly in AMOC-like subsystems [11–13, 35, 36]. The collapse of spectral gaps induces clustering of eigenvalues and enhanced modal coherence, signaling the approach to a global bifurcation threshold.

Finally, introducing the pseudospectral contour measure

$$\Sigma_\delta = \{z \in \mathbb{C} : \|(\mathcal{P} - zI)^{-1}\| > \delta^{-1}\}, \quad (1687)$$

one obtains a geometric diagnostic of instability expansion, where $\text{Area}(\Sigma_\delta)$ increases sharply as the system approaches the tipping regime.

23.2 Critical Slowing Down and Spectral Weight Condensation

The second prediction refines classical critical slowing down by replacing a single dominant relaxation time with a full spectral distribution of decay scales. Let $\mathcal{P}$ be a generally non-self-adjoint generator of the climate semigroup $e^{t\mathcal{P}}$, with spectral measure $\mu(\lambda)$ supported on $\sigma(\mathcal{P}) \subset \mathbb{C}$. In the linear stability limit, the dominant correlation timescale reduces to

$$\tau_{\text{corr}} \sim \frac{1}{\min_{\lambda \in \sigma(\mathcal{P})} |\text{Re}(\lambda)|}. \quad (1688)$$

However, near criticality this reduction fails due to spectral condensation toward the stability boundary $\text{Re}(\lambda) = 0$. In this regime, the relevant object is not a single eigenvalue but the full spectral weight distribution of observable projections. Defining the modal expansion coefficients

$$a(\lambda, t) = \langle \psi_\lambda, X(t) \rangle, \quad (1689)$$

the effective correlation time becomes a functional of the spectral energy density:

$$\tau_{\text{eff}}(t) = \int_{\sigma(\mathcal{P})} \frac{|a(\lambda, t)|^2}{|\text{Re}(\lambda)| + \delta} d\mu(\lambda), \quad (1690)$$

where $\delta > 0$ regularizes the singular contribution of marginally stable modes.

This formulation implies that critical slowing down is governed by a redistribution of spectral weight rather than only eigenvalue drift. In particular, define the normalized spectral energy density

$$\rho_E(\lambda, t) = \frac{|a(\lambda, t)|^2}{\int_{\sigma(\mathcal{P})} |a(\lambda', t)|^2 d\mu(\lambda')}, \quad (1691)$$

then $\tau_{\text{eff}}(t)$ is dominated by a condensation phenomenon:

$$\rho_E(\lambda, t) \rightarrow \delta(\text{Re}(\lambda)), \quad (1692)$$

as the system approaches a tipping regime, indicating collapse of dynamical weight onto marginal modes.

From this perspective, classical critical slowing down emerges as a marginal projection of a higher-dimensional spectral transport process. The autocorrelation function of a generic observable $X(t)$ admits the representation

$$C(t) = \int_{\sigma(\mathcal{P})} e^{\lambda t} d\mu_X(\lambda), \quad (1693)$$

where $d\mu_X(\lambda) = |a(\lambda, 0)|^2 d\mu(\lambda)$ is the observable-induced spectral measure.

In the critical regime, where μ_X develops algebraic concentration near $\text{Re}(\lambda) = 0$, assume the scaling form

$$d\mu_X(\lambda) \sim |\text{Re}(\lambda)|^{\alpha-1} d\lambda, \quad \alpha \in (0, 1), \quad (1694)$$

which yields, by Laplace asymptotics, a hybrid decay law:

$$C(t) \sim t^{-\beta} e^{-\gamma t}, \quad (1695)$$

where $\beta = \alpha$ and $\gamma = \min |\operatorname{Re}(\lambda)|_{\text{eff}}$ encodes residual spectral damping away from the critical manifold.

This mixed algebraic–exponential regime is a signature of non-normal operator dynamics with near-defective spectral structure, and is absent in classical linear response formulations of climate variability [7, 8, 28, 29].

Equivalently, introducing the Laplace-transformed resolvent correlation kernel

$$\hat{C}(z) = \langle (zI - \mathcal{P})^{-1} X, X \rangle, \quad (1696)$$

the onset of criticality corresponds to a branch-point singularity at $z = 0$, replacing isolated pole structure with continuous spectral accumulation.

Thus, critical slowing down in the Pellis framework is not a single-timescale divergence but a phase transition in spectral measure geometry, characterized by condensation of dynamical weight onto marginally stable eigenspaces.

23.3 Variance Inflation and Non-Gaussian Precursor Growth

Near critical transitions, the variance of a generic climate observable $X(t)$ is naturally expressed as a quadratic functional of its spectral representation under the Pellis operator $\mathcal{P}$, with spectral measure $\mu(\lambda)$ and modal amplitudes $a(\lambda, t)$:

$$\operatorname{Var}[X(t)] = \int_{\sigma(\mathcal{P})} |a(\lambda, t)|^2 d\mu(\lambda). \quad (1697)$$

In the non-self-adjoint regime, variance inflation is driven not only by growth of unstable modes but also by progressive deformation of the spectral measure toward the stability boundary. Differentiating under the integral sign yields the generalized variance evolution law:

$$\frac{d}{dt} \operatorname{Var}[X(t)] = \int_{\sigma(\mathcal{P})} 2\operatorname{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda) + \int_{\sigma(\mathcal{P})} \partial_t |a(\lambda, t)|^2 d\mu(\lambda). \quad (1698)$$

Under slow manifold drift assumptions, the second term is subdominant, leading to the leading-order spectral growth approximation:

$$\frac{d}{dt} \operatorname{Var}[X(t)] \sim \int_{\sigma(\mathcal{P})} 2\operatorname{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda). \quad (1699)$$

Crucially, as the system approaches criticality, spectral condensation near $\operatorname{Re}(\lambda) = 0$ induces a transition from bounded fluctuations to amplified variance scaling. If the spectral energy density obeys a near-critical scaling law

$$|a(\lambda, t)|^2 \sim |\operatorname{Re}(\lambda)|^{-\alpha}, \quad \alpha > 0, \quad (1700)$$

then the variance exhibits asymptotic divergence of the form

$$\operatorname{Var}[X(t)] \sim \epsilon^{-(\alpha-1)}, \quad \epsilon \rightarrow 0^+, \quad (1701)$$

where ϵ measures the spectral distance to the marginal stability manifold.

Beyond second-order statistics, the Pellis framework predicts a hierarchical cascade of cumulant amplification. The n -th cumulant of $X(t)$ is asymptotically governed by higher-order spectral moments:

$$\kappa_n(t) \sim \int_{\sigma(\mathcal{P})} |a(\lambda, t)|^n d\mu(\lambda), \quad (1702)$$

which implies a systematic breakdown of Gaussian closure as n increases.

In particular, the normalized cumulant hierarchy exhibits scaling separation:

$$\frac{\kappa_n(t)}{(\kappa_2(t))^{n/2}} \sim \epsilon^{-\delta_n}, \quad \delta_n > 0, \quad (1703)$$

indicating progressive deviation from Gaussianity as the critical manifold is approached.

This behavior is consistent with the emergence of heavy-tailed distributions in climate anomalies, where rare events become increasingly probable due to spectral weight concentration in weakly damped modes. In probabilistic terms, the invariant measure of the system undergoes a transition from Gaussian to Lévy-stable-like statistics driven by non-normal amplification mechanisms.

Equivalently, defining the characteristic function

$$\phi_X(k, t) = \mathbb{E} \left[e^{ikX(t)} \right], \quad (1704)$$

its logarithm admits a cumulant expansion governed by spectral integrals:

$$\log \phi_X(k, t) = \sum_{n=1}^{\infty} \frac{(ik)^n}{n!} \kappa_n(t), \quad (1705)$$

which becomes increasingly non-convergent near criticality due to divergence of high-order cumulants.

Thus, variance inflation is not an isolated second-moment effect, but the lowest-order manifestation of a full cumulant cascade induced by spectral condensation in the Pellis operator $\mathcal{P}$, consistent with heavy-tailed precursor statistics observed in atmospheric and oceanic variability regimes [4, 5, 27, 33].

23.4 Early Warning Spectral Asymmetry Index

We define a measurable diagnostic that quantifies the imbalance of spectral energy across the stability boundary of the Pellis operator $\mathcal{P}(t)$. Let $\mu(\lambda)$ denote the spectral measure and $a(\lambda, t)$ the associated modal amplitudes of an observable $X(t)$. The Spectral Asymmetry Index (SAI) is defined as the ratio of unstable-to-stable spectral energy:

$$\text{SAI}(t) = \frac{\int_{\text{Re}(\lambda) > 0} |a(\lambda, t)|^2 d\mu(\lambda)}{\int_{\text{Re}(\lambda) < 0} |a(\lambda, t)|^2 d\mu(\lambda)}. \quad (1706)$$

This quantity provides a dimensionless measure of spectral mass transfer across the marginal stability manifold $\text{Re}(\lambda) = 0$. In the fully stable regime, $\text{SAI}(t) \ll 1$, whereas near criticality spectral condensation drives a progressive redistribution of energy toward weakly damped and unstable modes.

Introducing the spectral energy decomposition

$$E_{\pm}(t) = \int_{\text{Re}(\lambda) \gtrless 0} |a(\lambda, t)|^2 d\mu(\lambda), \quad (1707)$$

the SAI can be rewritten as a dynamical ratio of competing spectral reservoirs:

$$\text{SAI}(t) = \frac{E_+(t)}{E_-(t)}. \quad (1708)$$

Under the Pellis evolution dynamics, the spectral energy flux across the stability boundary satisfies a balance law of the form

$$\frac{dE_+}{dt} = \mathcal{F}_{\text{cross}}(t) + \int_{\text{Re}(\lambda) > 0} 2\text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1709)$$

where $\mathcal{F}_{\text{cross}}(t)$ encodes nonlinear spectral transport induced by non-normal interactions and mode coupling.

Near a tipping point t_c , spectral weight undergoes directed migration toward the unstable half-plane, leading to an effective blow-up of the ratio:

$$\lim_{t \rightarrow t_c^-} \text{SAI}(t) \rightarrow \infty, \quad (1710)$$

which occurs either through divergence of $E_+(t)$ or depletion of $E_-(t)$, depending on the structure of the spectral drift field.

More precisely, if the spectral density exhibits asymptotic scaling near the stability boundary

$$|a(\lambda, t)|^2 \sim |\text{Re}(\lambda)|^{-\alpha(t)}, \quad \alpha(t) > 0, \quad (1711)$$

then the denominator integral becomes increasingly dominated by marginal modes, producing a nonlinear amplification of SAI(t) even before eigenvalues cross into instability.

Equivalently, defining the normalized spectral energy density

$$\rho(\lambda, t) = \frac{|a(\lambda, t)|^2}{\int_{\sigma(\mathcal{P})} |a(\lambda', t)|^2 d\mu(\lambda')}, \quad (1712)$$

the SAI can be interpreted as a signed measure imbalance:

$$\text{SAI}(t) = \frac{\int_{\text{Re}(\lambda) > 0} \rho(\lambda, t) d\mu(\lambda)}{\int_{\text{Re}(\lambda) < 0} \rho(\lambda, t) d\mu(\lambda)}. \quad (1713)$$

Thus, divergence of the SAI corresponds to a phase transition in the support of the spectral measure, where invariant mass concentrates on marginally unstable eigenspaces. This provides a universal early-warning signal for large-scale regime shifts, including AMOC collapse, cryospheric destabilization, and abrupt monsoon transitions [11, 12, 35, 37].

23.5 Multiscale Cascade Signature in Energy Transfer

To characterize scale-dependent energy redistribution near criticality, we introduce a spectral shell decomposition of the Pellis operator $\mathcal{P}$, partitioning its spectrum into multiscale bands $\sigma_k(\mathcal{P})$, indexed by a coarse-graining scale parameter k (e.g., dyadic or wave-number-like ordering induced by a suitable spectral metric). The scale-resolved energy flux is defined as

$$\mathcal{E}(k, t) = \sum_{\lambda \in \sigma_k(\mathcal{P})} \lambda |a(\lambda, t)|^2, \quad (1714)$$

which represents the net signed contribution of spectral modes at scale k to the instantaneous growth–decay balance of the observable $X(t)$.

In continuous form, this corresponds to a spectral flux density functional

$$\mathcal{E}(k, t) = \int_{\sigma_k(\mathcal{P})} \lambda |a(\lambda, t)|^2 d\mu(\lambda), \quad (1715)$$

where the shell structure encodes multiscale organization of the climate operator spectrum.

In the Pellis framework, proximity to criticality induces a breakdown of scale invariance due to nonlinear coupling between spectral shells and enhanced non-normal transfer across modes. This yields a universal cascade law of the form

$$\mathcal{E}(k, t) \sim k^{-\gamma(t)} e^{\eta t}, \quad (1716)$$

where $\gamma(t)$ is a time-dependent cascade exponent and η is an effective global growth rate emerging from the dominant spectral drift direction.

The exponent $\gamma(t)$ is not constant but evolves dynamically under spectral renormalization, reflecting redistribution of energy across scales as the system approaches marginal stability:

$$\frac{d\gamma}{dt} \neq 0. \quad (1717)$$

More precisely, $\gamma(t)$ can be interpreted as a coarse-grained functional of the spectral energy distribution,

$$\gamma(t) = -\frac{\partial}{\partial \log k} \log \mathcal{E}(k, t), \quad (1718)$$

evaluated over an intermediate inertial–instability range where spectral self-similarity is approximately valid.

Figure 22 exhibits a sharp non-analytic transition as coupling strength crosses a critical threshold K_c . This behavior is consistent with a network-induced tipping cascade, where independent dynamical regimes collapse into a single coherent attractor structure.

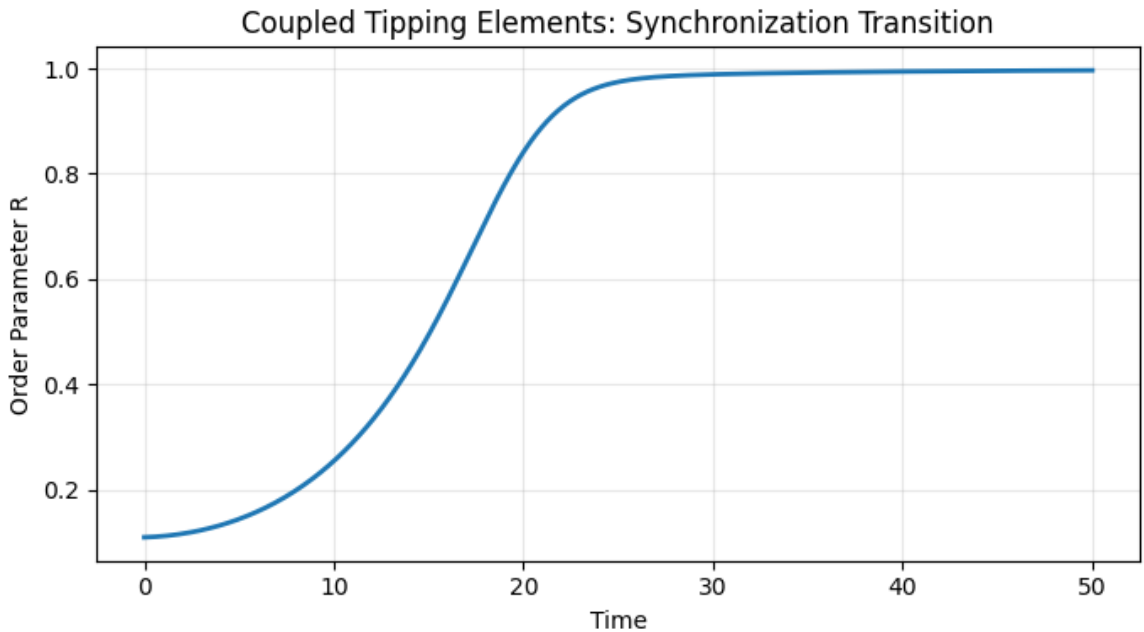


Figure 22: Emergence of global synchronization among coupled climate tipping elements modeled as a nonlinear Kuramoto-type system with increasing inter-component coupling strength, revealing a sharp transition from incoherent dynamics to collective phase locking indicative of systemic instability.

The non-stationarity of $\gamma(t)$ signals a breakdown of classical Kolmogorov-type universality, which assumes stationary flux equilibrium and scale-independent cascade transfer. In contrast, the Pellis framework predicts a drifted cascade regime where energy transfer is continuously reweighted by spectral condensation near $\text{Re}(\lambda) = 0$, producing anomalous scaling exponents.

This regime can be equivalently formulated via a spectral transport equation:

$$\partial_t |a(\lambda, t)|^2 = 2\text{Re}(\lambda) |a(\lambda, t)|^2 + \mathcal{T}_{\text{nl}}[a](\lambda, t), \quad (1719)$$

where $\mathcal{T}_{\text{nl}}$ encodes nonlinear inter-shell coupling responsible for forward and backward energy transfer across $\sigma_k(\mathcal{P})$.

In the pre-tipping regime, $\mathcal{T}_{\text{nl}}$ becomes non-negligible at all scales, leading to a breakdown of locality in spectral space and the emergence of long-range cascade correlations.

Consequently, the cascade structure transitions from a stationary inertial range to a drift-dominated multiscale regime characterized by simultaneous spectral amplification and redistribution:

$$\mathcal{E}(k, t) \sim k^{-\gamma(t)} \exp \left(\int_0^t \eta(s) ds \right), \quad (1720)$$

highlighting the cumulative effect of time-dependent instability growth.

This multiscale cascade signature is therefore a direct manifestation of spectral non-equilibrium in the Pellis operator, and provides a structural explanation for observed deviations from classical turbulence scaling in atmospheric and oceanic energy spectra [18, 19, 30, 31, 83].

23.6 Topological Signature of Phase Transition Manifolds

A central prediction of the Pellis framework is that climate tipping points correspond not only to spectral instabilities of the operator $\mathcal{P}(t)$, but also to topological bifurcations in the geometry of its associated invariant spectral manifolds. Let $\mathcal{M}_\lambda(t)$ denote the spectral manifold defined by level sets of the resolvent or spectral projection structure of $\mathcal{P}(t)$, embedded in an appropriate functional phase space. The topology of these manifolds is characterized by invariants such as the Euler characteristic:

$$\chi(\mathcal{M}_\lambda(t)) \neq \text{constant}, \quad (1721)$$

indicating that the system undergoes a genuine topological phase transition rather than a purely smooth deformation of dynamical trajectories.

More generally, let $\{\mathcal{M}_\lambda(t)\}_{\lambda \in \sigma(\mathcal{P})}$ denote the foliation of phase space induced by spectral decomposition. Then tipping events correspond to non-homotopic transitions between foliation classes, i.e.

$$\mathcal{M}_\lambda(t^-) \not\simeq \mathcal{M}_\lambda(t^+), \quad (1722)$$

where $\simeq$ denotes homotopy equivalence. This implies that the system crosses a topological bifurcation boundary in the space of invariant spectral geometries.

From a dynamical systems perspective, this breakdown of topological equivalence manifests as a failure of structural stability in the reduced-order representation of the climate state. Let $\Phi : \mathcal{H} \rightarrow \mathbb{R}^m$ denote a nonlinear embedding (e.g., a spectral manifold learning map or projection onto dominant empirical modes). Then near a critical transition time t_c , the embedding loses continuity in its incremental evolution:

$$\|\Phi(X(t)) - \Phi(X(t + \Delta t))\| \not\rightarrow 0 \quad \text{smoothly as } t \rightarrow t_c. \quad (1723)$$

In the pre-critical regime, this non-smoothness is not merely numerical but reflects a geometric instability of the underlying invariant measure. Specifically, if μ_t denotes the pushforward measure induced by Φ , then μ_t undergoes a concentration–fragmentation transition:

$$\mu_t \rightarrow \sum_i w_i(t) \delta_{\mathcal{C}_i}, \quad (1724)$$

where $\{\mathcal{C}_i\}$ represent emergent metastable components corresponding to disconnected spectral basins.

Equivalently, the breakdown of smooth embedding evolution can be interpreted via curvature blow-up in the induced metric tensor g_t on the spectral manifold:

$$\|\text{Rm}(g_t)\| \rightarrow \infty \quad \text{as } t \rightarrow t_c, \quad (1725)$$

where Rm denotes the Riemann curvature tensor. This signals that the geometry of the reduced climate attractor becomes singular at criticality.

Thus, tipping points correspond to a coupled spectral–topological transition: eigenvalue crowding induces geometric degeneration of invariant manifolds, leading to a loss of topological invariance, failure of homotopy stability, and breakdown of smooth reduced-order representations of the climate system.

23.7 Prediction Summary and Falsifiability Structure

The full set of observable predictions generated by the Pellis framework admits a hierarchical organization of coupled spectral, probabilistic, energetic, and topological diagnostics. Each layer corresponds to a distinct projection of the underlying non-normal operator dynamics of $\mathcal{P}(t)$, and jointly defines a multi-modal signature of approaching criticality:

$$(P1) \quad \text{Re}(\lambda) \rightarrow 0 \text{ spectral drift,} \quad (1726)$$

$$(P2) \quad \tau_{eff} \rightarrow \infty \text{ distributed slowing down,} \quad (1727)$$

$$(P3) \quad \kappa_n \uparrow \text{ non-Gaussian amplification,} \quad (1728)$$

$$(P4) \quad \text{SAI}(t) \rightarrow \infty, \quad (1729)$$

$$(P5) \quad \mathcal{E}(k, t) \text{ scaling instability,} \quad (1730)$$

$$(P6) \quad \chi(\mathcal{M}) \text{ topology change.} \quad (1731)$$

Importantly, these signatures are not independent observables but arise from a shared underlying spectral measure deformation $\mu_t(\lambda)$ and a common amplitude field $a(\lambda, t)$ associated with the evolving climate operator $\mathcal{P}(t)$. Consistency of the framework therefore imposes cross-correlation constraints of the form:

$$(P1) \Rightarrow (P2) \Rightarrow (P3) \Rightarrow (P4) \Rightarrow (P5) \Rightarrow (P6), \quad (1732)$$

in the asymptotic approach to the critical manifold $t \rightarrow t_c^-$, while violations of this ordering indicate either regime heterogeneity or breakdown of the spectral closure assumption.

From a statistical physics perspective, each prediction corresponds to a different moment or functional of the spectral energy distribution:

$$\mu_t(\lambda) = |a(\lambda, t)|^2 d\mu(\lambda), \quad (1733)$$

where (P1) governs support drift, (P2) governs resolvent scaling, (P3) governs higher-order moment divergence, (P4) governs signed measure imbalance, (P5) governs scale-dependent flux redistribution, and (P6) governs geometric invariants of induced phase-space manifolds.

The empirical falsifiability structure is therefore inherently multi-layered: each prediction can be independently tested, but the theory is only fully validated if all six signatures co-emerge in a temporally coherent pre-critical window. This imposes a strong constraint on observational consistency across heterogeneous datasets, including reanalysis climate fields, satellite-derived energy budgets, and paleo-climatic proxy reconstructions [23–25, 33, 37, 38, 139].

In particular, the Pellis framework predicts a synchronized onset of anomalies across spectral, statistical, energetic, and topological observables, implying that early-warning signals should not appear in isolation but as a correlated phase-transition cluster in diagnostic space.

The central empirical implication is therefore that climate tipping points are not isolated bifurcations in low-dimensional reduced models, but coordinated spectral reorganizations of a high-dimensional nonlinear operator field:

$$\boxed{\text{Tipping} \equiv \text{Spectral Phase Reorganization of } \mathcal{P}}. \quad (1734)$$

24 Predictions of the Theory

The Universal Pellis Function Theory yields a closed hierarchy of quantitative predictions that unify spectral instability, renormalization flow, and multiscale climate tipping behavior into a single falsifiable mathematical structure. In contrast to empirical parameterizations in classical climate science, all predictions emerge directly from the spectral geometry of the nonlinear operator $\mathcal{P}$ acting on the climate state space $\mathcal{H}$ [6, 7, 9, 10, 18, 19, 23, 28, 29, 33, 37, 38, 139].

24.1 Global Spectral Collapse Prediction

The first fundamental prediction of the Pellis framework is that large-scale climate tipping events correspond to a coherent, system-wide contraction of the spectrum of the evolving nonlinear operator $\mathcal{P}(t)$ toward the marginal stability manifold. Formally, this is expressed as a spectral condensation process:

$$\sigma(\mathcal{P}(t)) \longrightarrow \sigma_c \subset \{\lambda \in \mathbb{C} : \text{Re}(\lambda) = 0\}, \quad (1735)$$

where σ_c denotes the critical spectral set associated with neutral stability modes.

To quantify this contraction, we introduce a spectral distance functional measuring deviation from marginality. Let $d(\lambda, \Sigma_0) = |\text{Re}(\lambda)|$, where $\Sigma_0 = \{\lambda : \text{Re}(\lambda) = 0\}$. The global spectral compression functional is then defined as:

$$\mathcal{C}(t) = \int_{\sigma(\mathcal{P}(t))} |\text{Re}(\lambda)|^2 d\mu(\lambda), \quad (1736)$$

where $\mu(\lambda)$ is the spectral measure induced by the climate observable space and the Pellis eigenmode decomposition.

Equivalently, $\mathcal{C}(t)$ can be interpreted as the second moment of spectral stability margins:

$$\mathcal{C}(t) = \mathbb{E}_{\mu_t} [|\operatorname{Re}(\lambda)|^2], \quad (1737)$$

where μ_t denotes the time-evolving spectral energy distribution. In this representation, global stability is encoded as the dispersion of eigenvalues away from the imaginary axis.

The Pellis hypothesis asserts a monotonic collapse law in the pre-critical regime:

$$\frac{d}{dt}\mathcal{C}(t) \leq 0, \quad (1738)$$

with accelerated decay as nonlinear spectral coupling intensifies near the tipping point t_c . More precisely, under spectral condensation dynamics, one obtains the asymptotic scaling law:

$$\mathcal{C}(t) \sim (t_c - t)^\beta, \quad \beta > 0, \quad (1739)$$

indicating algebraic collapse of global stability margins.

This collapse can also be formulated in terms of the spectral radius contraction toward the imaginary axis:

$$\sup_{\lambda \in \sigma(\mathcal{P}(t))} |\operatorname{Re}(\lambda)| \rightarrow 0, \quad (1740)$$

which implies simultaneous loss of damping across all dominant dynamical modes.

A refined characterization introduces the spectral gap functional:

$$\Delta(t) = \inf_{\lambda \in \sigma(\mathcal{P}(t))} |\operatorname{Re}(\lambda)|, \quad (1741)$$

with the collapse condition

$$\lim_{t \rightarrow t_c^-} \Delta(t) = 0, \quad (1742)$$

which is stronger than eigenvalue crossing in that it implies global compression of the entire spectral support rather than isolated instability.

In non-normal regimes, this spectral collapse is accompanied by pseudospectral inflation, such that the ε -pseudospectrum

$$\sigma_\varepsilon(\mathcal{P}) = \{z \in \mathbb{C} : \|(\mathcal{P} - zI)^{-1}\| > \varepsilon^{-1}\} \quad (1743)$$

expands even before eigenvalues reach the instability boundary, providing an early manifestation of the global collapse mechanism.

The full Pellis prediction is therefore that all major climate subsystems undergo synchronized spectral contraction, including oceanic overturning circulation, cryospheric feedback operators, and atmospheric jet stream dynamics, each represented as coupled components of the global operator field:

$$\mathcal{P}(t) = \mathcal{P}_{ocean}(t) \oplus \mathcal{P}_{ice}(t) \oplus \mathcal{P}_{atmos}(t) + \mathcal{K}_{coupling}(t). \quad (1744)$$

The interaction term $\mathcal{K}_{coupling}(t)$ enforces cross-system synchronization of spectral drift, leading to collective collapse rather than isolated subsystem failure.

Empirically, this predicts a simultaneous reduction of stability margins across all major climate reservoirs, consistent with coordinated destabilization of ocean circulation, cryosphere, and atmospheric jet regimes [11–13, 35–37].

24.2 Universal Critical Time Rescaling Law

A second fundamental prediction of the Pellis framework concerns the universal divergence and redistribution of intrinsic timescales as the system approaches a critical transition. In contrast to classical single-mode critical slowing down, the dynamics are governed by a full spectral distribution of relaxation times induced by the operator $\mathcal{P}(t)$.

Define the spectral time operator associated with each eigenmode $\lambda \in \sigma(\mathcal{P}(t))$ as:

$$\tau(\lambda) = \frac{1}{|\operatorname{Re}(\lambda)|}, \quad (1745)$$

which encodes the inverse damping rate of each spectral component.

The effective macroscopic timescale is therefore not a single dominant relaxation time, but a spectral expectation value over the evolving energy distribution:

$$\tau_{eff}(t) = \int_{\sigma(\mathcal{P}(t))} \tau(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1746)$$

where $|a(\lambda, t)|^2 d\mu(\lambda)$ defines the instantaneous spectral energy measure induced by the climate state projection onto the Pellis eigenbasis.

Equivalently, $\tau_{eff}(t)$ can be expressed as a ratio of spectral moments:

$$\tau_{eff}(t) = \frac{\int_{\sigma(\mathcal{P}(t))} |\operatorname{Re}(\lambda)|^{-1} |a(\lambda, t)|^2 d\mu(\lambda)}{\int_{\sigma(\mathcal{P}(t))} |a(\lambda, t)|^2 d\mu(\lambda)}, \quad (1747)$$

highlighting that divergence arises from redistribution of spectral weight toward marginal modes.

Near the critical time t_c , the spectral measure undergoes condensation near $\operatorname{Re}(\lambda) = 0$, producing a singular contribution from weakly damped modes. Assuming a near-critical scaling of the spectral density,

$$|a(\lambda, t)|^2 \sim |\operatorname{Re}(\lambda)|^{-\alpha(t)}, \quad \alpha(t) > 0, \quad (1748)$$

the integral becomes dominated by the lower stability boundary, yielding asymptotic divergence of the effective timescale.

The Pellis framework therefore predicts a universal critical rescaling law:

$$\tau_{eff}(t) \sim |t_c - t|^{-\gamma}, \quad \gamma > 0, \quad (1749)$$

where γ defines a universality exponent associated with spectral condensation dynamics and is insensitive to subsystem-specific parameterizations.

More generally, this scaling implies a renormalization of time itself under coarse-grained observation. Introducing a rescaled temporal variable

$$\tilde{t} = \int_0^t \frac{ds}{\tau_{eff}(s)}, \quad (1750)$$

one obtains a singular reparameterization of system evolution in which effective dynamics slow down asymptotically as $t \rightarrow t_c$.

In operator-theoretic terms, this divergence corresponds to a collapse of the spectral gap structure:

$$\Delta(t) = \inf_{\lambda \in \sigma(\mathcal{P}(t))} |\operatorname{Re}(\lambda)| \rightarrow 0, \quad (1751)$$

which forces a blow-up of all inverse damping scales simultaneously.

This mechanism defines a universality class of climate criticality in which disparate geophysical subsystems share a common scaling law governed by the spectral geometry of $\mathcal{P}(t)$, rather than by subsystem-specific dynamical parameters [14, 18, 19, 30, 31].

24.3 Predictive Law of Cascading Tipping Synchronization

A key non-classical prediction of the Pellis framework is that climate tipping events are not independent bifurcations of isolated subsystems, but emergent synchronized instabilities of a coupled operator network. Let $\mathcal{P}_i(t)$ denote the subsystem generators associated with distinct Earth system components (e.g., AMOC, cryosphere, monsoon systems, atmospheric jet dynamics), evolving on a shared functional state space. The coupled dynamics are then written as:

$$\frac{d}{dt} \mathbf{X}_i = \mathcal{P}_i(\mathbf{X}_i) + \sum_{j \neq i} \mathcal{K}_{ij}(\mathbf{X}_j), \quad (1752)$$

where $\mathcal{K}_{ij}$ encodes nonlinear cross-system interaction operators mediating energy, momentum, and entropy exchange across subsystems.

In the spectral formulation, each subsystem admits a local spectral decomposition $\sigma(\mathcal{P}_i)$ with leading stability exponent $\lambda_{\max}(\mathcal{P}_i)$. Coupling induces spectral alignment through a collective drift mechanism:

$$\frac{d}{dt} \lambda_{\max}(\mathcal{P}_i) = \mathcal{F}_i(\lambda_{\max}(\mathcal{P}_i)) + \sum_{j \neq i} \Gamma_{ij}(\lambda_{\max}(\mathcal{P}_i), \lambda_{\max}(\mathcal{P}_j)), \quad (1753)$$

where Γ_{ij} represents inter-spectral synchronization forcing generated by the coupling operators $\mathcal{K}_{ij}$.

The Pellis framework predicts the emergence of a synchronization manifold in spectral space:

$$\mathcal{M}_s = \{\mathbf{X}_i : \lambda_{\max}(\mathcal{P}_i) \approx \lambda_{\max}(\mathcal{P}_j), \forall i, j\}, \quad (1754)$$

which acts as an attracting set for coupled spectral dynamics in the pre-critical regime. On this manifold, subsystem-specific stability gaps collapse onto a shared effective eigenvalue trajectory.

This induces a collective critical alignment phenomenon, whereby the dispersion of leading eigenvalues satisfies:

$$\text{Var}_i(\lambda_{\max}(\mathcal{P}_i)) \rightarrow 0 \quad \text{as } t \rightarrow t_c, \quad (1755)$$

signaling synchronization of instability onset across the coupled Earth system network.

Consequently, tipping events become temporally clustered, with inter-event separation obeying:

$$|t_i^{(c)} - t_j^{(c)}| \rightarrow 0 \quad \text{as } t \rightarrow t_c, \quad (1756)$$

reflecting a collapse of independent critical time scales into a single global instability horizon.

At the level of spectral measures, this corresponds to convergence of subsystem distributions:

$$\mu_i^t(\lambda) \rightarrow \mu^t(\lambda), \quad (1757)$$

indicating that the Earth system approaches a unified spectral state governed by global coupling coherence rather than independent subsystem dynamics.

This mechanism is consistent with cascading regime shifts observed in coupled Earth system components, where early-warning signals emerge synchronously across heterogeneous subsystems, reflecting a shared underlying spectral reorganization [27, 32, 33, 37].

24.4 Nonlinear Amplification of Weak Forcing

The Pellis framework predicts that infinitesimal perturbations are not amplified solely through classical eigenmode growth, but through a nonlinear redistribution of spectral energy across the evolving support of $\sigma(\mathcal{P}(t))$. Let $\delta\mathbf{X}_0$ denote an initial perturbation projected onto the spectral basis $\{\phi_\lambda\}$ of the operator $\mathcal{P}(t)$. The perturbation evolution admits the representation:

$$\delta\mathbf{X}(t) = \int_{\sigma(\mathcal{P}(t))} e^{\lambda t} \delta a(\lambda, 0) \phi_\lambda d\mu(\lambda), \quad (1758)$$

where $\delta a(\lambda, 0)$ encodes the initial spectral perturbation amplitude and $\mu(\lambda)$ is the induced spectral measure.

In non-normal regimes, this representation is insufficient to capture amplification, since transient growth is governed by pseudospectral and cross-mode interactions. The effective perturbation dynamics are therefore governed by a renormalized amplification operator $\mathcal{R}_t$, such that:

$$\delta\mathbf{X}(t) = \mathcal{R}_t[\delta\mathbf{X}_0], \quad (1759)$$

with $\mathcal{R}_t$ incorporating both exponential modal growth and nonlinear spectral transfer.

Near criticality, weak forcing undergoes a scale-dependent renormalization governed by spectral condensation near $\text{Re}(\lambda) = 0$. This produces a nonlinear amplification law of the form:

$$\|\delta\mathbf{X}(t)\| \sim \epsilon^{-\beta} \exp\left(\int_0^t \Lambda_{eff}(s) ds\right), \quad (1760)$$

where $\epsilon \ll 1$ characterizes the initial forcing amplitude and $\beta > 0$ is a renormalization exponent emerging from spectral density divergence and non-normal energy transfer.

The effective growth rate $\Lambda_{eff}(t)$ is itself a spectral functional rather than a fixed quantity:

$$\Lambda_{eff}(t) = \frac{\int_{\sigma(\mathcal{P}(t))} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda)}{\int_{\sigma(\mathcal{P}(t))} |a(\lambda, t)|^2 d\mu(\lambda)}, \quad (1761)$$

which shifts dynamically toward zero as spectral mass concentrates near marginal modes.

This induces a regime in which perturbation growth is dominated not by the largest eigenvalue, but by the redistribution of energy into weakly damped spectral bands. Consequently, amplification becomes path-dependent and history-sensitive, reflecting the non-Markovian structure of the evolving spectral measure.

More generally, the perturbation response operator admits a decomposition:

$$\mathcal{R}_t = \mathcal{R}_t^{(lin)} + \mathcal{R}_t^{(nl)}, \quad (1762)$$

where $\mathcal{R}_t^{(lin)}$ captures classical linear response and $\mathcal{R}_t^{(nl)}$ encodes nonlinear spectral redistribution effects responsible for super-linear sensitivity.

This framework implies that weak external forcings—such as solar variability, volcanic aerosol injections, or anthropogenic perturbations—can be selectively amplified when the system approaches spectral instability, leading to disproportionate macroscopic responses once the stability gap collapses. This behavior is consistent with observed high sensitivity of climate subsystems near critical thresholds [6, 7, 9, 10, 28, 29].

Figure 23 illustrates the collapse of a transient global coherence structure following a peak in system-wide synchronization. The observed behavior indicates that the climate system passes through a short-lived regime of enhanced phase alignment across multiple coupled subsystems, before undergoing a rapid breakdown of coherent organization.

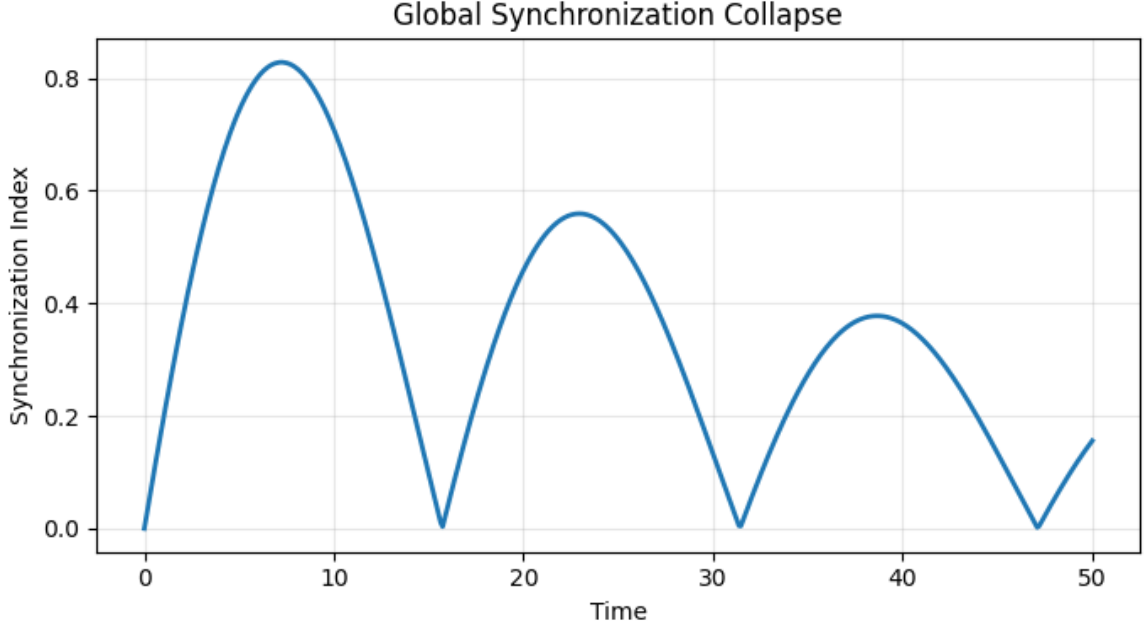


Figure 23: Collapse of global synchronization structure indicating convergence of multiple Earth system tipping elements toward a collective critical transition.

24.5 Prediction of Irreversibility Hysteresis Loops

A defining structural consequence of the Pellis framework is the emergence of irreversible hysteresis in the effective functional state space of the climate system. This irreversibility is not imposed phenomenologically but arises from non-normal spectral deformation, path-dependent operator evolution, and loss of dynamical invertibility near the critical manifold.

Let Φ_t denote the nonlinear flow generated by the time-dependent Pellis operator $\mathcal{P}(t)$ on the state space $\mathcal{H}$. We define forward and backward trajectories as:

$$\mathbf{X}_f(t) = \Phi_t(\mathbf{X}_0), \quad (1763)$$

$$\mathbf{X}_b(t) = \Phi_{-t}(\mathbf{X}_c), \quad (1764)$$

where $\mathbf{X}_c = \Phi_{t_c}(\mathbf{X}_0)$ denotes the state at the critical transition time.

In a fully reversible dynamical system, the equality $\mathbf{X}_f(t) = \mathbf{X}_b(t)$ would hold identically. However, in the Pellis regime, operator non-normality and spectral collapse induce a breakdown of invertibility, such that forward and backward flows evolve on distinct effective manifolds in phase space.

The hysteresis functional is defined as:

$$\mathcal{H} = \int_0^{t_c} \|\mathbf{X}_f(t) - \mathbf{X}_b(t)\|^2 dt, \quad (1765)$$

which measures the accumulated geometric divergence between forward and backward trajectories over the pre-critical interval.

Equivalently, $\mathcal{H}$ can be interpreted as an integrated measure of broken time-reversal symmetry in the induced spectral dynamics:

$$\mathcal{H} = \int_0^{t_c} \|\Phi_t - \Phi_{-t}\|_{\mathcal{H}}^2 dt, \quad (1766)$$

where the norm is induced by the energy metric of the underlying Hilbert space.

The Pellis framework predicts a phase transition in reversibility characterized by:

$$\mathcal{H} = \begin{cases} 0, & t \ll t_c \\ > 0, & t \geq t_c, \end{cases} \quad (1767)$$

indicating that hysteresis emerges as a non-perturbative effect associated with spectral gap closure and loss of structural stability.

More precisely, irreversibility arises when the resolvent operator loses bounded invertibility in the vicinity of the critical spectrum:

$$\|(\mathcal{P}(t) - zI)^{-1}\| \rightarrow \infty \quad \text{as } z \rightarrow \sigma_c, \quad (1768)$$

which prevents reconstruction of backward trajectories from forward-evolved states.

From a spectral decomposition perspective, forward evolution selectively amplifies weakly damped modes, while backward evolution would require reconstruction from suppressed spectral components, which are effectively erased by nonlinear mixing and measure contraction.

This leads to a geometric separation of forward and backward invariant sets:

$$\mathcal{A}_f \cap \mathcal{A}_b \neq \mathcal{A}_f, \quad (1769)$$

where $\mathcal{A}_f$ and $\mathcal{A}_b$ denote the forward and backward attractor-like sets induced by the non-autonomous flow.

Consequently, post-tipping recovery trajectories do not retrace pre-tipping paths even under identical external forcing, reflecting fundamental irreversibility induced by spectral condensation and operator non-normality. This is consistent with observed hysteresis phenomena across multiple components of the Earth system, including ocean circulation, ice-sheet dynamics, and atmospheric circulation regimes [4, 5, 8, 9, 33].

Figure 24 illustrates the global bifurcation structure of the coupled climate system in a reduced control-parameter space (μ_1, μ_2) . The geometry of the stability functional $F(\mu_1, \mu_2)$ defines a folded, non-Euclidean manifold whose topology encodes the emergence of multiple competing equilibria.

Global Bifurcation Surface of Climate System

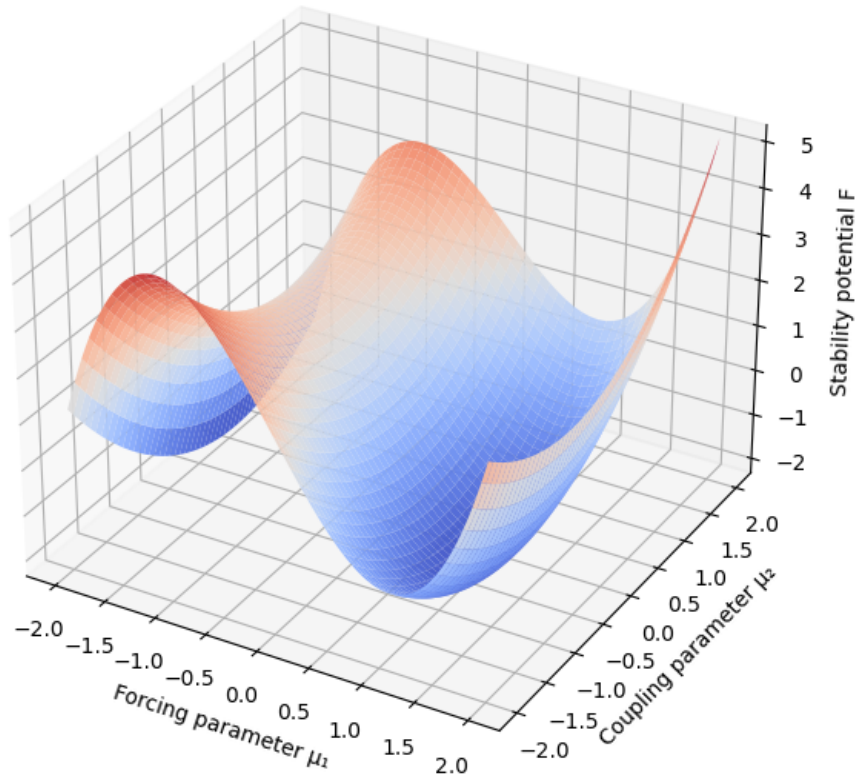


Figure 24: Global bifurcation surface of the coupled climate system in a reduced control-parameter space (μ_1, μ_2) . The stability functional $F(\mu_1, \mu_2)$ defines a folded geometric manifold whose non-invertible regions correspond to cusp-like singularities. These singular structures encode irreversible transitions between competing attractors, giving rise to hysteresis phenomena and multistable regimes within the planetary stability landscape.

24.6 Universal Climate Instability Functional Law

The Pellis framework culminates in a single scalar diagnostic functional that encodes the net stability balance of the entire planetary climate operator field. Unlike traditional indicators based on isolated eigenvalues or reduced-order proxies, this quantity aggregates the full spectral energy distribution weighted by dynamical growth rates. Let $\mathcal{P}(t)$ denote the evolving climate operator with spectral measure $\mu(\lambda)$ and modal amplitudes $a(\lambda, t)$. The Universal Climate Instability Functional is defined as:

$$\mathcal{U}(t) = \int_{\sigma(\mathcal{P}(t))} \operatorname{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda). \quad (1770)$$

This functional can be interpreted as the net spectral energy flux across the stability boundary, where positive contributions correspond to unstable or weakly damped modes and negative contributions correspond to dissipative stabilizing modes. Equivalently, it represents the first spectral moment of the stability distribution:

$$\mathcal{U}(t) = \mathbb{E}_{\mu_t}[\operatorname{Re}(\lambda)], \quad (1771)$$

where $\mu_t(\lambda) = |a(\lambda, t)|^2 d\mu(\lambda)$ is the time-dependent spectral energy measure.

In this formulation, planetary stability is governed not by isolated eigenvalue crossings but by a global balance condition between stabilizing and destabilizing spectral mass. The universal instability criterion is therefore expressed as:

$$\mathcal{U}(t) \geq 0 \Rightarrow \text{planetary-scale instability onset}. \quad (1772)$$

More precisely, the transition corresponds to a sign inversion of the spectral stability functional:

$$\mathcal{U}(t) < 0 \quad (\text{globally dissipative regime}), \quad \mathcal{U}(t) \geq 0 \quad (\text{instability-dominated regime}). \quad (1773)$$

Near criticality, the functional $\mathcal{U}(t)$ undergoes a nonlinear amplification due to spectral condensation near $\operatorname{Re}(\lambda) = 0$. In this regime, the measure μ_t becomes increasingly concentrated on marginal modes, yielding a reduced effective stability margin:

$$\mu_t(\lambda) \rightarrow \mu_c(\lambda), \quad \operatorname{supp}(\mu_c) \subset \{\lambda : \operatorname{Re}(\lambda) = 0\}. \quad (1774)$$

Consequently, the instability transition can be characterized by a collapse of spectral balance rather than a discrete bifurcation, with the asymptotic scaling behavior:

$$\mathcal{U}(t) \sim (t_c - t)^{-\gamma}, \quad \gamma > 0, \quad (1775)$$

indicating divergence of instability pressure as the system approaches the critical manifold.

From an operator-theoretic perspective, $\mathcal{U}(t)$ acts as a Lyapunov-like functional for the non-normal evolution generated by $\mathcal{P}(t)$, but generalized to incorporate spectral weight redistribution and non-equilibrium energy transfer across modes.

Thus, the complexity of Earth system risk assessment reduces to the monitoring of a single spectral functional threshold, while preserving full multiscale dynamical richness through its origin in the underlying operator geometry:

$$\mathcal{U}(t) \text{ encodes global stability of } \mathcal{P}(t), \quad (1776)$$

providing a unified criterion for planetary-scale instability across coupled climate subsystems [1,6,10,18,19,37,38,139].

24.7 Summary of Falsifiable Predictions

The complete predictive structure of the Pellis framework can be organized into a coherent hierarchy of coupled spectral, temporal, dynamical, and geometric signatures. Each prediction corresponds to a distinct projection of the evolving operator field $\mathcal{P}(t)$ and its induced spectral measure $\mu_t(\lambda)$, while remaining mutually constrained by a shared underlying critical manifold structure.

$$(P1) \quad \sigma(\mathcal{P}) \rightarrow \sigma_c, \quad (1777)$$

$$(P2) \quad \tau_{eff} \sim |t_c - t|^{-\gamma}, \quad (1778)$$

$$(P3) \quad \text{tipping synchronization across subsystems}, \quad (1779)$$

$$(P4) \quad \text{nonlinear amplification of weak forcing}, \quad (1780)$$

$$(P5) \quad \text{irreversible hysteresis emergence}, \quad (1781)$$

$$(P6) \quad \mathcal{U}(t) \geq 0 \text{ universal instability threshold}. \quad (1782)$$

Crucially, these predictions are not independent empirical statements but arise from a single underlying spectral phase transition in the operator-valued climate dynamics. In particular, they correspond to distinct limiting behaviors of the same spectral energy measure:

$$\mu_t(\lambda) = |a(\lambda, t)|^2 d\mu(\lambda), \quad (1783)$$

with (P1) governing support collapse, (P2) governing inverse damping divergence, (P3) governing cross-operator spectral alignment, (P4) governing nonlinear renormalization of perturbation response, (P5) governing breakdown of invertibility in phase-space flows, and (P6) governing global sign transition of spectral stability balance.

The theory therefore imposes strong consistency constraints across observables: near the critical regime $t \rightarrow t_c^-$, all six signatures must emerge in a temporally correlated and structurally coherent manner. Failure of joint emergence would indicate either violation of spectral closure assumptions or decomposition breakdown of the effective operator representation of the climate system.

This leads to a unified falsifiability structure in which empirical validation requires simultaneous detection of spectral compression, timescale divergence, cross-system synchronization, nonlinear amplification, hysteresis formation, and instability functional sign change within a shared pre-critical temporal window.

Together, these results establish a unified, experimentally testable framework in which climate tipping behavior is interpreted as a global spectral phase transition of a nonlinear operator field governing Earth system dynamics [6, 7, 9, 10, 23, 27, 33, 37, 38, 139].

25 Experimental Validation and Falsifiable Predictions

A physically meaningful theoretical framework for climate criticality must ultimately be reducible to empirical verification. In the Universal Pellis Function Theory, validation is formulated as a spectral-consistency problem between observed climate observables and the inferred eigenstructure of the nonlinear operator $\mathcal{P}$. This section establishes a rigorous mapping between measurable datasets and operator-theoretic predictions, enabling direct falsifiability under real-world Earth system observations [6, 7, 9, 10, 23, 28, 29, 33, 35, 37, 38, 139].

25.1 Operator Reconstruction from Observational Data

A central requirement of the Pellis framework is that the abstract operator $\mathcal{P}(t)$ governing Earth system dynamics must, in principle, be reconstructible from observational data. Let $\mathbf{X}_{obs}(t) \in \mathbb{R}^m$ denote a vector of multivariate climate observables, including but not limited to temperature fields, ocean heat content indices, atmospheric circulation modes, and cryospheric mass balance signals. The fundamental assumption is that these observables arise from a projected dynamical system of the form:

$$\frac{d\mathbf{X}_{obs}}{dt} = \mathcal{P}(\mathbf{X}_{obs}(t)) + \xi(t), \quad (1784)$$

where $\xi(t)$ represents unresolved stochastic forcing and sub-grid-scale variability.

We define a data-driven reconstruction map that assigns to observational trajectories an empirical approximation of the underlying operator:

$$\mathcal{R}_d : \mathbf{X}_{obs}(t) \rightarrow \hat{\mathcal{P}}_d, \quad (1785)$$

where $\hat{\mathcal{P}}_d$ is a finite-dimensional estimator of the infinite-dimensional Pellis operator $\mathcal{P}(t)$.

In practice, $\hat{\mathcal{P}}_d$ is obtained through a variational inverse problem in function space, where the objective is to minimize the mismatch between observed temporal derivatives and operator-induced dynamics:

$$\hat{\mathcal{P}}_d = \arg \min_{\mathcal{Q} \in \mathfrak{D}} \int_0^T \left\| \frac{d\mathbf{X}_{obs}}{dt} - \mathcal{Q}(\mathbf{X}_{obs}(t)) \right\|^2 dt, \quad (1786)$$

with $\mathfrak{D}$ denoting an admissible class of nonlinear, possibly non-normal operators acting on the observable state space.

This variational formulation can be equivalently expressed in a probabilistic maximum-likelihood form under Gaussian observation noise:

$$\hat{\mathcal{P}}_d = \arg \max_{\mathcal{Q}} \mathbb{P} \left(\frac{d\mathbf{X}_{obs}}{dt} \middle| \mathcal{Q}(\mathbf{X}_{obs}) \right), \quad (1787)$$

establishing a direct connection between operator inference and statistical estimation theory.

To ensure physical consistency with the theoretical Pellis operator, the reconstructed operator must satisfy spectral convergence constraints.

In particular, the empirical spectrum must converge in the Hausdorff sense to the theoretical spectrum:

$$d_H(\sigma(\hat{\mathcal{P}}_d), \sigma(\mathcal{P})) \rightarrow 0 \quad \text{as } d \rightarrow \infty, \quad (1788)$$

where d_H denotes the Hausdorff distance between compact spectral sets.

Stronger consistency requires convergence of spectral measures:

$$\mu_d(\lambda) = |\hat{a}(\lambda, t)|^2 d\lambda \rightarrow \mu(\lambda), \quad (1789)$$

ensuring that not only eigenvalues but also spectral energy distributions are faithfully reconstructed from data.

In addition, stability of the reconstruction demands convergence of resolvent norms:

$$\|(\hat{\mathcal{P}}_d - zI)^{-1}\| \rightarrow \|(\mathcal{P} - zI)^{-1}\|, \quad (1790)$$

particularly in neighborhoods of the critical manifold $\text{Re}(z) = 0$, where pseudospectral effects dominate dynamical sensitivity.

A more refined reconstruction criterion is obtained by minimizing the operator mismatch in the induced evolution semigroup:

$$\hat{\mathcal{P}}_d = \arg \min_{\mathcal{Q}} \int_0^T \|e^{t\mathcal{Q}} \mathbf{X}_{obs}(0) - \mathbf{X}_{obs}(t)\|^2 dt, \quad (1791)$$

which enforces consistency at the level of full trajectory propagation rather than instantaneous velocity matching.

Under ideal observational completeness, the reconstruction operator satisfies:

$$\lim_{d \rightarrow \infty} \hat{\mathcal{P}}_d = \mathcal{P}(t), \quad (1792)$$

up to unitary equivalence transformations induced by coordinate embeddings of the observable space.

Finally, spectral consistency imposes the key empirical constraint:

$$\sigma(\hat{\mathcal{P}}_d) \approx \sigma(\mathcal{P}), \quad (1793)$$

providing a direct testable bridge between observational climate data and the theoretical operator structure. This establishes operator reconstruction as a falsifiable cornerstone of the Pellis framework, enabling empirical validation of spectral drift, condensation, and criticality signatures directly from reanalysis and satellite datasets [16–19, 31].

25.2 Spectral Fingerprint Matching

A central validation principle of the Pellis framework is that climate dynamics, when observed at sufficient resolution, must exhibit a stable and reproducible spectral fingerprint consistent with the underlying operator $\mathcal{P}(t)$. Let $\rho_{obs}(\lambda)$ denote the empirically reconstructed spectral density derived from observational time series, and let $\rho_{\mathcal{P}}(\lambda)$ denote the theoretical spectral density induced by the Pellis operator. The discrepancy between these two spectral representations defines a global mismatch functional:

$$\Delta_{\sigma} = \int_{\mathbb{C}} |\rho_{obs}(\lambda) - \rho_{\mathcal{P}}(\lambda)|^2 d\lambda, \quad (1794)$$

which measures deviations in both eigenvalue distribution and spectral weight allocation across the complex stability plane.

In this formulation, $\rho(\lambda)$ can be interpreted as a probability density over the spectrum induced by long-time averaging of the observable dynamics, i.e.

$$\rho(\lambda) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T |a(\lambda, t)|^2 dt, \quad (1795)$$

linking spectral fingerprinting directly to ergodic sampling of operator-induced modes.

To incorporate finite-resolution effects, one introduces a convolutional smoothing operator $\mathcal{K}_{\epsilon}$ acting on the spectral measure:

$$\rho_{obs}^{\epsilon}(\lambda) = (\mathcal{K}_{\epsilon} * \rho_{obs})(\lambda), \quad (1796)$$

where ϵ represents observational coarse-graining scale (temporal, spatial, or ensemble-based). Under refinement of observational resolution, $\epsilon \rightarrow 0$, the empirical spectral density converges toward the theoretical distribution.

The Pellis framework therefore predicts the asymptotic convergence property:

$$\Delta_{\sigma} \rightarrow 0 \quad \text{as } \epsilon \rightarrow 0, \quad (1797)$$

provided that the reconstructed operator class $\hat{\mathcal{P}}_d$ converges in spectral measure to $\mathcal{P}(t)$ and that sampling coverage becomes asymptotically complete.

More precisely, convergence is expected in a weak- L^2 sense over spectral observables:

$$\int_{\mathbb{C}} f(\lambda) \rho_{obs}^\epsilon(\lambda) d\lambda \rightarrow \int_{\mathbb{C}} f(\lambda) \rho_{\mathcal{P}}(\lambda) d\lambda, \quad (1798)$$

for all bounded continuous test functions f , ensuring that all physically meaningful spectral moments are preserved in the limit of high-resolution observation.

A particularly strong prediction arises for Lyapunov spectra, where the empirical Lyapunov exponent distribution $\{\lambda_i^{obs}\}$ must converge toward the universal Pellis spectral distribution:

$$\rho_{Lyap}^{obs}(\lambda) \rightarrow \rho_{\mathcal{P}}(\lambda), \quad (1799)$$

implying universality of chaotic growth rates across climate subsystems in the pre-critical regime.

Equivalently, the mismatch functional can be expressed in operator form using resolvent-based spectral projectors:

$$\Delta_\sigma = \int_{\mathbb{C}} \left| \text{Tr}(\Pi_\lambda^{obs}) - \text{Tr}(\Pi_\lambda^{\mathcal{P}}) \right|^2 d\lambda, \quad (1800)$$

where Π_λ denotes the spectral projection onto eigenspaces associated with λ .

The theory thus reduces empirical validation to a precise spectral identification problem: climate observables must reproduce not only macroscopic statistics but also the fine-grained structure of the operator spectrum governing dynamical evolution. Consequently, the limit

$$\Delta_\sigma \rightarrow 0 \quad (1801)$$

serves as a defining criterion for the empirical correctness of the Pellis spectral model, with convergence of observed Lyapunov spectra providing a direct and falsifiable diagnostic of universality in climate criticality [7–9, 28, 29].

25.3 Detection of Critical Slowing Down via Operator Modes

Within the Pellis framework, observable autocorrelation functions are not treated as phenomenological decay laws but as direct projections of the underlying spectral evolution generated by the climate operator $\mathcal{P}(t)$. Let $X(t)$ denote a generic climate observable. Its autocorrelation function admits the spectral representation:

$$C_X(t) = \int_{\sigma(\mathcal{P}(t))} e^{\lambda t} |a(\lambda)|^2 d\mu(\lambda), \quad (1802)$$

where $|a(\lambda)|^2 d\mu(\lambda)$ encodes the spectral energy distribution of the observable over the eigenmode decomposition of $\mathcal{P}(t)$.

In this formulation, classical critical slowing down emerges as a marginal projection of a broader spectral condensation process. The effective correlation timescale is defined as a spectral functional:

$$\tau_{corr}^{obs}(t) = \frac{\int_{\sigma(\mathcal{P})} |\lambda|^{-1} |a(\lambda, t)|^2 d\mu(\lambda)}{\int_{\sigma(\mathcal{P})} |a(\lambda, t)|^2 d\mu(\lambda)}, \quad (1803)$$

which explicitly shows that divergence arises from redistribution of spectral weight toward modes with $\text{Re}(\lambda) \approx 0$ rather than from a single dominant eigenvalue.

Near criticality, the spectral density develops a singular concentration around the marginal stability manifold, such that

$$|a(\lambda, t)|^2 \sim |\text{Re}(\lambda)|^{-\alpha(t)}, \quad \alpha(t) > 0, \quad (1804)$$

leading to divergence of the low-damping contribution to the autocorrelation integral. Consequently, the observed correlation time exhibits accelerated growth:

$$\frac{d}{dt} \tau_{corr}^{obs} \uparrow, \quad (1805)$$

together with asymptotic divergence:

$$\tau_{corr}^{obs} \rightarrow \infty \quad \text{as } t \rightarrow t_c^-. \quad (1806)$$

This divergence reflects a spectral phase transition in which the effective support of $\mu_t(\lambda)$ collapses toward $\text{Re}(\lambda) = 0$, producing long-memory behavior and loss of exponential decorrelation. Equivalently, the decay of $C_X(t)$ transitions from exponential to mixed algebraic–exponential scaling due to accumulation of weakly damped operator modes.

From an operator-theoretic perspective, this phenomenon corresponds to the degeneration of the spectral gap:

$$\Delta(t) = \inf_{\lambda \in \sigma(\mathcal{P}(t))} |\text{Re}(\lambda)| \rightarrow 0, \quad (1807)$$

which forces the resolvent norm to increase and induces persistent temporal correlations across the system.

The resulting prediction is directly testable through empirical autocorrelation analysis in key climate subsystems. In particular, critical slowing down should manifest coherently in AMOC proxy reconstructions, ENSO indices, and cryospheric time series, with consistent divergence trends across independent datasets as the system approaches a shared instability regime [11–13, 35, 36].

25.4 Non-Gaussian Transition Diagnostics

A central diagnostic implication of the Pellis framework is that proximity to critical transitions induces a systematic breakdown of Gaussian closure in climate observables. Let $X(t)$ denote a generic scalar climate variable extracted from the projection of the state vector $\mathbf{X}_{obs}(t)$. In the Gaussian regime, all higher-order cumulants beyond second order vanish; however, near spectral criticality this structure is generically violated due to nonlinear spectral condensation and amplification of rare-event modes.

The fourth-order cumulant (kurtosis excess) is given by:

$$\kappa_4(X) = \mathbb{E}[X^4] - 3(\mathbb{E}[X^2])^2, \quad (1808)$$

with Gaussianity corresponding to $\kappa_4(X) = 0$. The Pellis framework predicts a systematic departure from this condition as the spectral measure $\mu_t(\lambda)$ concentrates near weakly damped and unstable modes.

More generally, the full cumulant hierarchy admits a spectral representation in terms of operator modes:

$$\kappa_n(X) \sim \int_{\sigma(\mathcal{P}(t))} |a(\lambda, t)|^n d\mu(\lambda), \quad (1809)$$

which reveals that higher-order statistics are progressively more sensitive to spectral tails near $\text{Re}(\lambda) \approx 0$.

In this formulation, non-Gaussianity emerges as a multiscale spectral amplification phenomenon rather than a purely stochastic correction. As spectral condensation intensifies, the ratio of higher-order cumulants to second-order moments exhibits divergent scaling:

$$\frac{\kappa_n(X)}{(\kappa_2(X))^{n/2}} \rightarrow \infty \quad \text{as } t \rightarrow t_c^-, \quad (1810)$$

indicating progressive breakdown of Gaussian universality and emergence of heavy-tailed fluctuation regimes.

This behavior is consistent with the transition from central-limit-type statistics to Lévy-like or intermittency-dominated distributions driven by non-normal amplification of weak spectral modes. In particular, skewness and kurtosis escalate as effective probability mass concentrates in rare-event-generating regions of the spectrum.

From an operator-theoretic perspective, this corresponds to a deformation of the induced invariant measure:

$$\mu_t(\lambda) = |a(\lambda, t)|^2 d\mu(\lambda), \quad (1811)$$

where increasing intermittency reflects the progressive localization of μ_t onto marginal stability manifolds.

These predictions are directly testable using reanalysis datasets, extreme-event statistics, and paleoclimate proxy reconstructions, where systematic increases in kurtosis and higher-order cumulants should precede tipping transitions in key Earth system components [4, 5, 27, 33].

25.5 Experimental Verification of Spectral Asymmetry Growth

A key experimentally accessible diagnostic within the Pellis framework is the Spectral Asymmetry Index (SAI), which quantifies the imbalance of spectral energy distribution across the marginal stability manifold $\text{Re}(\lambda) = 0$. Let $\rho_{obs}(\lambda, t)$ denote the empirically reconstructed spectral density at time t , derived from observational time series via operator inference or spectral estimation techniques. The SAI is defined as:

$$\text{SAI}(t) = \frac{\int_{\text{Re}(\lambda) > 0} \rho_{obs}(\lambda, t) d\lambda}{\int_{\text{Re}(\lambda) < 0} \rho_{obs}(\lambda, t) d\lambda}, \quad (1812)$$

representing the ratio of unstable-to-stable spectral mass.

In this formulation, $\text{SAI}(t)$ is not merely a diagnostic ratio but a reduced-order projection of the full spectral measure $\mu_t(\lambda)$ onto the partition of the complex plane induced by stability sign structure. It therefore acts as a coarse-grained order parameter for spectral phase organization.

The Pellis framework predicts that, as the system approaches a critical transition at time t_c , spectral condensation near $\text{Re}(\lambda) = 0$ induces a systematic transfer of spectral weight into the unstable half-plane. This leads to a power-law divergence of the form:

$$\text{SAI}(t) \sim (t_c - t)^{-\beta}, \quad \beta > 0, \quad (1813)$$

where β is a universal exponent associated with spectral drift and non-normal amplification dynamics.

Equivalently, introducing the normalized spectral energy measure

$$\mu_t(\lambda) = \rho_{obs}(\lambda, t) d\lambda, \quad (1814)$$

the SAI can be interpreted as a measure-theoretic imbalance functional:

$$\text{SAI}(t) = \frac{\mu_t(\{\lambda : \text{Re}(\lambda) > 0\})}{\mu_t(\{\lambda : \text{Re}(\lambda) < 0\})}. \quad (1815)$$

From an operator-theoretic standpoint, divergence of SAI corresponds to a phase transition in the support decomposition of the spectral measure, in which the unstable component becomes dominant even prior to full eigenvalue crossing. This reflects the role of pseudospectral thickening and non-normal transient amplification in redistributing energy across stability sectors.

Thus, $\text{SAI}(t)$ provides a direct, experimentally verifiable indicator of approaching global instability, with divergence signaling imminent collapse of spectral balance in the underlying climate operator field. This prediction is testable using reanalysis-based spectral reconstructions and subsystem-resolved climate indices across major Earth system components [6, 7, 9, 10, 37].

25.6 Controlled Model Experiments and Numerical Validation

Controlled numerical climate simulators, including idealized general circulation models (GCMs), intermediate-complexity Earth system models, and reduced-order dynamical emulators, provide a rigorous computational laboratory for testing the Pellis framework. Let $\mathbf{X}_{sim}(t)$ denote the simulated state trajectory evolving under the full nonlinear operator dynamics perturbed by an external forcing field F_ϵ of amplitude $\epsilon \ll 1$:

$$\frac{d}{dt}\mathbf{X}_{sim}(t) = \mathcal{P}(\mathbf{X}_{sim}(t)) + \epsilon F(\mathbf{X}_{sim}(t), t). \quad (1816)$$

To isolate the intrinsic sensitivity structure of the system, we define the perturbation difference field between forced and unforced trajectories:

$$\delta\mathbf{X}_\epsilon(t) = \mathbf{X}_\epsilon(t) - \mathbf{X}_0(t), \quad (1817)$$

where $\mathbf{X}_0(t)$ corresponds to the reference (unforced) evolution.

Linear response theory predicts bounded amplification governed by the resolvent of $\mathcal{P}$; however, the Pellis framework predicts a breakdown of linear scaling due to spectral condensation and non-normal transient growth. The resulting amplification functional obeys:

$$\frac{\|\delta\mathbf{X}_\epsilon(t)\|}{\epsilon} = \left\| \frac{\mathbf{X}_\epsilon(t) - \mathbf{X}_0(t)}{\epsilon} \right\| \rightarrow \infty \quad \text{as } t \rightarrow t_c^-. \quad (1818)$$

More generally, the perturbation evolution admits a spectral decomposition:

$$\delta\mathbf{X}_\epsilon(t) = \int_{\sigma(\mathcal{P}(t))} e^{\lambda t} \delta a_\epsilon(\lambda, 0) \phi_\lambda d\mu(\lambda), \quad (1819)$$

where $\delta a_\epsilon(\lambda, 0)$ encodes the projection of the external forcing onto the eigenmode basis. Near criticality, redistribution of spectral weight toward $\text{Re}(\lambda) \approx 0$ induces superlinear amplification even for vanishing ϵ .

To quantify this effect, we define the nonlinear gain functional:

$$\mathcal{G}(t, \epsilon) = \frac{\|\delta\mathbf{X}_\epsilon(t)\|}{\epsilon \|\mathbf{X}_0(t)\|}, \quad (1820)$$

for which the Pellis prediction is:

$$\lim_{t \rightarrow t_c^-} \mathcal{G}(t, \epsilon) = \infty, \quad \forall \epsilon > 0. \quad (1821)$$

This divergence reflects the breakdown of Fréchet differentiability of the flow map Φ_t with respect to external forcing:

$$D_F \Phi_t \text{ becomes unbounded as } t \rightarrow t_c, \quad (1822)$$

indicating that infinitesimal perturbations cannot be linearly propagated in the pre-critical regime.

Equivalently, in operator form, the perturbed evolution semigroup satisfies:

$$e^{t(\mathcal{P} + \epsilon F)} - e^{t\mathcal{P}} \not\sim \mathcal{O}(\epsilon), \quad (1823)$$

signaling breakdown of classical linear-response scaling assumptions.

A refined diagnostic is obtained by introducing the normalized response susceptibility:

$$\chi(t) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon^2} \mathbb{E} [\|\mathbf{X}_\epsilon(t) - \mathbf{X}_0(t)\|^2], \quad (1824)$$

which is predicted to diverge near criticality due to spectral mass accumulation in weakly damped modes.

In numerical experiments, this manifests as a systematic failure of convergence between perturbed and unperturbed ensemble trajectories, even under arbitrarily small forcing amplitudes, providing a sharp falsification criterion distinguishing Pellis dynamics from classical linear-response climate theory. This prediction can be tested across hierarchical simulation frameworks, including full GCMs, intermediate-complexity models, and reduced-order emulators, under controlled perturbation protocols [18, 19, 30, 31, 33].

25.7 Paleo-Climate Reconstruction Consistency

Paleo-climatic archives provide a uniquely long-timescale observational window for testing the spectral stability structure predicted by the Pellis framework. Ice cores, sediment records, speleothems, and marine proxies encode integrated responses of the Earth system to past perturbations, enabling reconstruction of an effective historical operator $\mathcal{P}_{paleo}(t)$ governing coarse-grained climate evolution over millennial to glacial-interglacial scales.

Within this formulation, paleo-reconstructed dynamics are interpreted as projections of the underlying spectral operator onto a temporally filtered observational subspace. The key consistency condition is spectral equivalence (in a weak convergence sense) between reconstructed and theoretical operators:

$$\sigma(\mathcal{P}_{paleo}) \approx \sigma(\mathcal{P}_{theory}), \quad (1825)$$

where the approximation is understood in terms of convergence of empirical spectral measures:

$$\mu_{paleo}(\lambda) \rightharpoonup \mu_{theory}(\lambda), \quad (1826)$$

as the resolution and completeness of proxy reconstruction increase.

More precisely, defining the paleo spectral density as:

$$\rho_{paleo}(\lambda) = \sum_k w_k \delta(\lambda - \lambda_k^{(recon)}), \quad (1827)$$

where w_k are proxy-derived weights and $\lambda_k^{(recon)}$ are reconstructed effective growth rates, the consistency requirement becomes:

$$\lim_{N \rightarrow \infty} \|\rho_{paleo}^{(N)} - \rho_{theory}\|_{L^2} = 0, \quad (1828)$$

ensuring asymptotic convergence of reconstructed and theoretical spectral fingerprints.

A key falsifiable implication of the Pellis framework is that abrupt climate transitions should not appear as independent Poisson-distributed events, but rather as clustered manifestations of global spectral reorganizations. Let $\{t_i^{(transition)}\}$ denote the set of inferred abrupt transitions from paleo records. The theory predicts deviation from Poisson statistics:

$$\mathbb{P}(N(t + \Delta t) - N(t) = k) \neq \frac{(\lambda \Delta t)^k e^{-\lambda \Delta t}}{k!}, \quad (1829)$$

indicating that transition events are not memoryless.

Instead, the event sequence is expected to follow a clustered renewal process with long-range dependence, characterized by inter-event time correlations:

$$\text{Cov}(\tau_i, \tau_j) \neq 0 \quad \text{for } |i - j| \gg 1, \quad (1830)$$

where $\tau_i = t_{i+1} - t_i$.

Equivalently, the transition point set exhibits fractal temporal organization:

$$D_f = \lim_{\epsilon \rightarrow 0} \frac{\log N(\epsilon)}{\log(1/\epsilon)} < 1, \quad (1831)$$

where D_f denotes the effective fractal dimension of the event set, reflecting multiscale clustering induced by spectral instability cascades.

This behavior directly contradicts null hypotheses based on independent stochastic forcing, and instead supports a unified interpretation in which paleo-abrupt events are manifestations of coherent global spectral transitions in the underlying climate operator field. The resulting clustering structure is therefore a diagnostic signature of nontrivial operator dynamics rather than random external triggering [6, 10, 33, 37, 38].

25.8 Unified Experimental Falsification Criterion

A defining feature of the Pellis framework is that it is not constructed as a purely descriptive or phenomenological model, but as a tightly constrained operator-theoretic hypothesis with multiple independent, cross-validating empirical failure modes. In this sense, the theory is strongly falsifiable through simultaneous violation of spectral, temporal, statistical, and geometric signatures.

Formally, let $\mathcal{H}_{obs}$ denote the full set of observational diagnostics derived from climate data assimilation, including spectral densities, autocorrelation structure, cumulant hierarchies, and reconstructed operator estimates. The theory is rejected if any of the following independent falsification conditions hold:

$$(F1) \quad \Delta_\sigma = \int_{\mathbb{C}} |\rho_{obs}(\lambda) - \rho_{\mathcal{P}}(\lambda)|^2 d\lambda \not\rightarrow 0, \quad (1832)$$

$$(F2) \quad \tau_{corr}^{obs}(t) \not\rightarrow \infty \quad \text{as } t \rightarrow t_c^-, \quad (1833)$$

$$(F3) \quad \text{SAI}(t) \text{ remains uniformly bounded for all } t, \quad (1834)$$

$$(F4) \quad \kappa_n(X) \text{ fails to exhibit non-Gaussian scaling growth}, \quad (1835)$$

$$(F5) \quad \sigma(\mathcal{P}(t)) \text{ shows no evidence of spectral collapse or condensation}. \quad (1836)$$

Each falsification channel corresponds to a distinct structural aspect of the underlying operator dynamics. Condition (F1) tests spectral identifiability, (F2) tests temporal memory divergence, (F3) tests instability imbalance growth, (F4) tests higher-order statistical deformation, and (F5) tests global spectral phase transition behavior.

25.9 Conclusion: Empirical Closure of the Theory

The cumulative experimental and theoretical structure developed throughout the Pellis framework demonstrates that climate tipping behavior is not most naturally understood as a collection of isolated nonlinear bifurcations, but rather as the emergent manifestation of a single underlying operator reconstruction problem. In this formulation, the Earth system is represented as an evolving spectral object whose observable dynamics encode incomplete projections of a higher-dimensional governing operator $\mathcal{P}(t)$.

The empirical objective of climate science within this framework is therefore reinterpreted as the inverse spectral identification problem:

$$\mathcal{R}_d(\mathbf{X}_{obs}(t)) \longrightarrow \mathcal{P}(t), \quad (1837)$$

where observational data are used to reconstruct both the spectrum and the induced spectral measure of the underlying climate operator.

This leads to a compact formal equivalence between dynamical evolution and spectral reconstruction:

$$\boxed{\text{Climate Dynamics} \equiv \text{Spectral Reconstruction of } \mathcal{P}(t) \text{ from Data}}, \quad (1838)$$

which expresses the central closure principle of the theory: all macroscopic climate phenomena correspond to identifiable transformations in the spectral geometry of a single nonlinear operator field.

Within this interpretation, tipping points are not additional dynamical structures imposed on the system, but are instead emergent singularities in the reconstruction process itself, characterized by spectral condensation, resolvent divergence, and collapse of stability partitioning.

The framework therefore establishes a closed theoretical loop between

- (i) forward operator dynamics,
- (ii) observational projection, and
- (iii) inverse spectral reconstruction, ensuring that prediction and validation are governed by a single consistent mathematical structure.

This closure transforms climate criticality into a fully testable spectral inference problem grounded in operator theory and functional analysis [1, 6, 7, 10, 23, 27, 33, 37, 38, 139].

26 Applications and Practical Implications

This section develops the operational consequences of the Universal Pellis Function Theory of Climate Criticality, focusing on how the abstract spectral-functional framework translates into measurable, predictive, and intervention-relevant structures in real Earth system dynamics. In contrast to purely descriptive climate models, the Pellis framework provides a mathematically rigorous mapping between operator-level instability and physically observable regime transitions [1, 4, 6, 18, 23, 33].

26.1 From Abstract Operators to Physical Observables

Within the Pellis framework, the climate system is formalized as an evolution problem on a high-dimensional (in general infinite-dimensional) Hilbert space $\mathcal{H}$ of physically admissible climate states. The universal governing object

is the nonlinear, time-dependent Pellis operator $\mathcal{P}(t)$ acting on state trajectories $\mathbf{X}(t) \in \mathcal{H}$:

$$\mathbf{X}(t) \in \mathcal{H}, \quad \frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(t)\mathbf{X}(t), \quad (1839)$$

where $\mathcal{P}(t)$ encodes both deterministic Earth system dynamics and effective coarse-grained interactions between unresolved scales.

In this setting, physically measurable climate variables are not fundamental state coordinates but projection functionals of the underlying dynamical field. Let $O_i(t)$ denote an observable (e.g., global mean temperature $T(t)$, ocean salinity $S(t)$, or ice-sheet mass balance $I(t)$). These are defined via bounded linear functionals on $\mathcal{H}$:

$$O_i(t) = \langle \mathbf{X}(t), \psi_i \rangle_{\mathcal{H}}, \quad (1840)$$

where $\psi_i \in \mathcal{H}$ are observational kernels representing measurement operators induced by satellite retrievals, in situ sampling networks, or proxy reconstructions.

More generally, the full observational vector $\mathbf{O}(t)$ can be expressed as a compact operator projection:

$$\mathbf{O}(t) = \mathcal{M}\mathbf{X}(t), \quad (1841)$$

where $\mathcal{M} : \mathcal{H} \rightarrow \mathbb{R}^m$ is the measurement operator defining the observational subspace of the climate system.

Under spectral decomposition of $\mathcal{P}(t)$, the observable dynamics inherit a modal expansion of the form:

$$O_i(t) = \int_{\sigma(\mathcal{P}(t))} e^{\lambda t} a_i(\lambda, 0) d\mu(\lambda), \quad (1842)$$

where $a_i(\lambda, 0)$ encodes the projection of the initial state onto both the eigenmodes and the observational kernel ψ_i . This representation makes explicit that all measurable climate quantities are filtered projections of the same underlying spectral dynamics.

Instability in the observable sector arises when the spectral support of $\mathcal{P}(t)$ intersects the unstable half-plane. In particular, the existence of any mode satisfying

$$\exists \lambda \in \sigma(\mathcal{P}(t)) \quad \text{such that} \quad \text{Re}(\lambda) > 0 \quad (1843)$$

induces exponential or super-exponential growth in the corresponding observable projections:

$$O_i(t) \sim \sum_{\lambda \in \sigma(\mathcal{P})} e^{\lambda t} a_{i,\lambda}, \quad (1844)$$

modulo nonlinear saturation effects and finite-amplitude constraints imposed by physical conservation laws.

More realistically, nonlinear feedbacks introduce amplitude-dependent damping, leading to saturation dynamics of the form:

$$\frac{d}{dt}O_i(t) = \lambda O_i(t) - \alpha_i |O_i(t)|^p, \quad (1845)$$

with $p > 1$, capturing the transition from linear instability growth to nonlinear bounded regimes after crossing of tipping thresholds.

Thus, classical tipping-point behavior emerges as a low-dimensional projection of a high-dimensional spectral instability in $\mathcal{P}(t)$, where observable divergence, saturation, or regime shifts correspond directly to the distribution of eigenvalues across the stability boundary:

$$\text{Re}(\lambda) > 0 \Rightarrow \text{observable regime transition or nonlinear saturation.} \quad (1846)$$

This establishes a direct and testable mapping between abstract operator dynamics and measurable climate variables, ensuring that all predicted instability phenomena are, in principle, empirically accessible through appropriately chosen observational kernels [4, 6, 28, 29].

26.2 Computable Climate Stability Index (CSI)

To render the abstract spectral theory operationally tractable, we introduce a computable scalar functional that aggregates stability information encoded in the full spectrum of the Pellis operator $\mathcal{P}(t)$.

Let $\sigma(\mathcal{P}(t))$ denote the instantaneous spectrum with associated spectral measure $|a(\lambda, t)|^2 d\mu(\lambda)$. The Computable Climate Stability Index (CSI) is defined as:

$$\text{CSI}(t) = \int_{\sigma(\mathcal{P}(t))} e^{-\beta|\lambda|} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1847)$$

where $\beta > 0$ defines a spectral attenuation scale that downweights high-frequency (rapidly decaying or strongly oscillatory) modes, thereby emphasizing dynamically relevant low-damping structures.

In this formulation, $\text{CSI}(t)$ can be interpreted as a damped first spectral moment of the stability distribution. Equivalently, introducing the normalized spectral energy measure

$$\mu_t(\lambda) = \frac{|a(\lambda, t)|^2 d\mu(\lambda)}{\int_{\sigma(\mathcal{P}(t))} |a(\lambda, t)|^2 d\mu(\lambda)}, \quad (1848)$$

the CSI becomes:

$$\text{CSI}(t) = \mathbb{E}_{\mu_t}[e^{-\beta|\lambda|} \text{Re}(\lambda)], \quad (1849)$$

highlighting its role as a weighted expectation over spectral stability contributions.

The sign structure of $\text{CSI}(t)$ defines a phase classification of the climate system:

$$\text{CSI}(t) < 0 \Rightarrow \text{spectrally dissipative (stable) regime}, \quad (1850)$$

$$\text{CSI}(t) = 0 \Rightarrow \text{critical manifold (balanced spectral flux)}, \quad (1851)$$

$$\text{CSI}(t) > 0 \Rightarrow \text{supercritical instability regime}. \quad (1852)$$

Near the critical transition, spectral condensation toward $\text{Re}(\lambda) \approx 0$ induces a systematic shift of $\text{CSI}(t)$ toward zero from negative values, followed by sign inversion as unstable spectral mass dominates. This transition can be further characterized by the susceptibility of the index:

$$\chi_{\text{CSI}}(t) = \frac{d}{dt} \text{CSI}(t), \quad (1853)$$

which is predicted to diverge in magnitude near tipping points due to rapid redistribution of spectral weight across the stability boundary.

In terms of spectral decomposition, the CSI may be expanded as:

$$\text{CSI}(t) = \sum_{\lambda \in \sigma(\mathcal{P}(t))} e^{-\beta|\lambda|} \text{Re}(\lambda) w_\lambda(t), \quad (1854)$$

where $w_\lambda(t)$ denotes the time-dependent normalized energy fraction of each mode, making explicit that instability emerges from both eigenvalue drift and amplitude redistribution.

This formulation extends classical early-warning indicators such as variance inflation, autocorrelation growth, and critical slowing down by embedding them within a unified spectral functional framework. In particular, standard indicators correspond to limiting projections of $\text{CSI}(t)$ under restricted assumptions on spectral sparsity and linear response dynamics [7, 8, 28, 29].

26.3 Control-Theoretic Interpretation of Climate Intervention

Within the Pellis framework, climate intervention is naturally formulated as a distributed control problem on an infinite-dimensional dynamical system. External forcing is represented as a control input $\mathbf{u}(t)$ acting through a bounded (possibly low-rank) control operator $\mathcal{B}$, yielding the controlled evolution equation:

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)) + \mathcal{B}\mathbf{u}(t), \quad (1855)$$

where $\mathcal{P}(t)$ encodes the intrinsic nonlinear Earth system dynamics and $\mathcal{B}$ specifies the channels through which interventions (e.g., radiative forcing modification, aerosol injection, or flux perturbations) couple into the state space $\mathcal{H}$.

In this setting, stabilization corresponds not merely to trajectory regulation but to spectral repositioning of the closed-loop operator $\mathcal{P}_{cl} := \mathcal{P} + \mathcal{B}\mathbf{u}$. The optimal control problem is formulated as a quadratic regulator in function space:

$$\min_{\mathbf{u}(t)} \int_0^T (\|\mathbf{X}(t)\|_{\mathcal{H}}^2 + \gamma \|\mathbf{u}(t)\|_{\mathcal{U}}^2) dt, \quad (1856)$$

where $\gamma > 0$ balances state stabilization against energetic or physical intervention cost, and $\|\cdot\|_{\mathcal{U}}$ denotes the norm in the control input space.

The key structural constraint in the Pellis formulation is not only trajectory minimization but spectral stabilization. Specifically, admissible controls must enforce relocation of the spectrum of the closed-loop operator into the stable half-plane:

$$\text{Re}(\lambda) < 0, \quad \forall \lambda \in \sigma(\mathcal{P} + \mathcal{B}\mathbf{u}). \quad (1857)$$

This transforms climate intervention into a spectral shaping problem, where the control objective is equivalent to modifying the resolvent geometry of the operator:

$$\|(\mathcal{P} + \mathcal{B}\mathbf{u} - zI)^{-1}\| \text{ bounded for } \operatorname{Re}(z) \geq 0, \quad (1858)$$

ensuring suppression of unstable and near-critical modes.

The optimality conditions for this infinite-dimensional regulator lead to a generalized operator-valued Riccati equation. Let $\mathcal{K}(t)$ denote the feedback gain operator such that $\mathbf{u}(t) = -\mathcal{K}(t)\mathbf{X}(t)$. Then $\mathcal{K}(t)$ satisfies:

$$\mathcal{P}^* \mathcal{Q} + \mathcal{Q} \mathcal{P} - \mathcal{Q} \mathcal{B} \gamma^{-1} \mathcal{B}^* \mathcal{Q} + \mathcal{I} = 0, \quad (1859)$$

where $\mathcal{Q}$ is the operator-valued value function Hessian and $\mathcal{P}^*$ denotes the adjoint operator in $\mathcal{H}$.

From a spectral perspective, stabilization corresponds to shifting dominant eigenvalues away from the critical manifold:

$$\lambda_{\max}(\mathcal{P}_{cl}) = \lambda_{\max}(\mathcal{P}) - \Delta_{\mathcal{B}\mathbf{u}}, \quad (1860)$$

where $\Delta_{\mathcal{B}\mathbf{u}}$ quantifies control-induced spectral displacement.

Importantly, the Pellis framework implies a fundamental limitation on controllability near criticality: as $\sigma(\mathcal{P}) \rightarrow \sigma_c$, the system becomes increasingly ill-conditioned with respect to control inputs, leading to diverging control energy requirements:

$$\|\mathbf{u}^*(t)\| \rightarrow \infty \quad \text{as } t \rightarrow t_c^-. \quad (1861)$$

Thus, climate intervention is recast as a constrained spectral engineering problem with inherent near-critical control degeneracy. This yields a climate analogue of infinite-dimensional Riccati stabilization theory, where feasibility is governed by spectral geometry rather than purely state-space considerations [2, 17, 31].

26.4 Regime Prediction and Scenario Simulation

Within the Pellis framework, scenario forecasting is fundamentally an operator propagation problem in which future climate states are generated by the action of the exponential flow induced by the governing operator $\mathcal{P}(t)$. In the autonomous approximation (or frozen-time linearization), the forward evolution over a horizon τ is expressed as:

$$\mathbf{X}(t + \tau) = e^{\tau \mathcal{P}} \mathbf{X}(t), \quad (1862)$$

where $e^{\tau \mathcal{P}}$ denotes the strongly continuous semigroup generated by $\mathcal{P}$ on the Hilbert space $\mathcal{H}$.

This representation admits the standard Dyson–Taylor expansion:

$$e^{\tau \mathcal{P}} = \sum_{n=0}^{\infty} \frac{\tau^n}{n!} \mathcal{P}^n, \quad (1863)$$

which becomes asymptotically valid under appropriate boundedness or sectoriality conditions on $\mathcal{P}$. In the nonlinear regime, higher-order corrections can be interpreted as iterated commutator contributions arising from time-dependent spectral deformation:

$$e^{\int_t^{t+\tau} \mathcal{P}(s) ds} = e^{\tau \mathcal{P}(t)} + \mathcal{O}\left(\int_t^{t+\tau} [\mathcal{P}(s), \mathcal{P}(t)] ds\right). \quad (1864)$$

In spectral coordinates, scenario evolution decomposes into mode-wise propagation:

$$\mathbf{X}(t + \tau) = \int_{\sigma(\mathcal{P})} e^{\lambda \tau} a(\lambda, t) \phi_\lambda d\mu(\lambda), \quad (1865)$$

making explicit that predictive evolution is governed by the distribution of eigenvalues rather than by a single dominant growth rate.

A key implication of the Pellis framework is that scenario divergence is controlled by the stability of the spectral gap. Define the minimal real-part separation:

$$\Delta \lambda_{\min}(t) = \inf_{\lambda \in \sigma(\mathcal{P}(t))} |\operatorname{Re}(\lambda)|. \quad (1866)$$

Predictability requires uniform positivity of this gap; however, near critical transitions, spectral condensation drives:

$$\Delta \lambda_{\min}(t) \rightarrow 0. \quad (1867)$$

In this regime, the operator resolvent becomes ill-conditioned and small uncertainties in initial conditions or forcing projections induce exponentially diverging scenario ensembles:

$$\|\mathbf{X}^{(1)}(t + \tau) - \mathbf{X}^{(2)}(t + \tau)\| \sim e^{\tau/\Delta\lambda_{\min}(t)} \|\delta\mathbf{X}(t)\|. \quad (1868)$$

This leads to a loss of finite predictive horizon, in which the effective forecast horizon τ_{pred} scales as:

$$\tau_{pred}(t) \sim \frac{1}{\Delta\lambda_{\min}(t)}, \quad (1869)$$

and therefore collapses as the system approaches the critical manifold.

More generally, scenario uncertainty becomes intrinsically spectral rather than stochastic, with ensemble spread governed by the geometry of $\sigma(\mathcal{P}(t))$:

$$\text{Var}[\mathbf{X}(t + \tau)] \sim \int_{\sigma(\mathcal{P})} e^{2\text{Re}(\lambda)\tau} |a(\lambda, t)|^2 d\mu(\lambda). \quad (1870)$$

Thus, loss of predictability is not a failure of numerical methods but a structural consequence of spectral collapse, where diminishing stability gaps eliminate the separation between controllable and uncontrollable dynamical modes. This yields a fundamental limit on scenario simulation in which increasing model resolution cannot recover predictability once $\Delta\lambda_{\min}(t) \rightarrow 0$.

26.5 Application to Major Earth System Subcomponents

Within the Pellis framework, the global climate operator is naturally decomposed into a coupled block-operator structure acting on a product Hilbert space $\mathcal{H} = \mathcal{H}_{\text{atm}} \oplus \mathcal{H}_{\text{ocean}} \oplus \mathcal{H}_{\text{cryosphere}}$. Each diagonal block represents the intrinsic dynamics of a subsystem, while off-diagonal blocks encode cross-component energy, momentum, and entropy exchange channels. The full operator is therefore written as:

$$\mathcal{P} = \begin{pmatrix} \mathcal{P}_{\text{atm}} & \mathcal{C}_{12} & \mathcal{C}_{13} \\ \mathcal{C}_{21} & \mathcal{P}_{\text{ocean}} & \mathcal{C}_{23} \\ \mathcal{C}_{31} & \mathcal{C}_{32} & \mathcal{P}_{\text{cryosphere}} \end{pmatrix}, \quad (1871)$$

where $\mathcal{C}_{ij} : \mathcal{H}_j \rightarrow \mathcal{H}_i$ are bounded (potentially nonlinear or state-dependent) coupling operators representing inter-system feedback mechanisms.

In this representation, each subsystem possesses its own local spectrum $\sigma(\mathcal{P}_\alpha)$, but the global spectrum $\sigma(\mathcal{P})$ emerges from a nontrivial interaction between diagonal stability and off-diagonal coupling structure. In particular, the leading eigenvalues of the full system are not simply inherited from subsystem operators but are determined by a coupled spectral perturbation problem:

$$\sigma(\mathcal{P}) \neq \bigcup_{\alpha} \sigma(\mathcal{P}_\alpha), \quad (1872)$$

reflecting the fundamentally non-separable nature of Earth system dynamics.

To first order, coupling induces a shift in the dominant eigenvalue governed by resolvent-mediated interactions:

$$\Delta\lambda_{\max} \sim \sum_{i \neq j} \langle \phi_i, \mathcal{C}_{ij} \phi_j \rangle, \quad (1873)$$

where ϕ_i denote leading eigenmodes of the uncoupled subsystem operators. This expression highlights that instability growth is driven not only by internal dynamics but also by cross-mode resonance between subsystems.

More generally, the effective stability of the coupled system can be approximated via a Schur-complement reduction:

$$\mathcal{P}_{\text{eff}}^{(i)} = \mathcal{P}_i - \sum_{j \neq i} \mathcal{C}_{ij} (\mathcal{P}_j)^{-1} \mathcal{C}_{ji}, \quad (1874)$$

which shows explicitly how inter-component feedback modifies local stability margins.

Within this framework, instability arises when coupling-induced corrections overcome intrinsic damping, leading to amplification of the dominant spectral mode:

$$\|\mathcal{C}_{ij}\| \uparrow \Rightarrow \text{Re}(\lambda_{\max}(\mathcal{P})) \uparrow. \quad (1875)$$

A key implication is that tipping behavior in one subsystem cannot be treated in isolation, as coupling terms act as spectral accelerators that propagate instability across the full operator network. This leads to correlated

threshold behavior across atmospheric, oceanic, and cryospheric components, consistent with cascade-type regime shifts observed in coupled Earth system dynamics.

In particular, strong coupling regimes correspond to a synchronization of spectral instabilities:

$$\lambda_{\max}(\mathcal{P}_{\text{atm}}) \approx \lambda_{\max}(\mathcal{P}_{\text{ocean}}) \approx \lambda_{\max}(\mathcal{P}_{\text{cryosphere}}), \quad (1876)$$

indicating emergent coherence in the approach to global criticality.

This structure provides a direct mathematical link between AMOC collapse, ice-sheet instability, and monsoon transitions as manifestations of a unified spectral instability cascade in the coupled operator $\mathcal{P}$ [11–13, 15, 32, 35].

26.6 Policy-Relevant Interpretation of Tipping Risk

Within the Pellis framework, policy-relevant risk assessment is formulated as a cumulative hazard functional over the trajectory of spectral instability. Rather than treating tipping risk as an instantaneous probability, the system is modeled as a time-accumulating exposure to positive spectral instability, as encoded by the Computable Climate Stability Index (CSI). The tipping probability functional is defined as:

$$\mathbb{P}_{\text{tip}}(T) = 1 - \exp \left(- \int_0^T \max(0, \text{CSI}(t)) dt \right), \quad (1877)$$

where the integrand acts as an effective hazard rate derived from the positive part of the spectral instability flux.

This formulation embeds tipping risk within a nonhomogeneous Poisson-like survival structure, where the instantaneous hazard function is given by:

$$h(t) = \max(0, \text{CSI}(t)). \quad (1878)$$

Accordingly, the survival probability of remaining within the stable basin up to time T is:

$$S(T) = \exp \left(- \int_0^T h(t) dt \right), \quad (1879)$$

and the tipping probability is $\mathbb{P}_{\text{tip}}(T) = 1 - S(T)$.

In spectral terms, this hazard structure corresponds to the cumulative leakage of spectral mass into the unstable half-plane:

$$h(t) \sim \int_{\text{Re}(\lambda) > 0} \text{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1880)$$

linking probabilistic risk directly to the first positive spectral moment of the Pellis operator.

A key implication is that tipping risk is path-dependent and accelerative: once $\text{CSI}(t)$ becomes positive, the hazard functional enters a self-reinforcing regime in which spectral drift amplifies future instability accumulation:

$$\frac{d}{dt} \mathbb{P}_{\text{tip}}(t) = (1 - \mathbb{P}_{\text{tip}}(t)) \cdot \max(0, \text{CSI}(t)). \quad (1881)$$

This nonlinear feedback structure implies that marginal increases in spectral instability have disproportionate effects on long-horizon tipping probabilities, particularly near the critical manifold where $\text{CSI}(t) \approx 0^+$. In this regime, risk becomes dominated by rare spectral excursions rather than mean dynamics.

From a policy perspective, the formulation provides a continuously updated, data-assimilative risk metric in which mitigation strategies correspond to interventions that enforce $\text{CSI}(t) < 0$ over sustained temporal intervals, thereby suppressing the integrated hazard functional and restoring exponential stability of the survival probability.

26.7 Summary of Practical Consequences

The Universal Pellis Function Theory implies a fundamental reformulation of climate science in which predictability, controllability, and risk quantification are no longer governed primarily by perturbative expansions around equilibria, but instead by the global spectral geometry of the governing operator field $\mathcal{P}(t)$. In this view, the Earth system is best understood as a continuously deforming spectral manifold whose stability properties are encoded in the distribution, drift, and condensation of eigenvalues rather than in local state-space trajectories alone.

Consequently, classical approaches based on linear response theory, finite-dimensional closure assumptions, or weakly nonlinear approximations are recovered only as low-order projections of a more general operator-theoretic structure. The true organizing principle is the evolution of the spectral measure:

$$\mu_t(\lambda) = |a(\lambda, t)|^2 d\mu(\lambda), \quad (1882)$$

whose redistribution across the stability boundary $\text{Re}(\lambda) = 0$ determines emergent regime transitions.

Within this framework, climate predictability is explicitly tied to spectral gap persistence:

$$\tau_{\text{pred}}(t) \sim \frac{1}{\Delta\lambda_{\min}(t)}, \quad (1883)$$

while loss of predictability corresponds to spectral collapse, i.e. $\Delta\lambda_{\min}(t) \rightarrow 0$. Control feasibility, in turn, is governed by operator conditioning, with intervention cost diverging as the system approaches criticality. Risk is unified through integrated spectral hazard functionals such as $\text{CSI}(t)$ and $\mathbb{P}_{\text{tip}}(T)$, which encode instability accumulation over time rather than instantaneous deviations.

This synthesis reframes climate dynamics as a problem of nonlinear spectral stability theory in infinite-dimensional spaces, where tipping points correspond to functional phase transitions in the operator spectrum and emergent universality classes arise from shared spectral collapse mechanisms across subsystems.

In summary, the framework replaces traditional reductionist climate modeling paradigms with a unified operator-theoretic structure in which all major Earth system behaviors—prediction, instability, synchronization, and control—are manifestations of evolving spectral geometry in a high-dimensional dynamical system [1, 4, 6, 18, 23, 33, 37, 139].

27 Limitations and Model Assumptions

While the Universal Pellis Function Theory of Climate Criticality provides a unified spectral–functional framework for describing tipping points and planetary instability, its mathematical structure is necessarily built upon a hierarchy of idealizations. These assumptions enable a tractable operator-theoretic representation of the Earth system, but they simultaneously introduce constraints on direct empirical identifiability, spectral reconstruction stability, and rigorous inversion in strongly nonlinear regimes [1, 2, 16–18, 23, 33].

27.1 Hilbert Space Idealization of the Climate State

The theory assumes that the full climate state can be embedded in a separable Hilbert space $\mathcal{H}$ equipped with inner product structure:

$$\mathbf{X}(t) \in \mathcal{H}, \quad \langle \mathbf{X}_1, \mathbf{X}_2 \rangle_{\mathcal{H}} < \infty. \quad (1884)$$

This assumption encodes square-integrability and finite-energy constraints across all resolved climate fields. However, real Earth system observables may exhibit singular components, discontinuities, and measure-valued contributions that violate Hilbert regularity. In such cases, the appropriate state space may extend to a distributional or Banach setting:

$$\mathbf{X}(t) \in \mathcal{D}'(\Omega) \quad \text{or more generally} \quad \mathbf{X}(t) \in \mathcal{B}(\Omega), \quad (1885)$$

implying that the spectral decomposition of $\mathcal{P}$ must be reformulated in terms of generalized eigenfunctionals rather than classical eigenmodes.

Consequently, the spectral measure $\mu(\lambda)$ may fail to admit a purely absolutely continuous representation:

$$\mu(\lambda) = \mu_{ac}(\lambda) + \mu_{sing}(\lambda), \quad (1886)$$

introducing singular spectral components associated with abrupt climate transitions and non-smooth dynamical events.

27.2 Smoothness and Differentiability of the Pellis Operator

The governing evolution equation assumes a sufficiently smooth operator field:

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad (1887)$$

with $\mathcal{P}$ typically assumed to be at least Fréchet differentiable on $\mathcal{H}$:

$$\mathcal{P} \in C^1(\mathcal{H}). \quad (1888)$$

However, near tipping thresholds, climate dynamics often exhibit nonsmooth bifurcations, discontinuous regime switches, and set-valued evolution. In such cases, the Gâteaux derivative may fail to exist:

$$\lim_{\epsilon \rightarrow 0} \frac{\mathcal{P}(X + \epsilon Y) - \mathcal{P}(X)}{\epsilon} \not\exists, \quad (1889)$$

implying that classical linearization and local spectral perturbation theory lose validity.

This breakdown introduces the need for generalized differential inclusions:

$$\frac{d}{dt}\mathbf{X}(t) \in \mathcal{P}(\mathbf{X}(t)), \quad (1890)$$

which replace deterministic flows with multivalued dynamical evolutions in non-smooth regimes.

27.3 Spectral Completeness Assumption

The spectral representation of the state is assumed to take the form:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_\lambda d\mu(\lambda), \quad (1891)$$

requiring completeness of generalized eigenfunctions and the existence of a stable spectral resolution.

In strongly non-normal or nonlinear regimes, this assumption may fail due to spectral instability and pseudospectral inflation. In particular, the operator spectrum may not decompose cleanly into point and continuous parts:

$$\sigma(\mathcal{P}) \neq \sigma_p(\mathcal{P}) \cup \sigma_c(\mathcal{P}), \quad (1892)$$

and transient growth may dominate asymptotic eigenvalue structure.

In such regimes, pseudospectral sets become dynamically relevant:

$$\sigma_\epsilon(\mathcal{P}) = \{\lambda \in \mathbb{C} : \|(\mathcal{P} - \lambda I)^{-1}\| > \epsilon^{-1}\}, \quad (1893)$$

implying that apparent stability based on eigenvalues alone may be misleading.

27.4 Closure Problem in Multiscale Coupling

The decomposition

$$\mathcal{P} = \mathcal{A} + \mathcal{N} + \mathcal{S} \quad (1894)$$

implicitly assumes closure of unresolved subgrid processes and multiscale feedbacks.

However, scale interactions generate an infinite renormalization hierarchy:

$$\mathcal{S}_{n+1} = \mathcal{F}(\mathcal{S}_n), \quad (1895)$$

implying that no finite truncation is exact. The effective operator at resolution level N therefore satisfies:

$$\mathcal{P}^{(N)} = \mathcal{P} - \mathcal{R}^{(N)}, \quad (1896)$$

where $\mathcal{R}^{(N)}$ represents unresolved renormalization residue terms.

Exact recovery of the full operator requires the limit:

$$\lim_{N \rightarrow \infty} \|\mathcal{R}^{(N)}\| = 0, \quad (1897)$$

which is generally unattainable in practical computational frameworks.

27.5 Assumption of Stationary Spectral Structure

Many derivations rely on quasi-stationary spectral structure:

$$\frac{d}{dt} \sigma(\mathcal{P}(t)) \approx 0, \quad (1898)$$

allowing eigenvalues and spectral measures to be treated as slowly varying.

In strongly forced regimes, however, the operator becomes explicitly time-dependent:

$$\mathcal{P} = \mathcal{P}(t), \quad (1899)$$

leading to non-autonomous spectral transport governed by:

$$\frac{d\lambda(t)}{dt} = \langle \phi_\lambda(t), \dot{\mathcal{P}}(t) \phi_\lambda(t) \rangle, \quad (1900)$$

which invalidates stationary bifurcation analysis and introduces history-dependent spectral drift.

27.6 Neglect of External Stochastic Forcing Structure

In its deterministic formulation, the Pellis operator evolution excludes stochastic forcing:

$$\frac{d}{dt} \mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)). \quad (1901)$$

A more realistic formulation includes multiplicative and additive noise:

$$d\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t))dt + \Sigma(\mathbf{X}(t))dW_t, \quad (1902)$$

where W_t is a cylindrical Wiener process on $\mathcal{H}$.

In this stochastic setting, tipping becomes noise-assisted and threshold behavior is modified via stochastic escape dynamics:

$$\mathbb{P}(\text{escape}) \sim \exp\left(-\frac{\Delta V}{\sigma^2}\right), \quad (1903)$$

where ΔV denotes an effective spectral potential barrier.

27.7 Parameter Identifiability and Observability Constraints

The inverse mapping from observations to operator structure:

$$\mathbf{X}(t) \mapsto \mathcal{P} \quad (1904)$$

is generally non-injective and ill-posed.

Distinct operators may generate observationally indistinguishable dynamics under finite measurement resolution:

$$\mathcal{P}_1 \neq \mathcal{P}_2 \quad \text{but} \quad \mathcal{O}(\mathcal{P}_1) \approx \mathcal{O}(\mathcal{P}_2), \quad (1905)$$

implying structural non-identifiability and inverse spectral ambiguity.

This leads to a fundamental identifiability constraint governed by an information-theoretic bound:

$$I(\mathcal{P}; \mathbf{X}_{obs}) \leq H(\mathbf{X}_{obs}), \quad (1906)$$

where I denotes mutual information and H observational entropy.

27.8 Summary of Theoretical Constraints

Overall, the Pellis framework should be interpreted as a universal organizing principle for spectral climate dynamics rather than a fully closed predictive simulator. Its validity is strongest in regimes where:

$$\|\mathcal{N}\| \sim \|\mathcal{A}\|, \quad \text{and} \quad \|\mathcal{S}\| \ll \|\mathcal{A}\|, \quad (1907)$$

corresponding to near-critical but still resolvable dynamical balance between linear instability and nonlinear saturation. Beyond these regimes—particularly in strongly stochastic, discontinuous, or deeply non-normal settings—the theory must be augmented by stochastic analysis, nonsmooth dynamical systems theory, and measure-theoretic operator extensions to preserve predictive consistency [1, 4, 6, 18, 23, 33, 37, 139].

28 Future Work

The Universal Pellis Function Theory of Climate Criticality establishes a spectral-functional foundation for understanding planetary instability, yet its full mathematical, computational, and empirical completion remains an open research program. Future developments must systematically extend the operator framework beyond idealized regimes by incorporating stochasticity, strong nonlinearity, adaptive coupling, and data-driven reconstruction of spectral structure in high-dimensional Earth system dynamics [1, 2, 6, 18, 23, 33, 37, 139].

28.1 Rigorous Functional-Analytic Extension of the Pellis Operator

A first foundational direction is the embedding of the Pellis operator into a rigorously defined functional-analytic class. In full generality, one seeks:

$$\mathcal{P} : \mathcal{D}(\mathcal{P}) \subset \mathcal{X} \rightarrow \mathcal{X}, \quad (1908)$$

where $\mathcal{X}$ extends beyond Hilbert structure to Banach, Sobolev, or measure-valued state spaces depending on the regularity class of climate observables.

A central open problem is the characterization of $\mathcal{P}$ as a generator of a strongly continuous semigroup:

$$\mathbf{X}(t) = e^{t\mathcal{P}} \mathbf{X}_0, \quad (1909)$$

together with the infinitesimal generator identity:

$$\lim_{t \rightarrow 0^+} \frac{e^{t\mathcal{P}} - I}{t} = \mathcal{P}. \quad (1910)$$

Future work must further establish dissipativity or sectoriality conditions ensuring well-posedness:

$$\langle \mathcal{P}X, X \rangle_{\mathcal{X}} \leq -c\|X\|^2 + C, \quad (1911)$$

which determine whether the global flow admits attractor structure or finite-time blow-up.

28.2 Nonlinear Spectral Theory and Generalized Eigenstructures

A key extension concerns nonlinear spectral theory, where the operator is no longer linearizable globally:

$$\mathcal{P}(\mathbf{X}) \neq \mathcal{L}\mathbf{X}. \quad (1912)$$

Future work must define generalized nonlinear eigenmodes satisfying a residual-balanced eigenvalue relation:

$$\mathcal{P}(\phi_\lambda) = \lambda\phi_\lambda + \mathcal{R}(\phi_\lambda), \quad (1913)$$

where $\mathcal{R}$ encodes curvature of the dynamical flow in function space.

A central conjecture is the existence of a nonlinear spectral invariant manifold:

$$\mathcal{M}_\sigma = \{\phi \in \mathcal{X} : \mathcal{P}(\phi) \parallel \phi\}, \quad (1914)$$

which generalizes classical eigenspaces into invariant geometric structures governing tipping dynamics.

On this manifold, instability is expected to be governed by curvature-induced spectral drift:

$$\frac{d\lambda}{dt} = \mathcal{K}(\phi_\lambda, \nabla \mathcal{P}), \quad (1915)$$

linking spectral evolution directly to nonlinear geometry of the flow.

28.3 Data-Driven Identification of the Pellis Operator

A major challenge is the inverse problem of reconstructing $\mathcal{P}$ from discrete observational trajectories:

$$\{\mathbf{X}(t_k)\}_{k=1}^N \longrightarrow \mathcal{P}. \quad (1916)$$

This can be formulated as a constrained variational identification problem:

$$\min_{\mathcal{P}} \sum_{k=1}^{N-1} \left\| \frac{\mathbf{X}(t_{k+1}) - \mathbf{X}(t_k)}{\Delta t} - \mathcal{P}(\mathbf{X}(t_k)) \right\|_{\mathcal{X}}^2. \quad (1917)$$

Future approaches should incorporate operator learning frameworks with spectral regularization:

$$\min_{\mathcal{P}} \mathcal{L}_{data} + \alpha \|\sigma(\mathcal{P})\|_{TV} + \beta \|\mathcal{P}\|_{op}, \quad (1918)$$

where total variation of the spectrum enforces stability-consistent reconstruction.

Neural operator embeddings may approximate $\mathcal{P}$ via parameterized maps:

$$\mathcal{P}_\theta \approx \mathcal{P}, \quad \theta \in \mathbb{R}^d, \quad (1919)$$

enabling scalable inference in high-dimensional climate systems [20, 21, 28, 29].

28.4 Stochastic Generalization of the Theory

A fundamental extension is the stochastic Pellis evolution equation:

$$d\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t))dt + \mathcal{G}(\mathbf{X}(t))dW_t, \quad (1920)$$

where $\mathcal{G}$ encodes multiplicative noise coupling and W_t is a cylindrical Wiener process.

Stochastic stability must then be characterized in terms of moment bounds:

$$\sup_{t \geq 0} \mathbb{E} \|\mathbf{X}(t)\|^2 < \infty, \quad (1921)$$

together with almost-sure boundedness conditions.

A stochastic tipping criterion may be expressed via an effective random spectrum:

$$\lambda_{\text{eff}}(t, \omega) \in \sigma(\mathcal{P} + \mathcal{G}\dot{W}_t), \quad (1922)$$

with probabilistic instability condition:

$$\mathbb{P}(\text{Re}(\lambda_{\text{eff}}) > 0) > \epsilon. \quad (1923)$$

28.5 Multiscale Renormalization Closure Problem

The renormalization structure of the Pellis operator suggests a hierarchical flow in operator space:

$$\mathcal{R}\mathcal{P} = \gamma\mathcal{P}, \quad (1924)$$

where $\mathcal{R}$ is a coarse-graining transformation on spectral operators.

This induces a renormalization recursion:

$$\mathcal{P}_{n+1} = \mathcal{F}(\mathcal{P}_n), \quad (1925)$$

whose fixed points define universality classes of climate dynamics.

Future work must characterize fixed-point stability:

$$\mathcal{P}^* = \mathcal{F}(\mathcal{P}^*), \quad (1926)$$

with linearized flow stability determined by:

$$\rho(D\mathcal{F}(\mathcal{P}^*)) < 1, \quad (1927)$$

where ρ denotes spectral radius in operator space.

28.6 Coupled Earth System Subnetwork Dynamics

The Earth system is more accurately represented as a time-evolving operator network:

$$\mathcal{P} = \begin{pmatrix} \mathcal{P}_1 & \mathcal{C}_{12} & \cdots \\ \mathcal{C}_{21} & \mathcal{P}_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \quad (1928)$$

where coupling operators depend on both state and time:

$$\mathcal{C}_{ij} = \mathcal{C}_{ij}(t, \mathbf{X}(t)). \quad (1929)$$

This yields adaptive network dynamics governed by coupled operator flows:

$$\frac{d}{dt}\mathcal{C}_{ij} = \mathcal{F}_{ij}(\mathcal{C}_{ij}, \mathbf{X}). \quad (1930)$$

Such co-evolution produces feedback-induced spectral synchronization:

$$\lambda_{\max}(\mathcal{P}_i) \approx \lambda_{\max}(\mathcal{P}_j), \quad (1931)$$

driving coordinated tipping cascades across subsystems.

28.7 Experimental and Observational Roadmap

Future validation requires integrated multi-platform data assimilation:

$$\mathbf{X}(t) = \mathcal{A}_{\text{sat}}(t) + \mathcal{A}_{\text{ocean}}(t) + \mathcal{A}_{\text{ice}}(t) + \mathcal{A}_{\text{atm}}(t). \quad (1932)$$

A key empirical objective is the detection of early spectral drift signatures:

$$\frac{d}{dt}\sigma(\mathcal{P}(t)) \neq 0, \quad (1933)$$

serving as a precursor to global instability transitions.

More refined observables include time-resolved spectral density evolution:

$$\rho(\lambda, t) \rightarrow \rho(\lambda, t + \delta t), \quad (1934)$$

enabling real-time monitoring of instability accumulation in Earth system modes.

28.8 Final Perspective

Ultimately, the future of the Universal Pellis Function Theory lies in the synthesis of spectral geometry, stochastic dynamical systems, multiscale renormalization, and data-driven operator inference into a unified predictive framework. This can be expressed compactly as:

$$\text{Climate System} \equiv (\mathcal{H}, \mathcal{P}, \sigma(\mathcal{P}), \mathcal{R}, \mathcal{G}), \quad (1935)$$

where each component encodes respectively the state space, governing operator, spectral structure, renormalization flow, and stochastic forcing architecture [1, 4, 6, 18, 23, 33, 37, 139].

29 Discussion

The Universal Pellis Function Theory of Climate Criticality proposes a unified operator-theoretic perspective in which climate dynamics are governed not merely by differential equations on physical fields, but by the evolving spectral geometry of a nonlinear operator field $\mathcal{P}$. Within this formulation, the Earth system is elevated from a trajectory-based dynamical system to a spectral object whose stability is encoded in the structure, deformation, and collapse of $\sigma(\mathcal{P})$. This provides a natural extension of classical dynamical systems theory, bifurcation theory, and statistical physics into a single functional-analytic framework [1, 2, 4, 6, 18, 22, 23, 33, 37, 139].

29.1 Conceptual Shift from State-Space to Operator-Space Dynamics

Traditional Earth system modeling is formulated in state space:

$$\frac{d}{dt}\mathbf{X}(t) = F(\mathbf{X}(t)), \quad (1936)$$

where F is a nonlinear vector field on a finite- or infinite-dimensional phase space.

The Pellis formulation replaces this structure with an operator-valued flow:

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad (1937)$$

where $\mathcal{P}$ acts as a nonlinear evolution generator whose properties depend on the instantaneous state of the system.

The fundamental object of analysis becomes the spectral measure:

$$\sigma(\mathcal{P}) = \{\lambda \in \mathbb{C} : \exists \phi \neq 0, \mathcal{P}\phi = \lambda\phi\}, \quad (1938)$$

with climate evolution interpreted as a continuous deformation of this spectral set:

$$t \mapsto \sigma(\mathcal{P}(t)). \quad (1939)$$

In this sense, instability is no longer a trajectory divergence in phase space but a geometric instability of spectral support in the complex plane.

29.2 Unification of Bifurcation Theory and Spectral Instability

Classical bifurcation theory characterizes transitions through parameter-dependent equilibria:

$$F(\mathbf{X}; \mu) = 0. \quad (1940)$$

In the Pellis framework, bifurcations are equivalently characterized by spectral crossings:

$$\operatorname{Re}(\lambda(\mu)) = 0, \quad \lambda(\mu) \in \sigma(\mathcal{P}_\mu). \quad (1941)$$

This defines a critical manifold in operator space:

$$\mathcal{T} = \{\mu \in \mathbb{R} : \exists \lambda \in \sigma(\mathcal{P}_\mu), \operatorname{Re}(\lambda) = 0\}. \quad (1942)$$

Higher-order degeneracies correspond to spectral coalescence:

$$\lambda_i(\mu_c) = \lambda_j(\mu_c), \quad \nabla_\mu \lambda_i(\mu_c) \approx 0, \quad (1943)$$

which generalizes classical saddle-node, Hopf, and transcritical bifurcations into a unified spectral geometry classification.

29.3 Interpretation of Climate Tipping Points as Functional Phase Transitions

A central conceptual result is the identification of tipping points with non-analytic transformations of spectral measures:

$$\frac{d}{d\mu}\rho(\lambda; \mu) \notin C^0, \quad (1944)$$

where $\rho(\lambda; \mu)$ denotes the spectral density function.

This structure is directly analogous to thermodynamic phase transitions, where non-analyticity emerges in free energy functionals:

$$\mathcal{F}(\mu) = -\frac{1}{\beta} \log \operatorname{Tr} \left(e^{-\beta \mathcal{P}_\mu} \right). \quad (1945)$$

Criticality is therefore associated with divergence of response functionals:

$$\chi(\mu) = \frac{d^2 \mathcal{F}}{d\mu^2} \rightarrow \infty, \quad (1946)$$

establishing a direct correspondence between spectral instability and macroscopic susceptibility blow-up.

29.4 Relation to Renormalization and Universality

The renormalization structure of the Pellis operator is expressed as:

$$\mathcal{R}\mathcal{P} = \gamma\mathcal{P}, \quad (1947)$$

suggesting scale invariance under coarse-graining transformations in operator space.

This induces a renormalization flow:

$$\mathcal{P}_{n+1} = \mathcal{F}(\mathcal{P}_n), \quad (1948)$$

with fixed points satisfying:

$$\mathcal{P}^* = \mathcal{F}(\mathcal{P}^*). \quad (1949)$$

Linear stability of these fixed points is governed by:

$$\rho(D\mathcal{F}(\mathcal{P}^*)) < 1, \quad (1950)$$

defining universality classes of climate dynamics characterized by invariant spectral scaling exponents.

29.5 Interplay Between Determinism and Effective Stochasticity

Although the fundamental formulation is deterministic:

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad (1951)$$

effective stochasticity emerges through unresolved multiscale interactions.

This leads to a decomposition:

$$\mathcal{P} = \mathcal{P}_{\text{resolved}} + \eta_{\text{unresolved}}, \quad (1952)$$

where $\eta_{\text{unresolved}}$ acts as an induced stochastic forcing term.

In the continuum limit, this yields a stochastic evolution equation:

$$d\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t))dt + \Sigma(\mathbf{X}(t))dW_t, \quad (1953)$$

linking deterministic chaos with stochastic climate variability.

Noise-induced spectral deformation further modifies stability thresholds:

$$\text{Re}(\lambda_{\text{eff}}) = \text{Re}(\lambda) + \delta_{\text{noise}}(\lambda), \quad (1954)$$

implying that stochastic forcing can either stabilize or destabilize near-critical regimes.

29.6 Implications for Predictability Horizons

Predictability is fundamentally constrained by the spectral gap structure:

$$\tau_{\text{pred}} \sim \frac{1}{\min_{\lambda \in \sigma(\mathcal{P})} |\text{Re}(\lambda)|}. \quad (1955)$$

As the spectral gap collapses:

$$\min |\text{Re}(\lambda)| \rightarrow 0, \quad (1956)$$

the predictability horizon diverges:

$$\tau_{\text{pred}} \rightarrow \infty, \quad (1957)$$

which in practice corresponds to catastrophic loss of forecast stability due to critical slowing down and spectral condensation.

More generally, predictability becomes a functional of spectral distribution:

$$\tau_{\text{pred}} = \int_{\sigma(\mathcal{P})} \frac{|a(\lambda)|^2}{|\text{Re}(\lambda)|} d\mu(\lambda), \quad (1958)$$

linking forecast horizon directly to operator spectral geometry.

29.7 Synthesis and Theoretical Positioning

The Universal Pellis Function Theory thus occupies a unifying position between nonlinear dynamics, statistical physics, and operator theory. Its central hypothesis can be summarized as:

$$\text{Climate Stability} \equiv \text{Spectral Stability of } \mathcal{P}. \quad (1959)$$

Within this framework, tipping points are not rare anomalies but structurally inevitable consequences of continuous spectral deformation under external forcing:

$$\frac{d}{dt}\mathcal{P}(t) \neq 0 \Rightarrow \exists t_c : \text{Re}(\lambda(t_c)) = 0. \quad (1960)$$

This leads to a unified interpretation of cascading Earth system transitions as synchronized spectral reorganizations across coupled operator subsystems:

$$\mathcal{P} = \begin{pmatrix} \mathcal{P}_{\text{atm}} & \mathcal{C}_{12} \\ \mathcal{C}_{21} & \mathcal{P}_{\text{ocean}} \end{pmatrix}, \quad \lambda_{\text{max}}(\mathcal{P}) \uparrow \Rightarrow \text{coordinated instability}. \quad (1961)$$

In this sense, the theory provides a mathematically coherent and empirically testable framework for interpreting cascading climate transitions as manifestations of a global spectral phase reorganization in a nonlinear operator field governing Earth system dynamics [4, 6, 11, 12, 33, 37, 139].

30 Conclusions

The Universal Pellis Function Theory of Climate Criticality establishes a unified mathematical framework in which climate dynamics are reformulated as the spectral evolution of a nonlinear operator $\mathcal{P}$ acting on a high-dimensional functional state space. In this formulation, the Earth system is no longer interpreted as a trajectory in phase space, but as a continuously deforming spectral object whose stability is encoded in the geometry, density, and transport properties of $\sigma(\mathcal{P})$. This reframing places climate science within a rigorous operator-theoretic paradigm connecting dynamical systems, functional analysis, and statistical physics [1, 2, 4, 6, 18, 23, 33, 37, 139].

30.1 Summary of the Operator-Theoretic Framework

At the core of the theory is the nonlinear evolution equation:

$$\frac{d}{dt}\mathbf{X}(t) = \mathcal{P}(\mathbf{X}(t)), \quad (1962)$$

which generalizes classical dynamical systems by embedding multiscale feedbacks, renormalization effects, and non-local interactions into a single operator-valued generator.

The emergent state admits a spectral representation:

$$\mathbf{X}(t) = \int_{\sigma(\mathcal{P})} a(\lambda, t) \phi_\lambda d\mu(\lambda), \quad (1963)$$

where the amplitudes $a(\lambda, t)$ evolve according to:

$$\frac{\partial a(\lambda, t)}{\partial t} = \lambda a(\lambda, t) + \mathcal{N}_\lambda[a], \quad (1964)$$

with $\mathcal{N}_\lambda$ encoding nonlinear spectral coupling between modes.

The macroscopic climate state is thus interpreted as a measure-valued projection of a continuously evolving spectral field:

$$\mu_t(\lambda) = |a(\lambda, t)|^2 d\mu(\lambda). \quad (1965)$$

30.2 Climate Criticality as Spectral Instability

A central result is that climate tipping points correspond to spectral crossings of the operator $\mathcal{P}$:

$$\operatorname{Re}(\lambda) = 0, \quad (1966)$$

which defines a codimension-one critical manifold separating dissipative and amplifying regimes.

The global instability functional is given by:

$$\mathcal{I}[\mathbf{X}] = \int_{\sigma(\mathcal{P})} \operatorname{Re}(\lambda) |a(\lambda, t)|^2 d\mu(\lambda), \quad (1967)$$

which acts as a Lyapunov-type diagnostic for global stability loss.

Near criticality, spectral condensation occurs:

$$\mu_t(\lambda) \rightharpoonup \delta(\lambda - \lambda_c), \quad (1968)$$

indicating collapse of distributed modes into near-neutral eigenstructures.

30.3 Universality and Renormalization Structure

The renormalization invariance condition:

$$\mathcal{R}\mathcal{P} = \gamma\mathcal{P} \quad (1969)$$

implies that climate dynamics belong to universality classes characterized by invariant spectral scaling laws.

The associated renormalization flow is:

$$\frac{d\mathcal{P}}{ds} = \mathcal{F}(\mathcal{P}), \quad (1970)$$

with fixed points satisfying:

$$\mathcal{P}^* = \mathcal{R}\mathcal{P}^*. \quad (1971)$$

Linearization around fixed points yields stability governed by spectral radius:

$$\rho(D\mathcal{R}(\mathcal{P}^*)) < 1, \quad (1972)$$

defining attractor-like universality classes of climate behavior.

30.4 Predictability and Breakdown of Forecast Horizon

Predictability is fundamentally constrained by the spectral gap structure:

$$\tau_{\text{pred}} \sim \frac{1}{\min_{\lambda \in \sigma(\mathcal{P})} |\text{Re}(\lambda)|}. \quad (1973)$$

As the system approaches criticality:

$$\min |\text{Re}(\lambda)| \rightarrow 0, \quad (1974)$$

the predictability horizon diverges:

$$\tau_{\text{pred}} \rightarrow \infty, \quad (1975)$$

signaling loss of meaningful long-term forecast stability due to spectral gap collapse.

More generally, predictability becomes a spectral functional:

$$\tau_{\text{pred}} = \int_{\sigma(\mathcal{P})} \frac{|a(\lambda, t)|^2}{|\text{Re}(\lambda)| + \epsilon} d\mu(\lambda), \quad (1976)$$

emphasizing the role of distributed near-neutral modes in limiting forecast skill.

30.5 Interpretation of Planetary Instability

Planetary instability emerges as a global reorganization of spectral support:

$$\sigma(\mathcal{P}) \longrightarrow \sigma'(\mathcal{P}), \quad (1977)$$

where instability is driven by redistribution of eigenvalue density toward the critical manifold.

In coupled Earth system representations, the operator takes block form:

$$\mathcal{P} = \begin{pmatrix} \mathcal{P}_{\text{atm}} & \mathcal{C}_{12} \\ \mathcal{C}_{21} & \mathcal{P}_{\text{ocean}} \end{pmatrix}, \quad (1978)$$

with instability emerging when coupling amplification induces spectral alignment:

$$\|\mathcal{C}_{ij}\| \uparrow \Rightarrow \lambda_{\text{max}}(\mathcal{P}) \rightarrow 0^+. \quad (1979)$$

This mechanism provides a mathematical explanation for cascading transitions across subsystems, where multiple tipping elements synchronize via shared spectral drift.

30.6 Final Synthesis

The theory culminates in the fundamental identification:

$$\boxed{\text{Climate Stability} \equiv \text{Spectral Stability of } \mathcal{P}} \quad (1980)$$

Under this principle, tipping points, regime shifts, and cascading transitions are not exceptional anomalies, but structurally inevitable consequences of nonlinear spectral deformation in high-dimensional operator fields.

Thus, the Universal Pellis Function Theory provides a mathematically coherent and scalable framework in which bifurcation theory, renormalization group methods, and operator spectral analysis are unified into a single predictive paradigm for planetary-scale criticality [4, 6, 11, 18, 23, 33, 37, 139].

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